



VANDERBILT UNIVERSITY
2301 VANDERBILT PL NASHVILLE, TN 37235



EXTRATERRESTRIAL LANDER FOR DATA
COLLECTION AND RF TRANSMISSION

CRITICAL DESIGN REVIEW

NASA STUDENT LAUNCH COMPETITION
2024 - 2025

JANUARY 8, 2025

Table of Contents

List of Figures	4
List of Tables	10
List of Acronyms	12
1 Summary	13
1.1 Team Summary	13
1.2 Launch Vehicle Summary	13
1.3 Payload Summary	13
2 Changes Since PDR	14
2.1 Vehicle Criteria Changes	14
2.2 Payload Criteria Changes	15
2.3 Project Plan Changes	15
3 Vehicle Criteria	16
3.1 Design and Verification of Launch Vehicle	16
3.1.1 Mission Statement and Success Criteria	16
3.1.2 Launch Vehicle Design Overview	17
3.1.3 Nose Cone	19
3.1.4 Payload Bay	22
3.1.5 Lander Separation Bay	26
3.1.6 Drogue Bay	28
3.1.7 Avionics Bay	29
3.1.8 Main Bay	32
3.1.9 Tail Section	34
3.1.10 Airframe Material Properties	39
3.2 Subscale Flight Results	42
3.2.1 Flight Objectives	42
3.2.2 Vehicle Scaling	43
3.2.3 Vehicle Fabrication	45
3.2.4 Flight Results	50
3.2.5 Flight Analysis	54
3.2.6 Implications for Full-scale Vehicle Design	66
3.3 Recovery Subsystem	67
3.3.1 Recovery System CONOPS	67
3.3.2 Recovery Hardware	68
3.3.3 Shock Cord Selection	70
3.3.4 Avionics	73
3.3.5 Blast Charges	75
3.3.6 GPS Tracker	77
3.4 Mission Performance Predictions	77

3.4.1	Flight Profile Simulations and Relevant Requirements	78
3.4.2	Numerical Integration Methodology	85
3.4.3	Parachute and Descent Dynamics	86
3.4.4	Launch Rail Dynamics	87
3.4.5	Environmental Modeling	88
3.4.6	Vehicle Verification	91
3.4.7	Vehicle Ascent Methodology	93
4	Payload Criteria	97
4.1	Design and Verification of Payload Experiment	97
4.1.1	Payload Mission CONOPS	98
4.1.2	Payload Design Overview	98
4.1.3	Payload Verification Overview	101
4.2	Mechanical Subsystems	106
4.2.1	Lander Separation Mechanism	106
4.2.2	Leg Deployment Mechanism	111
4.2.3	Parachute Detachment Mechanism	117
4.2.4	Antenna Extension Mechanism	121
4.2.5	STEMnaut Crew Module (SCM)	126
4.3	Electrical Subsystems	127
4.3.1	System Overview	128
4.3.2	Software Architecture	129
4.3.3	Redundancy in Landing Detection Strategy	130
4.3.4	Velocity Estimation	131
4.3.5	RF Transmission	133
4.3.6	Prototype Board and Power System	135
4.3.7	GPS Selection	137
4.3.8	Failure Case Study and Plans	137
4.3.9	Custom PCB Design	139
5	Safety	141
5.1	Safety Overview	141
5.1.1	VADL Safety Structure	141
5.1.2	Safety Personnel	142
5.2	Safety Matrix Design	142
5.2.1	Severity Classifications	143
5.2.2	Probability Levels	143
5.2.3	Risk Assessment Matrix	144
5.2.4	Hazard and Failure Mode Key	145
5.3	Personnel Hazard Analysis	147
5.4	Analysis of Failure Modes	152
5.5	Environmental Hazard Analysis	156
5.6	Launch Concerns and Operations Procedures	159
6	Project Plan	187

6.1	Testing	187
6.1.1	Required Testing	187
6.1.2	Completed Testing	192
6.2	Requirements Compliance	220
6.2.1	NASA Requirements	220
6.2.2	VADL Derived Requirements	228
6.3	Budgeting	236
6.3.1	Funding Plan	236
6.3.2	Budget Summary	236
6.3.3	To-Date Spending	238
6.3.4	Vehicle and Payload Cost Analysis	243
6.3.5	Expense Summary	245
6.4	Timeline	246
A	References	251
B	VADL Safety Agreement	252
C	VUSE Shop Safety Guidelines	253
D	Material Safety Data Sheets	285
E	FAA Regulations	287
F	NAR High Power Safety Code	297

List of Figures

1	Flight sequence depicting the vehicle mission.	16
2	Full-scale rocket SOLIDWORKS 3D rendering. The figure shows the sections, their lengths, and the separation points.	18
3	Full-scale rocket design from OpenRocket showing the center of gravity (CG) and the center of pressure (CP).	18
4	Full-scale rocket wet mass breakdown.	19
5	Full-scale nose cone 3D rendering.	19
6	Full-scale nose cone design drawing (top) and subscale design drawing (bottom) . .	20
7	Full-scale nose cone with the payload	21
8	Exploded assembly view (top) and assembled view (bottom) of the nose cone, retention ring, and payload bay	21
9	Full-scale nose cone retention ring SOLIDWORKS CAD drawing.	22
10	Full-scale payload bay.	22
11	Full-scale payload bay drawing.	23
12	Full-scale payload bay coupler drawing. This coupler attaches the payload bay to the lander separation bay.	24
13	Full-scale nose cone retention ring Finite Element Analysis.	26
14	Full-scale lander separation bay.	26
15	Full-scale lander separation bay drawing.	27
16	Full-scale lander separation bay bulk plate Finite Element Analysis.	27
17	Full-scale drogue bay.	28
18	Full-scale drogue bay drawing.	28
19	Full-scale avionics section assembly with the separation points indicated on either side (drogue on the left, main on the right).	29
20	Full-scale avionics housing.	30
21	Full-scale avionics assembly.	30
22	Engineering drawings for the avionics components (in order from top to bottom): fore baseplate, housing, and aft baseplate.	31
23	Full-scale avionics bay body tube engineering drawing.	32
24	Full-scale main bay location on the rocket.	32
25	Aft avionics bulk plate FEA.	33
26	Exploded diagram of the avionics aft baseplate, main bay body tube, and the coupler connecting it to the tail section (top) and main bay full-scale rocket body tube design drawing (bottom).	34
27	Full-scale tail section location on the rocket.	34
28	Full-scale fin profile engineering drawing.	35
29	Full-scale tail section assembly without body tube. Shown components (left to right) are the parachute hook plate, motor tube, centering ring, and motor retention ring.	36
30	The load path of the aft section of the rocket upon motor thrust.	37
31	FEA on the parachute hook plate.	37
32	Full-scale tail section assembly with the body tube.	38

33	Full-scale boat tail 3D rendering (left) and engineering drawing of the boat tail (right) detailing dimensions.	38
34	9 in long fiberglass airframe section in load frame during hoop stress test. The left shows the airframe at the start of the test, and the right shows the airframe at the end.	40
35	Beginning (left) and after (middle and right) Bolt tearout test.	41
36	Airframe section during axial loading test.	42
37	Full-scale and subscale vehicle comparison.	44
38	From left to right: nose cone mold; cutting fiberglass cloth pieces; nose cone upon removal from hot box; sanded nose cone.	46
39	From left to right: motor tube, fins, and centering rings in alignment jig; inner layer of fiberglass fillets curing in the hot box; motor tube, fins, and centering rings slid into airframe prior to outer fillet application.	47
40	3D-printed avionics housing, with mounting locations indicated.	48
41	Subscale payload fully assembled. The left shows the payload without an airframe, right shows it within.	49
42	Subscale vehicle fully assembled.	49
43	Subscale vehicle on the launch rail.	50
44	Timelines for (a) first subscale launch and (b) second subscale launch.	51
45	Landed configurations of the tail section (left) and fore section.	52
46	Full landed layout of the vehicle sections and recovery equipment.	52
47	Snapshots and time stamps for key stages from the second subscale flight.	53
48	Coefficients of Drag of the Subscale Vehicle CFD Model	55
49	Pressure Contours of the Subscale Vehicle CFD Model.	56
50	Streamlines of the Subscale Vehicle CFD Model.	56
51	Raw data of the main avionics altimeter.	57
52	Raw data of the backup avionics altimeter.	58
53	Raw data of the main avionics altimeter versus simulated altitude.	62
54	Raw data of the backup avionics altimeter versus simulated altitude.	62
55	Representative visualization of the parachute's descent dynamics captured during Flight Two, showing the velocity vector of the system's center of gravity ($\Theta = 50^\circ$) and the swirling motion of the parachute in the horizontal plane (Φ).	65
56	Flight Track of the Adjusted Side Drag Model.	66
57	Recovery sequence CONOPS.	68
58	Location of the drogue parachute in the full-scale vehicle.	68
59	Descent configuration under the drogue parachute.	69
60	Location of the main parachute in the launch vehicle.	69
61	Launch vehicle descent configuration under the main parachute, prior to payload jettison.	70
62	Location of the payload parachute retention tube in the launch vehicle.	72
63	Payload side shock cord configuration, showing how single large loop is used to connect to a single release latch in the center.	73
64	Avionics circuit diagram.	74
65	BigRedBee GPS/APRS Transmitter.	77
66	Scenario A (Baseline) Simulated Flight Track.	81
67	Scenario A (Baseline) Simulated Altitude.	81

68	Scenario A (Baseline) Simulated Velocity.	81
69	Scenario A (Baseline) Simulated Acceleration.	82
70	Flight Tracks of Select Wind Conditions for 5-Degree Launch Angle of Full-Scale Vehicle.	83
71	Maximum Theoretical Drift Conditions for 5-Degree Launch Angle of Full-Scale Vehicle.	85
72	L1720 Adjusted Thrust Curve.	88
73	L1720 Adjusted Thrust-to-Weight Ratio.	88
74	Full-scale rocket design from OpenRocket showing the center of gravity (CG) and the center of pressure (CP).	92
75	Flight Dynamics of Rocket Flight During Ascent.	93
76	Boundary conditions used in ANSYS Fluent simulation.	94
77	Side view of full-scale mesh.	95
78	Front view of full-scale mesh; cross section measured 150 cm from tip of nose cone.	95
79	Full Scale coefficient variation as a function of vertical velocity for 0 degree angle of attack.	96
80	Static pressure contour plot for full-scale travelling at 175 m/s and 0 deg angle of attack.	96
81	Shock cord loading vs length for 4 ft and 5 ft parachutes.	97
82	Payload mission concept of operations.	98
83	Payload design overview with major components highlighted.	99
84	Lander Separation Mechanism (LSM).	99
85	Leg Deployment Mechanism (LDM).	100
86	Parachute Detachment Mechanism (PDM).	100
87	Antenna Extension Mechanism (AEM).	101
88	Lander during separation, parachute opening and leg deployment, descent, and landing. Antenna extension can be seen in landing configuration.	102
89	Summary of results for each of the four drops as they pertain to key payload objectives.	103
90	APRS ground receiver screenshot.	103
91	Flight telemetry for the 3 successful drops.	103
92	LSM location in the launch vehicle. The LSM remains with the launch vehicle throughout the entire flight and does not jettison with the lander.	107
93	CAD rendering of LSM assembly.	107
94	Exploded view of LSM assembly.	108
95	FEA result for bottom LSM bulk plate.	109
96	FEA result for payload hook plate attaching to the LSM latches. It is made of aluminum and has a safety factor of 9.8.	110
97	LSM Approval System with Amplification Circuit.	110
98	LDM components.	111
99	Assembled LDM.	112
100	Single LDM bracket and leg.	112
101	LDM bracket FEA result.	113
102	Single LDM bulk plate clamped for testing.	113
103	Full view of test setup while loaded.	114

104	Leg deforming under loading from applied weight. Total deformation distance can be seen.	115
105	Suspension prototype.	116
106	Closeup of leg bracket for suspension prototype.	117
107	Full-scale lander prototype post drop. The PDM successfully actuated, however the parachute shock cord caught on the latch and did not detach properly. This can be seen on the right where the shock cord loops are stuck on the sides of the latch, even when open.	118
108	PDM with new latch cover to prevent shock cord loops from catching on latch during detachment. The latch cover is made of PLA and adds negligible mass (6 grams). The hook bulk plates have not yet been re-manufactured. The previous iteration is visualized.	118
109	PDM assembly drawing. The smooth rods making up the lander skeleton are not visualized for clarity.	119
110	PDM hook bulk plate FEA results. Red faces represent fixed regions, while yellow arrows indicate force of magnitude 129 lb (574 N) pointing at a 45 degree angle in the YZ plane. The minimum safety factor on this part was calculated to be 3. . . .	119
111	PDM latch bulk plate FEA results. Red faces represent fixed regions, while yellow arrows indicate force of magnitude 129 lb (574 N) pointing perpendicular to the YX plane. The minimum safety factor on this part was calculated to be 5.7. . . .	120
112	CAD rendering of AEM assemblies on lander bulk plate. Antenna/tape springs are shortened for visual clarity.	121
113	CAD rendering of individual AEM sub-assembly. Antenna/tape spring is shortened for visual clarity.	122
114	Individual AEM sub-assembly	122
115	Exploded view of high-level AEM assembly drawing. All dimensions are in inches. Antenna/tape spring is shortened for visual clarity.	123
116	Exploded individual AEM drawing with call outs.	123
117	Antenna cover drawing. All dimensions are in inches. This cover was designed to extend less than 0.25 in above the airframe surface in accordance with NASA requirement 4.2.2.	124
118	Full antenna extension in full-scale drop test from 300 ft AGL. During this test, the lander successfully transmitted telemetry via RF to the team ground station at a distance of 150 ft.	126
119	STEMnaut crew module 3D rendering and exploded view showing the seats. . . .	126
120	Assembly of the nose cone in an isometric view showing the location of the STEMnaut Crew Module.	127
121	Assembled subscale payload. subscale and full-scale payloads share the same electronics.	128
122	System Block Diagram.	128
123	Software High-level Data Flow Diagram	129
124	Landing Detection Algorithm with Built-in Redundancy. Height values are specific to the drone drop test; terrain height is exaggerated.	130
125	Initial test with MATLAB, using simulation data.	132
126	Test performed on the elevator, with the safety office. Real-time velocity estimation. 133	

127	Screenshots from the receiver test in one of the drop tests.	134
128	Assembled prototype board.	135
129	Assembled prototype board.	136
130	Disorganized wiring made it difficult to check for poor solder joints.	137
131	Schematics of Custom PCB.	139
132	4-Layer PCB Layout.	140
133	3D Model of PCB	140
134	Safety flowchart.	141
135	Status of testing as of CDR cycle. Additional tests shall be done and re-done in order to maximize the chances of mission success.	192
136	VADL funding plan for the 2024-2025 competition year.	236
137	Summary of VADL’s budget for the 2024-2025 competition year.	237
138	Total to-date spending, as of the CDR milestone.	238
139	Spending over time for the 2024-2025 competition year, projected for remaining months.	245
140	Key project dates and milestones.	246
141	Pre-phase A and Phase A of the project.	247
142	Phase B of the project.	247
143	Subscale portion of Phase C of the project.	248
144	Full-scale portion of Phase C of the project.	248
145	Subscale portion of Phase D of the project.	249
146	Full-scale portion of Phase D of the project.	249
147	Phases E and F of the project.	250

List of Tables

1	Acronyms and acronym definitions	11
2	Vehicle Criteria Changes	14
3	Payload Criteria Changes	15
4	Changes made to the project plan.	15
5	Vehicle mission success criteria and safety requirements.	17
6	FEA Conditions for Components.	25
7	Subscale flight objectives and corresponding requirements.	42
8	Comparison of full-scale and subscale vehicle parameters.	45
9	Launch conditions, and the relevant requirements that justified their measurements, for both launches.	54
10	Subscale Baseline Flight Simulations.	55
11	Predicted Coefficients of Drag of the Subscale Vehicle.	55
12	Subscale Baseline Maximum Drift Simulations.	57
13	Flight One Launch Rail Exit Velocity Comparisons.	61
14	Flight Two Prediction Benchmarks against Actual Data.	63
15	Flight Two Landing Characteristics.	64
16	Flight Two Drift Characteristics.	64
17	Adjusted Drift Modeling Results.	65
18	Full-scale design modifications informed by the subscale flight.	67
19	Parameters for the drogue parachute.	69
20	Parameters for the main parachute.	70
21	Vehicle shock cord lengths.	71
22	Anchoring hardware components and their associated ratings, loadings, and factors of safety.	71
23	Specifications for the payload parachute	72
24	Parameter table for nylon shear pins and parachute bay calculations.	76
25	Parameter table for nylon shear pins and parachute bay calculations.	76
26	Relevant NASA and VADL Requirements.	78
27	Flight profiles of select scenarios. Values that meet their respective NASA/VADL requirements are shown in green.	80
28	Sample comparison of predicted performance metrics using DRSS vs. Open- Rocket.	82
29	Drifts at landing using the DRSS model.	84
30	Maximum theoretical drifts.	85
31	Simulated parachute deployment loading.	86
32	Predicted Apogee vs. Wind.	91
33	Comparison of Rocket CP Measurement Methods.	92
34	Comparison of Rocket CP Measurement Methods.	92
35	Predicted Coefficients of Drag of the Fullscale Vehicle.	93
36	Payload testing summary.	101
37	DRI calculations for the 3 successful drop tests. Drop 1 impacted a tree which lowered the overall acceleration experienced by the STEMnauts.	105

38	Actual vs adjusted predictions of parachute accelerations. Initial conditions were set to match actual data.	105
39	Actual vs adjusted predictions of parachute accelerations. Initial conditions were set to match actual data.	106
40	LSM Testing Summary	111
41	Summarized AEM testing results.	125
42	Payload Power Consumption	136
43	Severity classifications.	143
44	Probability levels.	144
45	Color key.	144
46	Risk assessment matrix.	145
47	Safety matrix subsections.	146
66	Itemized summary of the VADL budget.	237

Table 1: Acronyms and acronym definitions

Acronym	Acronym Meaning
ABS	Acrylonitrile Butadiene Styrene
AGL	Above Ground Level
AIAA	American Institute of Aeronautics
AOA	Angle of Attack
BMS	Battery Management System
CAD	Computer-Aided Design
CFD	Computational Fluid Dynamics
CFR	Code of Federal Regulations
CG	Center of Gravity
CNC	Computer Numerical Control
CP	Center of Pressure
DRI	Dynamic Response Index
ESB	Engineering and Science Building
FAA	Federal Aviation Administration
FEA	Finite Element Analysis
FOV	Field of Vision
FPV	First Person View
FRR	Flight Readiness Review
GPS	Global Positioning System
IMU	Inertial Measurement Unit
KE	Kinetic Energy
LiPo	Lithium Polymer (Battery)
LRE	Launch Rail Exit
MECO	Main Engine Cut Off
MSDS	Material Safety Data Sheet
NAR	National Association of Rocketry
NFPA	National Fire Protection Association
PC	Personal Computer
LSM	Lander Separation Mechanism
PID	Proportional, Integration, Derivative (Control)
PLA	Polylactic Acid
PPE	Personal Protective Equipment
RF	Radio Frequency
RSO	Range Safety Officer
R/C	Remote Control
SDR	Software Defined Radio
SCM	STEMnaut Crew Module
SL	Student Launch
PDM	Parachute Detachment Mechanism
Continued on next page	

Table 1 – continued from previous page

Acronym	Acronym Meaning
LDM	Leg Deployment Mechanism
AEM	Antenna Extension Mechanism
SSM	Static Stability Margin
SSO	Student Safety Officer
STEM	Science, Technology, Engineering, and Math
T/W	Thrust to Weight Ratio
USLI	University Student Launch Initiative
UTM	Universal Testing Machine
VADL	Vanderbilt Aerospace Design Lab
VUSE	Vanderbilt University School of Engineering
WBS	Work Breakdown Structure

1 Summary

1.1 Team Summary

Vanderbilt Aerospace Design Laboratory

Mailing Address

Dr. Amrutur V. Anilkumar
VU Department of Mechanical Engineering
VU Station B 351592
2301 Vanderbilt Place
Nashville, TN 37235-1592

Rocketry Mentor

Chris Dondanville
Phoenix Missile Works, NAR Level III
NAR #92729
931.446.4370
chris.dondanville@gmail.com

The team intends to travel to Huntsville, Alabama for the competition launch on May 3, 2025. Since the PDR milestone, the team has spent 1834 hours combined working on the project.

1.2 Launch Vehicle Summary

Parameter	Specification	Parameter	Specification
Target Apogee	4200 ft	Motor Selection	Primary: Cesaroni L1720, Secondary: N/A
Rocket Length	104 in	Burnout mass without ballast	37.8 lb
Length/Mass (nose cone, payload bay, and lander separation bay)	50.5 in, 14.6 lb	Recovery system	Double deployment
Length/Mass (avionics and recovery)	16.97 in, 12.98 lb	Drogue parachute	Fruity Chutes Elliptical 18 in, 1.5 CD
Length/Mass (tail section)	36 in, 14.05 lb	Drogue deployment	Primary: Apogee, Backup: Apogee + 1s
Landing Mass (vehicle/lander)	29.7 lb, 8.0 lb	Main Parachute	Fruity Chutes Toroidal 96 in, CD 2.2
Dry mass without ballast	37.8 lb	Main deployment	Primary: 600 ft, Backup: 550 ft
Dry mass with ballast	37.8 lb	Lander separation mechanism	Electromagnetic latches, 300 ft
Wet mass with ballast	41.6 lb	Rail size	144 in

1.3 Payload Summary

The Extraterrestrial Lander for Data Collection and RF Transmission is a parachute-arrested lander that acts as the nose cone of the launch vehicle and jettisons at 300 ft AGL during main parachute descent. Jettison is enabled by electromechanical latches which release the payload, allowing gravity to pull it from the foresection of the launch vehicle. Once jettisoned, the payload will deploy a parachute and landing legs that double as the ground plane for the antenna system. The payload will land horizontally (nose cone parallel to the ground) upon which the parachute will be detached and the antenna raised to a height of 24 in above the ground. Once fully deployed, the payload will begin RF transmission of flight telemetry. The payload is constructed out of aluminum and 3D printed bulk plates attached to rods to transfer load throughout the structure. The airframe and nose cone are made of fiberglass to reduce RF interference. The legs and antenna are deployed through slots cut into the airframe.

2 Changes Since PDR

2.1 Vehicle Criteria Changes

The structure of the vehicle remains generally similar to that declared in PDR besides a few changes in material selection and geometry. These changes are detailed below:

Table 2: Vehicle Criteria Changes

Change	Justification
The fin material for the full-scale rocket will be DragonPlate carbon fiber [5].	The team has dragon plate available in the lab and decided to utilize this resource and ensure that materials are not overlooked. Another reason was to control the mass of the tail section so as to not destabilize the center of gravity of the rocket. Since, fiberglass is significantly lighter than carbon fiber, it was an advantageous decision. One potential concern with the use of carbon fiber fins was the bond strength of the material to fiberglass; this was proven to be a non-issue through the successful subscale manufacturing process and flight, which saw no structural damage. Lastly, since the tail section does not interfere with the AEM or any RF functionalities of the payload, the material of the fins does not impact the success of the mission. This was further confirmed through the subscale flight where carbon fiber fins were implemented and no disruptions in the RF capabilities were observed.
The nose cone was redesigned from a rounder cross-section to an elliptical shape.	To enhance the aerodynamic performance of the full-scale rocket, the team has shifted to a more standard, elongated elliptical shape for the nose cone. A 67% scaled version of this design was flown successfully on the subscale rocket. This design is still sufficiently round to store the STEMnaut Crew Module and other crucial elements.
The length of the rocket has increased from 93 inches to approximately 104 inches.	This was done in order to accommodate for the changes in payload design, detailed in Section 2.2, and a lengthening of the LSM bay to accommodate the switch to a larger parachute.

2.2 Payload Criteria Changes

The payload functionality and high-level structure in the CDR design cycle remains consistent with the PDR version. Changes made to the payload design were driven through testing results to improve the reliability and functionality of the payload.

Table 3: Payload Criteria Changes

Change	Justification
Landing legs were extended from 0 to 1.7 in below the lander belly to soften the impact of landing.	STEMnaut survivability in full-scale lander drop tests did not satisfy VADL requirement 4.1.6. (See Section 4.1.3). Legs that protrude further from the lander belly dampen landing forces and decrease likelihood of serious STEMnaut injury.
Payload parachute size was increased from 4 to 5 ft.	To further reduce landing impact, a slower descent velocity is needed.
Parachute shock cord length was shortened from 50 to 30 ft.	In order to maintain a similar parachute opening force with a larger parachute, the shock cord needed to be shortened.

Modifications that do not affect the payload functionality include modularized electronics and improved electronics sleds.

2.3 Project Plan Changes

Table 4: Changes made to the project plan.

Change	Justification
Added new, and modified existing, VADL derived requirements (Section 6.2.2).	As the design has matured, and the understanding of the project has deepened, new requirements have emerged.
Constructed a testing procedure for use in all tests (Section 6.1).	With the project moving into more sophisticated testing, the need for a standardized testing documentation framework presented itself.
Combined to-date spending and expense projections to graph spending over time for the competition year (Figure 139).	Having a clearer picture of the overall spending trends aids the team in ensuring that the budget is not exceeded.
Restructuring and rewriting of safety sections of report.	Following the feedback received at the PDR milestone, the team determined that a full rework of the safety sections was merited.

3 Vehicle Criteria

3.1 Design and Verification of Launch Vehicle

3.1.1 Mission Statement and Success Criteria

The specific mission of the 2024-2025 VADL competition vehicle is to launch to within 5% of its projected apogee of 4200 ft and successfully deploy both drogue and main parachutes at apogee and 600 ft, respectively, while securely retaining four STEMnauts and jettisoning them and the payload at 300 ft. **Most importantly, VADL understands the primary objective of vehicle design and verification is to ensure mission safety under NASA vehicle flight and recovery requirements.** The entire flight sequence is depicted in Figure 1 below.

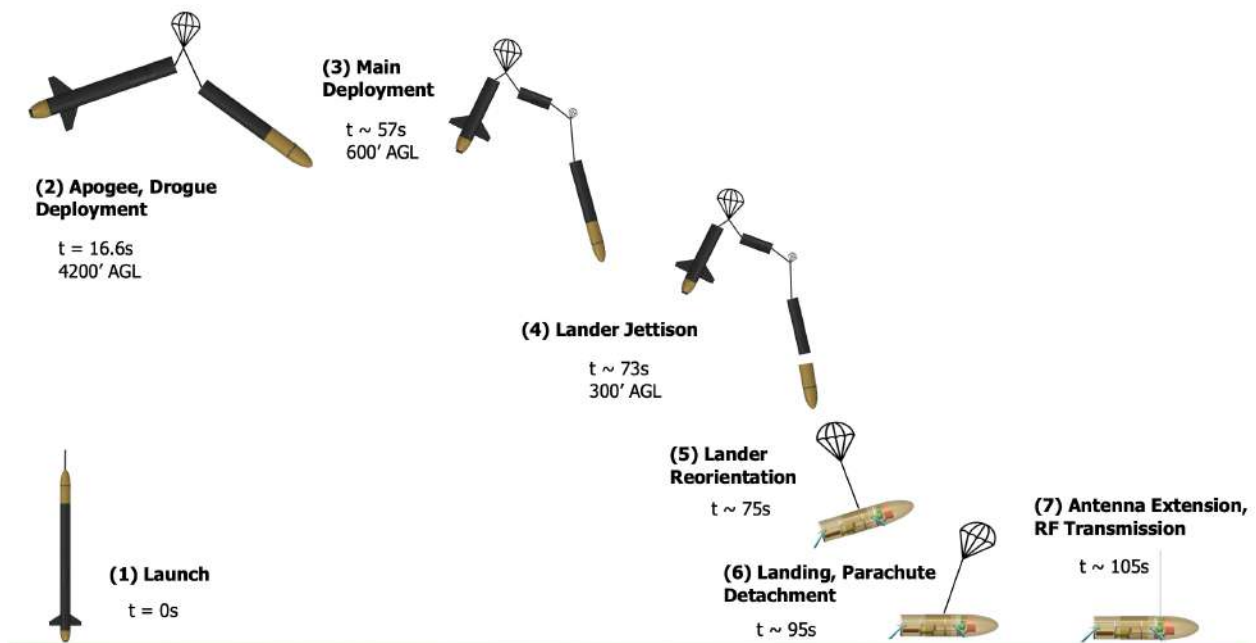


Figure 1: Flight sequence depicting the vehicle mission.

The vehicle mission has been decomposed into a list of objectives that define mission success. It is imperative that these success criteria are clearly defined, as they drive vehicle design requirements and design selection to ensure that the final vehicle can safely and functionally achieve mission success. The success criteria and their corresponding requirements are provided in Table 5 below.

Table 5: Vehicle mission success criteria and safety requirements.

Success Criteria	Relevant Requirements
Fly within 5% of the predicted apogee of 4200 ft	NASA 2.1; VADL 2.1.1
Deploy drogue and main parachutes at apogee and 600 ft, respectively	NASA 2.3, 3.1, 3.1.1-2
Safely retain four STEMnauts until jettison	NASA 4.1
Jettison the lander at 300 ft	VADL 4.2.3
Record altimeter flight data	NASA 3.4-6; VADL 3.7
Achieve L.R.E. velocity greater than 52 fps	NASA 2.17
Have a static stability margin between 2.0 and 2.5 calibers on the launch pad	NASA 2.14; VADL 2.1.2
Safely land all vehicle sections with L.K.E. below 75 ft-lbf	NASA 3.3
Drift less than 2500 ft from the launch pad	NASA 3.11
Descend in less than 90 s	NASA 3.12

3.1.2 Launch Vehicle Design Overview

The challenge for the 2024-2025 Student Launch competition requires the vehicle design to be built with careful consideration of the requirements of the payload. These mission requirements dictate the design of the vehicle including the material selection for RF compatibility. The overall length of the vehicle is 104 inches, the dry mass is 37.8 lbs, and the wet mass with the Cesaroni L1720 motor is 41.6 lbs. The airframe will be made of fiberglass with a 6.17-inch outer diameter. The nose cone will be constructed in-house with fiberglass. As mentioned, the Cesaroni L1720 motor will be the primary selection for the launch this year. Although the team did not explicitly declare a secondary motor at the PDR milestone, the team is confident in its inventory of L1720 motors to fulfill the team's rocket flight needs for the competition year.

The team is continuing with fiberglass as the material selection for the full-scale vehicle airframe, as declared at the PDR milestone and validated through the subscale flight. Previous Vanderbilt teams have used Blue Tube wrapped with carbon fiber; however, several factors have driven a switch to fiberglass for this competition year. Fiberglass, unlike carbon fiber, is compatible with RF transmission and allows the team to fulfill this year's mission regardless of payload jettison. Furthermore, Blue Tube has presented challenges in the past with warping when subjected to temperature changes in the field. This change in shape heightens the risk of failed jettison due to additional friction between the vehicle airframe and payload coupler, rendering Blue Tube an unsuitable airframe material for this year's mission. Fiberglass is a strong and reliable material, one which the team has successfully deployed in both the subscale flight demonstration (Section 3.2.4) and in-house strength testing (Section 3.1.10). For these reasons, VADL is confident in its

selection of fiberglass as the airframe material for the full-scale vehicle.

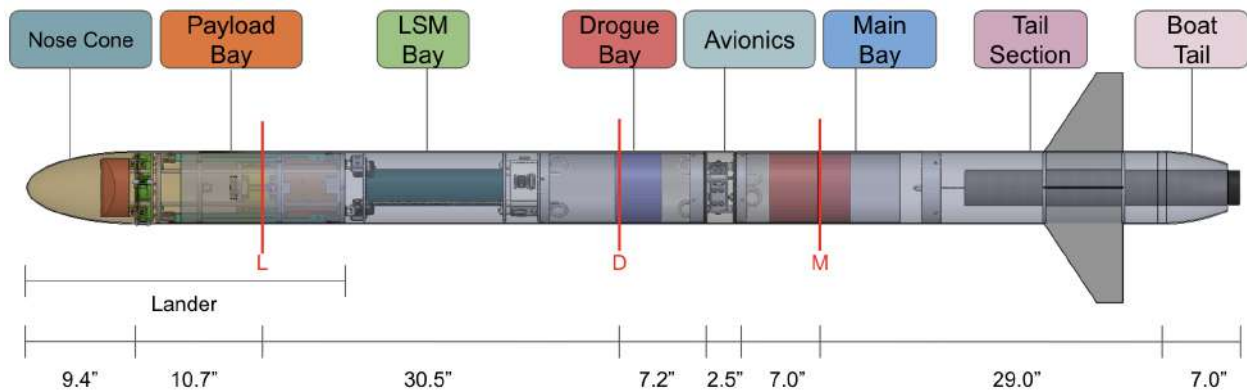


Figure 2: Full-scale rocket SOLIDWORKS 3D rendering. The figure shows the sections, their lengths, and the separation points.

In Figure 2, the transparent 3D rendering of the vehicle presents all mechanical subsystems and the airframe. It also displays the different sections of the vehicle along with both the drogue and main separation points marked appropriately with L for lander, D for drogue, and M for main separation points. The drogue (blue) and main (red) recovery packaging is also displayed in the figure which will utilize both the respective recovery bays but also utilize the coupler space. If the lander is allowed to separate with permission from the RSO, the appropriate separation point for the separation event is marked with L for lander separation. The lander is also marked and has a length of 27.125 inches.

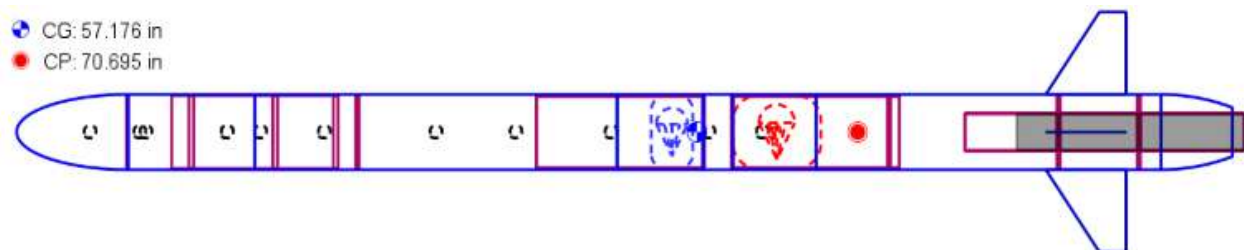


Figure 3: Full-scale rocket design from OpenRocket showing the center of gravity (CG) and the center of pressure (CP).

Figure 74 shows the OpenRocket file with both the CG and the CP which are at 57.18 inches and 70.70 inches respectively, see figure legend. This results in an SSM of 2.19 based on the fin design, refer to Section 3.1.9. The OpenRocket file was also constructed carefully with precise placement of the bulk plates and other mass to ensure the simulation produced accurate results.

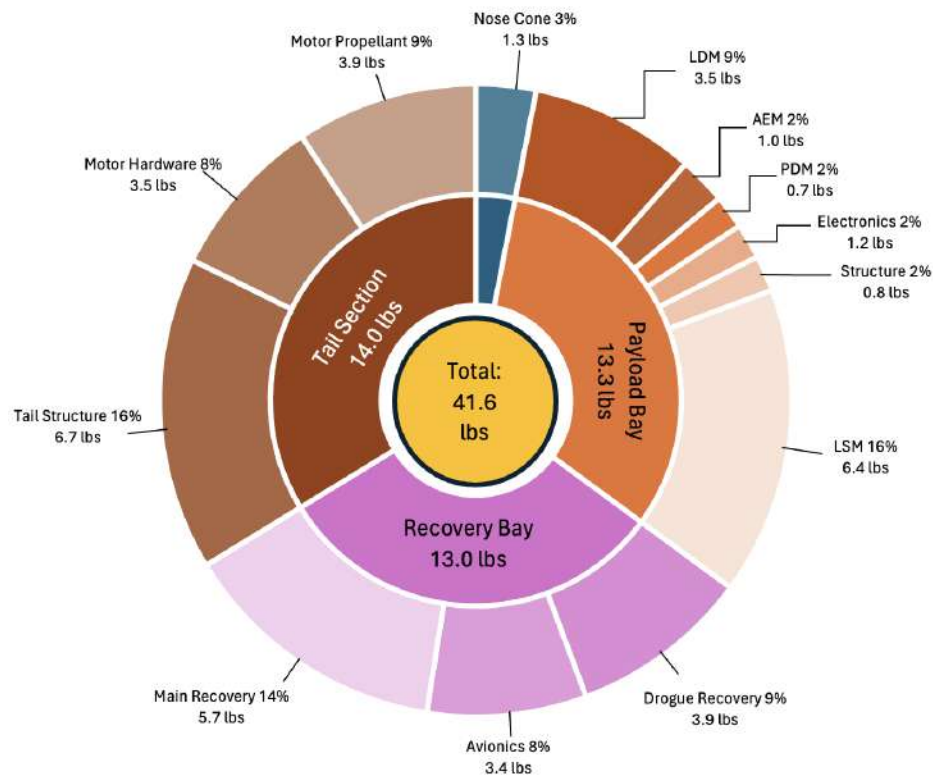


Figure 4: Full-scale rocket wet mass breakdown.

The above figure, Figure 4, shows the mass breakdown of the wet mass of the fully assembled rocket. The higher-level sections of Payload Bay and Recovery Bay are further broken down into the respective mechanical subsystems. The wet mass of the rocket on the pad is 41.6 lbs with the motor hardware and propellant loaded in. After motor burnout, the propellant is utilized completely and the dry mass of the rocket is 37.8 lbs. The massing was completed using the available inventory, SOLIDWORKS, and estimates for bulk plates that will be machined.

3.1.3 Nose Cone

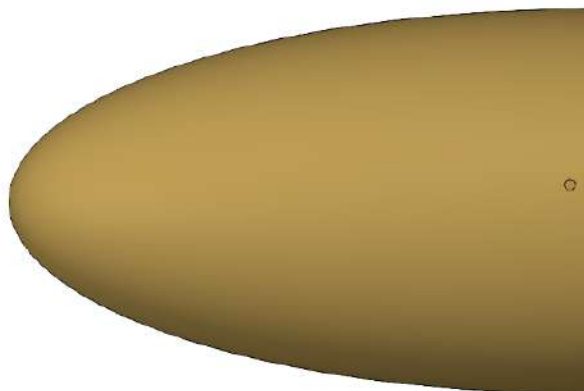


Figure 5: Full-scale nose cone 3D rendering.

Informed by the design and fabrication methods employed during the manufacturing of the subscale rocket and validated through subscale flight testing, the team will employ a similar manufacturing process with the full-scale nose cone. This process is detailed in Section 3.2.3. As illustrated in Figure 5, the nose cone is a geometrically scaled version of the subscale nose cone, which has been demonstrated to yield optimal performance through both subscale CFD analysis and flight results (Section 3.2.5). Given that the manufacturing process for this geometric configuration was well understood and thoroughly validated during subscale testing, the team determined that a direct scale-up would be sufficient for the full-scale implementation. The elliptical design of the nose cone facilitates efficient storage of the AEM assembly and the STEMnaut Crew Module. The AEM cutouts just behind the nose cone on the payload bay did not have a significant effect on aerodynamics, as is discussed in Section 3.2.4. The successful performance of the subscale flight, coupled with the nose cone's ability to function as intended without sustaining damage, has provided the team with the confidence to proceed with this fabrication approach for the full-scale vehicle.

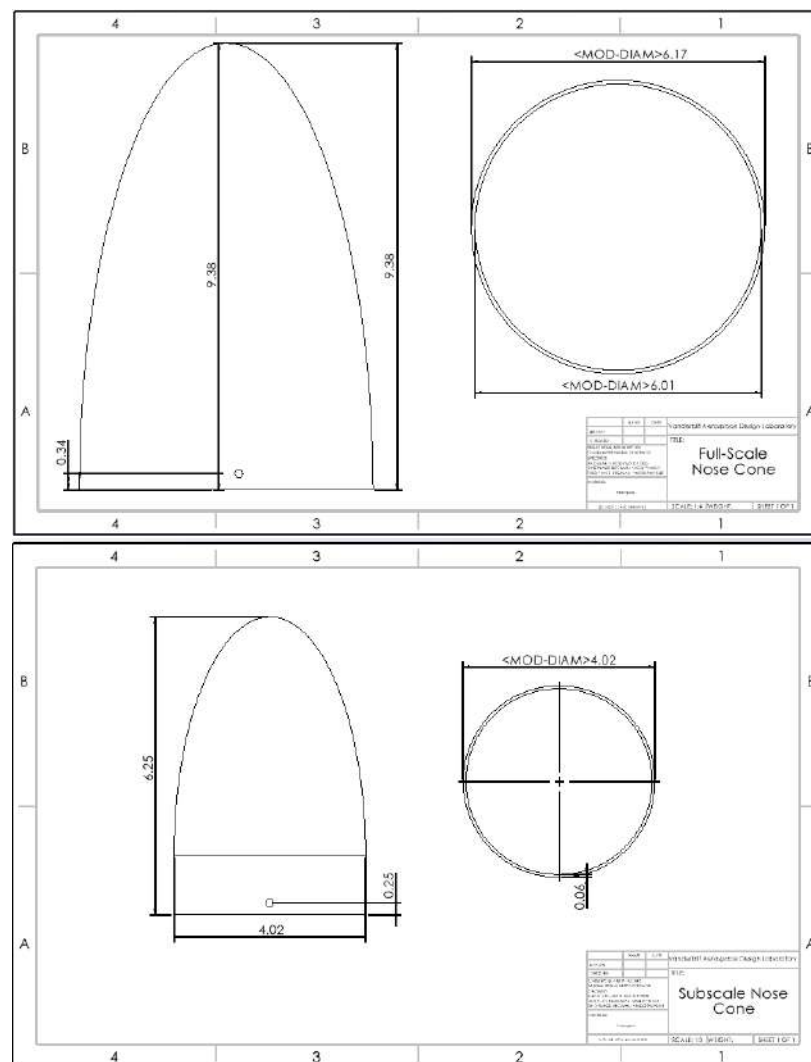


Figure 6: Full-scale nose cone design drawing (top) and subscale design drawing (bottom)

The above figure shows both the full-scale and the subscale nose cones in a side-by-side comparison. This comparison further confirms that the nose cone was scaled up from the subscale flight. Since the direct ratio generated by the base diameter of the nose cone is approximately 1.5 (4.02-inch diameter to 6.17-inch diameter), the same elliptical shape was then scaled up from a 6.25-inch length to a 9.375-inch nose cone. The ease of manufacturing and structural integrity of the nose cone after the successful subscale launch corroborated the decision to scale the nose cone.

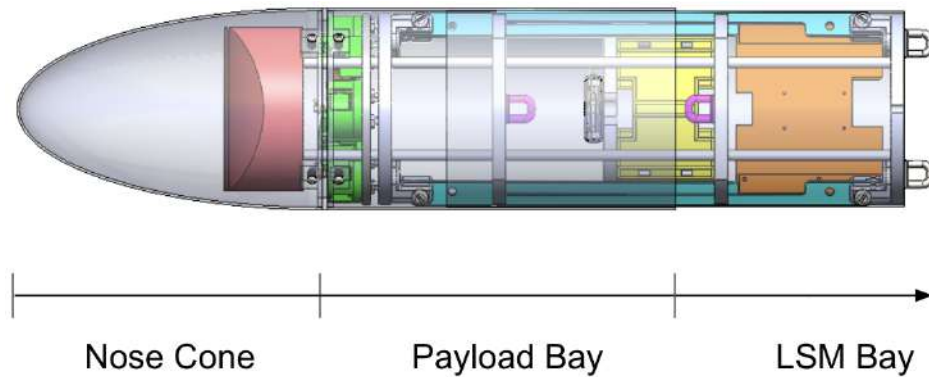


Figure 7: Full-scale nose cone with the payload

Figure 7 shows the nose cone integrated with the payload bay. The nose cone is attached to the payload bay via an onyx retention ring, shown below in Figure 8. The retention ring is optimized to ensure high-strength attachment while allowing space for routing the AEM electronics.

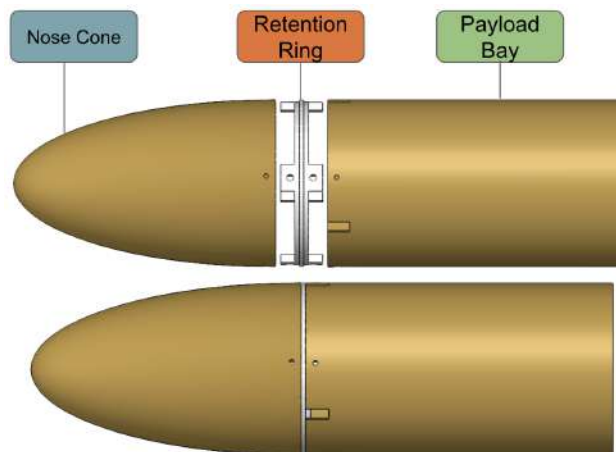


Figure 8: Exploded assembly view (top) and assembled view (bottom) of the nose cone, retention ring, and payload bay

The figure above shows the nose cone, retention ring, and payload bay body tube, in order. The exploded view of the assembly of the three parts displays the method of retention of the nose cone which further demonstrates the change from PDR where the nose cone with the coupler were merged to form one piece. This change was made to further simplify the manufacturing of the nose cone and ensure that a robust method of retaining the nose cone with the payload bay body tube was utilized.

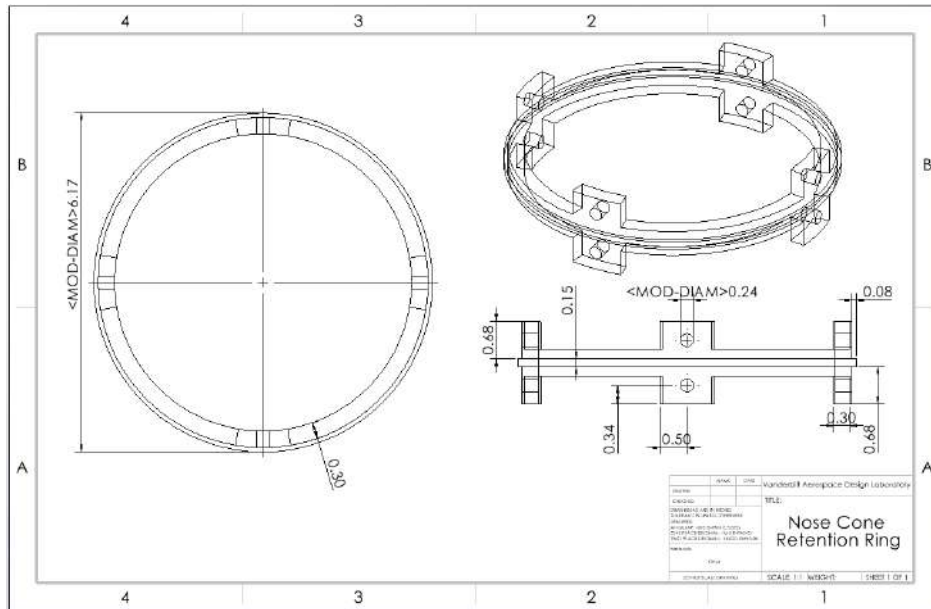


Figure 9: Full-scale nose cone retention ring SOLIDWORKS CAD drawing.

The figure above shows the engineering drawing of the retention ring for the full-scale nose cone. The retention ring is optimized for mass as well as for structural robustness to allow for a sturdy connection between the nose cone and the payload bay body tube section. This retention ring will be 3D printed with Onyx due to its structural reliability as well as ease of manufacturability. The airframe section and nose cone will be attached to the retention ring via 8-32 bolts fastening to heat inserts.

3.1.4 Payload Bay

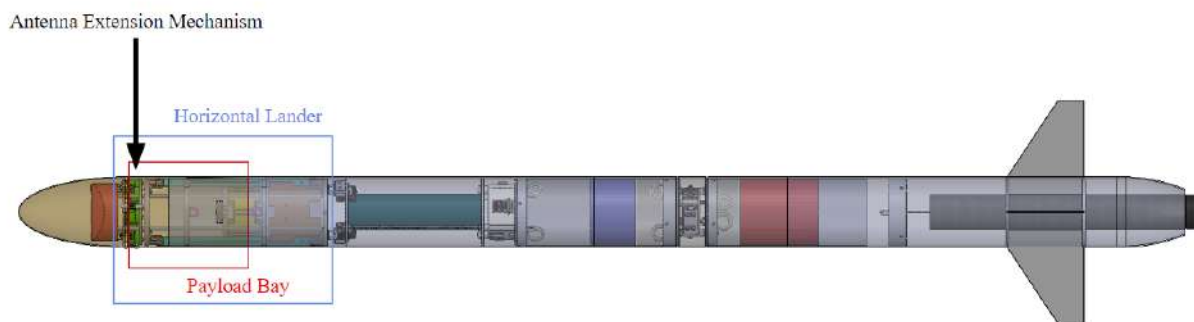


Figure 10: Full-scale payload bay.

The payload bay, which houses the horizontal lander structure and electronics, is located in the fore section of the vehicle, as indicated in Figure 10 above.

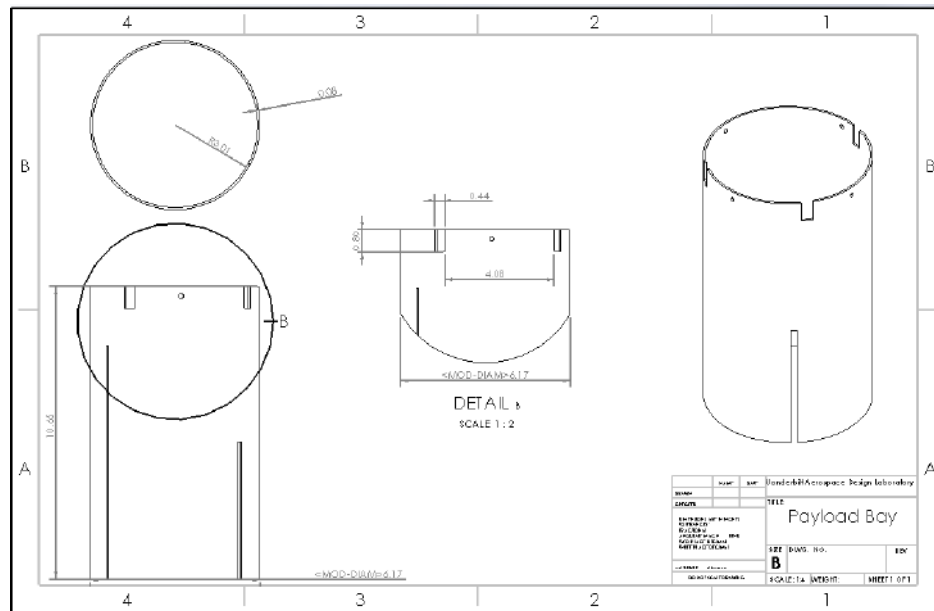


Figure 11: Full-scale payload bay drawing.

In order to ensure that the vehicle has RF transmission capabilities whether or not the payload can be jettisoned, AEM cutout slots have been designed in the payload bay. These slots will allow the antennas to extend in the case that RSO approval to jettison is not granted. The structural and aerodynamic viability of such cutouts was proved during the team's subscale launch, described in further detail in Section 3.2.4. RF transmission capabilities are further allowed for by the usage of fiberglass as the airframe material.

A coupler is located between the lander separation bay and payload bay to safely retain the payload bay to the rest of the vehicle prior to lander jettison. This coupler connects the two sections and ensures structural integrity during flight. The coupler will be epoxied to the payload bay and will be retained to the lander separation bay via the two rotary latches. The coupler is 14 inches long with a 7 inch overlap with each airframe section, thus satisfying NASA Requirement 2.4.1.

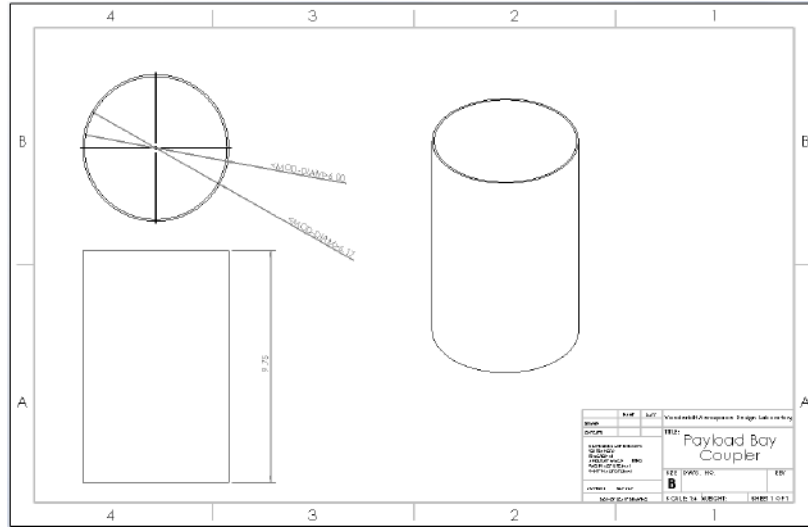


Figure 12: Full-scale payload bay coupler drawing. This coupler attaches the payload bay to the lander separation bay.

Structural Analysis

To ensure the airframe and the overall payload bay's structural integrity, FEA was performed on the retention ring which connects the nose cone to the payload bay. This analysis simulated the maximum force experienced at main parachute deployment, with fixed supports at each screw hole. For this analysis, and the analyses in subsequent sections, the following methodology is used. The maximum load experienced can be described using the following equations:

$$F_{shock} = C_D * \frac{\rho V_y^2 A}{2} \quad (1)$$

Where F_{shock} represents the **maximum instantaneous loading** experienced when the shock cord becomes fully taut, V_y is the terminal velocity under drogue, accounting for 25% of the vehicle's side drag due to nondeterministic vehicle swing under drogue descent (see Section 3.2.5), and C_D and A refer to the drag coefficient and cross-sectional area of the main parachute, respectively. Using $C_D = 2.2$, $\rho = 0.002377 \text{ slugs/ft}^3$, $A = 50.27 \text{ ft}^2$, and $V_y = 95.7 \text{ ft/s}$ yields the following result:

$$F_{shock} = 2.2 * \frac{0.002377 \text{ slugs/ft}^3 * 95.7(\text{ft/s})^2 * 50.27 \text{ ft}^2}{2} = 1204 \text{ lbf} \quad (2)$$

In this case, F_{shock} represents the maximum instantaneous loading acting on the entire rocket system. To determine the deceleration experienced by the vehicle, Newton's Second Law can be employed as follows:

$$a_{main} = \frac{F_{shock}}{m_{dry}} = \frac{1204 \text{ lbf}}{1.17 \text{ slugs}} = 1029 \text{ ft/s}^2 = 32.0 \text{ G} \quad (3)$$

It is not possible to solve for the exact forces exerted onto each rocket section (which are **internal forces** inside the rocket system) without accurate information on shock cord deformation. From here, the force experienced by each bulk plate is computed by multiplying the mass of the supported section a_{main} . These results are summarized in Table 6 below.

Table 6: FEA Conditions for Components.

Component	FEA Boundary Conditions	Loading Scenario	F.O.S.
Nose Cone Retention Ring	Fixed supports at screw holes	Force experienced at payload shock cord extension, running through rod holes	5.6
LSM Latch bulk plate	Fixed supports at the screw holes	Main deployment	11.3
PDM Latch bulk plate	Fixed supports at the mounting screw holes and smooth rod through holes	Force experienced at full payload shock cord extension, assuming full vertical force on bolts	5.7
Leg Bracket	Fixed supports at two mounting screw holes in base	Torsional load applied by lever arm of leg. Maximum should be result of landing speed, length of leg, and weight of payload.	7
PDM Hook Bulk plate	Fixed supports at smooth rod through holes and hook screw holes	Force applied through attached hook at 45 degree angle to plate	3
LSM Bay Bulk plate	Fixed supports at screw holes	Force experienced at main deployment running through eyebolt hole	7.4
Avionics Aft Bulk plate	Fixed supports at screw holes	Force experienced at main deployment running through eyebolt hole	5.1
Avionics Fore Bulk plate	Fixed supports at the screw holes	Force experienced at main deployment running through eyebolt hole	7.2
Tail Parachute Hook Plate	Fixed supports at the screw holes	Force experienced at main deployment running through eyebolt hole	5.4
Payload Hook Plate	Fixed supports at the screw holes	Force experienced at main deployment running through latch hook holes	9.8

In previous years, the acceleration experienced during main deployment was calculated simply by taking the velocities before and after deployment, divided by the time required for the parachute to open; this calculates the **average** acceleration during opening rather than **peak** loading. It has been discovered through empirical testing, however, that this is not representative of the true load experienced. Corroborated by drop test data, it is understood that the parachute opening time is significantly less than full shock cord extension; as such, the actual jerk comes from the shock cord itself becoming taut. This event results in a significantly greater deceleration than predicted using the previously used method. As described in Section 4.1.3, this finding was proven through IMU

data obtained during payload drop tests, which showed a maximum deceleration of 14.5 G. Using the old method, this deceleration was predicted to be 3.1 G, whereas the new method predicted 16.3 G, much closer to the experimental result. The load analysis confirmed that each bulk plate has a minimum safety factor of 5 and thus can withstand the maximum loads felt during main deployment. The nose cone retention ring, 3D printed with Onyx, as seen in the FEA in Figure 13, of 5.6.

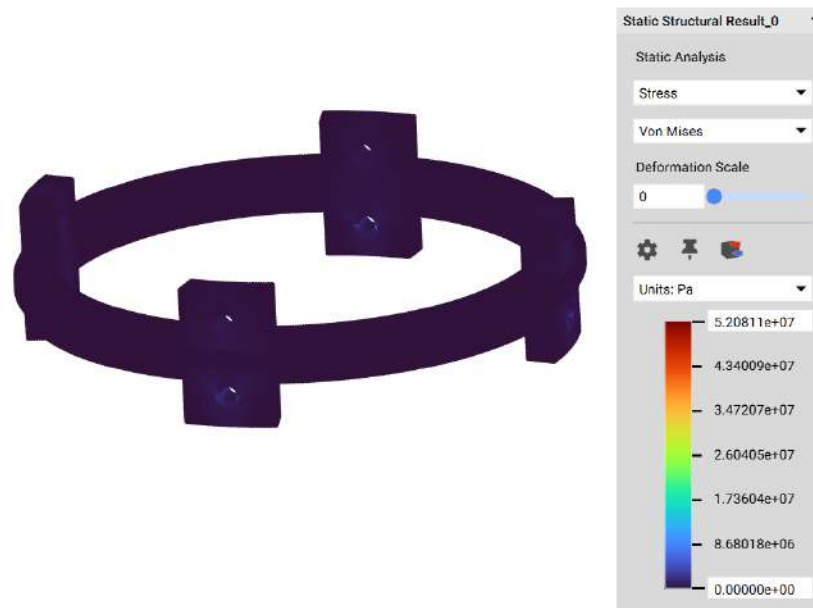


Figure 13: Full-scale nose cone retention ring Finite Element Analysis.

3.1.5 Lander Separation Bay

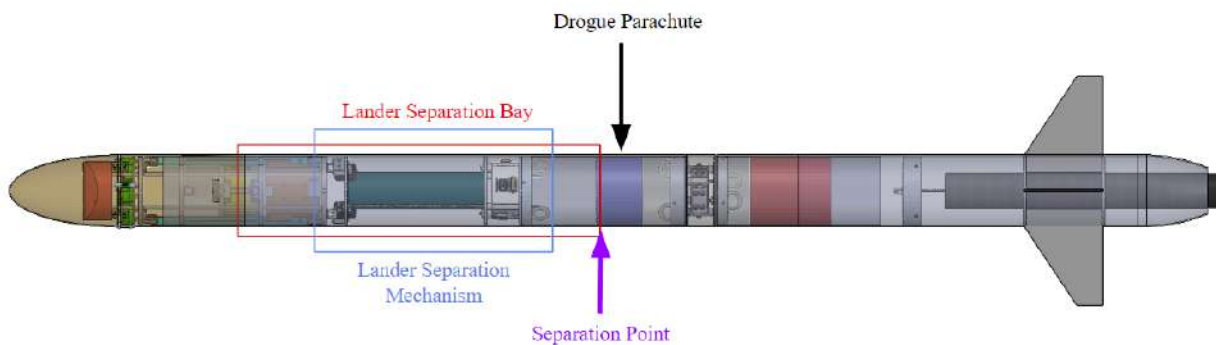


Figure 14: Full-scale lander separation bay.

The lander separation bay houses the lander separation mechanism and the fore drogue recovery hardware and is located aft of the payload bay and fore of the drogue bay, as indicated in Figure 14. Also depicted in Figure 14, the drogue parachute separation point is located at the aft end of the lander separation bay. The lander separation bay retains the payload via two electromagnetic rotary latches. The payload is released at an altitude of 300 ft, pending RSO approval, which allows for

the horizontal lander payload to successfully detach from the lander separation bay. Further details on the lander separation mechanism are provided in Section 4.2.1.

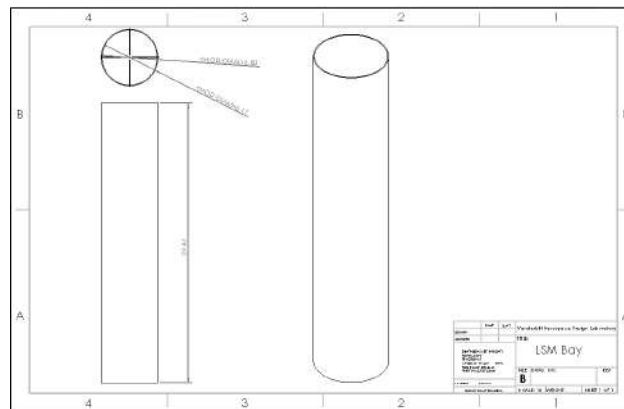


Figure 15: Full-scale lander separation bay drawing.

As previously noted, the lander separation bay separates from the drogue bay during drogue parachute deployment. To ensure the vehicle can withstand the loads experienced during parachute deployment, FEA was completed on the aluminum bulk plate connecting the drogue bay and parachute to the lander separation bay. This analysis simulated the maximum force experienced at main deployment through the eyebolt holes, with fixed supports at the screw holes. The load analysis confirmed that this bulk plate, fabricated with Aluminum, has a safety factor of 7.4 (see Figure 16); this ensures that the maximum loads felt during main deployment will not result in any structural damage to the lander separation bay.

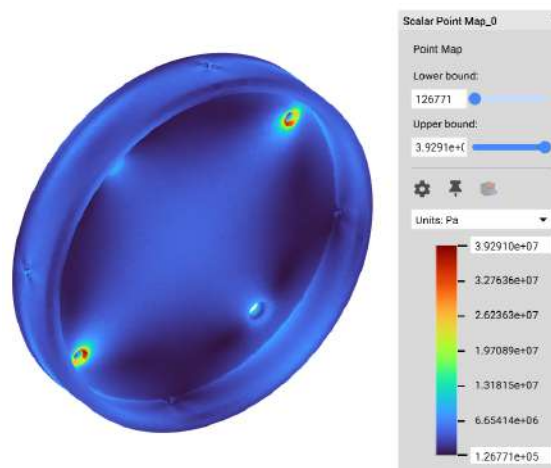


Figure 16: Full-scale lander separation bay bulk plate Finite Element Analysis.

3.1.6 Drogue Bay

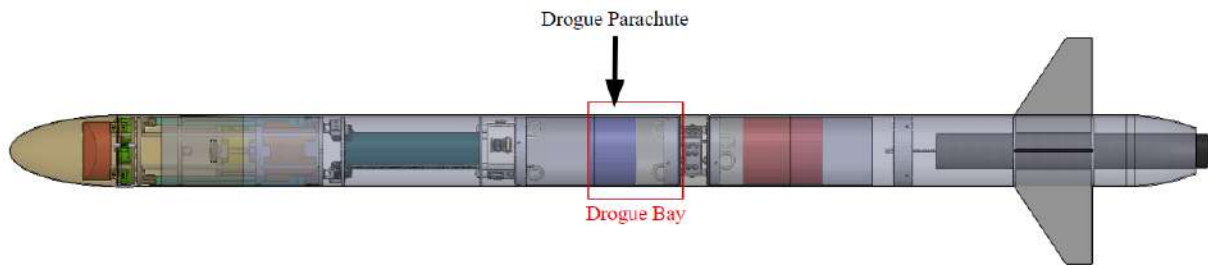


Figure 17: Full-scale drogue bay.

The drogue bay houses the drogue parachute, as indicated in Figure 17. The selected drogue parachute is the Fruity Chutes Elliptical 18” diameter parachute, which deploys at apogee as detected by the Altus Metrum EasyMini main altimeter. If deployment fails, the backup altimeter (of the same make and model) triggers deployment one second after apogee. Please refer to Section 3.3.2 for further details regarding the drogue parachute selection justification.

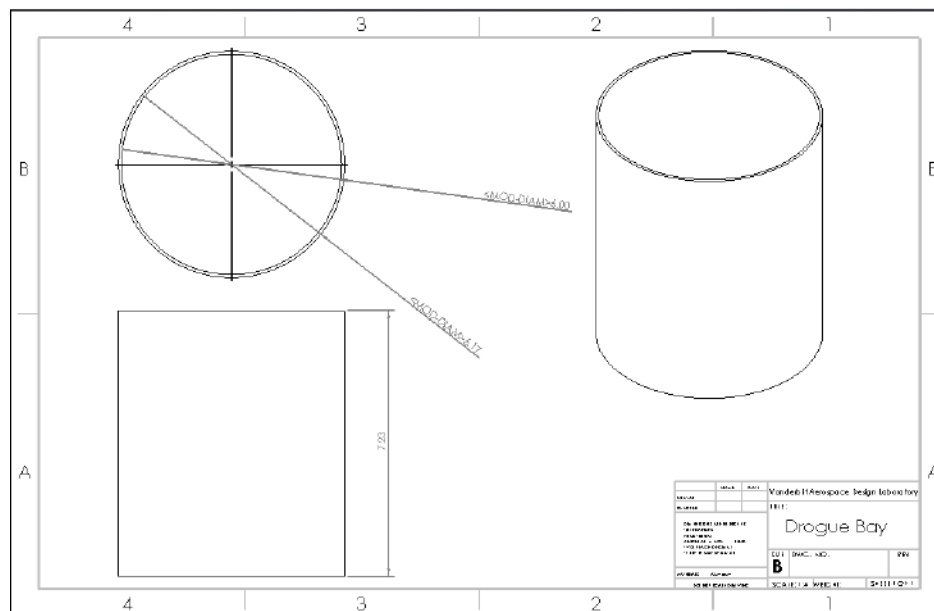


Figure 18: Full-scale drogue bay drawing.

To ensure structural integrity during flight and deployment, the fiberglass airframe is designed to withstand maximum expected loads, proven through testing detailed in Section 3.1.10. #2-56 nylon bolts are used as shear pins to secure the drogue bay until deployment; when the black powder charges ignite, this detonation will break the shear pins and allow the drogue parachute to deploy. The product data sheet for #2-56 bolts yields a maximum shear force of 32 lbf. To ensure that main deployment (following drogue) does not occur prematurely due to forces caused by drogue deployment, additional calculations were completed to confirm that the pins can withstand forces caused by lander inertia and parachute jerk. This analysis was performed by calculating the shear

force experienced by each shear pin during drogue parachute deployment. The usage of the below equations also assumed that the drogue parachute opened instantaneously, overestimating the shear force by assuming the maximum possible theoretical shock loading scenario since the parachute usually takes a bit more time to fully unfurl. These calculations, along with the maximum shear force of the #2-56 bolts, analytically prove the success of the pins in retaining the vehicle until the main separation.

$$F_{shear} = \frac{F_{shock}}{4} \quad (4)$$

$$F_{shock} = C_D * \frac{\rho V_{max}^2 A}{2} \quad (5)$$

$$V_{max} = \sqrt{V_{x,max}^2 + V_{y,max}^2} \quad (6)$$

$$V_{y,max} = \sqrt{2gL_{cord}} \quad (7)$$

$$(8)$$

F_{shear} refers to the load applied to each shear pin as a result of shock loading during parachute deployment (assuming equal load distribution on all 4 shear pins). F_{shock} refers to the shock load on each shear pin; in this case, the maximum velocity was found as a vector sum of the maximum horizontal velocity from simulations, from a 10-degree launch rail angle and 20 mph headwind, and the maximum vertical velocity due to the drogue shock cord length (10 ft). The following values were utilized and discerned from calculations: $C_D = 1.5$; $\rho = 1.225 \text{ kg/m}^3$; $A = 1.767 \text{ ft}^2$; $V_{y,max} = 35.90 \text{ ft/s}$; $V_{x,max} = 105 \text{ ft/s}$; $L_{cord} = 20 \text{ ft}$. Using these values, it was determined that the calculated force per shear pin is 10 lbs, with a safety factor of 3.2, ensuring reliable performance before and during drogue deployment.

3.1.7 Avionics Bay

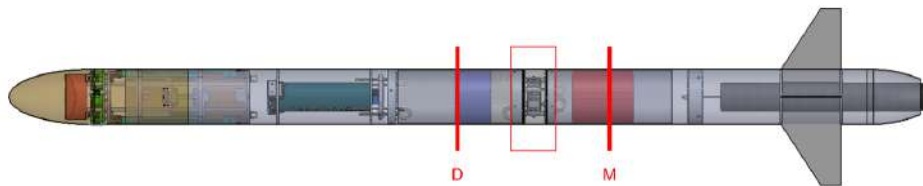


Figure 19: Full-scale avionics section assembly with the separation points indicated on either side (drogue on the left, main on the right).

As shown above in Figure 19, the avionics section is located aft of the drogue bay and fore of the main bay and contains the avionics housing and electronics. The separation events will be triggered via redundant EasyMini altimeters for both the drogue and the main. The avionics body tube will have pressure holes that are drilled to ensure proper data collection by the altimeters, the proper sizing of which has been informed by the subscale flights (Section 3.2.4) where a successful data collection of altimeter reading and apogee detection have validated pressure hole design.

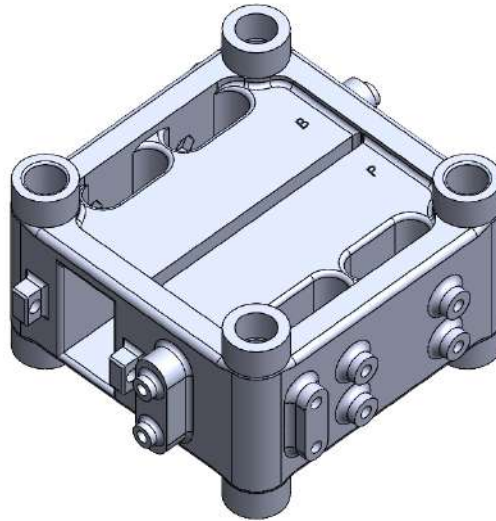


Figure 20: Full-scale avionics housing.

Figure 20 shows the avionics housing where the redundant altimeters, screw switches, and batteries will be secured. The avionics housing has been condensed and optimized to perform full functionality while being sandwiched between the drogue and main bays in a narrow space. Figure 21 shows the avionics housing assembled with the eyebolts, and fore and aft bulk plates that will also hold the blast charges. For detailed information on blast charges, the functionality of the avionics section, and recovery, refer to Section 3.3.

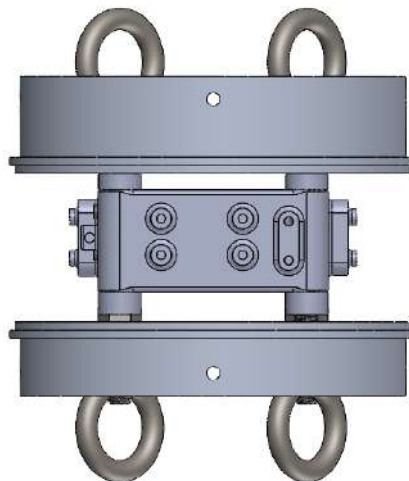


Figure 21: Full-scale avionics assembly.

In the figure above, the full-scale avionics assembly shows the avionics housing, which hosts all the necessary electronics for the blast charges and the altimeters, sandwiched between the fore and the aft base plates. The fore and aft base eye bolts are connected to the drogue or main parachutes. The base plates also host the blast charge caps which comprise of both a primary and a backup blast charge.

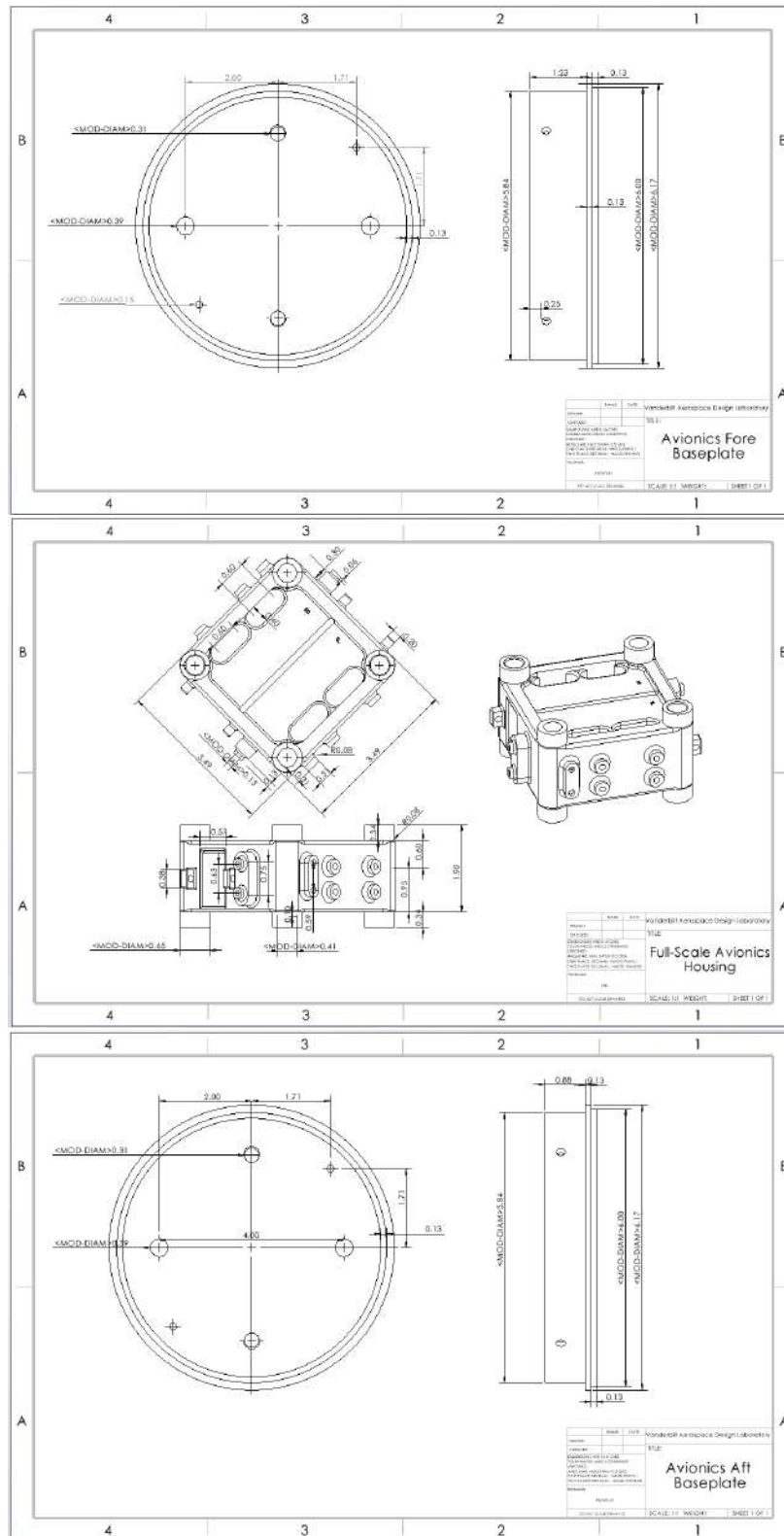


Figure 22: Engineering drawings for the avionics components (in order from top to bottom): fore baseplate, housing, and aft baseplate.

In Figure 22, the critical values of the three components from Figure 21 have been displayed through engineering drawings of each component. The generated drawings are for the avionics section of the full-scale rocket where both baseplates will be manufactured using aluminum and the housing will be 3D printed using ABS.

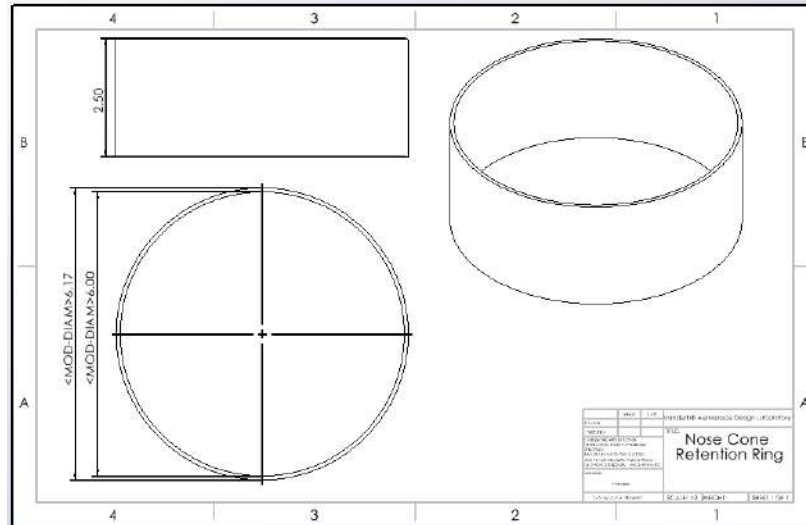


Figure 23: Full-scale avionics bay body tube engineering drawing.

The above figure shows the body tube that will be encasing the avionics section. This section will also contain drilled holes for pressure sensing through the altimeters. As learned through field testing and the data recorded through the subscale launch, decently sized holes are required to avoid incorrect altimeter sensing and consequently premature separation.

3.1.8 Main Bay

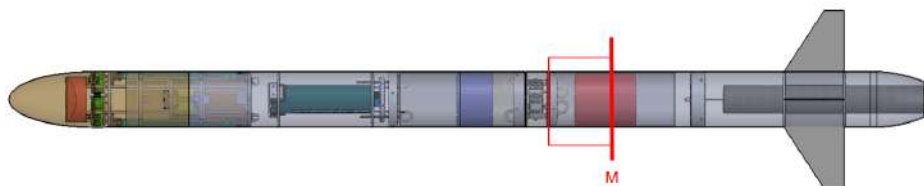


Figure 24: Full-scale main bay location on the rocket.

Figure 24 shows the location of the main bay on the vehicle. The main bay is located aft of the drogue bay so that once the main parachute is deployed, the nose cone, which hosts the lander, will be the lowest section during descent. This will allow for a smooth and unobstructed deployment of the lander, as compared to if the position of the drogue and main bays were switched. As with the drogue bay, the coupler is attached to the main bay so that the parachute is pushed out of the coupler, thus mitigating the risk of the parachute getting stuck inside the rocket. Further, the main parachute chamber is sufficiently large to ensure that requisite space is available to package the main parachute and recovery equipment.

For the separation of the main bay, as with the drogue bay, #2-56 shear pins are utilized which have a maximum shear force of 32-lbf. The team has verified this separation method and the use of these shear pins through the subscale flight, where nominal separation was observed.

Further, the methodology used in Section 3.1.4 can be applied here to ensure a sufficient safety factor for the aft avionics bulk plate. By multiplying the main parachute deployment deceleration by the mass of the supported section, it is found that the bulk plate must withstand a maximum load of 419 lbf. Results from FEA on this bulk plate, fabricated with Aluminum, are visualized below in Figure 25, show a Factor of Safety of 5.1, instilling confidence that the bulk plate will successfully survive rocket flight.

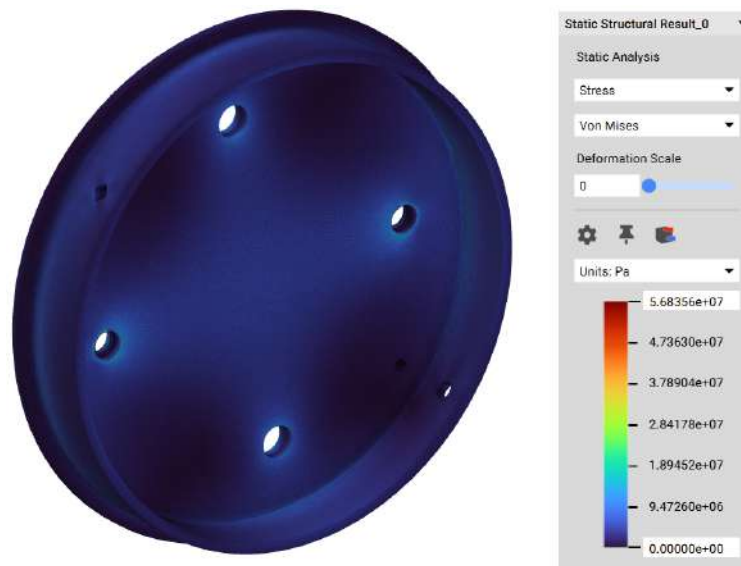


Figure 25: Aft avionics bulk plate FEA.

The figure displays the maximum stress the aft avionics bulkplate is able to withstand which then delivers a safety factor of 5.1. This also shows that the bulkplate will withstand maximum loading.

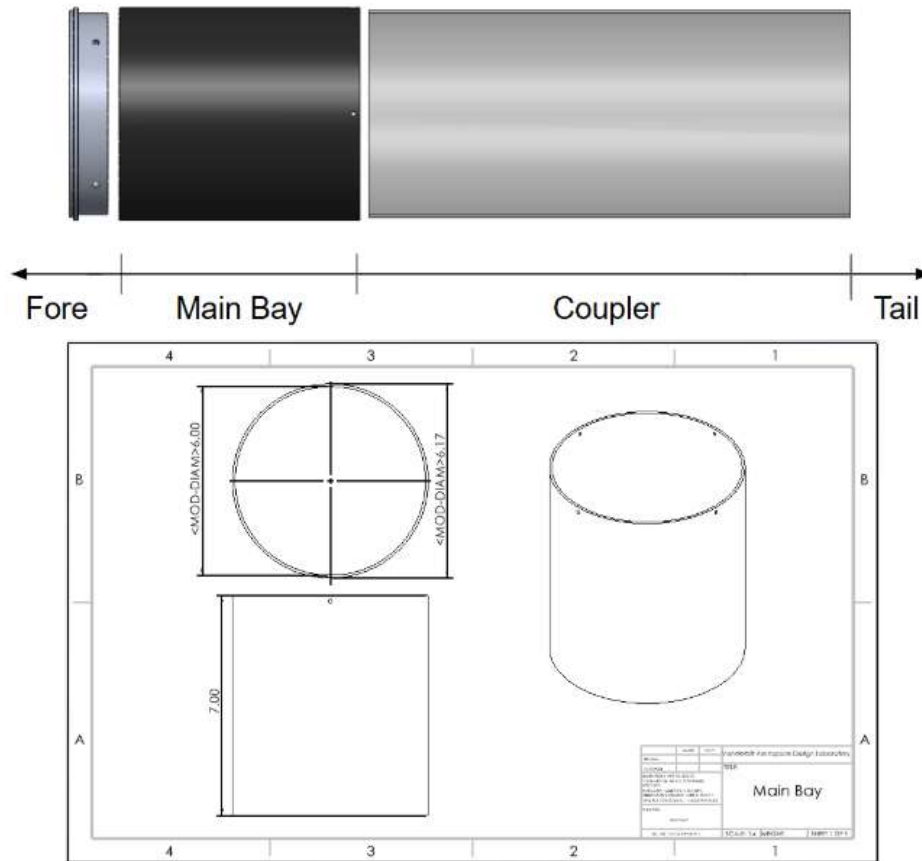


Figure 26: Exploded diagram of the avionics aft baseplate, main bay body tube, and the coupler connecting it to the tail section (top) and main bay full-scale rocket body tube design drawing (bottom).

The above figure shows the main bay body tube which will consist of holes for the shear pins. The shear pins will be inserted to hold the main bay and the tail section together during ascent and with the blast charges those shear pins shall break to allow for separation. The other side of the main bay is attached to the avionics section and stays intact upon separation.

3.1.9 Tail Section



Figure 27: Full-scale tail section location on the rocket.

In Figure 27, the tail section of the rocket is highlighted. As also mentioned in PDR, the team is continuing with a four-fin design—all fins are geometrically similar. The fin design, a clipped delta

shape as seen in Figure 28, was optimized and sized to ensure an SSM of approximately 2.2. The team previously proposed using waterjet-cut fiberglass sheets for the fins with fiberglass fabric fillets, but the team is moving forward with carbon fiber fins with fiberglass fillets. This switch has been made to reduce the mass of the tail section in addition to complying with the already available inventory. From subscale manufacturing, the team proved that fiberglass fiber can still be applied as fillets despite the fins being made from carbon fiber. The successful subscale flights and recovery instill confidence in the team to replicate the subscale fin material and retention method for the full-scale vehicle.

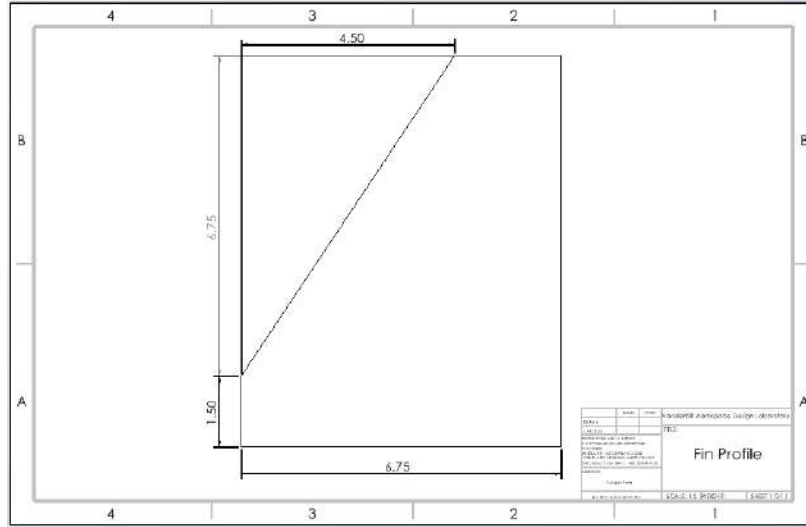


Figure 28: Full-scale fin profile engineering drawing.

The tail section includes the motor hardware, the parachute hook plate with an eyebolt, the motor tube, centering rings, and the fins. The motor is encased within the motor tube and is fixed in place using the motor retention rings. The centering rings help to stabilize the fins and play a role in ensuring structural integrity during flight. The centering rings also allow for effective load distribution as high pressure and drag forces are faced by the fins. This then ensures that there's no structural failure in one part of the tail section by efficiently transferring the high loads to the body tube. As with the subscale vehicle, outer fiberglass fillets will further retain the motor tube and centering rings to the airframe. See Figure 29 for a detailed drawing of all parts the tail section comprises including the parachute hook plate, centering rings, and the motor tube.

Per VADL Requirement 2.2.1.5, the fins must be designed to avoid fin fluttering. This phenomenon can result in structural damage to the fins, affecting the stability of the vehicle.

To calculate the critical fin flutter velocity, V_f , the following equation can be used:

$$V_f = a \sqrt{\frac{G}{\frac{1.337 A_R^3 P (\lambda + 1)}{2(A_R + 2)(t/c)^3}}} \quad (9)$$

In the above equation, a is the speed of sound, G is the shear modulus of the material, A_r is the aspect ratio of the wing, P is the local ambient pressure, λ (λ) is the taper ratio of the fin, t

is the thickness of the fin, and c_r is the root length of the fin.

Using the following, the aspect ratio can be calculated. Based on the fin profile from Figure 28, the cord tip length is 2.25 inches and the cord root length is 6.75 inches.

$$A_R = \frac{(2)(6.75)}{6.75 + 2.25} = 1.5 \quad (10)$$

Additionally, λ (λ), which is the ratio between the cord tip length and the cord root length, can be calculated using the following:

$$\lambda = \frac{c_t}{c_r} = \frac{2.25}{6.75} = 0.33 \quad (11)$$

With all the variables sorted, the critical flutter velocity is calculated as follows:

$$V_f = 1125 \sqrt{\frac{5.22E6}{\frac{1.337(1.5)^3(14.7)(0.33+1)}{2(1.5+2)(0.125/6.75)^3}}} = 1825 \text{ ft/s} \quad (12)$$

From the team's flight simulations, the maximum velocity of the rocket does not exceed 600 ft/s. Therefore, according to the calculations above, the rocket will not undergo any fin fluttering. This analysis strengthens the team's confidence in the fin design and material selection to provide a stable and successful rocket flight.

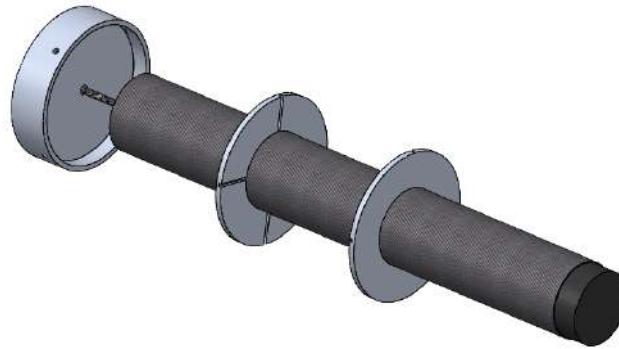


Figure 29: Full-scale tail section assembly without body tube. Shown components (left to right) are the parachute hook plate, motor tube, centering ring, and motor retention ring.

Using Figures 29 and 32, the visualization of the motor thrust load path becomes clearer. Since the source of the load is the motor thrust, this loading is primarily an axial load. This means that all components in the load path must be able to withstand the appropriate compressive loads. To further elaborate:

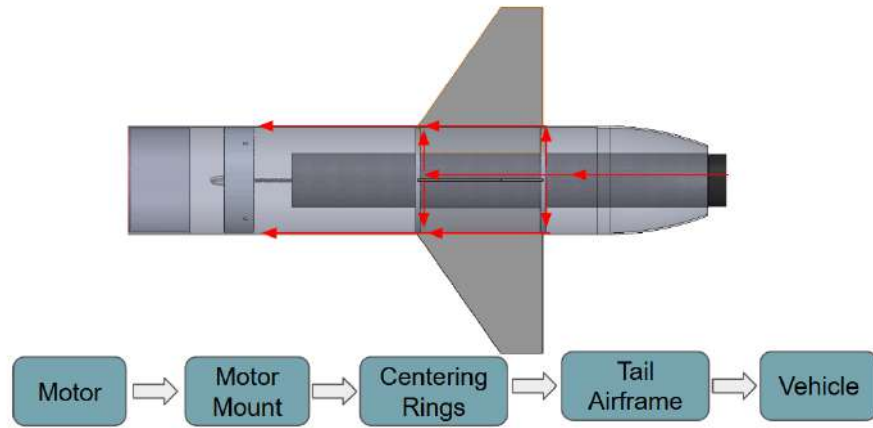


Figure 30: The load path of the aft section of the rocket upon motor thrust.

In this particular order, the load path is developed for the motor thrust load. The motor thrust exerts a load on the motor mount. Since the centering rings prevent any axial or radial movement, the load is transferred to the rings. From here, the centering rings evenly distribute this load to the body tube, which the centering rings use as support to retain the motor tube in place. The load is then transferred from the local body tube section to the rest of the rocket's structure and evenly distributed out to ensure no localized stress deformations result in fracturing or other damages.

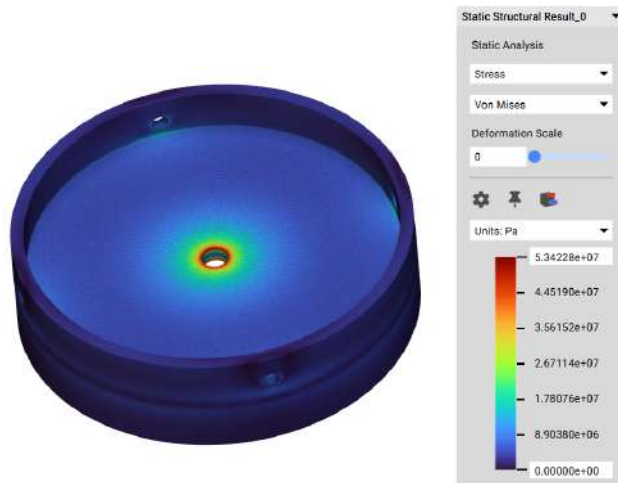


Figure 31: FEA on the parachute hook plate.

The FEA on the parachute hook plate which will be fabricated with Aluminum, Figure 31, shows that the bulkplate has a safety factor of 5.4 and therefore survives maximum loading.

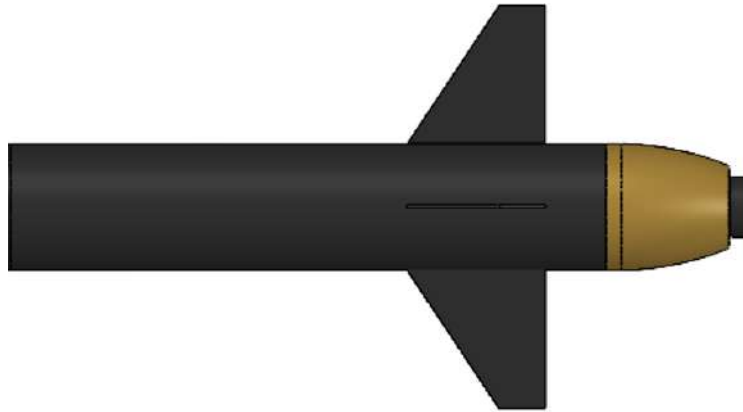


Figure 32: Full-scale tail section assembly with the body tube.

The aft-most component of the tail section is the boat tail. The boat tail helps reduce the drag faced by the vehicle during flight. The boat tail was designed and optimized to reduce the pressure drag and allow a smoother flow separation, which further enhances its aerodynamic performance. The tangential ogive boat tail design has a smoother flow separation as compared to a conical boat tail, which is a major reason why the team is proceeding with the former option as opposed to the latter. Further, the performance of this boat tail design was validated through the subscale flights. The fore and aft diameters of the boat tail are 6.17 and 4.00 inches, respectively.

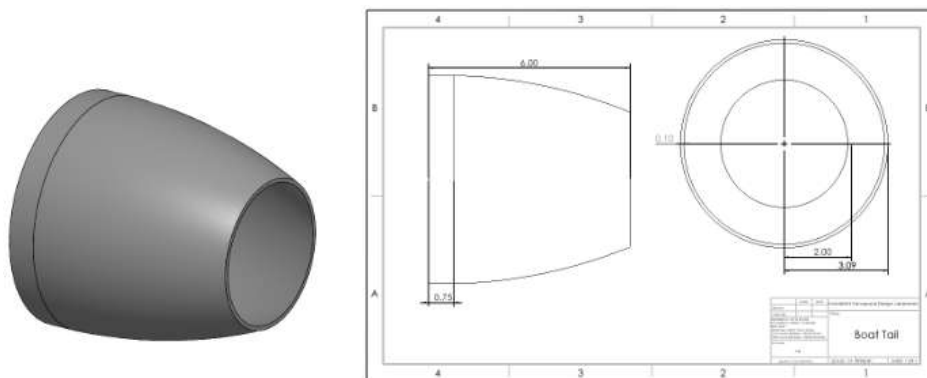


Figure 33: Full-scale boat tail 3D rendering (left) and engineering drawing of the boat tail (right) detailing dimensions.

The boat tail will be printed with ABS using a high-resolution 3D printer, replicating the process used for the subscale boat tail. This reduces manufacturing time, thus allowing the team to devote more effort to higher-intensity fabrication tasks. Since the boat tail in this vehicle design essentially acts as a shell to smooth the airflow and reduce drag, the boat tail is considered to be non-load-bearing. The material choice for this, then, purely depends on the structural integrity of the material and heat resistance. As observed in the subscale flight, ABS was able to withstand the rigors of flight and landing without any damage to the boat tail. This result instills confidence in the team's boat tail shape and material selections for the full-scale vehicle.

3.1.10 Airframe Material Properties

The team conducted a series of 3 load tests on sections of fiberglass airframe in order to determine its material properties and verify its suitability as the airframe material selection. The fiberglass sections were placed into a pneumatic load frame capable of exerting up to 5000 kgf. This load frame was capable of imposing tensile and compressive loads on the airframe sections. Data was directly collected into an Excel file during runs. Procedures for testing are further outlined in Section 6.1.1. Fiberglass airframe sections used had a length of 9 in with a diameter of 4 in. 4 in airframe sections were used prior to subscale launch vehicle fabrication to validate the selected fiberglass material. The material that shall be used to manufacture the full-scale vehicle is the same material as the subscale vehicle.

Critical Hoop Stress (Test Code 1.1.5)

There are two events that cause significant hoop stress within the airframe material: drogue parachute deployment, and landing. During drogue parachute deployment the internal pressure of the chamber rises to 4.82 psi, or 33.2 kPa. During landing, the lander experiences upwards of 20 G of acceleration. With a payload mass of about 7.63 lbs, the lander will experience a load of 152.6 lbf on its airframe upon landing. These two loads exert two different hoop stresses on the airframe. For the pressure-induced hoop stress, the stress can be calculated as such:

$$\sigma = \frac{Pd}{2t}, \sigma = \frac{4.82 \text{ psi} * 6.17 \text{ in}}{2 * 0.079 \text{ in}} = 188.2 \text{ psi} / 1.3 \text{ MPa} \quad (13)$$

In the above equation, P is the internal pressure, d is the diameter of the airframe, and t is the thickness of the airframe walls. For the full-scale vehicle, the airframe has a diameter of 6.17 in and a thickness of 0.079 in. The radial compression-induced hoop stress can be calculated in the following equation:

$$\sigma = \frac{F}{tL}, \sigma = \frac{152.6 \text{ lbf}}{0.079 \text{ in} * 17.8 \text{ in}} = 108.5 \text{ psi} / 0.8 \text{ MPa} \quad (14)$$

For the purposes of this analysis, the 1.30 MPa value shall be taken as the maximum expected hoop stress experienced by the airframe. The cylinder was loaded in radial compression beneath a flat metal plate to prevent the load from concentrating into the side of the airframe and fracturing it. Two 3 lbm weights were placed on either end of the airframe to keep it from sliding during the initial stages of the load test.

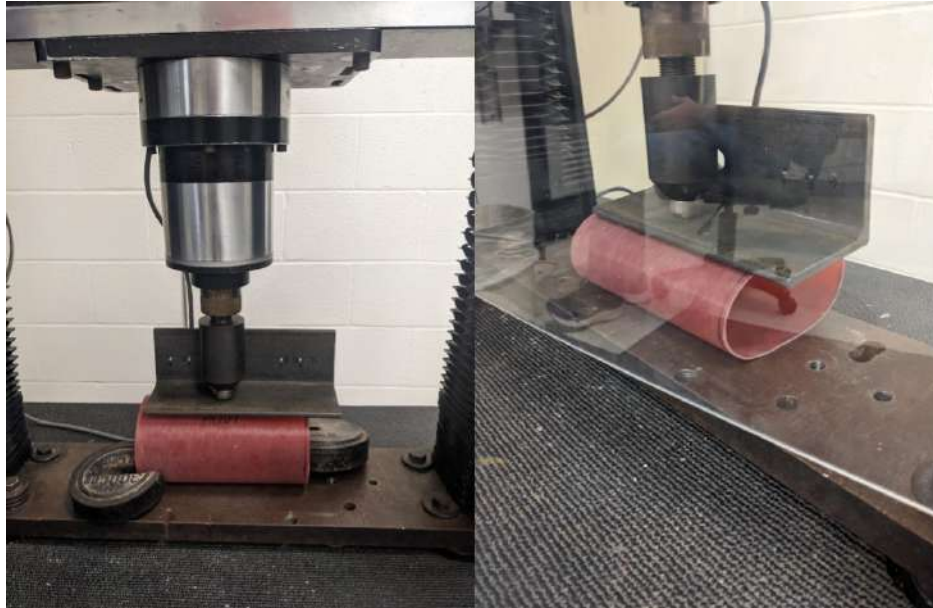


Figure 34: 9 in long fiberglass airframe section in load frame during hoop stress test. The left shows the airframe at the start of the test, and the right shows the airframe at the end.

The airframe section was compressed until the load frame hit its limit switch and was unable to continue further. The load frame reached a maximum load of 22.2 kN, corresponding to 67.2 MPa.

The fiberglass material did not fail during this test and returned to its original shape without any visible cracks or damage of any sort. The load of 67.2 MPa compared to the maximum expected loading of 1.3 MPa provides a **factor of safety of 51.7**, which greatly exceeds the safety factor of 2.5 (VADL 2.2.2) imposed by the team.

Critical Bolt Tear Out Stress (Test Code 1.1.6)

The main event that causes bolt tear-out stress is the main parachute opening. During this, the main parachute bulk plates experience their highest tensile forces due to the acceleration of up to 32 G. These bulk plates are attached to the airframe radially with bolts, which can damage the airframe to the point it is not airworthy, or able to be flown again, within 2 hours of a completed launch, or even shear through the airframe. The bulk plate experiencing the highest load is the recovery bay bulk plate connecting to the main parachute. The combined mass it is supporting is 18.5 lbs. At 32 G, the bulk plate experiences a tensile force of 592 lbf.

Bolt tear out stress is dependent on the airframe thickness, distance between the hole to edge, bolt hole diameter, and number of bolts. The equation is as follows:

$$\sigma = \frac{F}{n(e - \frac{d}{2}) * t}, \sigma = \frac{592 \text{ lbf}}{4(7.25 \text{ in} - \frac{.25}{2}) * 0.079 \text{ in}} = 262.9 \text{ psi} / 1.8 \text{ MPa} \quad (15)$$

Where F is the tensile load, n is the number of bolts, e is the distance between the center of the bolt hole to the edge, d is the diameter of the bolt, and t is the airframe thickness. The bulk plate in question is located at 7.25 in from the edge of the airframe and has four 1/4-20 bolts attaching it

to the airframe. This stress assumes that all load passes through these 4 bolts. In reality, there is another bulk plate with 4 bolts attached to the same airframe with threaded rods passing through both to transfer the load. For the sake of analysis, all load will be treated as if it passes through the airframe. Therefore, the maximum expected stress is 1.8 MPa.



Figure 35: Beginning (left) and after (middle and right) Bolt tearout test.

During this test, two bulk plates were attached to the airframe using four #10-32 bolts each. The top bulk plate holes were located 2.5 in from the edge, and the bottom bulk plate holes were located 3 in from the edge. During this test, the load frame was run all the way to 14.2 kN of tensile force, which cut long slots into the airframe, and severely bent the bolts. For the sake of analysis, the team decided to treat the point where the strain reached 10 mm to be the failure point, as that is about 2 diameters of the #10-32 holes. Additionally, at this point, the bolts themselves did not deform either. Because reusability is vital to the project, the point when bolts are bent is also a failure point.

The critical bolt tear out stress was determined to be 8 MPa, which provides a factor of safety of **4.4**. This exceeds the minimum safety factor of 2.5 imposed by the team in VADL requirement 2.2.2.

Axial Compression Stress (Test Code 1.1.7)

The main event that imposes compression loading on the airframe is launch. During launch, the entire mass of the rocket is compressed onto the airframe closest to the motor. At an acceleration of up to 12 G, and a rocket wet mass minus motor hardware of 34.3 lb, the lowest airframe section experiences about 411.6 lbf.

Compressive stress is represented as the compressive force divided by the cross-sectional area.

$$\sigma = \frac{F}{\frac{\pi}{4}(d_o^2 - d_i^2)}, \quad \sigma = \frac{411.6 \text{ lbf}}{\frac{\pi}{4}(6.17^2 - 6.012^2)} = 272.3 \text{ psi} / 1.9 \text{ MPa} \quad (16)$$



Figure 36: Airframe section during axial loading test.

Where d_o and d_i are the airframe's outer and inner diameters respectively. To verify that the airframe can withstand this force, the airframe was placed into the load frame. The test run went all the way to 22.6 kN of compressive force. This corresponds to a stress of 49.5 MPa. This value provides a factor of safety of **26**, which is greater than the imposed safety factor of 2.5 imposed by VADL requirement 2.2.2.

3.2 Subscale Flight Results

3.2.1 Flight Objectives

In compliance with NASA Requirement 2.18, VADL completed two subscale flights on December 8, 2024. These demonstration flights sought to validate several aspects of the full-scale vehicle and payload, as outlined in Table 7 below.

Table 7: Subscale flight objectives and corresponding requirements.

Subscale Flight Objective	Relevant Requirement(s)
Successfully launch and recover the vehicle	NASA 2.1, 2.18, 3.1.2; VADL 2.2.1.4, 3.3, 3.4, 3.5, 3.6, 3.7, 3.9, 3.10
Refly the vehicle within 2 hours	NASA 2.3, 2.5; VADL 2.1.4
Achieve an apogee within 5% of the projected value.	VADL 2.1.1

Table 7 continued from previous page

Subscale Flight Objective	Relevant Requirement(s)
Verify payload data collection	NASA 4.2.1; VADL 4.7.1, 4.7.2, 4.7.4, 4.7.5, 4.7.6, 4.7.7, 4.7.8
Verify in-rocket antenna extension	VADL 4.5.5
Verify payload RF transmission	NASA 4.2.1, 4.2.3, 4.2.4, 4.2.6; VADL 4.6.1, 4.6.2
Benchmark flight simulations and CFD framework to inform full-scale flights	NASA 2.2, 3.3, 3.11, 3.12; VADL 2.3.7, 2.4.1

The team intended to successfully launch and recover the subscale vehicle; accomplishing this goal would prove out the manufacturing process and instill confidence in the team to move forward with full-scale fabrication and flight. Additionally, the team wished to impose the requirement of reflying the vehicle within two hours. This objective would prove the reusability of the vehicle, as mandated by NASA Requirement 2.3, along with the requirement that the vehicle be assembled within two hours (NASA Requirement 2.5). The goal was also made to reach an apogee within 5% of the predicted value. In accomplishing this objective, the team would validate their flight simulation methodologies. Along with these vehicle goals, several payload objectives were placed. Essentially, the team wished to prove out the ability to complete the payload mission sequence in an in-rocket configuration, as would be necessary in the case that jettison of the payload were denied. The outcomes of these objectives are covered in Section 3.2.4.

3.2.2 Vehicle Scaling

A dimensional comparison of the full-scale and subscale vehicles is shown in Figure 37. The mission elements that are replicated with the subscale flight are also depicted. To fulfill NASA Requirement 2.18.5, the subscale vehicle is 61 in long and has a diameter of 4 in, both scaled roughly equally to yield an aspect ratio approximately the same as that of the full-scale vehicle. Beyond these dimensions, however, the subscale vehicle was designed and built with the intention of maintaining as many parameters constant as possible. The full-scale and subscale vehicle specifications are compared in Table 8.

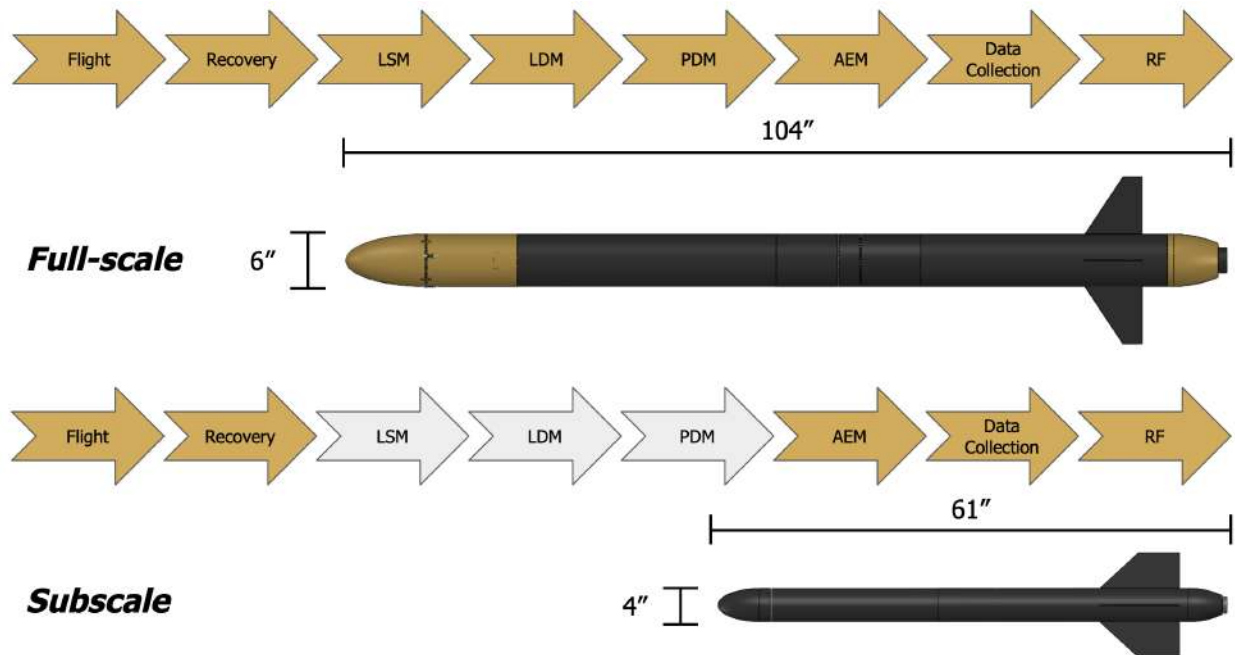


Figure 37: Full-scale and subscale vehicle comparison.

Table 8: Comparison of full-scale and subscale vehicle parameters.

Parameter	Subscale	Full-scale	Scaling Factor
Length	61 in	104 in	59%
Diameter	4.02 in	6.17 in	65%
Aspect Ratio	15.17	16.86	90%
Airframe Material	Fiberglass	Fiberglass	N/A
Wet Mass	13.1 lbm	41.6 lbm	31%
Motor / Retention Method	Aerotech I366 / Motor tube epoxied to fins and centering rings	Cesaroni L1720 / Motor tube epoxied to fins and centering rings	N/A
Thrust-to-Weight Ratio	5.8	9.1	64%
SSM (On Pad)	2.21	2.20	~1
L.R.E. Velocity	76.8 ft/s	77.5 ft/s	~1
Nose Cone Shape	Elliptical	Elliptical	N/A
Fin Shape / Material	Clipped Delta	Clipped Delta	N/A
Bulk Plate Material / Retention Method	Aluminum / Radial bolts	Aluminum / Radial bolts	N/A
Avionics Altimeter	Entacore AIM	EasyMini	N/A
Parachute / Deployment	48" Iris Ultra / At apogee	18" Classic Elliptical / At apogee; 96" Main / 600 ft	N/A
Shock Cords	6000 lb, ½ in Kevlar	6000 lb, ½ in Kevlar	1
Quick Links	4000 lb steel	4000 lb steel	1
Payload Configuration	In-rocket	Jettisoned and independently landed	N/A

By fixing variables such as airframe material, nose cone and fin shape, motor and bulk plate retention, L.R.E. velocity, and SSM, the team was able to produce a rocket whose flight would be a true proof of concept for the full-scale vehicle. Meanwhile, length, diameter, and thrust-to-weight ratio were all scaled roughly the same. As for the payload itself, it was decided to instead test the case of in-rocket RF transmission due to difficulties with scaling down the lander payload to fit inside a 4-in airframe. The team's drop test protocol continued to test the prototypes of the full-scale lander, covered more in Section 4.

3.2.3 Vehicle Fabrication

Nose Cone

A detailed manufacturing process was followed in the construction of the subscale vehicle, unfolding over three weeks from November 1 to November 22. The first step was to manufacture the nose cone. With the use of fiberglass for the vehicle airframe, a novel selection for VADL, the team decided to use fiberglass cloth to construct the nose cone to maintain material uniformity. The full-scale nose cone will also be made with fiberglass. This material selection during subscale testing allowed the team to verify its nose cone manufacturing process to inform full-scale manu-

facturing. It was experimentally verified that 12 layers would be sufficient to attain a thickness that would match the outer diameter of the airframe.

First, a mold was printed out of ABS to match the inner diameter of the nose cone (Figure 38). The mold was then sanded using increasing grits, up to 2000 grit, to ensure a smooth working surface. Three coats of TR-104 Hi-Temp Mold Release paste were then applied to the mold, waiting 30 minutes between each coat. While waiting, four sets of 12 fiberglass cloth pieces were cut to match the length of the nose cone, with a 1 in offset, and the width of approximately a 90 degree sector of the nose cone (Figure 38). Once all coats had been applied, the fiberglass cloth pieces were epoxied and laid on the mold. After all cloth pieces were laid, the fiberglass was covered with rice paper and cotton to absorb excess epoxy. The nose cone was enclosed in a vacuum seal bag and placed in the hot box to cure for 14 hrs at 140F. Upon removal from the hot box, the nose cone was then sanded up to 2000 grit for a clean surface finish (Figure 38). The end result is an aerodynamic and structurally robust nose cone.



Figure 38: From left to right: nose cone mold; cutting fiberglass cloth pieces; nose cone upon removal from hot box; sanded nose cone.

Tail Section

The next step in the manufacturing process was to fabricate the tail section. Arguably both the most difficult and important part of the fabrication process, it is essential to construct an aligned and secure tail section. As such, the motor is retained at two levels: the first is a set of fiberglass fillets securing the motor tube to the fins and centering rings, and the second is another set of fiberglass fillets securing the airframe to the fins.

First, a 16 in motor tube was cut using 38 mm Blue Tube, selected such that the motor casing forms a snug fit within the tube. The fins were waterjet cut out of DragonPlate carbon fiber, and the centering rings were 3D printed using Markforged Onyx for its high strength-to-weight ratio. To ensure proper fin and centering ring alignment, an alignment jig was assembled using laser cut MDF plates and ½ in threaded rods (Figure 39). 2-ton epoxy was applied to the fins and centering rings and allowed to cure for 24 hours. Next, four layers of fiberglass cloth fillets were cut to cover the motor tube space in between each set of fins, with 1 in of overlap with each fin. These cloth pieces were epoxied onto the motor tube and fins, covered with rice paper and cotton, enclosed in a vacuum seal bag, and placed in the hotbox at 140F to cure for 14 hours (Figure 39). While

waiting for the epoxy to cure, the tail section fiberglass airframe was measured and cut. Slots were measured and cut for the fins, and four sets of four pieces of fiberglass cloth were cut for the outer fillets. The epoxying and curing process was then repeated for the outer fillets. The end result is a well-aligned and structurally secure tail section, suitable for withstanding the high loads of motor burn. The various stages are depicted in Figure 39 below.

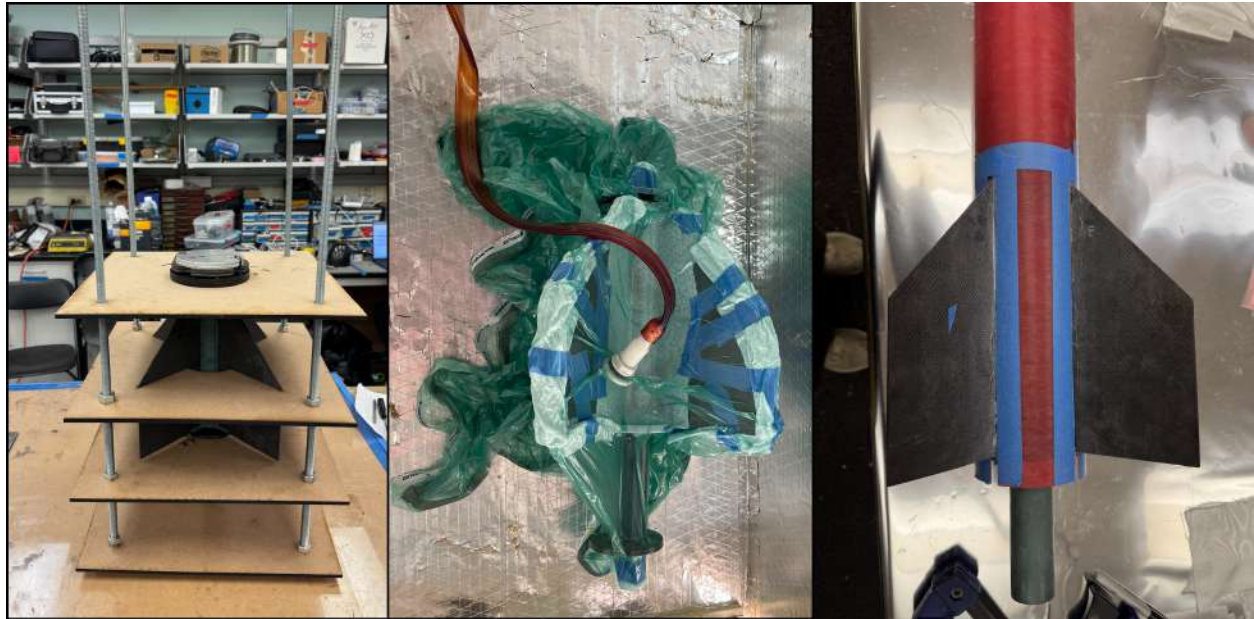


Figure 39: From left to right: motor tube, fins, and centering rings in alignment jig; inner layer of fiberglass fillets curing in the hot box; motor tube, fins, and centering rings slid into airframe prior to outer fillet application.

After completing the tail section, the separation and boat tail couplers were cut. Since the blast charge would be sent from the tail section to the fore section, it was decided to epoxy the coupler to the tail section to minimize the risk of the parachute getting stuck in the rocket after separation. The portions of the couplers being epoxied were first sanded and scored, then a thin layer of 2-ton epoxy was applied to the couplers. The couplers were then placed inside the airframe to cure for 24 hours. Additionally, the motor retainer was secured to the motor tube using JB Weld. While waiting for these to cure, the fore section fiberglass airframe was cut, thus completing the shell of the vehicle.

Avionics

The avionics section was contained in the tail section of the vehicle. A housing for the altimeters, batteries, and screw terminals was designed and 3D printed, shown below in Figure 40. This housing was connected to the avionics bulk plate via $\frac{1}{4}$ "-20 aluminum threaded rods and retained in place via locknuts. Following the fabrication process, testing was conducted to ensure proper altimeter calibration. This testing procedure is outlined in further detail in Section 6.1.

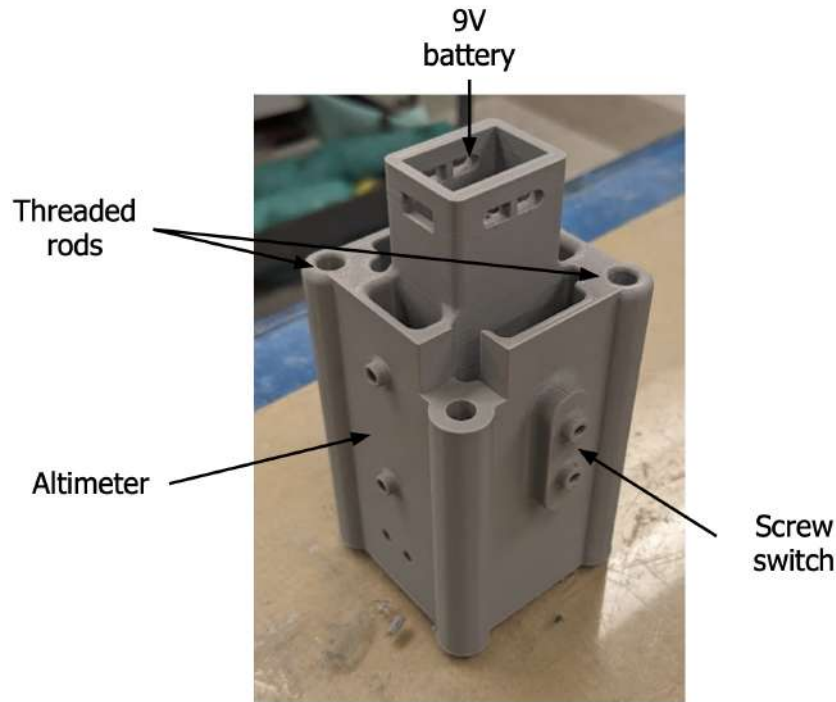


Figure 40: 3D-printed avionics housing, with mounting locations indicated.

Payload

The payload was constructed using several 3D-printed pieces sandwiched between two aluminum bulk plates. 5/8-16 threaded rods were used to fix all pieces together. The payload contains 3 antennas, a Raspberry Pi, and a Light APRS board. The electronics sit at the fore section of the payload closest to the nose cone, while antennas sit aft. The antennas were wired in series with a coaxial connector to connect it to the Light APRS board. To ground the antenna, the connector was jumped to one of the threaded rods which grounds the entire metal skeleton. The antennas are coiled within the Antenna Extension Mechanisms (AEMs) and holes were cut out of the airframe to allow the antennas to extend outwards. 3D-printed covers were printed to prevent aerodynamic forces from striking the antennas during flight.

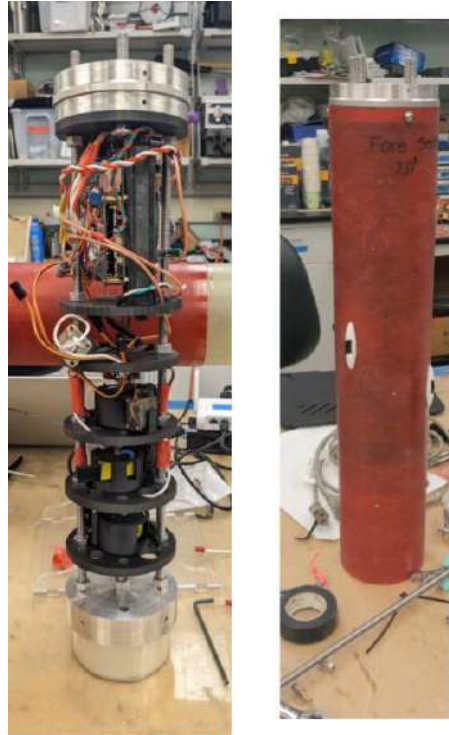


Figure 41: Subscale payload fully assembled. The left shows the payload without an airframe, right shows it within.

The purpose of the payload is to verify mission success without payload jettison. In order to fulfill VADL requirement 4.5.5, three antennas ensure that regardless of landing orientation 1 antenna will extend away from the ground.

Integrated Vehicle

With all airframe sections and internal components manufactured, it was time to fully assemble the vehicle. At this point, bulk plate and shear pin holes were carefully measured to ensure alignment and drilled. The fully assembled vehicle is shown in Figure 42 below. Following full vehicle assembly, a deployment test was conducted to verify the blast charge sizing and section separation. The full procedure for this test is outlined in Section 6.1.



Figure 42: Subscale vehicle fully assembled.

3.2.4 Flight Results



Figure 43: Subscale vehicle on the launch rail.

Flight Summaries

VADL flew two subscale flights on December 7, 2024, at Neal Farms in Lebanon, TN. The team arrived at the field at 9:00 am, began preparing the vehicle and payload at 9:30 am, and flew at 11:30 am and 1:00 pm. The launch day timeline is summarized below in Figure 44. Detailed launch day checklists were prepared leading up to the flights, thus ensuring that the preparation and assembly protocols were carried out in a safe and organized manner. Each checklist detailed the necessary materials, personnel, and procedure, and required the signatures of the responsible team member(s), safety officer, and rocketry mentor. These checklists informed the development of the full-scale operations procedures and checklists detailed in Section 5.6.

9:00-9:30 AM	9:30-10:00 AM	10:00-10:30 AM	10:30-11:00 AM	11:00-11:30 AM (a)
Arrived at launch field @ 9:00 AM - Unpacked equipment - Reviewed launch day checklists and procedures	Began vehicle and payload preparations @ 9:30 AM - Assembled recovery equipment - Performed preliminary payload checks	- Inspected altimeters - Wired blast charges	- Assembled motor - Powered on payload and performed final tests	- Assembled vehicle sections - Obtained wet mass and CG - Loaded vehicle on launch rail Launch 1 @ 11:30 AM
11:35 AM-12:00 PM	12:00-12:30 PM	12:30-1:00 PM	1:05-1:30PM	1:30-2:00 PM (b)
- Disarmed altimeters - Obtained pictures of as-landed sections - Returned vehicle sections to tent - Performed inspections for structural damage	- Assembled motor - Wired blast charges - Drilled additional pressure taps for altimeters	- Assembled vehicle sections - Obtained wet mass and CG - Loaded vehicle on launch rail Launch 2 @ 1:00 PM	- Disarmed altimeters - Obtained pictures of as-landed sections - Returned vehicle sections to tent - Performed inspections for structural damage	- Packed equipment - Inspected launch site to ensure no trash or debris was left Departed @ 2:00 PM

Figure 44: Timelines for (a) first subscale launch and (b) second subscale launch.

As shown in the timeline, VADL assembled and flew the rocket within two hours and then flew it again in under two hours, thus satisfying NASA Requirements 2.3 and 2.5 and VADL Requirement 2.1.4 for assembly time and reusability of the launch vehicle. In both flights, the vehicle was successfully launched and recovered, fulfilling NASA Requirements 2.1, 2.18, and 3.1.2 and VADL Requirements 2.2.1.4, 3.3, 3.4, 3.5, 3.6, 3.7, 3.9, and 3.10. Further, no structural damage was incurred to any vehicle section through both flights, as seen in Figure 45 and Figure 46 of the landed sections and full landed layout.



Figure 45: Landed configurations of the tail section (left) and fore section.



Figure 46: Full landed layout of the vehicle sections and recovery equipment.

Images and time stamps from the second flight are shown below in Figure 47. The first flight experienced a premature parachute deployment due to insufficient pressure vent hole sizing for the altimeters, resulting in an apogee of 969 ft. This issue was corrected for the second flight, which saw an apogee of 1076 ft. Further analysis of the flights will be detailed in Section 3.2.5.

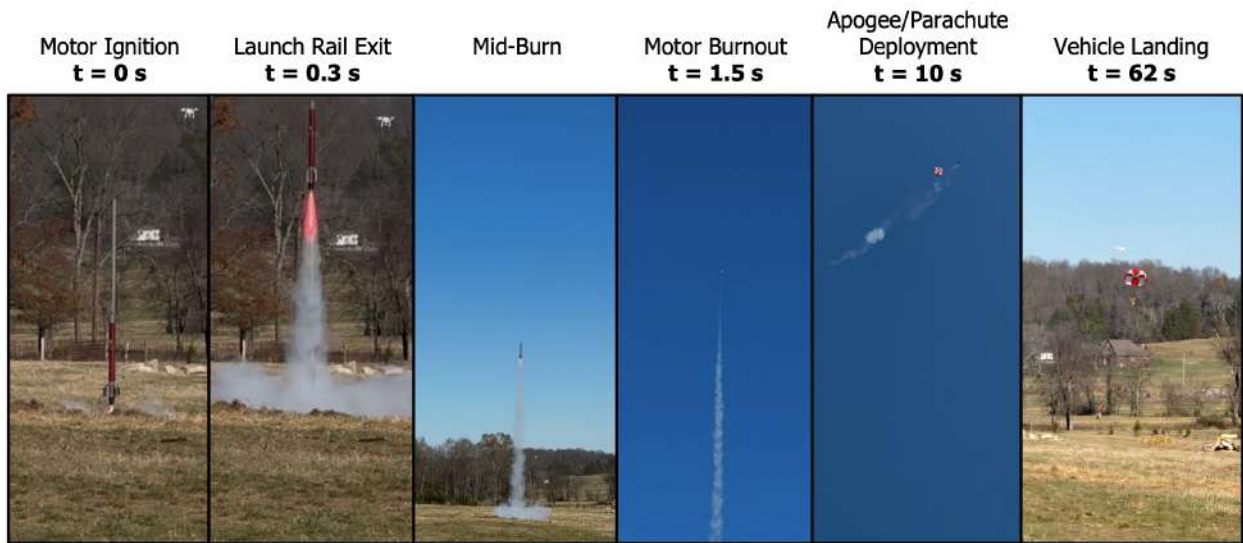


Figure 47: Snapshots and time stamps for key stages from the second subscale flight.

Data

Launch condition measurements were conducted in the field prior to each launch. Temperature and average wind speed readings were taken using a thermometer and an anemometer, respectively. Other environmental conditions were taken from public weather data from the National Weather Service (weather.gov). Vehicle massing and CGs were measured using a hand-held digital scale. Table 9 presents the launch conditions for both subscale test flights.

Table 9: Launch conditions, and the relevant requirements that justified their measurements, for both launches.

Subscale Launch Conditions	Value	Justifying Requirements
Cloud presence	None	FAA 14 CFR §101.25; NASA 5.4, 5.5
Dry bulb temperature	Flight 1: 42 °F Flight 2: 45 °F	VADL 2.3.7, 2.3.1
Humidity	<20%	VADL 2.3.7, 2.3.1
Air Pressure (mean sea level)	103 kPa	VADL 2.3.7, 2.3.1
Elevation	528 ft	VADL 2.3.7, 2.3.1
Average Wind Speed	5 mph	NASA 3.11, 5.4
Cardinal Wind Direction	230 ° (SW)	NASA 3.11, 5.4
Measured Dry Mass	11.9 lb	NASA 2.1, 2.14, 2.15, 3.3, 5.4, 5.5; VADL 2.1.2, 2.1.3; FAA 14 CFR §101.22, §101.23, §101.25
Measured Dry-Mass CG	29.3 in (2.8 cal SSM)	NASA 2.14, 5.4; VADL 2.1.2
Measured Dry Mass	13.1 lb	NASA 2.1, 2.14, 2.15, 3.3, 5.4, 5.5; VADL 2.1.2, 2.1.3; FAA 14 CFR §101.22, §101.23, §101.25
Measured Wet-Mass CG	31.5 in (2.2 cal SSM)	NASA 2.14, 5.4; VADL 2.1.2
Launch Rail Length	12 ft	NASA 1.12, 5.4
Launch Rail Angle	Flight 1: 2° Flight 2: 4°	NASA 3.11, 5.4

Due to a power failure in the onboard payload electronics, followed by subsequent handling errors that resulted in the unintentional short of the electronics, as detailed in Section 4.3.8, the IMU data could not be reliably retrieved nor presented.

3.2.5 Flight Analysis

Using the launch conditions measured in the field (9), VADL updated and ran the pre-flight subscale simulation models in the field to generate the below flight performance predictions prior to each launch to verify compliance with relevant NASA and VADL requirements. Flight simulation softwares, frameworks and methodologies are consistent with 3.4 to better inform full-scale flight. Table 10 shows that baseline predictions meet relevant requirements (NASA 2.15, 3.3, 3.11; VADL 2.1.2, 3.2) to ensure safe vehicle flight and recovery.

In DRSS, VADL supplies the mean coefficient of drag of 0.403 for the subscale vehicle using the Computational Fluid Dynamics (CFD) software ANSYS Fluent (Canonsburg, Pennsylvania, U.S). The subscale and full-scale CFD models are further detailed in Section 3.4.6. The marginal divergence between the two simulation frameworks allows the team to benchmark using subscale flight

Table 10: Subscale Baseline Flight Simulations.

	Apogee (ft)	Drift from Pad (ft)	Max Sectional Landing KE (ft-lbf)	Descent Velocity (fps)	Velocity at LRE (fps)	Avg. Thrust-to-Weight Ratio
Flight One						
CFD-DRSS	1141	-180	37.8	19	76.8	5.8
OpenRocket	1014	-230	38.6	19.2	73.2	6.2
% Diff	-12.5%	21.7%	2.1%	1.1%	-4.9%	6.6%
Flight Two						
CFD-DRSS	1133	-80	37.8	19.0	76.8	5.8
OpenRocket	1007	-150	38.6	19.2	73.2	6.2
% Diff	-12.5%	46.7%	2.1%	1.1%	-4.9%	6.6%

Table 11: Predicted Coefficients of Drag of the Subscale Vehicle.

	Predicted Coefficient of Drag
OpenRocket	0.48
ANSYS Fluent	0.40

data to better inform full-scale vehicle flight simulations. During vehicle ascent, since inertial and motor dynamics are modeled similarly between the two frameworks, the differences in predicted apogee primarily stem from the estimated drag coefficients as seen below:

Figures 48, 49, and 50 present the drag characteristics as provided by CFD.

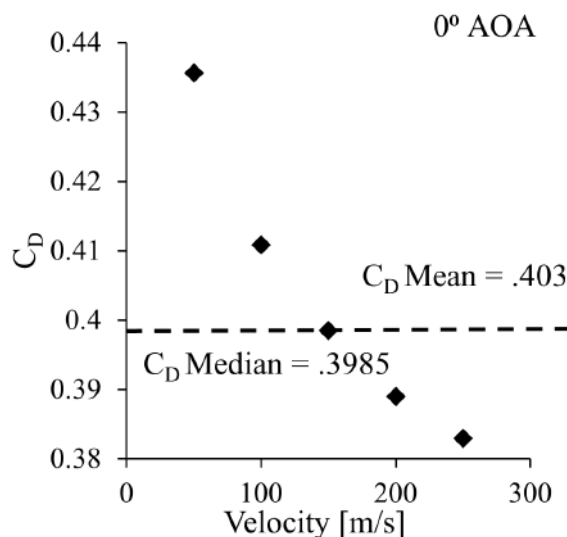


Figure 48: Coefficients of Drag of the Subscale Vehicle CFD Model

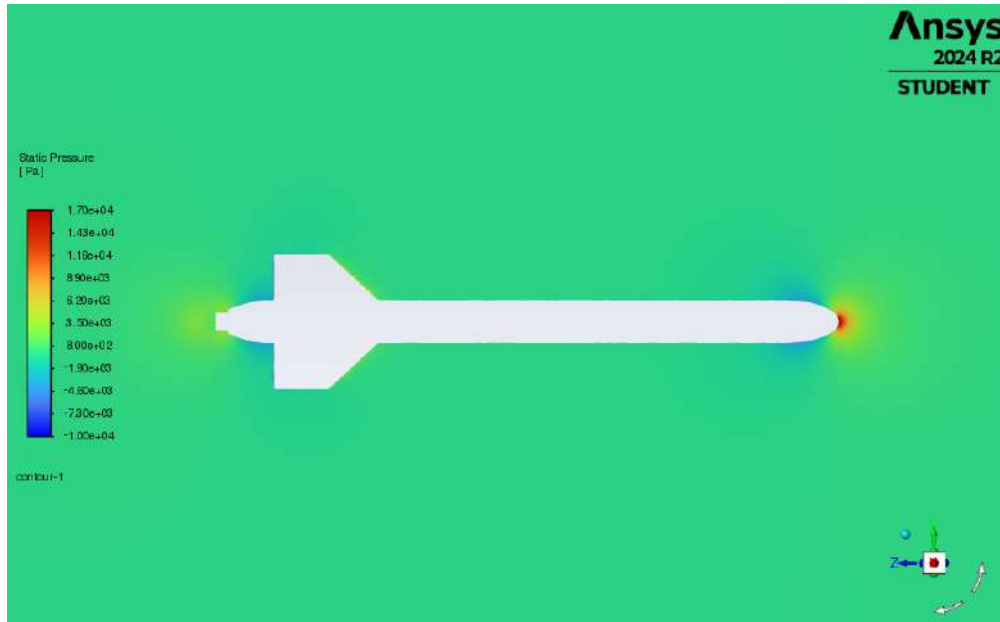


Figure 49: Pressure Contours of the Subscale Vehicle CFD Model.

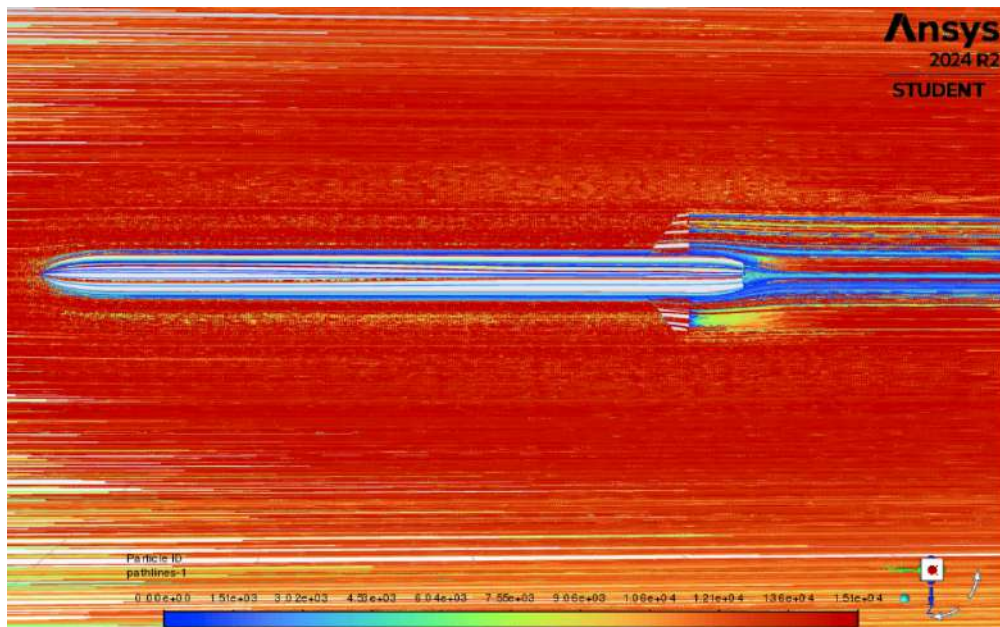


Figure 50: Streamlines of the Subscale Vehicle CFD Model.

To further verify NASA Requirement 3.11 on maximum vehicle drift, the team obtains a worst-case scenario drift by using a simpler method: VADL assumes the rocket is above the launch rail at apogee, then immediately catches full wind speed. The expected maximum drift for the highest apogee simulations (and therefore maximum drift) are shown in Table 12, which remain well within the 2500 ft recovery radius imposed by NASA.

Table 12: Subscale Baseline Maximum Drift Simulations.

	Descent Time (s)	Descent Time $\times$ Wind Speed (ft)
CFD-DRSS	57.9	425
OpenRocket	52.8	387
% Diff	-9.7%	

Flight One

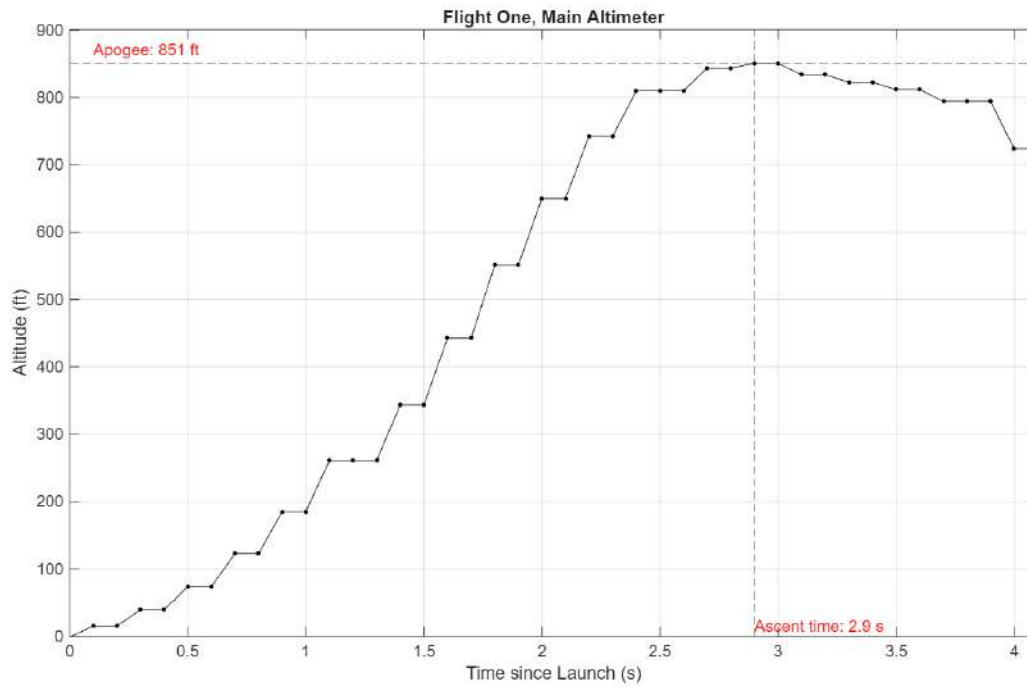


Figure 51: Raw data of the main avionics altimeter.

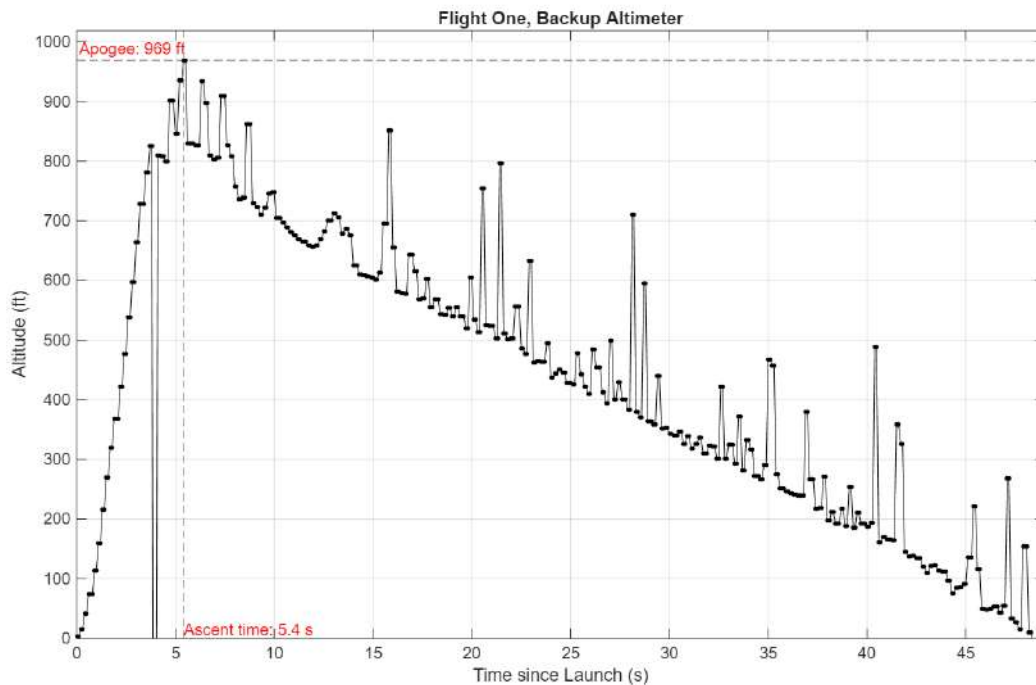


Figure 52: Raw data of the backup avionics altimeter.

During Flight One, the main altimeter stopped recording data approximately 4.1 seconds into the flight. While the exact cause is not fully confirmed, it is believed that vibrations during ascent may have loosened the screw terminal that completes the altimeter’s circuit. This would have caused a brief disconnection of the terminal, interrupting data collection. While post-flight inspection revealed the screw terminal was connected and the altimeter remained powered, even a brief disconnection during ascent could have caused the sudden stop in data recording. The team recognized this and secured critical connections more effectively in Flight Two, which presented no similar issues, and will continue to do so in future launches.

The main altimeter data (Figure 51) also experienced local pressure fluctuations during rocket ascent, which diverges from the backup altimeter data in Figure 52. Consistent with visual observations on launch day, these fluctuations resulted in an *apparent* inflection point in the altitude readings at 851 ft; this triggered the premature apogee detection and detonation of the parachute deployment charges, occurring approximately 150 feet before the rocket reached its intended apogee. Due to the rocket’s residual kinetic energy, the premature parachute deployment led to a highly unpredictable and un-modelable descent trajectory. For this reason, **flight analysis and comparison against predicted performance metrics for Flight One were unable to be conducted**. During preparations for the second launch, it was determined that these local pressure fluctuations were caused by Bernoulli effects from the transient responses of the rocket’s angle of attack during wind cocking. Modifications were subsequently made to the airframe’s pressure tap holes to address this issue, as detailed in Sections 3.2.4 and 3.3.4). For full-scale flight, VADL will use upgraded avionics hardware to increase sampling rate from 10 Hz to 50 Hz (3.3.4). In addition to improvements to pressure tap holes, the larger diameter of the full-scale airframe relative to the subscale (6 in vs 4 in) and, consequently, a larger airframe volume inside the recovery bays are expected to further

diminish the effects of this local pressure fluctuation. IMU data was unavailable in this flight due to issues described in Section 4.3.8.

In addition to improvements to avionics hardware as mentioned above, it is shown below using simplified physics scaling laws that for the full-scale rocket: (1) the rotational kinetic energy (and thus the induced angular velocity) immediately after separation during main deployment is proportionally smaller compared to the subscale, and (2) the longer and heavier rocket sections, the longer shock cords, and the increased wind damping from larger vehicle side profiles further reduce the amplitude of any swing and increase aerodynamic damping. Thus, the pressure disturbances in the avionics bay during descent should be significantly less than in the subscale test, resolving the noise issues observed.

A simplified expression for the moment of inertia of a rocket section about the separation plane can be approximated by:

$$I \propto M \times L_{\text{char}}^2,$$

where M is the mass of the section and L_{char} is a characteristic length (e.g., distance from the pivot point, on the order of half the section length for a rod-like approximation).

The sudden “kick” imparted by the release of chemical potential energy from the blast charges (the product of mass flow and exit velocity of the expanding gases) is the source contributing to angular kinetic energy and swinging of the rocket sections, which scales roughly with the **volume** that must be pressurized:

$$J \propto (\text{Pressure}) \times (\text{Cross-sectional area}) \times (\text{Length of ejection bay}).$$

Strictly, impulse J results from the integration of force $F = p \cdot A$ applied over the duration (Δt) of black powder burn and gas expansion in filling and pressurizing the ejection chamber. If the ejection bay has some characteristic length L and the hot gas expands at roughly some constant speed c , then $\Delta t \propto L/c$, and:

$$J \approx p A \left(\frac{L}{c} \right).$$

Therefore, the $J \propto p \times A \times L$ is a first-order approximation that captures how impulse scales with ejection-bay volume.

When increasing rocket diameter from 4” to 6”, the area grows by roughly a factor of $\left(\frac{6}{4}\right)^2 = 2.25$. Length changes (e.g., from 12 in to 14–16 in for the full-scale vehicle parachute bays) also affect the required ejection charge, but **not as aggressively** as the rocket’s overall mass and moment of inertia grow.

Immediately after separation,

$$\omega_{\text{max}} \sim \frac{J}{I}.$$

If the moment of inertia I **grows faster** than J , then ω_{max} (the maximum spin rate) **decreases**—thus reducing the “whipping” or swinging motion.

Going from subscale to full-scale, a very rough measure of the **mass ratio** is $\frac{M_{\text{full}}}{M_{\text{sub}}} \approx \frac{28.5}{12.8} \approx 2.2$. A rough measure of the **linear-dimension ratio** is $\frac{L_{\text{full}}}{L_{\text{sub}}} \approx \frac{99}{56} \approx 1.8$. Hence, the moment of inertia of the full-scale system about the separation plane can scale by roughly:

$$I_{\text{full}} \sim \left(\frac{M_{\text{full}}}{M_{\text{sub}}} \right) \times \left(\frac{L_{\text{full}}}{L_{\text{sub}}} \right)^2 \approx 2.2 \times (1.8)^2 \approx 2.2 \times 3.24 \approx 7.1,$$

i.e., on the order of magnitude of approximately **7 times** larger than subscale.

By contrast, the **ejection impulse** scales more closely to the bulk plate area (factor of ~ 2.25 from 4" to 6" diameter) and a modest increase in length of the parachute bay (e.g., from ~ 12 in to ~ 14 - 16 in, say around the order magnitude of 1.3x).

Putting this all together:

$$\frac{\omega_{\text{full}}}{\omega_{\text{sub}}} = \frac{J_{\text{full}}/I_{\text{full}}}{J_{\text{sub}}/I_{\text{sub}}} \approx \frac{(2.5 \sim 3)}{(7 \sim 8)} \approx 0.3 \sim 0.4.$$

Hence the expected **angular velocity** after main deployment for the full-scale rocket is substantially **lower** if everything else remains similar. Additionally, by doubling the shock cord lengths (e.g., from 10'/20' to 20'/40' for the main parachute shock cords) from subscale to full-scale, the rocket sections hang farther from the pivot point, further reducing the restoring torque for small swings and acting more like a "gentle pendulum." This is also expected to substantially decrease the amplitude of oscillations. Larger canopies, lines, and rocket bodies tend to increase damping effects, further reducing the amplitude of swinging motion over time.

When the rocket sections swing, local airflow around the avionics bay can accelerate or decelerate, leading to local pressure differentials and altimeter readings. A simple way to relate rotational motion to dynamic pressure is via the Bernoulli effect:

$$q_{\text{dynamic}} = \frac{1}{2} \rho v^2, \quad \text{where } v = \omega r.$$

Suppose the subscale rocket's angular velocity right after separation is ω_{sub} , the full-scale rocket's angular velocity is $\omega_{\text{full}} = k \omega_{\text{sub}}$ where $k < 1$ (as the full-scale vehicle rotates more slowly due to higher inertia). Let the radial length scale be r_{sub} for subscale and $r_{\text{full}} \approx \alpha r_{\text{sub}}$ for full scale ($\alpha > 1$). Then the ratio of **dynamic pressures** from rotation can be approximated:

$$\frac{q_{\text{full}}}{q_{\text{sub}}} = \frac{\frac{1}{2} \rho (r_{\text{full}} \omega_{\text{full}})^2}{\frac{1}{2} \rho (r_{\text{sub}} \omega_{\text{sub}})^2} = \left(\frac{r_{\text{full}}}{r_{\text{sub}}} \right)^2 \left(\frac{\omega_{\text{full}}}{\omega_{\text{sub}}} \right)^2.$$

Using the previous estimated ratios, the following is arrived:

$$\frac{q_{\text{full}}}{q_{\text{sub}}} \approx (1.77)^2 \times (0.43)^2 \approx 3.13 \times 0.1849 \approx 0.58.$$

The **dynamic pressure** (and therefore the Bernoulli-induced local pressure variation) in the full-scale event is only $\sim 58\%$ of that in the subscale. Without accounting for increased wind damping

due to larger vehicle side profiles, the **full-scale rocket is expected to experience at least 40+ % reduction** in rotationally induced pressure noise seen from the altimeters. Therefore, the large oscillations and local pressure fluctuations seen in the subscale test are **not** expected to reoccur at the same level in the full-scale flight. This justifies the team’s confidence that the altimeter measurements in the full-scale rocket will be **substantially less** susceptible to the noise caused by vehicle swing during descent.

As mentioned, apogee and other rocket descent predictions were unable to be compared against flight predictions. VADL computes the Launch Rail Exit (LRE) velocity from the average slopes of the two altimeters’ data at LRE and compares against predictions. Both simulation frameworks resulted in exceptionally close predictions against velocity at LRE, with VADL’s in-house DRSS software producing a marginally closer prediction relative to OpenRocket (Table 13).

Table 13: Flight One Launch Rail Exit Velocity Comparisons.

	Predicted Velocity at LRE (fps)	Actual Velocity at LRE (fps)	% Diff
CFD-DRSS	76.8	76.7	0.1%
OpenRocket	73.2		-4.8%

Flight Two

IMU data was unavailable in this flight due to issues described in 4.3.8; as a result, transient vehicle kinetics during motor ascent and parachute deployments, and drag coefficient estimation were unable to be analyzed. The altimeter data in comparison against predictions of the two simulation frameworks are presented below.

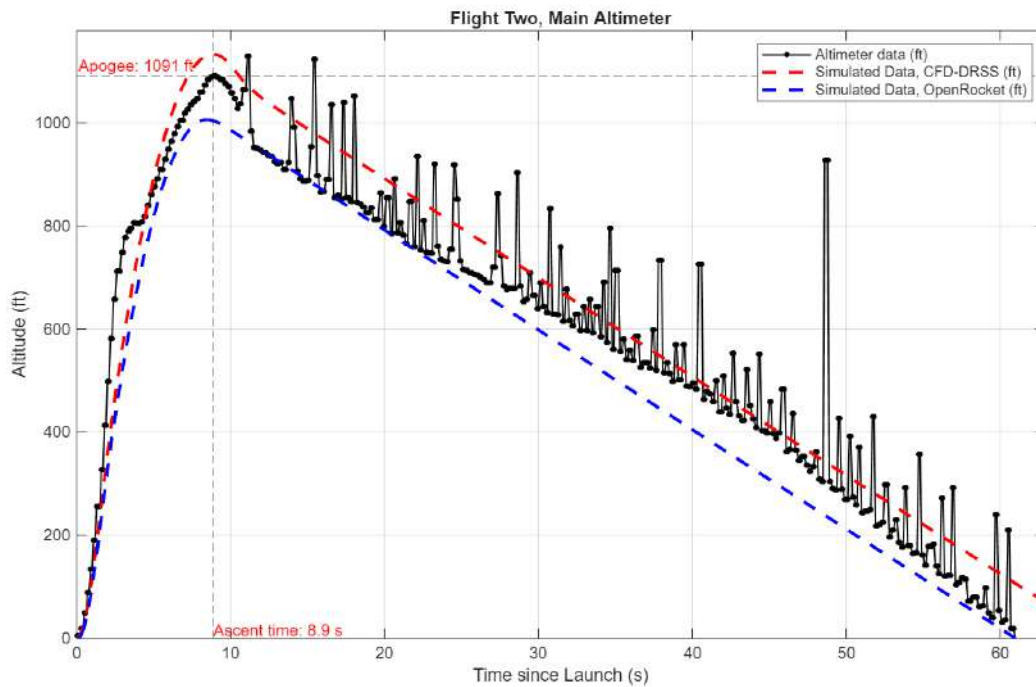


Figure 53: Raw data of the main avionics altimeter versus simulated altitude.

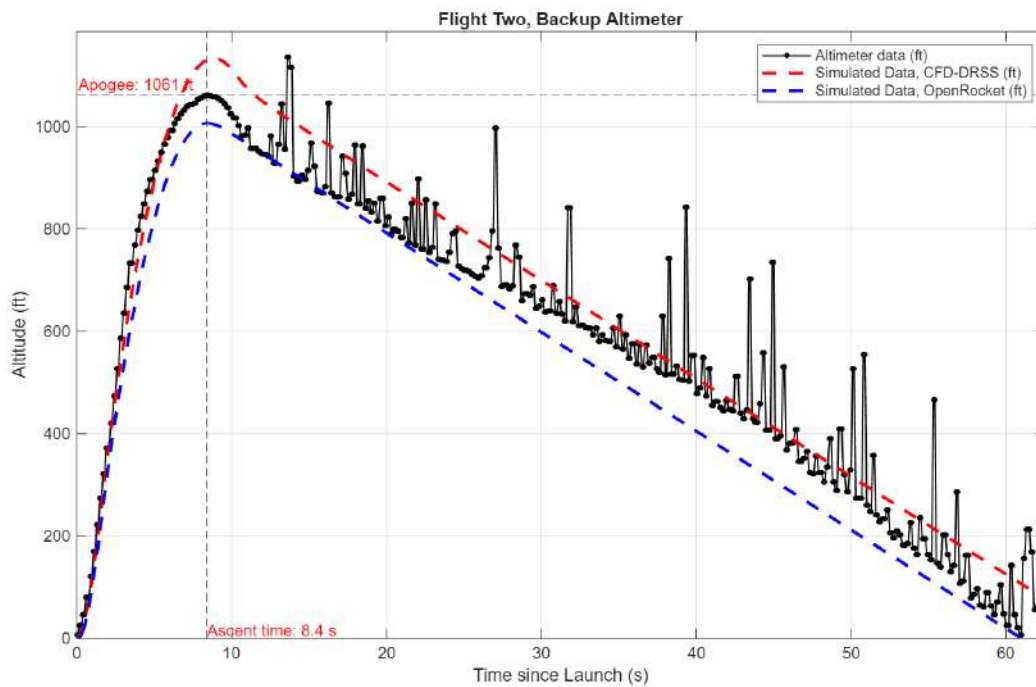


Figure 54: Raw data of the backup avionics altimeter versus simulated altitude.

Figures 53 and 54 show both altimeters report similar altitude data, which aligns with theoretical predictions by both DRSS and OpenRocket. VADL again verifies the actual versus predicted LRE velocity as well as other flight performance metrics in Table 14. The average descent velocity was computed using linear interpolation of the entire descent flight while the velocity at landing was estimated using the same technique limited to the final 4-5 seconds of descent flight. The actual flight metrics were generated by averaging the data from the two altimeters.

Table 14: Flight Two Prediction Benchmarks against Actual Data.

	CFD-DRSS	OpenRocket
Apogee (ft)	1076	
<i>Predicted Apogee (ft)</i>	1133	1007
% Error	5.3%	-6.9%
<i>Ensemble Avg. of Model Predictions (OpenRocket & DRSS; ft)</i>	1070	
% Error	-0.6%	
Ascent Time (s)	8.65	
<i>Predicted Ascent Time (s)</i>	8.9	8.4
% Error	2.9%	-2.5%
Avg Descent Velocity (fps)	19.0	
<i>Predicted Descent Velocity (fps)</i>	19.0	19.4
% Error	0.0%	2.1%
Velocity at Landing (fps)	24.9	
LRE Velocity (fps)	75.3	
<i>Predicted LRE Velocity (fps)</i>	76.8	73.2
% Error	2.1%	-2.8%

Table 14 shows high alignment of actual flight performance metrics against theoretical predictions made by VADL's flight simulation frameworks and methodology. Specifically, the average descent velocity was successfully predicted with <0.1% error by VADL's in-house DRSS software, which takes into account modified vehicle drag profiles due to non-deterministic swinging of the rocket body and improves upon OpenRocket's straight rocket descent assumption. This directly validates both the methodology behind VADL's flight simulation frameworks and the 2.2 drag coefficient of the subscale recovery parachute, which instills additional confidence that the team will deliver safe and effective recovery systems for the full-scale vehicle. The predicted apogees of both frameworks performed reasonably well against actual flight data; notably, the *average* value of the predicted apogees of the two frameworks matches exactly the actual recorded apogee; this consideration is taken into account for full-scale vehicle apogee predictions (Section 3.4). During vehicle ascent, other than environmental, inertial and motor dynamic, which are deterministically accounted for in DRSS, the relatively low deviation in predicted apogee from the actual apogee indirectly validates the CFD methodology that is applied to both subscale and full-scale vehicles. As required by the NASA USLI Handbook, estimation of the drag coefficient of the full-scale vehicle was therefore *enabled* by the subscale flight data in the context that the subscale performance of the same simulation and CFD frameworks has instilled the team's confidence to apply them to

the full-scale vehicle (Section 3.4.6). Due to the lack of available IMU kinetics data during subscale flight and due to the fact that subscale vehicle is not a direct scaling of the exact full-scale geometry, direct calculation of full-scale vehicle from subscale flight data was not possible.

Despite accounting for launch rail friction dynamics (as further detailed in 3.4), temperature fluctuations and inherent heterogeneity in motor manufacturing may have resulted in the relatively low errors in predicted apogee, which aligns most closely with the *average* of the two framework predictions. Due to transient and non-deterministic swinging of the rocket body and parachute, the oscillation of the drag profile of the parachute immediately before landing resulted in higher than predicted final landing velocity of 24.9 fps. Due to the high safety factor on landing KE imposed during the design of the subscale vehicle, the team was able to verify that this variation in landing velocity remains within NASA requirements in Table 15.

Table 15: Flight Two Landing Characteristics.

Vehicle Section	Dry Mass Without Parachute (lb)	Avg Descent Velocity (fps)	Landing Velocity (fps)	Landing KE (lb-ft)	NASA 3.3
Fore	4.6	19.0	24.9	44.9	75 lb-ft
Aft	6.9			66.8	

Coordinates of the launch pad and landing sites were geolocated using the built-in GPS module inside a Samsung S22+ mobile telephone (accuracy: $\pm 1\text{m}$; Samsung Electronics Co., Ltd., Suwon-si, South Korea); the Euclidean norm between the coordinates were calculated using Google Maps (Alphabet Inc., Mountain View, California, U.S) to determine final vehicle drift. The pre-flight models are compared against this measured drift data below.

Table 16: Flight Two Drift Characteristics.

	Actual Drift (ft)	Predicted Drift (ft)	% Error
DRSS OpenRocket	-177	-80	-54.8%
		-150	-15.3%

The OpenRocket's drift prediction is in alignment with the actual drift; however, in VADL's DRSS model, despite for accounting the rocket's side drag profile and coefficient of drag of the parachute's side, the predicted drift was underestimated, which suggests that the actual time constant / duration at which the rocket catches the wind was much faster than the theoretical model. Consequently, adjustments to the analytical model in DRSS were necessary to more accurately reflect the rocket's descent behavior.

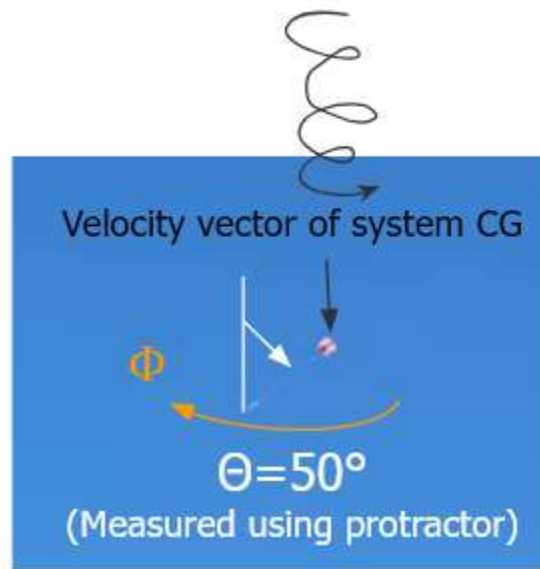


Figure 55: Representative visualization of the parachute's descent dynamics captured during Flight Two, showing the velocity vector of the system's center of gravity ($\Theta = 50^\circ$) and the swirling motion of the parachute in the horizontal plane (Φ).

The new approach accounts for the non-deterministic dynamics that exist between the rocket and parachute system's velocity vector, the angle of descent (Θ), and the swirling motion of the parachute in the horizontal plane (Φ). The side drag adjustment incorporates the following considerations: (1) the maximum exposed area of the parachute canopy, $A_{\text{parachute}}(\theta)$, varies with the angle of descent as $A_{\text{parachute}}(\theta) = \pi \left(\frac{D}{2}\right)^2 \cdot \sin(\theta)$ (D is the diameter of the parachute); (2) the effective drag coefficient is estimated by weighting contributions from the rocket and parachute components based on their respective drag coefficients and areas as $C_{D,\text{eff}} = \frac{C_{D,\text{rocket}}A_{\text{rocket}} + C_{D,\text{parachute}}(\theta)A_{\text{parachute}}(\theta)}{A_{\text{eff}}}$ where $A_{\text{eff}} = A_{\text{rocket}} + A_{\text{parachute}}(\theta)$; (3) the swirling motion of the parachute in the horizontal plane introduces a time-averaged exposure factor, which is derived from the integration of the cosine of the swirl angle (ϕ) over a full 360° rotation give by Average exposure $= \frac{1}{2\pi} \int_0^{2\pi} |\cos(\phi)| d\phi = \frac{2}{\pi}$; (4) to account for the swirling motion and descent angle, an average of $\Theta = 50^\circ$, estimated using protractor on the subscale flight video footage, is used to compute the average projected area factor as $\sin(50^\circ) \times \frac{2}{\pi}$. The adjusted model captures the combined effects of side drag and swirling motion more accurately, and the adjusted predictions below confirm this notion:

Table 17: Adjusted Drift Modeling Results.

	Actual Drift (ft)	Predicted Drift (ft)	% Error
Old Model, DRSS	-177	-80	-54.8%
New Model, DRSS		-170	-4.0%
OpenRocket		-150	-15.3%

The effects of this model adjustment decrease the simulated duration required for the rocket-parachute system to catch onto wind after apogee, which is visualized in the below simulation

using DRSS:

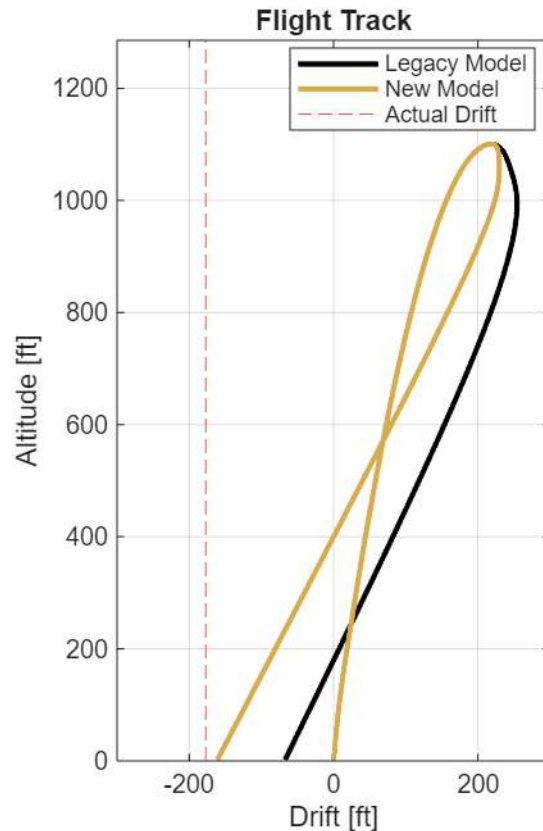


Figure 56: Flight Track of the Adjusted Side Drag Model.

3.2.6 Implications for Full-scale Vehicle Design

The subscale design, build, and fly cycle taught the team a number of lessons. The team learned a great deal about the manufacturing process and best practices; **the successful subscale flights with mission performance metrics in high alignment with theoretical predictions instills confidence in the team’s ability to construct a well-built and safe full-scale vehicle and fly it to its intended apogee in the spring semester.** The process of maintaining a fabrication and testing timeline was also educational, and the team saw the importance of incorporating margin for unexpected setbacks. VADL also gained valuable field experience through the first demonstration flight of the competition year, preparing the team for full-scale demonstration flights in the spring. Beyond these learnings, the subscale flight also informed several design changes to the full-scale vehicle. These modifications are summarized in Table 18 below.

Table 18: Full-scale design modifications informed by the subscale flight.

Design Modification	Justification	Relevant Requirement(s)
Modified nose cone shape to be 1.5x scale version of the subscale nose cone.	Subscale nose cone manufacturing process and performance was verified through successful flights; maximizing similarity between subscale and full-scale vehicles increases reliability of the full-scale vehicle.	NASA 2.18.1
Switched to EasyMini for full-scale vehicle altimeter.	Entacore AIM altimeter presented issues resulting from low sampling rate.	NASA 3.1.1-2, 3.4
Custom PCB with wires managed by Molex Connectors. A larger battery is also being selected. Improved active cooling.	Prototype Electronics experienced a structural failure during the subscale test (optional). More robust electronic hardware is needed.	NASA 2.3, 2.6

3.3 Recovery Subsystem

The recovery subsystem is responsible for the safe and controlled descent and landing of the vehicle sections and payload. As defined by NASA Requirement 2.3, the vehicle must be recoverable and reusable. Vehicle section landing energy, recovery area, and descent time are constrained by NASA Requirements 3.3, 3.11, and 3.12, respectively. Therefore, it is of the utmost importance to ensure that the recovery sequence is appropriately defined and hardware properly selected in accordance with flight profile requirements, loading analysis, and safety considerations. The following sections outline the recovery system CONOPS, hardware, avionics, and GPS tracker.

3.3.1 Recovery System CONOPS

Figure 57 provides a visual overview of the recovery system CONOPS. The recovery sequence begins at apogee with the drogue parachute deployment. The drogue parachute, an 18 in Fruity Chutes Classic Elliptical, is located at the separation point between the drogue bay and tail section and allows the vehicle to descend in a controlled manner prior to main parachute deployment. At 600 ft, the 96 in Fruity Chutes Iris Ultra main parachute is deployed at the separation point between the main and lander separation bays. This event slows the vehicle sections down to 16.5 ft/s. Finally, the lander is jettisoned at 300 ft and descends under a 60 in Fruity Chutes Iris Ultra. Detailed recovery system performance outcome metrics are outlined in Section 3.4.

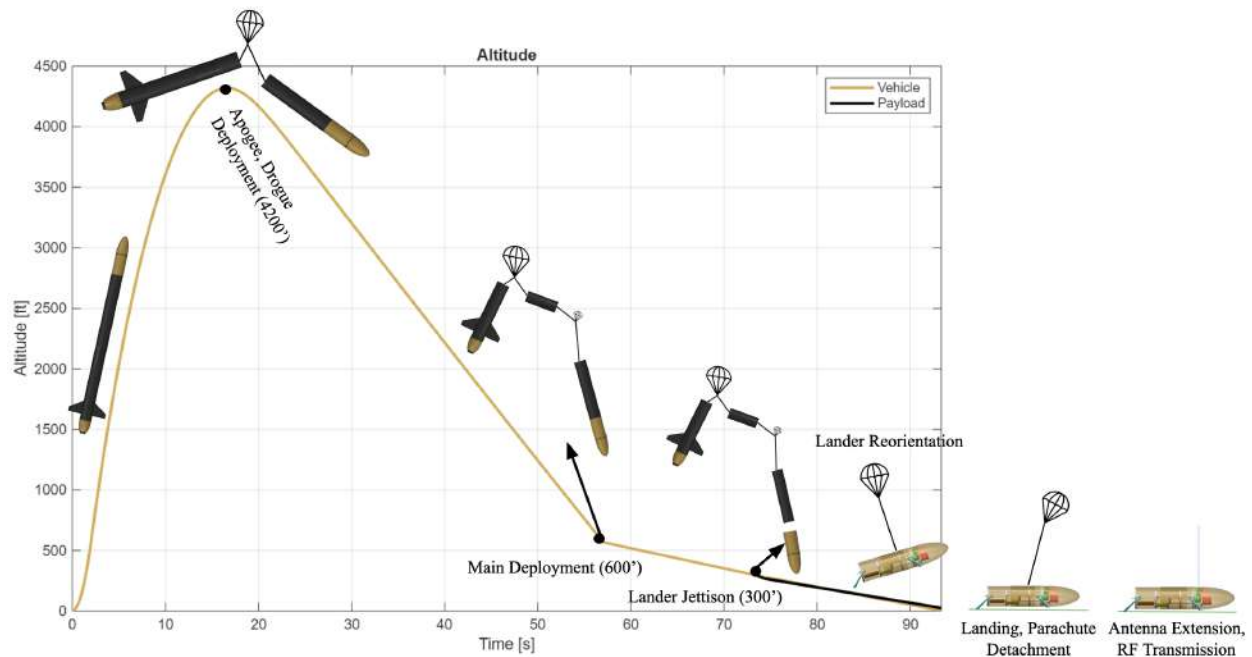


Figure 57: Recovery sequence CONOPS.

3.3.2 Recovery Hardware

Drogue Parachute

The location of the drogue parachute in the full-scale vehicle is indicated in Figure 58. As noted in Section 3.1.6, the drogue parachute bay is located fore of the main parachute bay to ensure that the nose cone is the lowest section during descent.

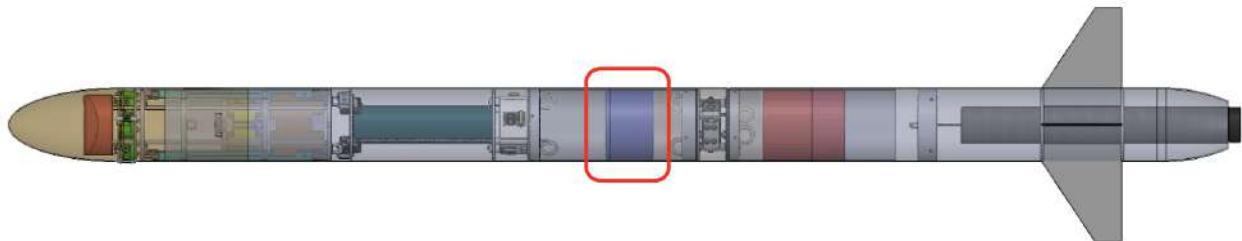


Figure 58: Location of the drogue parachute in the full-scale vehicle.

The drogue parachute is critical to minimizing descent time, per NASA Requirement 3.12, while still slowing the vehicle to a low enough velocity to ensure the g forces experienced during main deployment are not excessive. With these requirements in mind, the 18 in Fruity Chutes Classic Elliptical was selected for the drogue parachute. The descent configuration of the full-scale vehicle under drogue is depicted in Figure 59, and key parameters are summarized in Table 19.

Drogue Descent

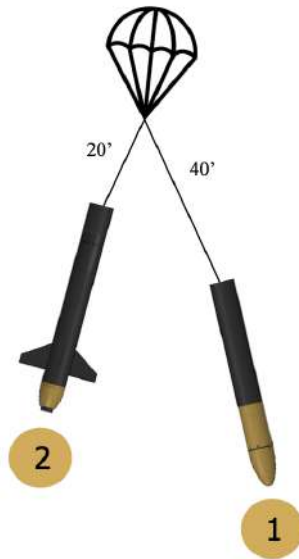


Figure 59: Descent configuration under the drogue parachute.

Table 19: Parameters for the drogue parachute.

Parameter	Drogue Parachute
Parachute	18 in Fruity Chutes Classic Elliptical
Parachute Material	Nylon
Brand	Fruity Chutes
Drag Coefficient	5.5
Deployment	Apogee
Backup Deployment	Apogee + 1s
Terminal Velocity	95.7 ft/s

Main Parachute

The main parachute location in the launch vehicle is shown in Figure 60.

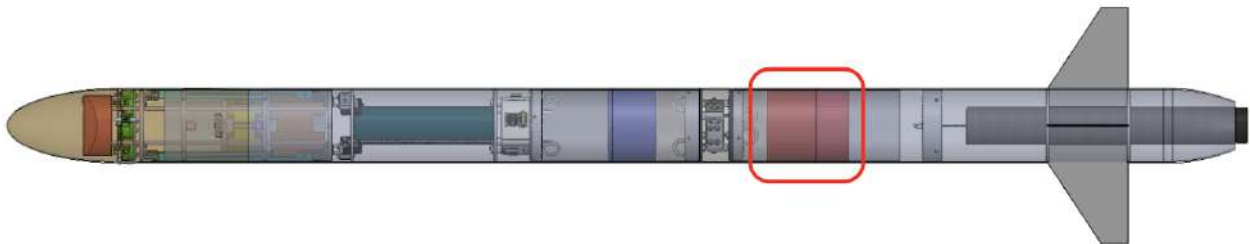


Figure 60: Location of the main parachute in the launch vehicle.

The main parachute serves to slow the vehicle to its final landing velocity. It is critical that the main parachute is large enough to comply with NASA Requirement 3.3 dictating maximum land-

ing kinetic energy, while being small enough to remain within maximum drift and descent time as mandated by NASA Requirements 3.11 and 3.12, respectively. As such, VADL has selected the 8 ft Fruity Chutes Iris Ultra for the main parachute. The descent configuration following main deployment and before payload jettison is shown in Figure 61, and the main parachute specifications are detailed in Table 20.

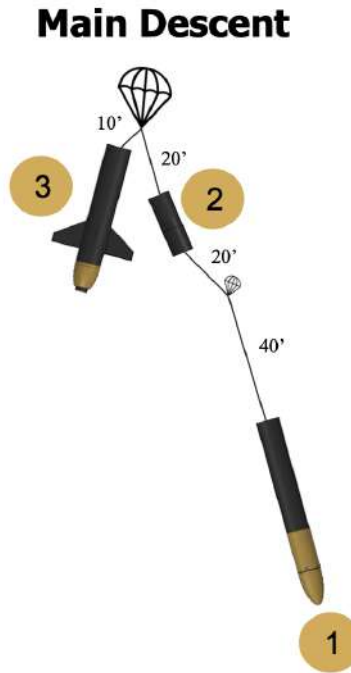


Figure 61: Launch vehicle descent configuration under the main parachute, prior to payload jettison.

Table 20: Parameters for the main parachute.

Parameter	Main Parachute
Parachute	96 in Iris Ultra
Parachute Material	Nylon
Brand	Fruity Chutes
Drag Coefficient	2.2
Deployment	600 ft
Backup Deployment	550 ft
Terminal Velocity (Pre/Post-Jettison)	16.5/14.6 fps

3.3.3 Shock Cord Selection

Shock cords serve to connect the parachutes to the anchoring hardware in the vehicle. As such, it is essential that they are properly selected to ensure a safe recovery of the vehicle. To inform the shock cord selection, the analysis presented in Section 3.1.4 is employed, which results in a parachute force of 1204 lbf for the two shock cords during main deployment. Specifically, the fore

shock cord must withstand a load of 879 lbf, while the aft shock cord is subjected to 325 lbf at cord extension. VADL has selected to use ½ in Kevlar shock cords rated for 6000 lbf, thus ensuring a **safety factor of 6.8 and 18.5 for the fore and aft main shock cords, respectively.**

Along with the load rating, the shock cord length must also be properly selected. The relative sizes must be chosen such that the individual sections do not collide with one another during descent, and the overall length must be long enough to dampen the loading of parachute deployments. The vehicle shock cord lengths are listed in Table 21.

Table 21: Vehicle shock cord lengths.

Parachute / Anchored Section	Shock Cord Length
Drogue / Lander Separation Bay	40 ft
Drogue / Avionics Bay	20 ft
Main / Avionics Bay	20 ft
Main / Tail Section	10 ft

Anchoring Hardware

Each parachute is attached to the shock cords via a swivel link rated for 3000 lbf. With the maximum expected parachute force of 1204 lbf, this ensures a safety factor of 2.5. The shock cords will be attached to 3/18”-16 eye bolts on the vehicle sections via quick links rated for 4000 lbf. The parachutes will be attached to the shock cords using swivel links to be decoupled from the angular momentum of the vehicle sections. The parachutes will be folded within fire blankets to provide protection from the blast charge events, and fireballs will be located at the ends of the shock cords, at the edges of the vehicle airframe and coupler, to prevent zippering. Table 22 summarizes the strength rating, expected load, and factor of safety for each anchoring hardware element. The eye bolts on the avionics, or fore, side were used for max loading expectation, as they each must support a greater weight than that of the tail side hardware. Loading for these components was found by multiplying the main parachute deployment deceleration calculated in Section 3.1.4 by the dry mass of the tail section.

Table 22: Anchoring hardware components and their associated ratings, loadings, and factors of safety.

Component	Strength Rating	Expected Max Load	Minimum F.o.S.
Swivel link	3000 lbf	1204 lbf	2.5
Quick link	4000 lbf	1204 lbf	3.3
3/8”-16 Eye bolt (Fore)	1300 lbf	440 lbf	3.0

Payload Parachute

The payload parachute is used to slow the lander down and provide a safe, smooth landing for the STEMnauts. To ensure compliance with NASA Requirement 3.12 for descent time, the parachute must not be too large. Nevertheless, landing velocity data from the most recent slate of drop tests

informed a switch to a 60 in Fruity Chutes Iris Ultra from the 48 in model declared at the PDR stage. This analysis is discussed in further detail in Section 3.3.4 and flight simulations provided in Section 3.4 show that predicted descent times still fall under the 90 s maximum. The payload parachute location is shown in Figure 62, and the relevant parameters are listed in Table 23.

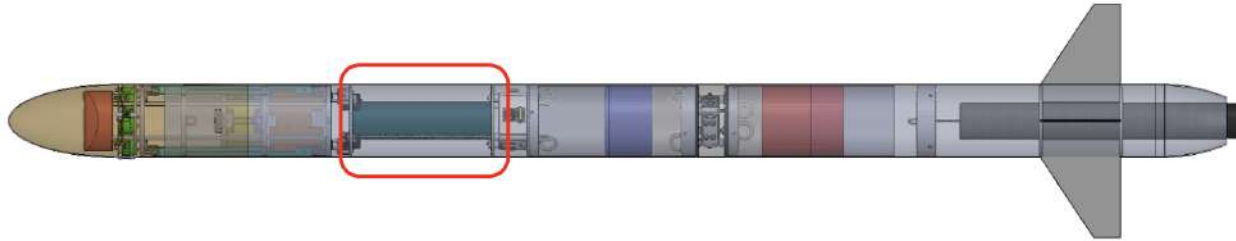


Figure 62: Location of the payload parachute retention tube in the launch vehicle.

Table 23: Specifications for the payload parachute

Parameter	Payload Parachute
Parachute	60 in Iris Ultra
Parachute Material	Nylon
Brand	Fruity Chutes
Drag Coefficient	2.2
Deployment	300 ft
Terminal Velocity	12.8 ft/s

Lander Recovery Hardware

As detailed in the PDR report, early drop tests motivated a switch to a 50 ft long, $\frac{1}{8}$ in thick Kevlar shock cord. Since then, however, further drop tests have presented complications with shock cord tangling and high loading from the delay in shock cord unfurling. As such, the team has decided to switch to a 30 ft, $\frac{1}{4}$ in shock cord to address these two issues. These results and design decisions are discussed in more detail in Sections 4.1 and 4.1.3. The parachute will be retained within the parachute retention tube in the lander separation bay and anchored to the payload detachment latch on the body of the payload. The payload end of the shock cord forms a large (12 in diameter) loop, which is passed through two metal brackets on the shock cord bulk plates down to the central EM latch. This loop was formed by sewing the loose end of the shock cord back onto itself, overlapping by about 3 in and using Kevlar thread in order to maximize strength. Figure 63 below shows this setup, with the step-by-step process for shock cord attachment. The ultimate purpose of this configuration is to maintain stability throughout descent by way of the triangular shock cord configuration while also allowing for a single point of detachment upon landing. This system will be discussed further in Section 4.2.3, along with some of the drawbacks of the current design.

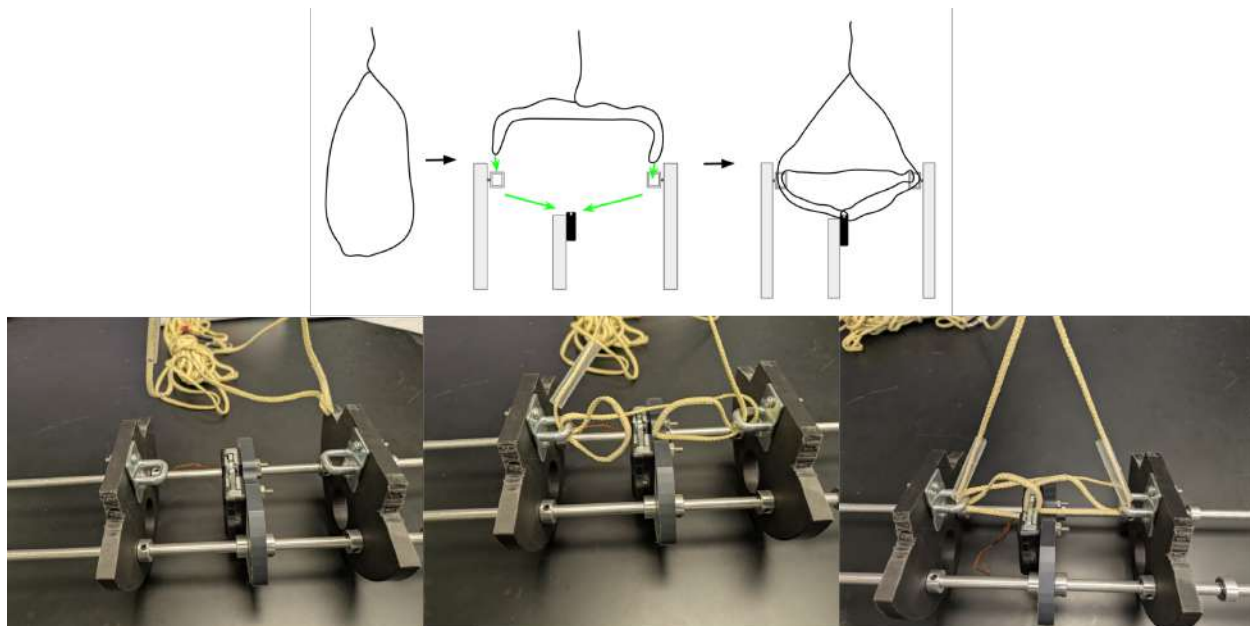


Figure 63: Payload side shock cord configuration, showing how single large loop is used to connect to a single release latch in the center.

3.3.4 Avionics

The avionics system is a critical component of the rocket, ensuring successful deployment, recovery, and data acquisition throughout the mission. This system includes two EasyMini altimeters, a 9 volt battery, and a screw terminal which is accessible from the outside of the rocket. The avionics bay is housed in a reinforced compartment to protect sensitive electronics from the mechanical and thermal stresses of flight. The altimeters serve dual roles: monitoring altitude and triggering the deployment of both drogue and main parachutes at the designated points of the flight. The bay also includes redundant systems to increase reliability.

A circuit diagram [64](#) is presented below, illustrating the interconnections between power sources, altimeters, and actuators. This design was chosen for its simplicity and fault tolerance, ensuring minimal signal interference. The diagram shows the connections to the altimeter in the order the connections are made on the terminal block on the physical device. Also shown are the fully independent primary and backup altimeters, which provide redundancy in the event of a failure with the primary altimeter.

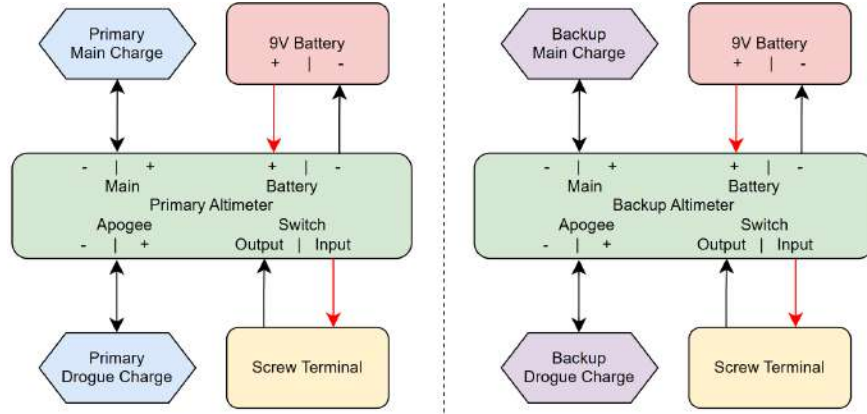


Figure 64: Avionics circuit diagram.

The EasyMini dual deployment altimeters were chosen following the results of the subscale launch, which revealed issues with the previously used Entacore AIM USB 4.0 altimeters which have been discussed in Section 3.2.4. During the subscale flights, the AIM altimeters demonstrated low sampling rates that contributed to premature deployment events and discrepancies in altitude data. These issues necessitated modifications to the airframe pressure vent hole sizes to mitigate Bernoulli effects and pressure fluctuations, as detailed in Section 3.2.4. During subscale flights, the AIM altimeters demonstrated low sampling rates that contributed to premature deployment events and discrepancies in altitude data. The EasyMini altimeters sample at 100 Hz and are expected to deliver better reliability and precision, particularly under dynamic flight conditions, aligning with NASA and VADL requirements.

Pressure vent holes are incorporated in the avionics bay to allow the altimeters to sense external pressures. The size of the hole is directly proportional to the volume of the chamber housing the altimeter and can be determined using the equation below.

$$d_n = (0.1)d\sqrt{\frac{l}{kn}} \quad (17)$$

Where d_n is the diameter of the hole, n is the number of holes, k is a constant, d is the diameter of the body tube and l is the length of the chamber. A k value of 500mm was initially used in the subscale vehicle, but the issues presented in the first flight necessitated modifications to the vent hole sizes to mitigate Bernoulli effects due to scaling and pressure fluctuations as fully detailed in Section 3.2.5. For subscale flight number two, k was decreased to 300mm, resulting in a vent hole diameter of 4.1mm. This resulted in a clear improvement in pressure readings by both altimeters, so this k value will be used moving forward. For the full-scale vehicle and avionics bay, $d = 152.4\text{mm}$, $l = 56.6\text{mm}$, and $n = 2$. Therefore, the vent whole size is 4.7mm as shown below.

$$d_n = (0.1) \cdot 152.4 \cdot \sqrt{\frac{56.6}{300 \cdot 2}} \quad (18)$$

$$d_n = 4.7 \text{ mm} \quad (19)$$

3.3.5 Blast Charges

The rocket employs nylon shear pins designed to withstand the maximum loads experienced during flight but shear under the pressure generated by black powder detonation. The force required to shear these pins is calculated using the following equation:

$$F = \sigma A = \sigma \frac{D_{\text{pin}}^2 \pi}{4} \quad (20)$$

where σ represents the material shear strength, and A is the cross-sectional area of the shear pins.

To account for the use of four shear pins for section separation, the total force is scaled as:

$$F = NF_{\text{shear}} \quad (21)$$

where N is the number of pins.

The ideal gas law is then applied to determine the amount of black powder required to generate the necessary pressure for bulkhead separation. The number of moles of gas generated by the black powder is calculated as:

$$n = \frac{pV}{RT_{\text{burn}}} \quad (22)$$

where p is the pressure needed to separate the sections, V is the internal volume of the rocket section, R is the universal gas constant, and T_{burn} is the burn temperature of the black powder.

Substituting the force equation into the pressure calculation:

$$p = \frac{F}{A_{\text{bulk plate}}} \quad (23)$$

and combining this with the ideal gas law, the number of moles becomes:

$$n = \frac{FV}{RT_{\text{burn}}A_{\text{bulk plate}}} \quad (24)$$

The mass of black powder required can then be calculated by multiplying the number of moles by the molar mass of the black powder:

$$m = \frac{\sigma D_{\text{pin}}^2 \pi V M_{4F}}{RT_{\text{burn}}A_{\text{bulk plate}}} \quad (25)$$

where M_{4F} is the molar mass of the black powder.

The computed mass of black powder ensures successful section separation. Specific values used for the parameters and calculations are included in Table 24.

Table 24: Parameter table for nylon shear pins and parachute bay calculations.

Parameter	Variable Name	Value
Amount of Nylon #2-56 Shear Pins	N	4
Shear Strength of Nylon #2-56 Shear Pins	σ	6.62×10^7 Pa
Diameter of Nylon #2-56 Shear Pins	D_{pin}	0.00218 m
Molar Mass of 4F Black Powder	M_{4F}	65 g/mol
Combustion Temperature of 4F Black Powder	T_{burn}	1837 K
Universal Gas Constant	R	8.314 J/mol K
Drogue Parachute Bay Length	$\frac{V_d}{A_b}$	14 in
Main Parachute Bay Length	$\frac{V_m}{A_b}$	14 in
Drogue Parachute Deployment Safety Factor	SF_{drogue}	3
Main Parachute Deployment Safety Factor	SF_{main}	3

Plugging in all parameters into the equation:

$$m = \frac{\sigma D_{\text{pin}}^2 \pi V M_{4F} SF}{RT_{\text{burn}}}$$

For both drogue and main parachutes (with bay lengths 14 in or 0.3556 m):

$$m_{\text{drogue}} = \frac{(6.62 \times 10^7)(0.00218)^2 \pi (0.3556)(65)(3)}{(8.314)(1837)}$$

$$m_{\text{drogue}} = 4.49 \text{ g}$$

Thus, the required mass of black powder is 4.49 g for the drogue and main parachutes. Both backup charges have an additional 1.2 factor of safety multiplied by the main charges to ensure separation. The summary of all blast charge parameters can be found in Table 25 below.

Table 25: Parameter table for nylon shear pins and parachute bay calculations.

Parameter	Drogue Parachute	Main Parachute
Primary Blast Charge (g)	4.49	4.49
Primary Deployment	Apogee	600 ft
Backup Blast Charge (g)	5.38	5.38
Backup Deployment	Apogee + 1s	550 ft

3.3.6 GPS Tracker

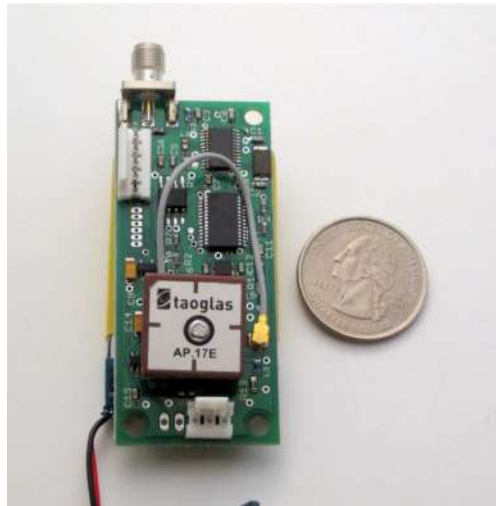


Figure 65: BigRedBee GPS/APRS Transmitter.

Since PDR, the GPS selection has changed to the BigRedBee 70 cm 100 mW GPS/APRS Transmitter. This design change was informed by the NASA recommendation to switch from the 33 cm band to the 70 cm band for transmitting data. The BigRedBee GPS APRS transmitter combines a u-blox Max 10 GPS module with an FM radio transmitter on an on-board microcontroller. The transmitter operates at a pre-set frequency of 433.920 MHz (configurable) with a data rate of 1200 baud. GPS data is converted into AX.25 data packets, which will be decoded using a software-based decoder. Initial tests will utilize the RTL-SDR method, which has successfully decoded 2M band APRS messages, to evaluate its performance for this purpose. The microcontroller also stores over 2 hours of GPS data sampled at 1 hz in non-volatile memory, which can be downloaded by USB to a computer as a kml file.

In order to comply with NASA requirement 3.13, two of these will be installed into the rocket, one in the fore section and one in the payload bay. These two transmitters have a time slotting functionality that allows multiple transmitters to share the same frequency by synchronizing their transmit times to the GPS clock. This will be used to allow both transmissions from both devices to be received without interfering. To power the fore section GPS, it will be connected to a 850 mah 3.7V LIPO battery, which is capable of powering the GPS and transmitter for over 8 hours, well above the NASA requirement 2.6 minimum of 3 hours of functionality.

Additionally, the Featherweight GPS, as suggested by NASA, will serve as a backup option if the BigRedBee system proves unsuccessful. The BigRedBee is prioritized due to its superior documentation and more user-friendly interface.

3.4 Mission Performance Predictions

Accurate flight simulation is fundamental to mission safety compliance and serves as a critical tool for informing design decisions in alignment with safety requirements imposed by NASA and VADL. Simulations play a role in evaluating load dynamics, selecting propulsion systems, determining parachute parameters, estimating apogee, and ensuring the overall success of the mission.

Validation of these simulations is achieved by cross-referencing results from custom MATLAB simulations with those generated by OpenRocket. **Extensive subscale flight data analysis (Section 3.2.5) obtained from launch testing has, to date, provided a definitive benchmark that validated VADL’s theoretical frameworks.** A high degree of alignment between predicted and actual subscale flight performance metrics has instilled the team’s confidence in the accuracy of VADL’s simulation and modeling for the full-scale vehicle as outlined in this section.

This year, VADL has introduced significant advancements in its in-house simulation framework to address limitations, innovate functionality and improve the accuracy of physics modeling. The [Declarative Rocket Simulation Software \(DRSS\)](#), introduced in this year’s PDR, represents a major leap forward. DRSS is a MATLAB-based domain-specific language to provide a modular, scalable, and solver-agnostic architecture. This transition from Euler’s method to advanced differential equations solvers has improved computational accuracy and reduced processing time, enabling higher simulation throughput for design optimization. New modeling methodologies innovated during this year’s PDR and validated during the subscale flight in CDR further increase the robustness of VADL’s simulations. By adopting an object-oriented design, DRSS allows for ergonomic integration of complex event-driven behaviors and alternative descent modeling. These innovations position DRSS as a robust and adaptable tool, both for current mission design and future applications, while adhering to the NASA and VADL performance requirements outlined in Section 6.2.

3.4.1 Flight Profile Simulations and Relevant Requirements

VADL understands the primary objective of flight trajectory simulations is to guide design decisions concerning safety, material and hardware selection, and overall vehicle design. A mission is deemed successful by VADL if it satisfies the requirements outlined by NASA and VADL, as detailed in Section 6.2. Table 26 provides specific performance predictions derived from flight simulations alongside the corresponding NASA/VADL requirements that must be achieved.

Table 26: Relevant NASA and VADL Requirements.

Requirement ID	Requirement
NASA 2.1	The vehicle shall deliver the payload to an apogee altitude between 4,000 and 6,000 feet above ground level (AGL). Teams flying below 3500 feet or above 6500 feet on their competition launch will receive zero altitude points toward their overall project score and will not be eligible for the Altitude Award.
NASA 2.14	The launch vehicle shall have a minimum static stability margin of 2.0 at the point of rail exit.
NASA 2.15	The launch vehicle shall have a minimum thrust to weight ratio of 5.0 :1.0.
NASA 2.17	The launch vehicle shall accelerate to a minimum velocity of 52 fps at the rail exit.
VADL 2.1.3	The full scale and subscale launch vehicles shall have an average Thrust-to-Weight ratio of 5.5 minimum.

Continued on the next page

Requirement ID	Requirement
VADL 2.3.5	Flight simulations in the case of denied jettison shall be conducted.
VADL 3.2	In the event of failed jettison, the descent velocity of the vehicle shall not exceed 20 fps upon landing.
VADL 3.11	The terminal descent velocity under drogue parachute of the rocket shall not exceed 115 fps.
NASA 3.3	Each independent section of the launch vehicle shall have a maximum kinetic energy of 75 ft-lbf at landing. Teams whose heaviest section of their launch vehicles, as verified by vehicle demonstration flight data, stays under 65 ft-lbf will be awarded bonus points.
NASA 3.11	The recovery area shall be limited to a 2500 ft radius from the launch pads.
NASA 3.12	Descent time of the launch vehicle shall be limited to 90 seconds (apogee to touch down). Teams whose launch vehicle descent, as verified by vehicle demonstration flight data, stays under 80 seconds will be awarded bonus points.

To evaluate safety compliance with the outlined rocket flight and recovery requirements, as well as to assess mission performance, three flight scenarios are considered. Unless specified otherwise, the predicted values provided below are based on the VADL-defined baseline conditions of an **11 mph headwind and a 5° launch rail angle**.

- **Scenario A: Official Launch Day Baseline.** This scenario represents the expected conditions on the day of launch in Huntsville, AL. The predicted apogee in this case is the team's **Official Competition Launch Target Altitude**.
- **Scenario B: Jettison Denied.** This scenario evaluates the implications of a denied jettison by the RSO, which necessarily leads to increased sectional mass and kinetic energies upon landing. The simulation ensures compliance with NASA/VADL safety and landing requirements under such conditions.
- **Scenario C: Calm Weather Conditions.** This scenario simulates the absence of wind to evaluate compliance with maximum descent time and other relevant requirements.
- **Scenario D: High Wind Conditions.** This scenario simulates the 20 mph wind case to evaluate compliance with recovery radius and other relevant requirements.

Table 27 outlines the performance outcome metrics of these scenarios; VADL verifies all scenarios are within the vehicle safety and recovery requirements set by NASA and VADL.

Table 27: Flight profiles of select scenarios. Values that meet their respective NASA/VADL requirements are shown in green.

Predicted Values and Relevant Requirements.	Scenario A (Baseline)	Scenario B	Scenario C	Scenario D
Static Stability Margin at LRE (NASA 2.14; cal)	2.27			
Static Stability Margin on Pad (VADL 2.1.2; cal)	2.20			
Thrust-to-Weight Ratio at LRE (NASA 2.15; VADL 2.1.3)	9.1 to 1			
Rail Exit Velocity (NASA 2.17; fps)	DRSS: 77.5 OpenRocket: 79.4			
Apogee (AGL; NASA 2.1; ft)	DRSS: 4319 OpenRocket: 4072 Average: 4196		DRSS: 4468 OpenRocket: 4172 Average: 4320	DRSS: 4088 OpenRocket: 3950 Average: 4019
Vehicle Landing Velocity / Descent Velocity Under Main Parachute (VADL 3.2; fps)	14.6	16.5	14.6	14.6
Descent Velocity Under Drogue Parachute (VADL 3.11; fps)	95.7		95.7	95.8
Payload Landing Velocity / Descent Velocity Under Payload Parachute (VADL 4.1.2; fps)	12.8	N/A	12.8	12.8
Vehicle Drift (descent time $\times$ wind speed; NASA 3.11; ft)	1237	1201	0	2192
Payload Drift (descent time $\times$ wind speed; NASA 3.11; ft)	1270	N/A	0	2253
Vehicle Descent Time after Apogee (NASA 3.12; s)	76.7	74.5	78.3	74.7
Payload Descent Time after Apogee (NASA 3.12; s)	78.7	N/A	80.4	76.8
Sectional Landing Kinetic Energies (NASA 3.3; lb-ft)	Fore: 21.8 Mid: 43.4 Aft: 33.8 Payload: 20.4	Fore: 61.5 Mid: 55.1 Aft: 42.9 Payload: N/A	Fore: 21.8 Mid: 43.4 Aft: 33.8 Payload: 20.4	Fore: 21.8 Mid: 43.4 Aft: 33.8 Payload: 20.3
Sectional Kinetic Energies under Drogue Parachute (lb-ft)	Fore: 2082 Aft: 3313		Fore: 2082 Aft: 3313	Fore: 2083 Aft: 3315

For the baseline **Scenario A**, the flight profile simulations for altitude, velocity, and acceleration versus time predictions are presented below; due to VADL's highly customized vehicle descent methodologies, the figures below were produced by the DRSS model rather than OpenRocket.

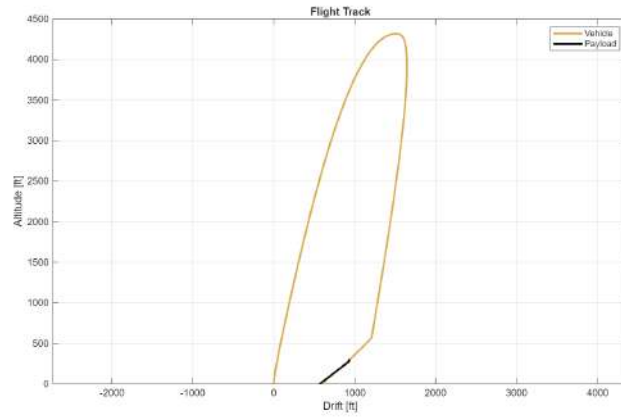


Figure 66: Scenario A (Baseline) Simulated Flight Track.

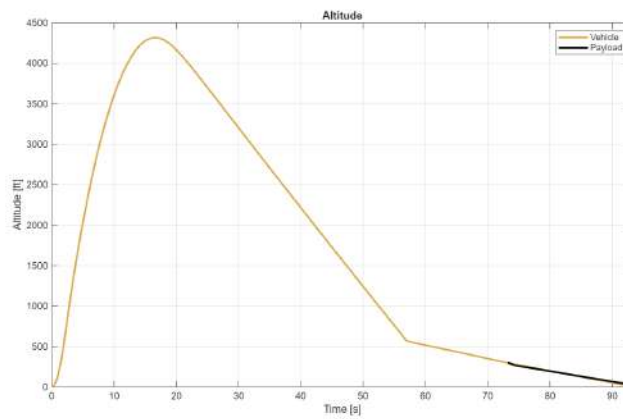


Figure 67: Scenario A (Baseline) Simulated Altitude.

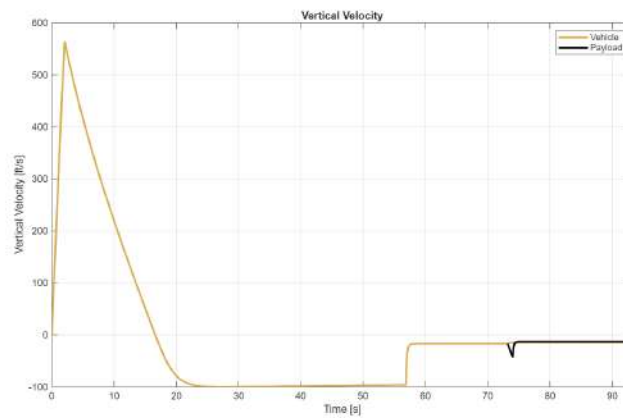


Figure 68: Scenario A (Baseline) Simulated Velocity.

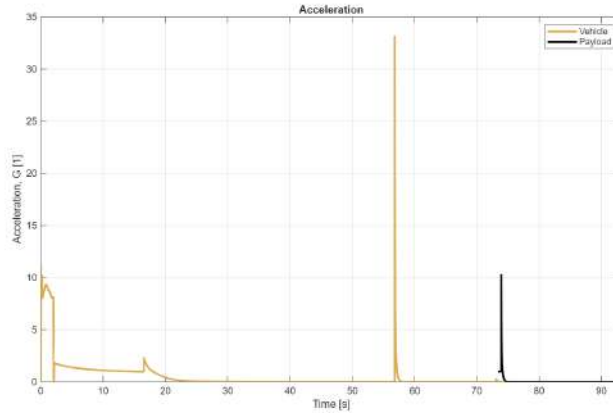


Figure 69: Scenario A (Baseline) Simulated Acceleration.

Comparison with OpenRocket

In the above flight profiles, OpenRocket results are included for flight ascent metrics (e.g., velocity at LRE and predicted apogee). As outlined in Section 3.4.3, this decision was primarily driven by VADL’s highly customized descent modeling, whose methodology inherently tends to produce increased drift and longer descent durations vis-à-vis standard OpenRocket outputs, thus providing a comparatively more conservative basis for demonstrating adherence to NASA Requirements 3.11 and 3.12. The VADL descent modeling, particularly for the full-scale vehicle, is anticipated to outperform OpenRocket in terms of alignment with real-world flight conditions — an expectation that is informed and supported by subscale flight analyses (e.g., Tables 14 and 17).

Because VADL was not able to modify the OpenRocket source code to account for these empirically substantiated model adjustments — implemented within DRSS — that impact simulation outcomes, the OpenRocket projections were conducted and are not graphically nor exhaustively presented here. VADL acknowledges that additional source-level modifications to the OpenRocket software would be required to bring the OpenRocket descent model into more precise alignment with the empirically validated refinements implemented in DRSS.

Table 28: Sample comparison of predicted performance metrics using DRSS vs. OpenRocket.

Framework	Scenario B Predicted Performance Metrics					
	Apogee (ft)	Drift from Pad (ft)	Max Vertical Velocity (fps)	Descent Velocity (fps)	Velocity at LRE (fps)	Thrust-to-Weight Ratio at LRE
CFD-DRSS	4319	636	560	16.5	77.5	9.1
OpenRocket	4065	323	555	16.9	79.4	9.2
% Diff	-6.2%	-96.9%	-0.9%	2.4%	2.5%	1.0%

Concurrent with subscale flight analysis and further discussed in later sections, the discrepancies in apogee predictions between DRSS and OpenRocket can largely be traced to divergent estimations of drag coefficients. In the case of DRSS, the drag estimations are supplied from Computational Fluid Dynamics (CFD) software, with detailed methodology further discussed in Sections 3.4.6 and 3.4.7.

The practice of taking an *average apogee value* from both OpenRocket and DRSS in Table 27 is similarly guided by subscale flight analysis, where the ensemble mean of these two modeling frameworks aligned more reliably with actual altimeter readings than either approach in isolation (Section 3.2.5, Table 14).

Further modeling differences between DRSS and OpenRocket—such as numerical integration methods, wind modeling, aerodynamic theory, and other simulation assumptions—are addressed in Sections 3.4.2 through 3.4.7.

Section 3.4.6 verifies key elements of vehicle design via simulated and computed values obtained from multiple software tools (Solidworks, DRSS, and OpenRocket), which converge to verify against relevant NASA and VADL requirements to ensure mission safety.

Recovery Radius

During subscale flight testing, drift simulation and methodology in VADL’s DRSS model have shown to predict vehicle drift **within 4% of actual flight data** (Table 17). The benefit of using DRSS over a simple multiplication of descent time and wind velocity is to assess the bounding of flight trajectory within recovery radius **at all times of flight**, in addition to landing location. The below figure shows the DRSS-generated flight tracks for wind conditions from 0 to 20 mph in both head and tail wind.

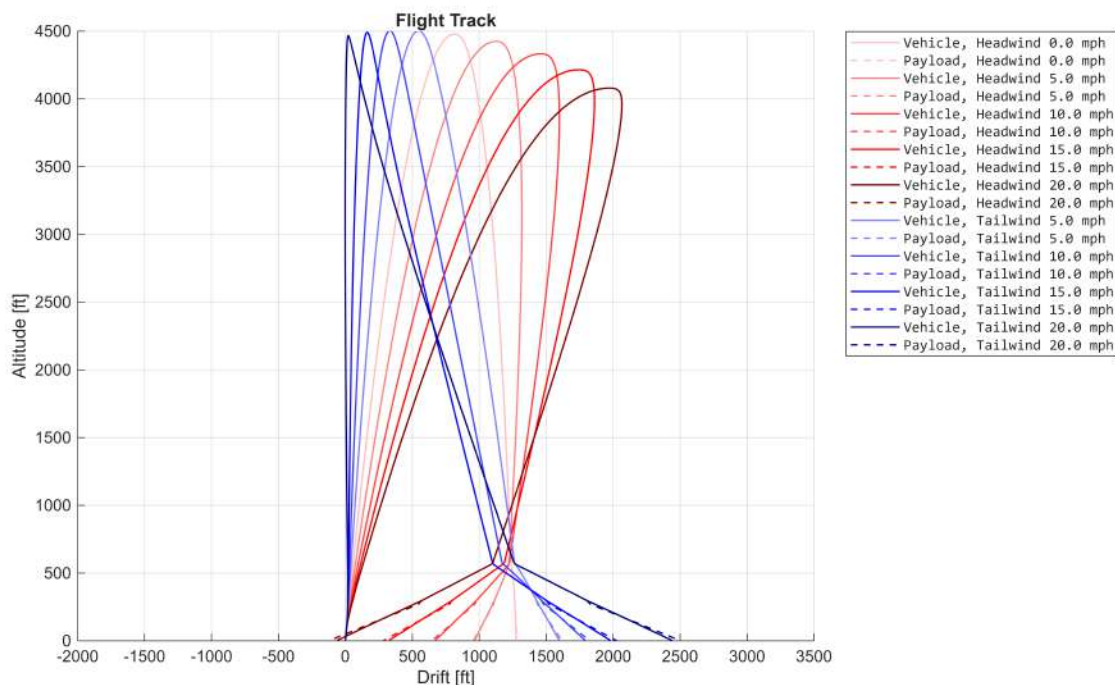


Figure 70: Flight Tracks of Select Wind Conditions for 5-Degree Launch Angle of Full-Scale Vehicle.

Table 29: Drifts at landing using the DRSS model.

Wind Speed (mph)	Payload Drift from Pad (ft)	Payload Drift from Apogee (ft)	Rocket Drift from Pad (ft)	Rocket Drift from Apogee (ft)	Rocket Descent Time (ceiled; s)
0	463	1282	458	1277	79
Headwind, 5	175	956	164	967	79
Headwind, 10	821	644	792	673	77
Headwind, 15	1470	282	1425	327	76
Headwind, 20	2097	124	2033	60	75
Tailwind, 5	1071	1618	1050	1597	79
Tailwind, 10	1494	1827	1457	1790	79
Tailwind, 15	1865	2028	1815	1979	79
Tailwind, 20	2484	2495	2417	2438	79

It is shown above that even under tailwind conditions the trajectories of both the vehicle and payload are strictly bounded within the 2500 ft recovery radius limit **at all times of flight and upon landing**. In all wind cases, the payload and vehicle descent times comply with NASA Requirement 3.12 on maximum descent time limit.

While the above model was able to bound the trajectories at any time during flight, such model still assumes a non-zero time constant for the rocket to catch onto wind speed after apogee. To obtain the maximum theoretical drift **upon landing** to ensure compliance against NASA 3.11 on maximum recovery radius, Table 27 and other results detailed in this section show drift values calculated using a simpler methodology of multiplying the total descent time by the wind speed. In essence, the simpler method assumes the rocket is directly above the launch rail at apogee before drifting to the ground at the velocity of wind (immediately catching the wind). The results are presented below.

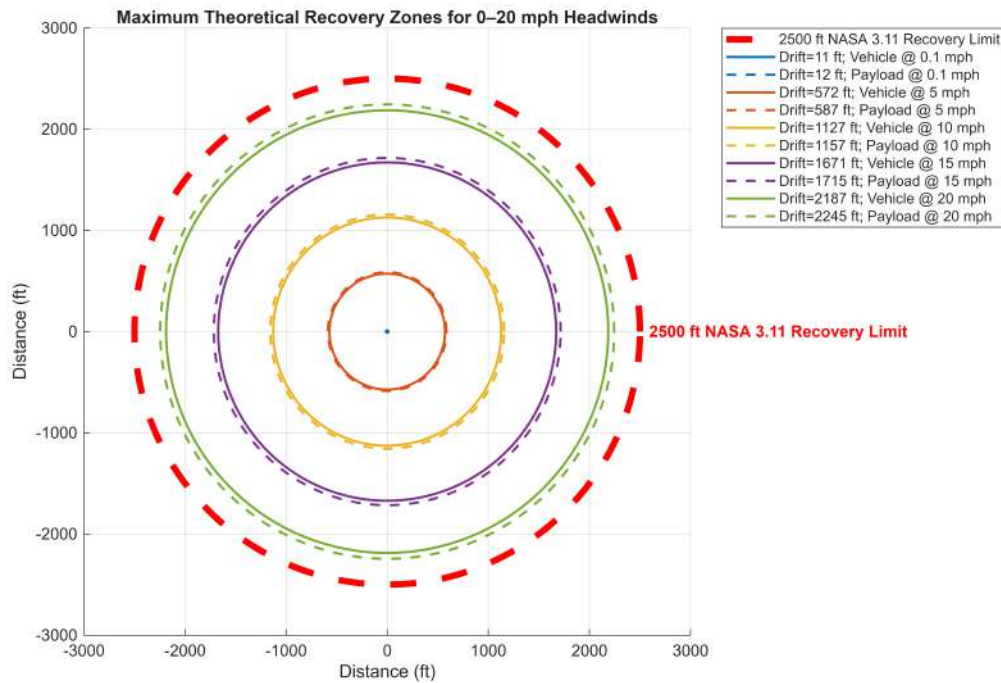


Figure 71: Maximum Theoretical Drift Conditions for 5-Degree Launch Angle of Full-Scale Vehicle.

Table 30: Maximum theoretical drifts.

Wind Speed (mph)	Vehicle Descent Time (s)	Vehicle Recovery Radius (ft)	Payload Descent Time (s)	Payload Recovery Radius (ft)
0	78.3	0	80.3	0
5	78.0	572	80.0	587
10	76.9	1128	78.9	1157
15	76.0	1671	78.0	1716
20	74.5	2187	76.5	2245

Using both DRSS and the maximum theoretical drift methods, the above results verify compliance against NASA 3.11 and NASA 3.12 to ensure safe vehicle and payload recovery.

3.4.2 Numerical Integration Methodology

To ensure a sufficiently high simulation resolution, the **variable-step, variable-order, Adams-Bashforth-Moulton PECE solver of orders 1 to 13** was utilized by VADL’s in-house DRSS to solve the differential equations describing vehicle kinematics and kinetics. As mentioned, this new solver was implemented this year during the PDR cycle to remove previous years’ reliance on Euler’s method of integration. The maximum allowed integration step was 100 ms, the **relative tolerance was limited to 0.0001%**, and the minimum integration step determined and utilized by DRSS to achieve this tolerance was 270 femtoseconds. Optimization of DRSS allows computation

of this low tolerance to be performed under 800 ms of runtime per flight simulation. In Open-Rocket, the time step of 10 ms was used under the default 6-th order Runge-Kutta solver. VADL is confident that simulation resolution does not realistically impose any numerical artefacts that would have otherwise negatively impacted the accuracy and precision of the team’s simulations and analyses.

3.4.3 Parachute and Descent Dynamics

Parachute Unfurling Model

As empirically confirmed through all payload drop tests conducted so far, shock cords greater than 20 ft take greater time to unravel and tighten than the time it takes for parachute unfurling of diameter approximately less than the order of magnitude of 6 ft. As opposed to relying on the exponential parachute unfurling model as utilized during PDR, the CDR DRSS-based models assume that parachute immediately opens after the rocket sections have traveled the unraveled length of the designed shock cords, which necessarily produces a maximum theoretical instantaneous loading during parachute deployment that is more useful for structural analyses, bulk plate and airframe design, and mission performance predictions relative to the previous model. This methodology was empirically and consistently validated through multiple instrumented payload drop tests, which have shown a **high degree of alignment** between theoretical prediction and recorded IMU shock loading Gs as detailed in Table 39. The application of this new model applies homogeneously throughout this report for **structural and safety analyses** in all relevant sections such as Section 3.1.4. The implementation of this model is reflected in DRSS by setting the parachute opening time offset to negative infinity. The estimated shock loadings of each parachute deployment are presented below.

Table 31: Simulated parachute deployment loading.

Parachute	Maximum Shock Loading G	Maximum Total Shock Cord Tension
Drogue	8.1	308 lbf
Main	32.0	1204 lbf
Payload (Table 39)	16.3	132 lbf

In addition to structural analysis of load bearing structures detailed in vehicle sections such as Section 3.1.4, the implications of such loading on recovery system hardware are further discussed in Section 3.3.

Drag Adjustments due to Swinging

As detailed extensively in Section 3.2.5, VADL’s subscale flight revealed that the rocket and parachute side drag must account for the nondeterministic swinging of the rocket body during descent. To address potential underestimations of drift and descent drag characteristics, the full-scale vehicle’s DRSS model has been updated using the new methodology described in Section 3.2.5. This approach partially exposes the top of the parachute canopy to side wind as a function

of the angular swinging of the rocket. The result of this adjusted side drag calculation has been verified in Section 3.2.5 against actual flight data and has been shown to **outperform OpenRocket’s drift predictions** (Table 17).

Additionally, similar adjustments have been made to the vehicle’s frontal drag characteristics where the vehicle’s side profile has been partially exposed in the vertical direction due to large angular swinging. OpenRocket models drag under the assumption that the rocket descends straight and deterministically, akin to the vehicle ascent stage, which does not accurately reflect actual flight conditions. Empirical validation from VADL’s subscale flight demonstrated that **this adjusted DRSS model produced a relative error of less than 0.1% against actual flight data** (Section 3.2.5), outperforming OpenRocket’s prediction, when compared to the actual average descent velocity obtained from altimeter readings. This result provides the team with a reasonable degree of confidence in applying the adjustment to the DRSS model.

As outlined in Section 3.2.5, theoretical scaling considerations on blast charge chemical potential energy, shock cord design, and rocket inertial characteristics point to approximately a **40% reduction** in angular swinging from subscale to full-scale vehicle; this has been applied accordingly to the above modeling.

3.4.4 Launch Rail Dynamics

Theoretical modeling and analyses conducted by Flanigan et al. [6] show that launch rail friction may impose a reduction up to **3.7% in the delivered impulse of a rocket motor**, particularly under crosswind conditions, where the normal forces between the rocket and the rail are increased. **NFPA regulations further allow for a total impulse variation of $\pm 20\%$** in manufactured rocket motors under standard room conditions, making precise apogee prediction challenging. Based on VADL’s historical IMU flight data of the L1720 motor and others, it has been consistently observed that losses in delivered motor impulse cause deviations of up to 5% from manufacturer-published values. To account for these factors, a relatively conservative 3.5% reduction in motor thrust has been applied to all VADL flight simulations in this report. This adjustment was empirically validated by a **high degree of alignment** between theoretical prediction and recorded apogee during VADL’s subscale flight testing (Section 3.2.5) under such methodology. The adjustment was implemented by scaling the nominal L1720 thrust curve down by 3.5% within both DRSS and OpenRocket simulations. Using the above methodology, the adjusted thrust curves for the L1720 motor are presented below.

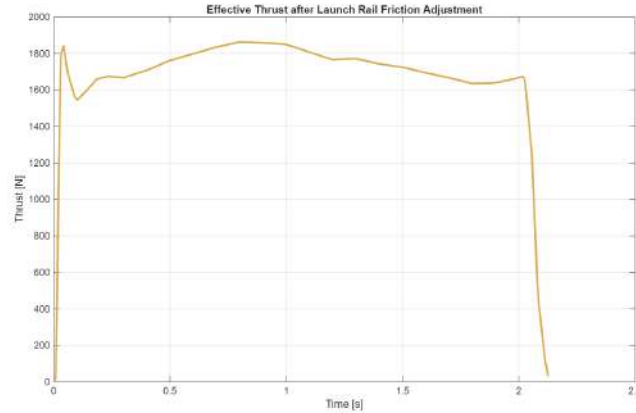


Figure 72: L1720 Adjusted Thrust Curve.

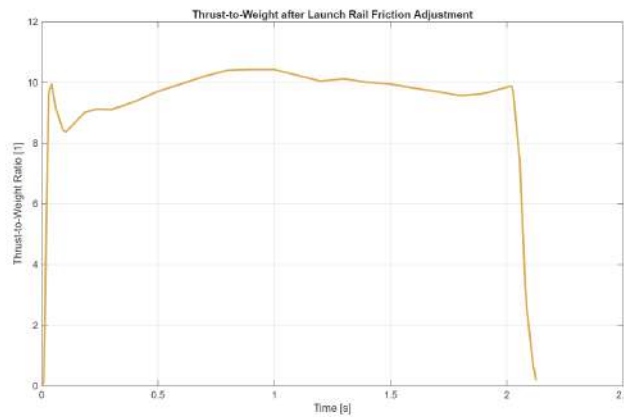


Figure 73: L1720 Adjusted Thrust-to-Weight Ratio.

3.4.5 Environmental Modeling

This section describes non-exhaustive and major aspects of environmental modeling implemented in VADL's in-house DRSS.

Planet Boundary Layer

The DRSS simulations model wind speed as a function of the rocket's altitude above surface level (ASL), accounting for the effects of viscous friction within the planetary boundary layer. The wind speed at a given altitude is computed using Equation 26:

$$w = w_r \left(\frac{z}{z_r} \right)^{\frac{1}{\xi}} \quad (26)$$

Here, w is the computed wind speed at altitude z above the surface, w_r is the reference wind speed at a reference altitude z_r , and ξ is a dimensionless parameter representing atmospheric stability. For these simulations, the reference altitude z_r corresponds to the launch site altitude of 800 feet

ASL, and w_r is the specified wind speed at this altitude. Under neutral stability conditions, ξ is set to 7.

As an example, for a base wind speed w_r of 10 mph at 800 feet and a target altitude z of 4800 feet, the computed wind speed is as follows:

$$w = 10 \left(\frac{4000}{800} \right)^{\frac{1}{7}} \approx 12.9 \text{ mph}$$

Air Density Variation

Variation in air density as a function of altitude ASL is modeled using Equation 27:

$$\rho(z) = \frac{\rho_0}{R(T_0 - \beta z)} \left(1 - \frac{\beta z}{T_0} \right)^{\frac{g_0}{R\beta}} \quad (27)$$

Here, T_0 is the reference temperature at the launch site, ρ_0 is the air density at the reference altitude, R is the specific gas constant for air, g_0 is the acceleration due to gravity, and z is the altitude. A linear temperature lapse rate, β , is assumed due to the flight occurring within the troposphere. For these calculations, $\beta = 6.50 \times 10^{-3} \text{ K/m}$, $R = 287 \text{ J/(kg}\cdot\text{K)}$, $T_0 = 288.15 \text{ K}$, and $\rho_0 = 1.225 \text{ kg/m}^3$.

Motor Exhaust Friction

During the powered ascent phase, the inertia of the propellant exhaust against the motor housing induces frictional damping that generates a moment opposing the tilting motion of the rocket. This phenomenon is characterized by Equation 28, where $\dot{m}$ represents the propellant mass flow rate, $\dot{\theta}$ is the angular velocity, L denotes the total rocket length, and L_{CG} is the distance from the nose cone tip to the center of gravity (CG). Thrust values are extracted from [3] as a CSV file, and linear interpolation is applied to compute thrust values for each simulation time step.

$$M = \dot{m}\dot{\theta}(L - L_{CG})^2 \quad (28)$$

For example, assume the following parameters for a powered ascent scenario:

- $\dot{m} = 2 \text{ kg/s}$ (propellant mass flow rate)
- $\dot{\theta} = 0.1 \text{ rad/s}$ (angular velocity)
- $L = 2.5 \text{ m}$ (rocket length)
- $L_{CG} = 1.5 \text{ m}$ (distance to CG)

Substituting these values into Equation 28:

$$M = 2 \cdot 0.1 \cdot (2.5 - 1.5)^2 = 0.2 \text{ Nm}$$

The resulting moment would be 0.2 Nm under this scenario, which would provide the thrust damping effect during powered ascent.

Air Compressibility

Compressibility effects on aerodynamic coefficients become non-negligible when the Mach number (M) exceeds approximately 0.3. These effects are modeled using Equation 29, which accounts for the adjustment of the baseline coefficient C due to air compressibility:

$$C' = \frac{C}{\sqrt{1 - M^2}} \quad (29)$$

Here, C' represents the adjusted coefficient, C is the baseline coefficient, and M is the Mach number.

For instance, assume the following parameters for a scenario where compressibility effects are considered:

- $C = 0.5$ (baseline aerodynamic coefficient)
- $M = 0.4$ (Mach number)

Substituting these values into Equation 29 yields:

$$C' = \frac{0.5}{\sqrt{1 - (0.4)^2}} = \frac{0.5}{\sqrt{0.84}} \approx \frac{0.5}{0.9165} \approx 0.595$$

The adjusted coefficient C' is approximately 0.595, which has non-negligible difference against the nominal $C = 0.5$ value due to air compressibility effects at $M = 0.4$.

Gravity Modeling

For the team's declared apogee of approximately 4200 ft, the acceleration due to gravity decreases non-negligibly with increasing altitude as the distance from the Earth's center grows. This variation is modeled using Equation 30, where g_0 is the standard gravitational acceleration at sea level, r_E is the Earth's radius, and z is the altitude above sea level:

$$g = g_0 \left(\frac{r_E}{r_E + z} \right)^2 \quad (30)$$

Assume the following parameters for a scenario considering altitude effects on gravity:

- $g_0 = 9.80665 \text{ m/s}^2$ (gravitational acceleration at sea level)
- $r_E = 6371 \text{ km}$ (Earth's radius)
- $z = 2 \text{ km}$ (altitude)

Substituting these values into Equation 30:

$$g = 9.80665 \left(\frac{6371}{6371 + 2} \right)^2 \approx 9.794 \text{ m/s}^2$$

, which yields an approximately -0.13% decrease in gravitational pull. Within DRSS and OpenRocket, the local gravity of the Alabama area of $9.7964m \cdot s^{-2}$ was applied (Department of Physics, Montana State University [1]).

3.4.6 Vehicle Verification

Mass Ballast Effects

The effects of mass ballast variation are not assessed as the team’s final vehicle design and massing at the CDR milestone reaches apogee within a reasonable margin of the declared altitude of 4200 ft. To comply with NASA Requirement 2.1, it is important for the rocket massing to be able to reach above 4000 ft; while subscale flight testing has shown that the ensemble average of model simulations accurately predicted the true apogee, since the lower end of the predictions made by the two simulation frameworks (i.e., OpenRocket) approach this limit for most wind conditions (10 mph to 20 mph), the team does not intend to use any method of mass ballasting on launch day to mitigate the risk of falling below the 4000 ft threshold.

Additionally, subscale flight testing indicates that anemometer wind readings are unreliable for fully characterizing wind effects, as wind velocities and gust strengths were highly time-variant even within the duration of a single flight; consequently, the team decision is conservative to forgo ballast adjustments in ensuring compliance with altitude requirements under varying wind conditions. The effects of wind on apogee are outlined below. As mentioned, the use of the ensemble average of model simulations were validated through subscale flight testing, which has shown that the predicted apogees are reasonably close to the declared 4200 ft altitude without massing adjustments (Table 32).

Table 32: Predicted Apogee vs. Wind.

	Predicted Apogee (ft)			
	5 mph	10 mph	15 mph	20 mph
CFD-DRSS	4425	4333	4215	4080
OpenRocket	4134	4083	4022	3954
% Diff	-7.0%	-6.1%	-4.8%	-3.2%
Ensemble Avg.	4280	4208	4119	4017
Target Apogee	4200			
% Diff	-1.9%	-0.2%	2.0%	4.6%

In the subsequent section, the stability margin of the full scale vehicle is shown to comply with relevant NASA and VADL requirements without any ballast, which again supports VADL’s decision to not rely on ballast adjustments for apogee control.

Stability Margin Verification

To ensure vehicle compliance against NASA 2.14 on minimum static stability margin (SSM), the vehicle Center of Gravity (CG) values are computed using OpenRocket, DRSS and Solidworks,

which all converge as shown below.

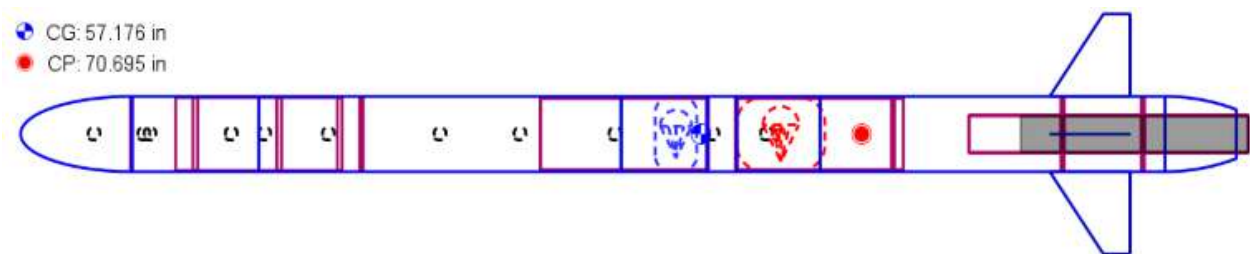


Figure 74: Full-scale rocket design from OpenRocket showing the center of gravity (CG) and the center of pressure (CP).

Table 33: Comparison of Rocket CP Measurement Methods.

Software	Overall Rocket CG from tip of Nose Cone (in)
SolidWorks	57.5
OpenRocket	57.2
DRSS	57.1

The vehicle Center of Pressure (CP) is independently computed using OpenRocket and DRSS from vehicle geometry, which all converge to verify compliance against SSM as shown below.

Table 34: Comparison of Rocket CP Measurement Methods.

Software	Rocket CP from tip of Nose Cone (in)	SSM on pad / Minimum SSM (cal.)
OpenRocket	70.7	2.19
DRSS	69.5	2.20

Vehicle Drag Calculations

To achieve accurate flight simulations, drag coefficients for both the subscale and full-scale vehicles were calculated using the CFD software ANSYS Fluent (Canonsburg, Pennsylvania, U.S.). For the full-scale vehicle, the resulting coefficient of drag of 0.3138 was subsequently incorporated into the DRSS model as reference values using Box’s method [2], which is grounded in the United States Air Force’s DATCOM aerodynamic theory. Adjustments were made to account for angle-of-attack variations and other environmental factors, with the drag coefficients further decoupled into normal and axial components for use within the DRSS simulation. The below table details the differences in drag characteristic predictions between OpenRocket and DRSS.

A similar divergence in predicted drag characteristics was observed when conducting subscale pre-flight simulations. As extensively analyzed in Section 3.2.5, the actual subscale apogee lies in-between the predicted apogees of the two frameworks (i.e., CFD-DRSS and OpenRocket) — both frameworks equally achieved a reasonable degree of accuracy with 5.3% and -6.9% relative

Table 35: Predicted Coefficients of Drag of the Fullscale Vehicle.

	Predicted Coefficient of Drag
OpenRocket	0.42
ANSYS Fluent	0.31

errors, respectively. As required by the NASA USLI Handbook, estimation of the drag coefficient of the full-scale vehicle was enabled by the subscale flight data in the context that the subscale performance of the same simulation and CFD frameworks has instilled the team's confidence to apply them to the full-scale vehicle. The CFD framework is likely under a more idealistic context and therefore produced a marginally decreased coefficient of drag relative to OpenRocket.

3.4.7 Vehicle Ascent Methodology

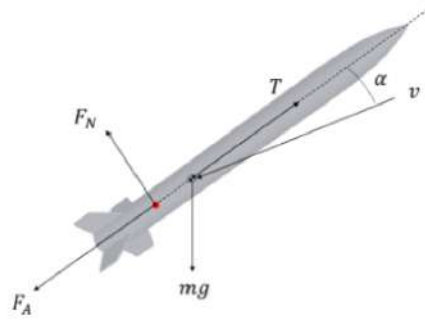


Figure 75: Flight Dynamics of Rocket Flight During Ascent.

In DRSS, the respective forces of the side wind F_N , air resistance F_A , thrust T , gravitational force mg , angle of attack α , and free-stream velocity v , are employed in simulating the rocket's trajectory. Using Equation 31, the lift and drag forces can be determined, where i represents N (normal) or A (axial), ρ is the fluid density, C_i is the aerodynamic coefficient, and A is the cross-sectional area of the rocket at the base of the nose cone.

$$F_i = \frac{1}{2} \rho v^2 C_i A \quad (31)$$

For C_N , the method of Box with reference value provided by CFD is utilized, which incorporates inputs such as the angle of attack, Reynolds number, and rocket geometry to determine the value of C_N . The coefficient of drag, C_D , is obtained through computational fluid dynamics (CFD) simulations as mentioned. Finally, C_A is calculated using Equation 32:

$$C_A = \frac{C_D \cos(\alpha) - 0.5 C_N \sin(2\alpha)}{1 - \sin^2(\alpha)} \quad (32)$$

In Equation 32, α , representing the rocket's angle of attack, is derived using Equation 33. Here, θ denotes the rocket's angle relative to the vertical, and ϕ represents the angle of the free-stream velocity relative to the vertical:

$$\alpha = \phi - \theta = \tan^{-1} \left(\frac{v_x + w}{v_z} \right) \quad (33)$$

Utilizing Equations 31–33, an example calculation for determining the normal and axial forces is provided below. A baseline case with a 10 mph wind and a predicted rail exit velocity of 60 fps is considered for a perpendicular launch rail. The angle of attack is first calculated:

$$\alpha = \phi - \theta = \tan^{-1} \left(\frac{v_x + w}{v_z} \right) = \tan^{-1} \left(\frac{15fps}{60fps} \right) = 14.0^\circ$$

Next, the method of Box yields a C_N value of 2.1 under these conditions. Assume CFD analysis provides an average C_D value of 0.32. Substituting these into Equation 32 results in a C_A value of 0.30. Finally, Equation 31 is used to compute the axial and normal forces. At rail exit, the normal force F_N , representing the force inducing wind-cocking, is determined as follows:

$$F_N = \frac{1}{2} \rho v^2 C_N A = (0.5)(1.2, kg/m^3)(18.3, m/s)^2(2.1)(0.0220, m^2) = 9.28, N = 2.09, lb$$

The above calculation illustrates the magnitude of forces present at the rail exit, which must be accounted for in trajectory modeling to ensure accuracy in predicting vehicle behavior

CFD Methodology

The VADL's CFD methodology involves first rendering the full-scale and subscale vehicle structures in Solidworks. Detailed meshing characteristics including inflation layers were then created around the rocket body and the Reynolds-Averaged Navier-Stokes (RANS) equations were applied using the $k-\omega$ turbulence model. Appropriate boundary conditions, derived from field observations, were applied to accurately define the simulation domains (e.g., walls, outlet pressure, and inlet velocity).

Since the rocket body encounters varying atmospheric conditions as it ascends, the boundary conditions that change with altitude were averaged to a constant value during the simulation convergence process. This approach simplifies simulation procedures, reduces the number of required studies, and maintains computational accuracy. The temperature was set at 288.16 K, reflecting the likelihood of a launch occurring within ± 5 K of this average atmospheric value. The average dynamic viscosity was specified as 1.6×10^{-5} Pa-s, and the average air density was set to 1.15 kg/m^3 . Additional parameters, such as the Y^+ value for the heat transfer coefficient, were maintained at their default settings, as they do not significantly influence the accuracy of drag or pressure studies. Other constants, including the ratio of specific heats ($\gamma = 1.4$) and gauge pressure ($P = 0 \text{ Pa}$), were based on experimentally validated values.

Reference Values	
Area [m ²]	0.0094689
Density [kg/m ³]	1.225
Enthalpy [J/kg]	0
Length [m]	1.44
Pressure [Pa]	0
Temperature [K]	288.16
Velocity [m/s]	50
Viscosity [kg/(m s)]	1.7894e-05
Ratio of Specific Heats	1.4
Yplus for Heat Tran. Coef.	300

Figure 76: Boundary conditions used in ANSYS Fluent simulation.

The $k - \omega$ turbulence model was selected due to its proven accuracy in predicting flow behavior in near-wall regions and its effectiveness in handling flows with adverse pressure gradients. Furthermore, the $k - \omega$ -SST (Shear Stress Transport) variation was utilized for its accuracy in simulating complex geometries, as it can effectively account for the significant curvature gradients present in rocket designs.

To ensure simulation accuracy, an optimized mesh configuration was developed. The number of mesh elements was increased incrementally until a balance between simulation convergence and accuracy was achieved. Additionally, inflation layers were incorporated to more accurately represent fluid behavior in the boundary layer near the rocket's surface. The thickness of the inflation layers was determined based on a prior literature review and calculations performed using online mesh-optimization tools.

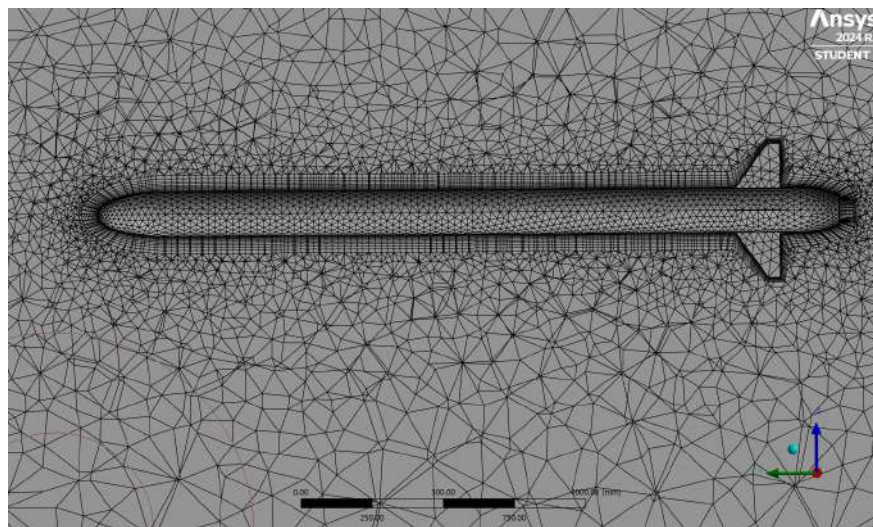


Figure 77: Side view of full-scale mesh.

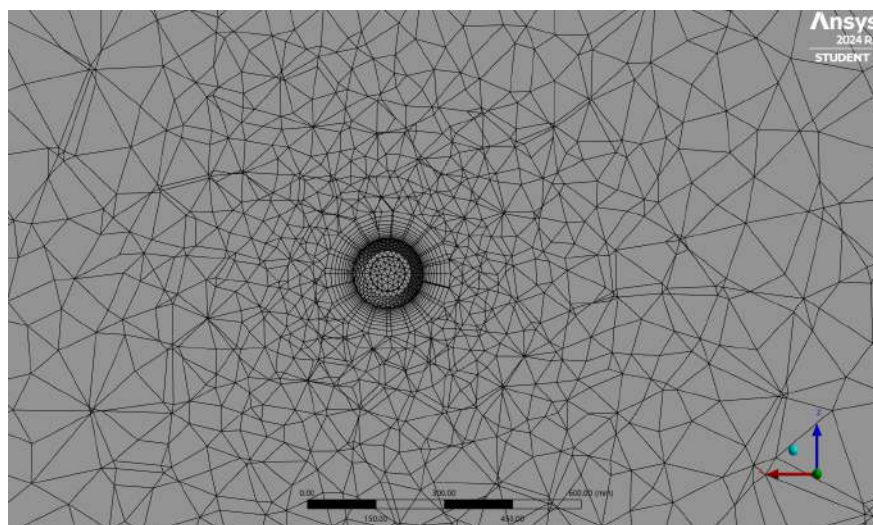


Figure 78: Front view of full-scale mesh; cross section measured 150 cm from tip of nose cone. As seen below, for the full scale model, the drag coefficient generally decreases with increasing

velocity and converges at 0.303 as velocity approaches infinity. The mean and median C_D was determined as .3138 and .3113, respectively.

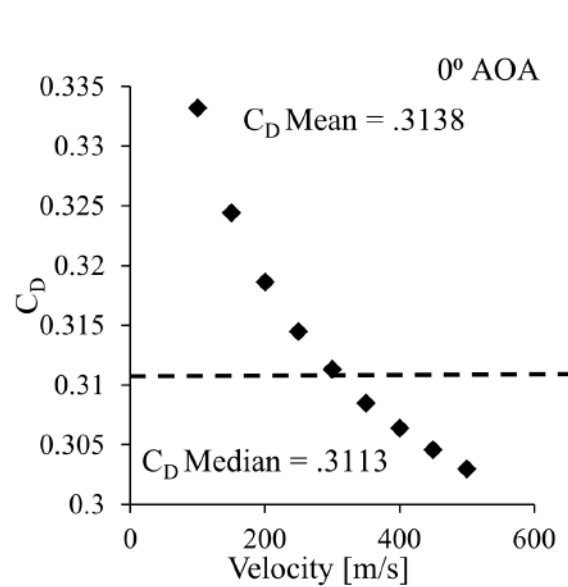


Figure 79: Full Scale coefficient variation as a function of vertical velocity for 0 degree angle of attack.

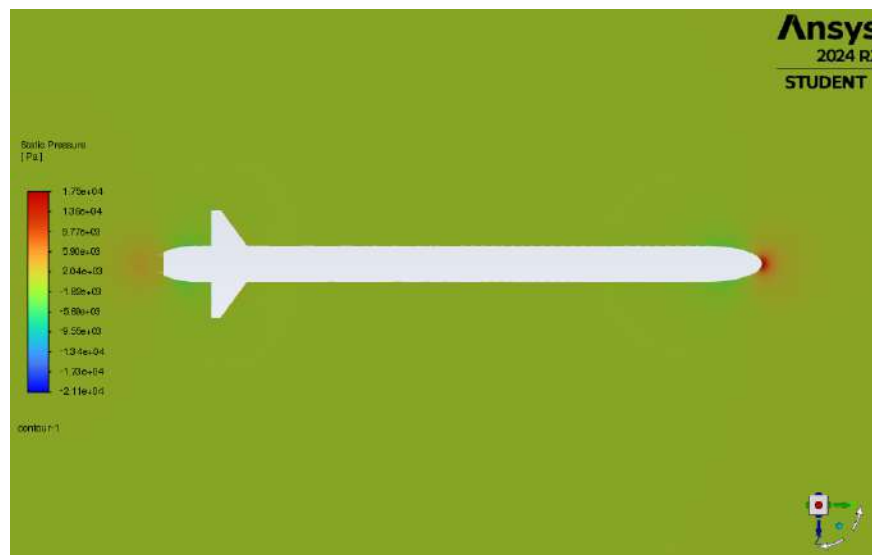


Figure 80: Static pressure contour plot for full-scale travelling at 175 m/s and 0 deg angle of attack.

4 Payload Criteria

4.1 Design and Verification of Payload Experiment

Lander Design Evolution

Previously during the PDR cycle, the team went through multiple lander design iterations in order to determine which would best satisfy VADL requirement 4.1.7. It was determined that in order to ensure an upright landing the lander needed a low CG and wide base. This was solved by utilizing a horizontal lander where the nose cone lands parallel to the ground instead of perpendicular as in the vertical lander versions. Three drop tests from 300 ft AGL were conducted, and horizontal lander designs successfully landed upright every single time.

The overall payload structure in the CDR cycle remains consistent with the PDR version. Components such as the LSM, LDM, PDM, and AEM contain small adjustments to improve reliability and account for slight changes in materials. The biggest change to the lander design is the switch to a larger parachute diameter and shorter shock cord. The parachute was increased from a diameter of 4 ft to 5 ft, and the shock cord was shortened from 50 ft to 30 ft. The larger parachute was chosen to reduce the landing velocity from 14 to 12.8 fps. The shock cord was shortened in order to reduce the loading experienced during parachute opening to match the previous 4 ft diameter parachute.

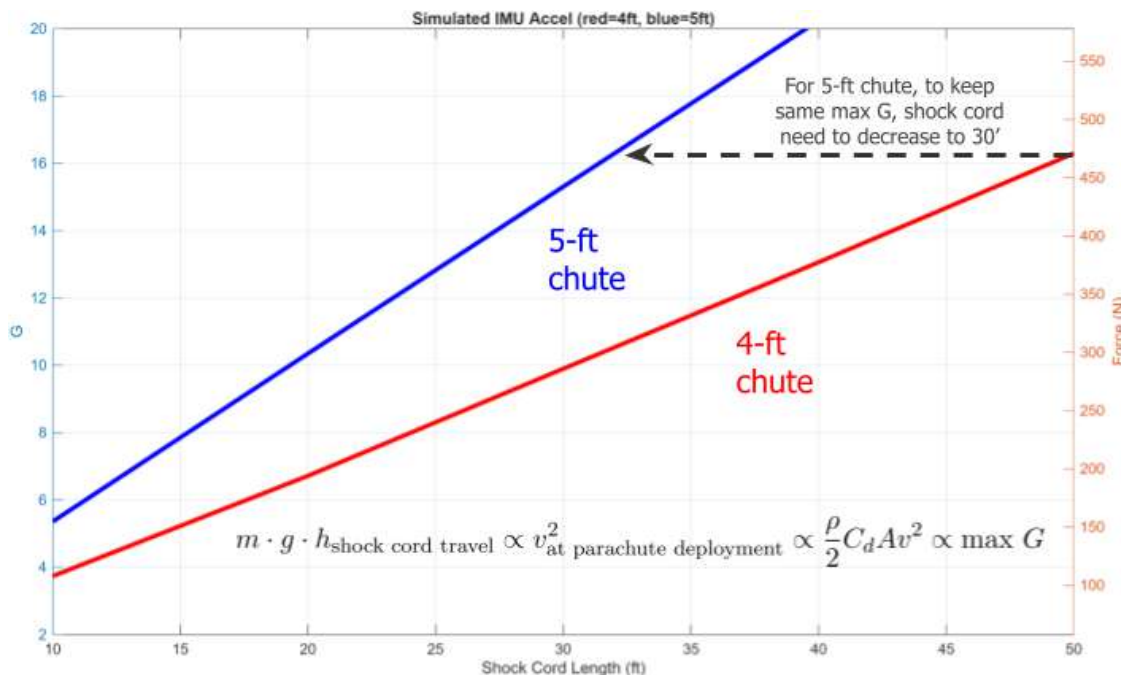


Figure 81: Shock cord loading vs length for 4 ft and 5 ft parachutes.

The shock cord loading calculation in Figure 81 uses the same methodology as set forth in Section 3.1.4 Equation 1.

4.1.1 Payload Mission CONOPS

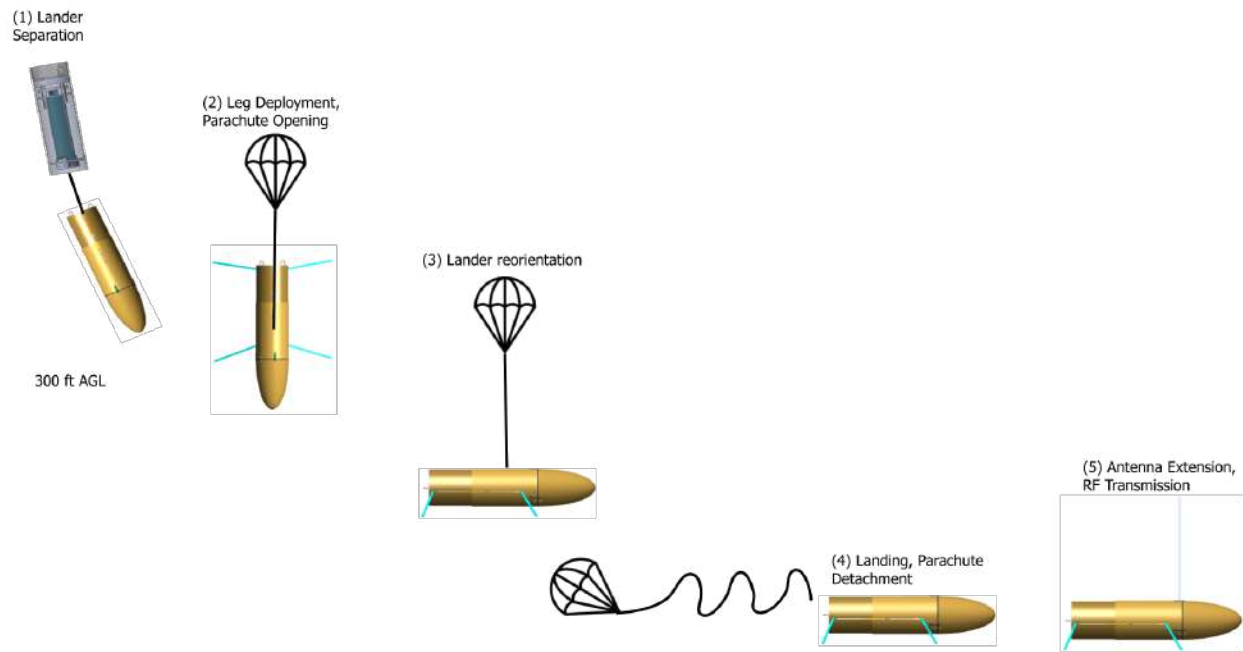


Figure 82: Payload mission concept of operations.

The payload consists of a parachute- arrested lander that separates from the fore-section of the launch vehicle during main parachute descent at an altitude of 300 ft AGL. The lander separation mechanism (LSM) checks for both altitude and RSO approval from a ground RF transmitter to keep consistent with NASA requirement 4.3.3. The leg deployment mechanism (LDM) releases the legs once they are not restrained by the launch vehicle body. These legs serve to reduce the landing impact on the structure of the lander, facilitate an upright landing, and double as the ground plane for the antenna system as they are constructed out of aluminum.

The lander reorients from a vertical (nose-cone towards ground) configuration to a horizontal (nose-cone parallel to ground) following parachute opening. Upon landing, the parachute detachment mechanism (PDM) releases the parachute, preventing it from toppling the payload and interfering with RF transmission by preventing the antennas from achieving maximum height.

The antenna extension mechanisms (AEMs) unfurl the antennas to a height of 24 in above the ground and begin transmitting telemetry back to the NASA ground station over the 2M band. The AEMs consist of tape springs wound around a damped cylinder that self deploy when released. The tape spring acts as the antenna itself as it is constructed out of steel. The pieces of data chosen to transmit are the temperature of the landing site, apogee, battery status (in progress), orientation of STEMnauts, landing time, maximum lander vertical velocity, landing velocity/G forces sustained, and STEMnaut crew survivability using Dynamic Response Index (in progress).

4.1.2 Payload Design Overview

The payload consists of the lander and the LSM that remains on the launch vehicle after separation. The overall design layout can be seen in Figure 83.

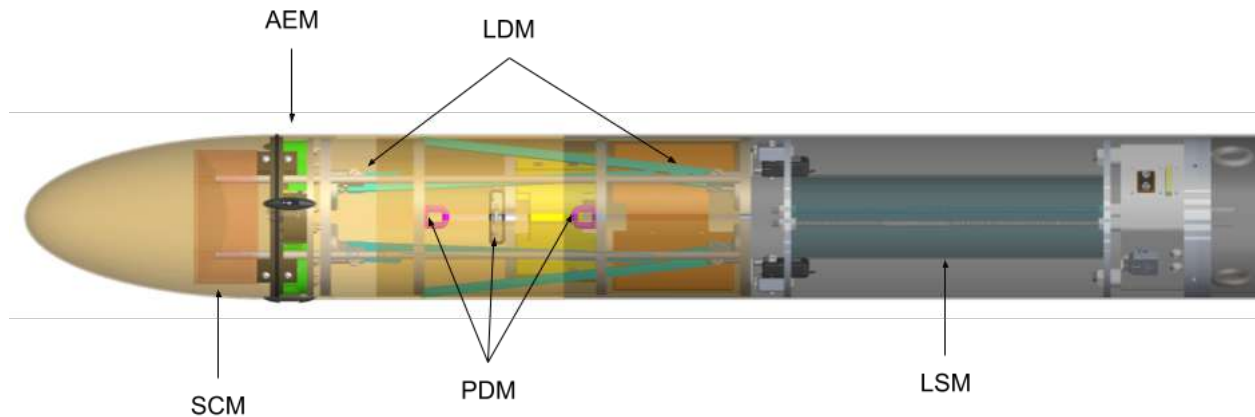


Figure 83: Payload design overview with major components highlighted.

The LSM consists of a Teensy microcontroller connected to an RF receiver and IMU/altimeter to ensure separation occurs at 300 ft AGL and with the approval of the RSO. Sandwiched between the electronics and the bulk plate holding the latches is an inner tube that ensures the parachute does not become entangled with the latches. The structure of the LSM can be seen in Figure 84.

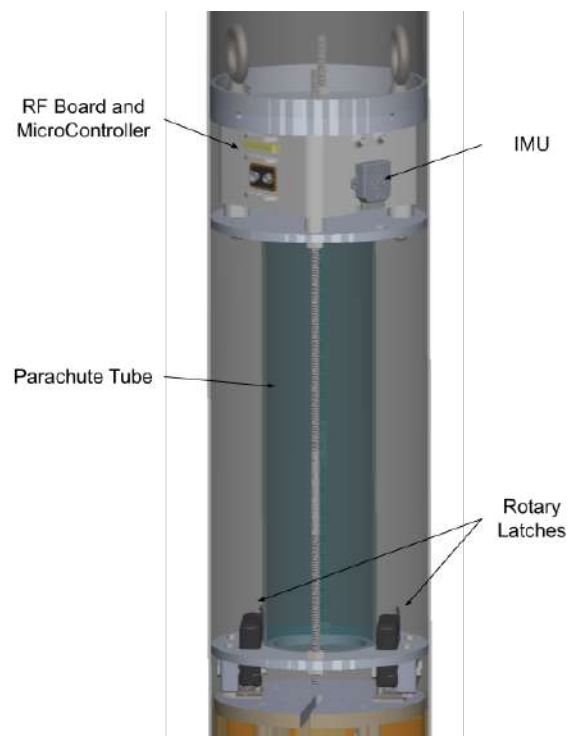


Figure 84: Lander Separation Mechanism (LSM).

The LDM serves two purposes: it softens the impact of landing, and ensures the lander lands upright and doesn't tip over afterwards. The LDM structure can be seen in Figure 85 below:

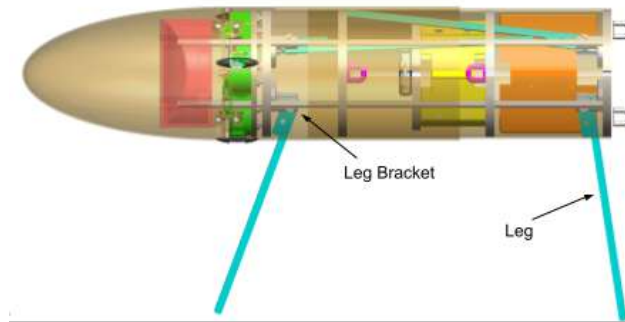


Figure 85: Leg Deployment Mechanism (LDM).

The PDM serves to ensure the parachute does not drag the payload and jeopardize RF transmission by damaging the antenna or tipping the lander over. The PDM is actuated upon landing detection, and its overall structure can be seen in Figure 86 below.

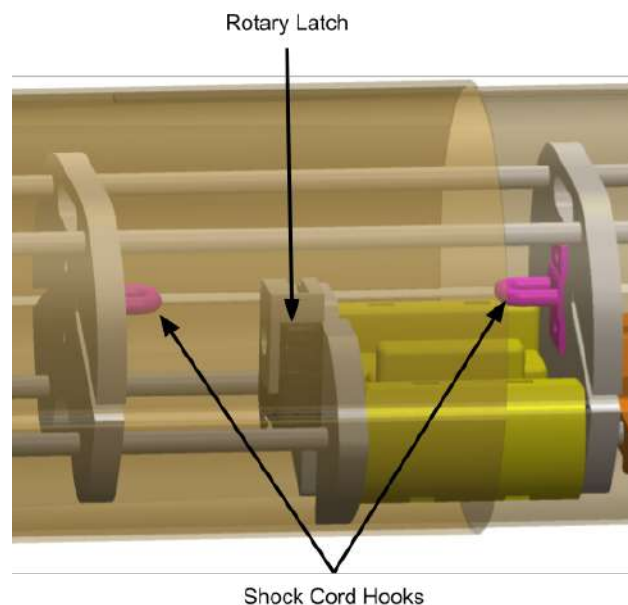


Figure 86: Parachute Detachment Mechanism (PDM).

The AEM serves to maximize the RF transmission distance by raising the antenna. There are 3 AEMs on the lander to ensure 1 extends away from the landing surface in the event of an in-rocket landing where the lander doesn't separate from the launch vehicle and the LDM doesn't deploy. The overall structure can be seen in Figure 87 below. The antenna covers were designed such that they do not extend further than 0.25 in from the airframe surface in accordance with NASA requirement 4.2.2.

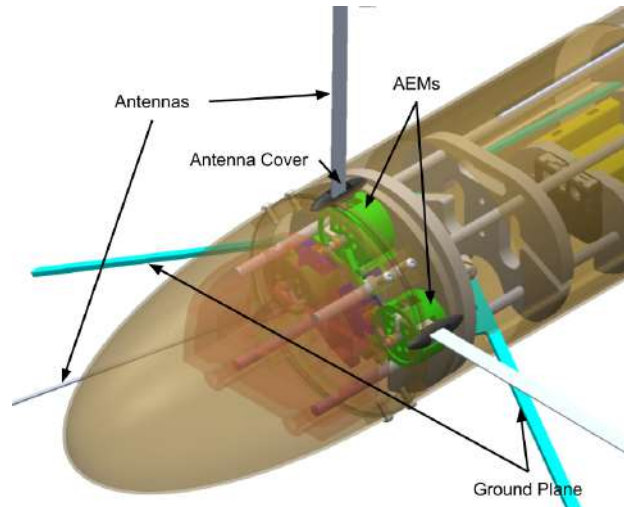


Figure 87: Antenna Extension Mechanism (AEM).

4.1.3 Payload Verification Overview

Prior to this drop test, the team conducted thorough subsystem testing on the electronics and mechanisms to ensure they would function properly under rigorous conditions. These tests included repeated drop testing from heights varying from 4-6 ft, and the drop test from 300 ft AGL verified the structural integrity once again, giving the team confidence in the design. Summaries of these tests are provided in Table 36 below, and further details regarding each test can be found in Section 6.1.2.

Table 36: Payload testing summary.

Payload Aspect	Goal	Verifying Test	Test Results
Integrated Payload functionality	Payload successfully recovers from a 300 ft AGL drop and transmits telemetry over RF to ground station	Full-scale lander drop test from 300 ft AGL	All mechanisms except PDM performed successfully; payload safely recovered and transmitted data during 3 out of 4 drops
Payload structural integrity	Payload structure withstands expected loads during all aspects of rocket flight including launch, parachute deployments, and landing.	In lab drop tests from 4 to 6 ft AGL Full-scale lander drop test from 300 ft AGL	Payload undamaged and ready to be re-flown immediately
STEMnaut Survivability	STEMnauts survive landing with low chance of injury in accordance with DRI modeling	Full-scale lander drop test from 300 ft AGL	DRI values for 2 of 3 drops corresponded to a moderate risk of injury, thus driving the design change to a larger parachute, and increasing leg extension below lander belly.

Integrated Payload Verification

The team repeated a 300 ft AGL drop test in similar fashion to the tests conducted during the PDR cycle. The lander was raised under hexacopter and repeatedly swung to simulate the non-quiescent conditions of rocket descent. The LSM was triggered from a ground RF signal and the lander was safely recovered under a 4 ft diameter parachute. The lander detected landing via acceleration and altitude checks, raised the antenna, and transmitted telemetry back to the team ground station using APRS on the 2M RF band.

Four drops were performed during this test. On the second drop, the lander landed on a hill above the altitude threshold for landing detection, so the flight computer did not trigger the landing and RF transmission sequence. The data was also not logged due to this error. This issue was resolved by raising the altitude threshold for the remaining drops. For the remaining 3 drops the lander successfully detected landing and transmitted data over RF using the 2M band to a ground receiver. The data was packaged using the APRS protocol, as shown in Figure 90. Snapshots depicting various stages of the descent and landing sequence are shown in Figure 88, and the results from the tests are summarized in Figure 89. Altitude over time for the three successful drops is shown in Figure 91.

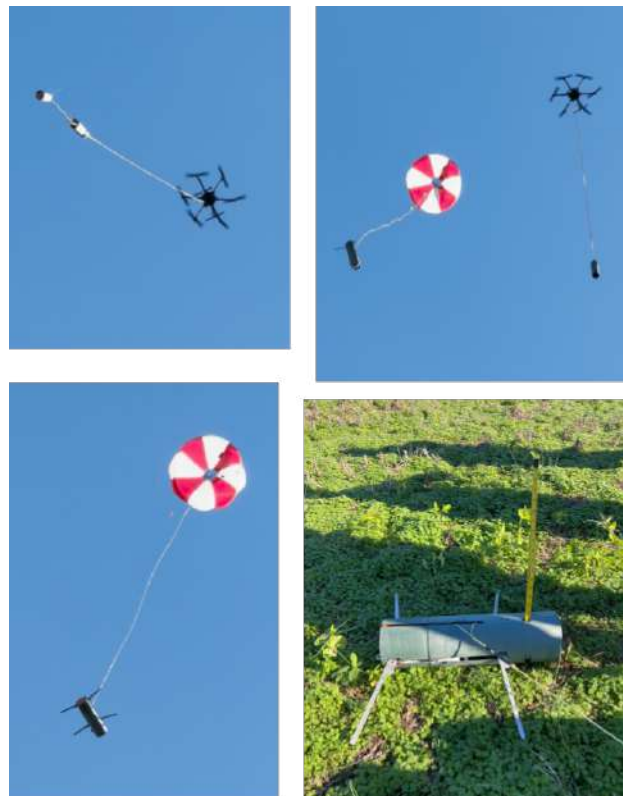


Figure 88: Lander during separation, parachute opening and leg deployment, descent, and landing. Antenna extension can be seen in landing configuration.

Payload Function	Drop 1	Drop 2	Drop 3	Drop 4
Lander Separation	Successful	Successful	Successful	Successful
Leg Deployment	Successful	Successful	Successful	Successful
Landing Detection	Successful	Fail	Successful	Successful
Data Collection	Successful	Fail	Successful	Successful
Parachute Detachment	Partially Successful	Fail	Partially Successful	Partially Successful
Antenna Extension	Successful	Fail	Successful	Successful
RF Transmission	Successful	Fail	Successful	Successful

Figure 89: Summary of results for each of the four drops as they pertain to key payload objectives.

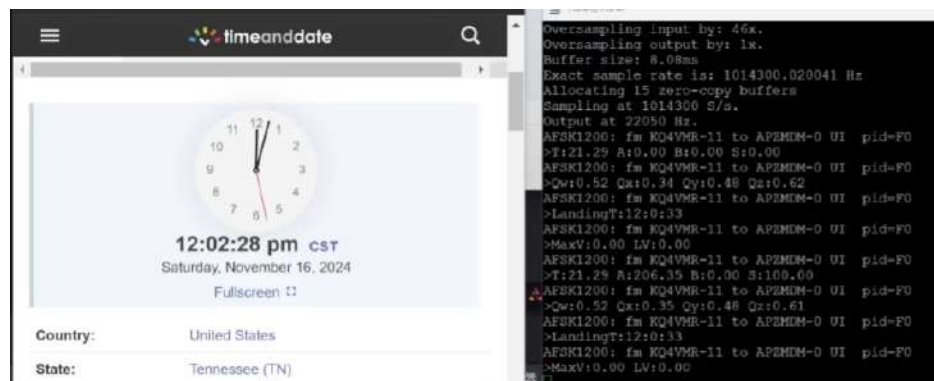


Figure 90: APRS ground receiver screenshot.

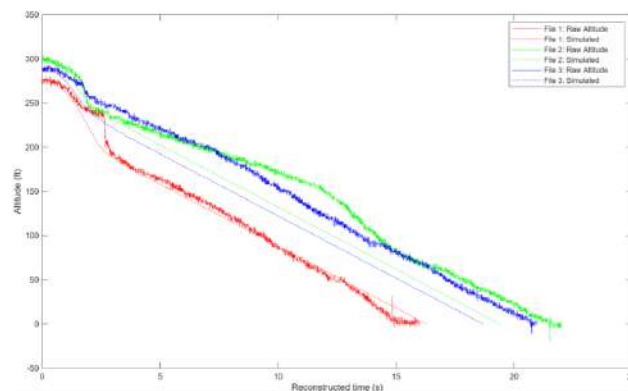


Figure 91: Flight telemetry for the 3 successful drops.

This test successfully verified the functionality of the LSM, LDM, and AEM. The PDM was the only mechanism that was not fully verified. The PDM did actuate properly and opened to release the parachute shock cord, however the shock cord was caught on the sides of the hook preventing proper release. Further discussion of this failure and further design improvements will be discussed in 4.2.3.

Payload Structure Verification

All components and the overall payload structure remained intact after all 4 drops. No visible damage was observed in the field, nor during disassembly after the completion of the drop test. The electronics likewise did not suffer any damage and were reused in the payload for the subscale vehicle.

As stated earlier, the only modification made to the payload structure from the final design is the use of a blue tube airframe and lack of nose cone. These changes were made to expedite field testing and provide practice for the final payload fabrication. The fiberglass airframe has a much higher tensile and compressive strength compared to the blue tube, and vehicle airframe testing has confirmed that it is unlikely that the material will deform or crack due to the landing forces. Likewise, the nose cone is also to be constructed out of fiberglass and is unlikely to be damaged by landing impact. The nose cone fabrication method was verified using the subscale vehicle launch, which uses a scaled down version of the final payload nose cone.

STEMnaut Survivability Verification

As discussed in the PDR report, the team is utilizing the Dynamic Response Index (DRI) that correlates spinal compression to chance of significant injury. DRI has been utilized for the Artemis, Gemini, and Apollo programs for ejection seat design dating back to 1966 when Stech and Payne first described DRI at the Air Force Research Laboratory [4]. The equation is as follows:

$$\ddot{X} + 2\zeta\omega\dot{X} + \omega_n^2x = \ddot{z}(t) \quad (34)$$

$\ddot{X}$ represents the relative acceleration of the dynamic system (the human spine) along the Z axis, $\dot{X}$ represents the relative velocity, x represents the relative deflection of the dynamic system, with a positive value representing spinal compression, $\ddot{z}(t)$ represents the measured time-dependent shock acceleration along the Z-axis, $\omega = 52.90\text{rad/s}$ represents the lumped spinal frequency, and $\zeta = 0.224$ represents the damping ratio.

During the landing event, vertical deceleration values from the onboard IMU will be fed as $\ddot{z}(t)$ into Equation 34, and solving the second-order differential equation will yield a value for X, representing the current spinal compression. Upon conclusion of the landing event, the maximum value of $\ddot{X}$ will be used in the following equation to determine the DRI.

$$DRI = \omega^2/G(X_{max}) \quad (35)$$

A DRI value of 17.7 correlates to a 10% chance of significant injury, and the team considers this to be “low”. A chance of significant injury between 10 and 30% is considered “moderate”, and a chance of significant injury above 30% is considered “high”. A 30% chance of significant injury correlates to a DRI value of 21.

Table 37: DRI calculations for the 3 successful drop tests. Drop 1 impacted a tree which lowered the overall acceleration experienced by the STEMnauts.

Drop #	Landing Velocity (fps)	Landing KE (lb-ft)	Landing G	Landing Force (N)	DRI	Injury Risk
Drop 1	20.9	43.9	14.5	418	16.2	Low
Drop 3	13.9	19.6	24.8	716	19.9	Moderate
Drop 4	12.8	16.5	25.5	735	20.4	Moderate

While the payload structure and components - with the exception of the PDM - performed as desired, the team discovered that the lander experienced a higher landing impact force than simulated. When calculating the DRI of the STEMnauts, only one drop had a “low” risk of injury. The reason for this was due to impacting a tree right before landing which softened the impact.

The velocities in Table 37 were calculated by linearly interpolating the altimeter data immediately prior to landing. This provides reliable velocity data without bringing in the errors associated with the acceleration spike upon landing. The cause for the high landing force for drops 3 and 4 is due to the lack of damping from the legs that causes the lander to bounce on impact. For Drop 1, due to the effect of the tree the lander did not bounce even though it hit at around 7 fps faster than Drops 3 and 4.

As can be seen in Table 37 the landing KE is well below the 75 lb-ft threshold imposed by NASA in requirement 3.3. However, the landing force is still quite high. The cause of this is due to the inadequate damping provided by the legs which causes the lander to bounce on impact. This issue is to be addressed in two ways: a larger parachute to soften the descent further, and legs that protrude further from the bottom of the lander belly to damp the impact more. The legs are to be lengthened from 10 in to 12 in, and will extend 1.7 in below the belly instead of the initial 0 in.

The team decided to change the parachute from a 4 ft diameter to a 5 ft diameter with the same CD of 2.2. It was discovered that during lander separation, the parachute unfurls before the shock cord tightens. This imposes a jerk greater than initially expected from DRSS simulations (Table 38). To account for this phenomenon, the adjusted model produced predictions much more in-line with the actual parachute shock loading data (Table 39).

Table 38: Actual vs adjusted predictions of parachute accelerations. Initial conditions were set to match actual data.

Drop #	Parachute G	Parachute Force (N)	<i>Initial Predicted G</i>
Drop 1	14.5	419	3.1
Drop 3	8.4	243	1.9
Drop 4	4.2	121	1.9

Table 39: Actual vs adjusted predictions of parachute accelerations. Initial conditions were set to match actual data.

File	Descent Time (s)	Predicted time	% Error	Effective Shock Cord Length	Parachute G	Adjusted Predicted G	% Error
Drop 1	~16	17.6	-9.0%	~50'	14.5	16.3	-11%
Drop 3	~22	20.8	5.8%	~20'	8.4	6.6	27%
Drop 4	~21	20.1	4.5%	~20'	4.2	6.6	-37%

During drop 2 and 3, the shock cord did not unravel completely, as it got tangled up. To prevent this from happening again, the team derived an improved method of shock cord folding combined with lightly scored masking tape to ensure no knots form during lander separation.

In order to reduce the parachute jerk load, the parachute shock cord had to be decreased from a length of 50 ft to a length of 30 ft as can be seen in Figure 81. Although the lander remained upright in all drops despite the parachute remaining attached, the switch to a larger parachute and shorter shock cord will result in a heightened risk of the parachute dragging the lander following landing. As such, the PDM is still being developed for the full-scale payload. The improvements made to the LDM and PDM will be discussed in further detail in Section 4.2.5.

4.2 Mechanical Subsystems

Mechanical subsystems were designed to meet and exceed NASA and VADL imposed requirements. Modularity, reliability, and low mass were key aspects that also constrained the design. Key subsystems include the Lander Separation Mechanism (LSM), Leg Deployment Mechanism (LDM), Parachute Detachment Mechanism (PDM), and Antenna Extension Mechanism (AEM). These mechanisms were optimized for a continually evolving integrated lander design to ensure that each mechanism did not interfere with the functionality of another, and to ensure that electronic integration would flow smoothly. The design and testing for individual components and mechanisms are discussed in further sections.

4.2.1 Lander Separation Mechanism

The Lander Separation Mechanism (LSM) is responsible for safely retaining the payload within the rocket during all loads experienced during launch and descent. Additionally, it ensures the jettison of the payload lander, upon Range Safety Officer (RSO) approval, when an altitude of 300 feet above ground level (AGL) is reached during descent. All mechanical and electronic components of the LSM are located at the LSM bay, as seen in Figure 92. It is important to note that the LSM is external to the payload and stays within the rocket body throughout the entirety of the mission, whether or not jettison is approved.

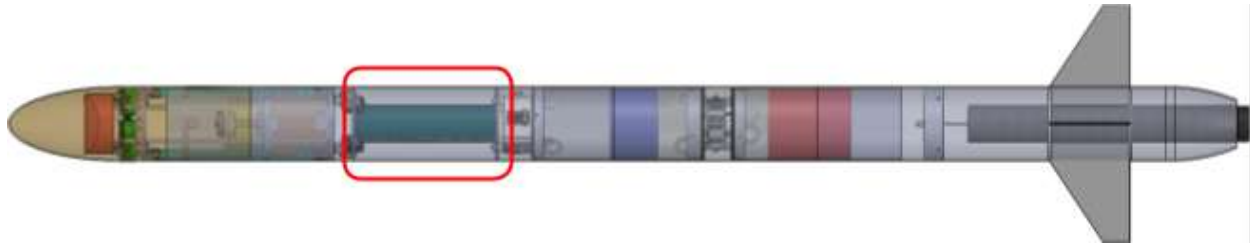


Figure 92: LSM location in the launch vehicle. The LSM remains with the launch vehicle throughout the entire flight and does not jettison with the lander.

The payload bay houses the lander prior to jettisoning, along with its parachute. It is critical to the mission that the 60 in Iris Ultra exits the payload bay smoothly and avoids entanglement with the shock cord. The LSM, shown in Figure 93, includes an electronics bay, retained between two aluminum plates, and an 11.7 in tall, 75 mm diameter parachute compartment fabricated from blue tube (see Figure 94). This compartment securely retains the folded parachute throughout the flight until the payload's gravity ejects it during descent. At the bottom of the LSM, there is a third aluminum plate, with a 75 mm center hole through which the parachute enters the tube where it will be stored prior to jettison. By isolating the parachute, this design ensures reliable deployment, free from entanglements. The overall assembly design can be seen in Figures 93 and 94.



Figure 93: CAD rendering of LSM assembly.

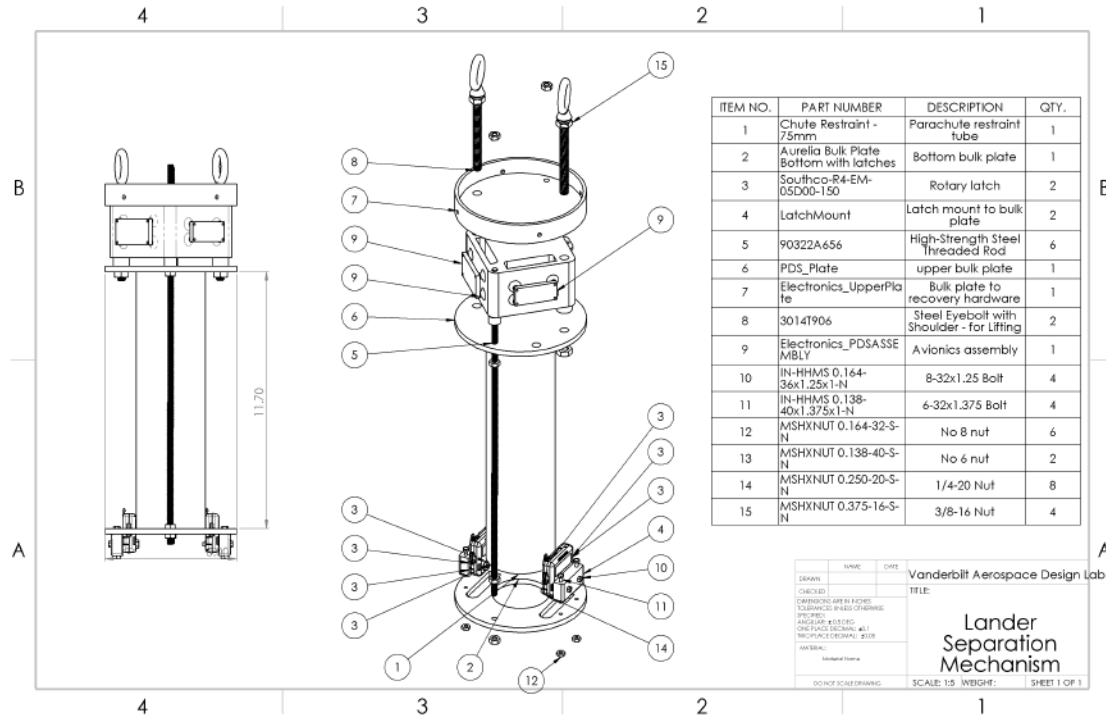


Figure 94: Exploded view of LSM assembly.

Payload Retention

The LSM retains the payload using two SouthCo R4 Rotary Latches, secured to the LSM's bottom plate via aluminum mounting structures fastened with 8-32 bolts. The latches operate through a voltage-driven mechanism: upon receiving a high-voltage signal, a small internal motor releases the latch arm. During flight, the latches remain in the closed position, securing the payload via rings located at the top of the lander.

The testing documentation for the R4 Rotary Latches was obtained from SouthCo, indicating a maximum tensile load of 1061 lbf before latch deformation affecting functionality may occur. As detailed in Section 3.4.3, the maximum load experienced by the vehicle occurs during main parachute deployment, with an associated acceleration of 32 G. The latches must support the payload mass of 8.05 lb. The worst-case loading can be calculated as:

$$\text{Load} = 0.25 \text{ slug} \times 32.2 \text{ ft/s}^2 \times 32 \text{ g} = 258.0 \text{ lbf}$$

With two latches sharing the load, each latch would experience 129.0 lbf. This results in a safety factor of:

$$\text{F.O.S.} = \frac{1061 \text{ lbf}}{129.0 \text{ lbf}} \approx 8.25$$

Similarly, the four 8-32 screws securing the latches to the mounting plate must withstand the same 258 lbf load. With a minor diameter of 0.120 inches and a cross-sectional area of 0.0113 in², the maximum shear stress per screw is:

$$\text{Shear Stress} = \frac{258 \text{ lbf}}{4 \times 0.0113 \text{ in}^2} = 5710 \text{ psi}$$

The selected high-strength stainless-steel screws have a shear strength of 80,000 psi, providing a safety factor of:

$$F.O.S = \frac{80,000 \text{ psi}}{5710 \text{ psi}} \approx 7.0$$

These calculations confirm that the latches and their attachment mechanisms have robust safety margins under the expected loading conditions. Finite Element Analysis simulations were run on the electronics upper plate where the eyebolts connect to the drogue parachute, under the load conditions corresponding to the 16 G acceleration expected at main deployment. The FEA results can be found in Section 3.1.5. FEA was also run on the bottom bulk plate containing the EM latches, with the results shown in Figure 95.

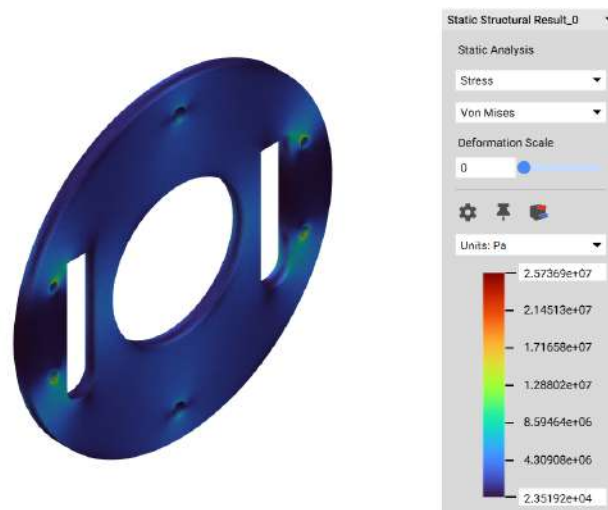


Figure 95: FEA result for bottom LSM bulk plate.

The factor of safety of this bulk plate can be calculated using these FEA results as such:

$$F.O.S. = \frac{290 MPa}{2.57 * 10^7 Pa} = 11.3 \quad (36)$$

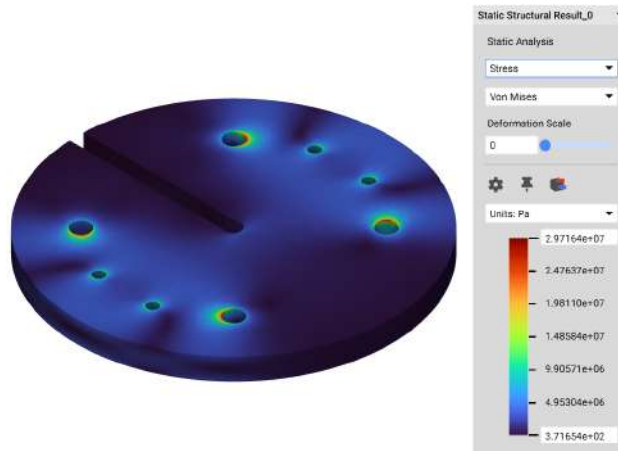


Figure 96: FEA result for payload hook plate attaching to the LSM latches. It is made of aluminum and has a safety factor of 9.8.

The RSO approval signal aspect of LSM was built and tested during drop tests conducted using the Aurelia drone. In these tests, the latches of the LSM were attached to the drone with the payload underneath and were released by a radio signal from a steady altitude of 300 feet. The mechanism performed its task during each trial, achieving consistent and reliable payload separation. The same mechanism was reused across multiple drop tests without requiring adjustments, highlighting its durability and repeatability. The LSM was not utilized during subscale launch, as no payload jettison was needed for that flight configuration.

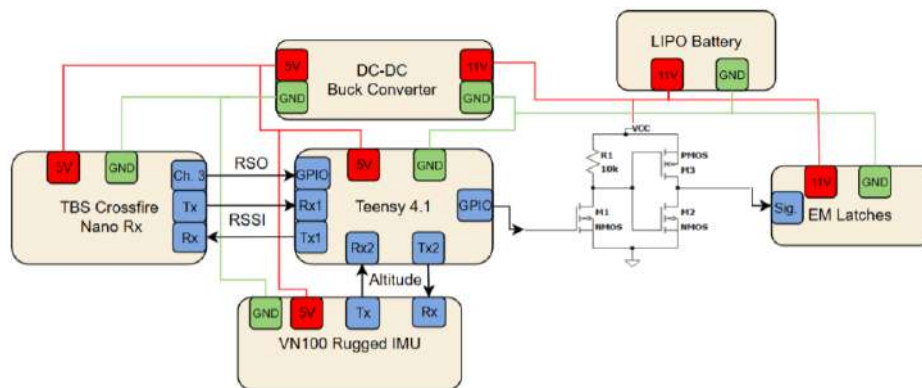


Figure 97: LSM Approval System with Amplification Circuit.

The electrical design of the LSM has remained unchanged since PDR. The amplification circuit used in the payload to trigger the EM latch for parachute detachment has proven to be effective in previous drop tests, so this circuit will be recycled to amplify the digital high output from the Teensy GPIO pin which triggers the LSM latches. This circuit, shown in the Figure 97 above, floats a high voltage at the drain of the initial NMOS, when the GPIO output sets the gate of that NMOS high, the drain falls to zero volts. This gets sent through an CMOS inverter so the EM latch signal flips from 0 V to the 11.1V supplied by the battery.

Table 40: LSM Testing Summary

Mechanism Aspect	Goal	Verifying Test	Test Results
Payload Retention	The LSM must securely retain the payload during all phases	Drop testing	Successfully retained the payload in all tests.
Payload Release	The LSM must release the payload within 1 second of receiving the command.	In-lab command tests	Instant command response and latch opening
		Drop testing	Payload successfully released within 1 second in all trials
RF Transmission	The LSM must receive commands reliably at both altitude and horizontal distance.	LSM RF transmission test	Successful transmission to a distance of 1300 ft
		Drop testing	Successful reception and response at 300-400ft AGL

4.2.2 Leg Deployment Mechanism

The landing legs serve a dual purpose; both facilitating an upright landing and serving as the ground plane for the quarter wave antenna selected by the team. Each of these is a mission critical component, making the LDM an extremely important subsystem. The LDM consists of the landing legs, leg brackets, and torsional springs for each leg. These parts are shown individually in Figure 99 below and assembled in Figure 98. The CAD drawing of the assembled leg and bracket is included as well in Figure 100.

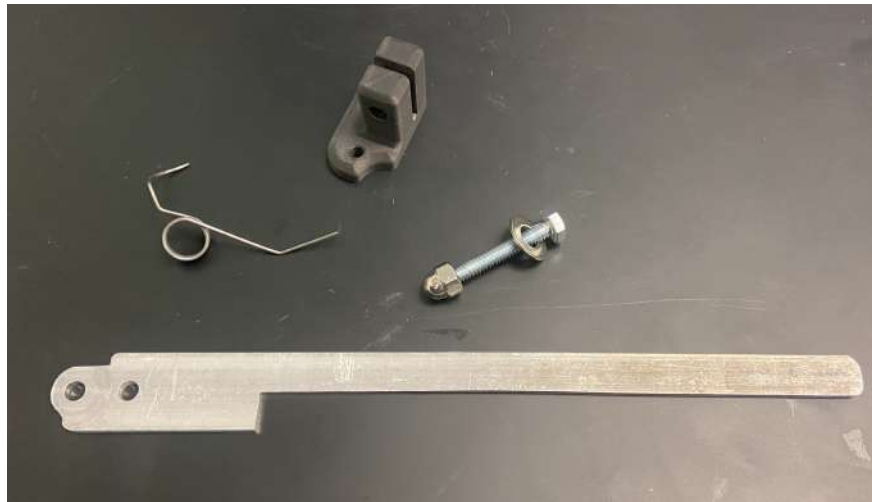


Figure 98: LDM components.

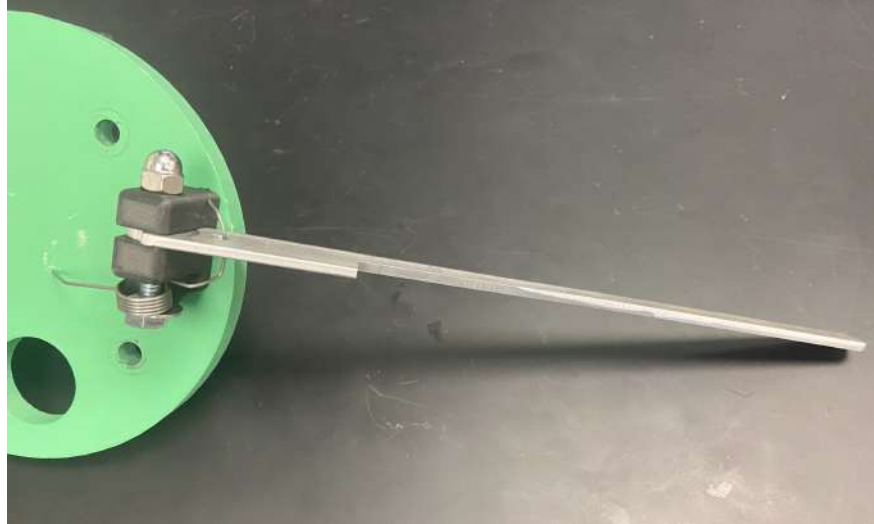


Figure 99: Assembled LDM.

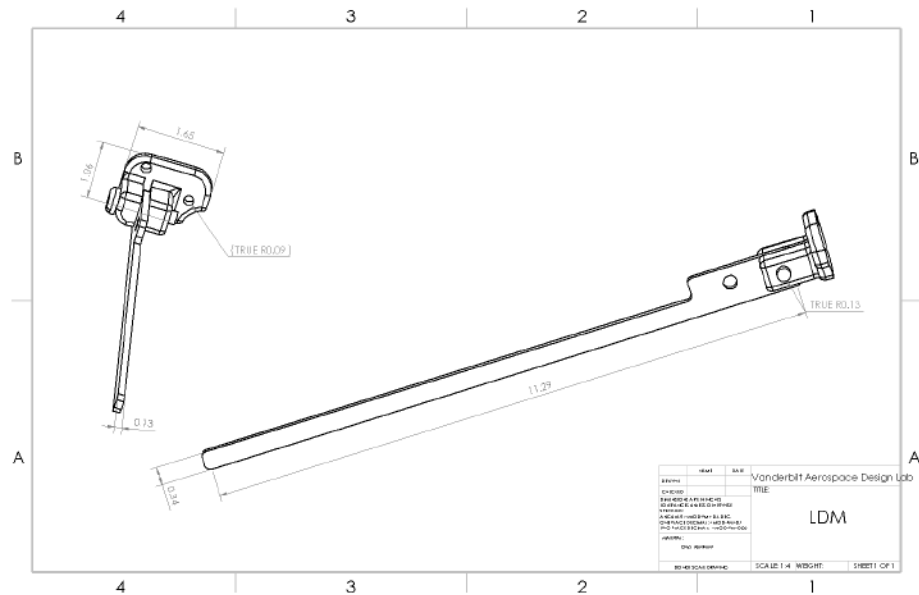


Figure 100: Single LDM bracket and leg.

The legs are 12 in long and are constructed from 2024-T4 aluminum, which was chosen for its strength, flexibility, and availability. The leg brackets are 3D printed using onyx filament, which is infused with carbon fiber flakes for added strength. These brackets, along with the legs, have proved to be of sufficient strength in each drop test thus far. This gives the team confidence in the material selection, and this confidence is supported by FEA analysis on the leg brackets, shown in Figure 101 below.

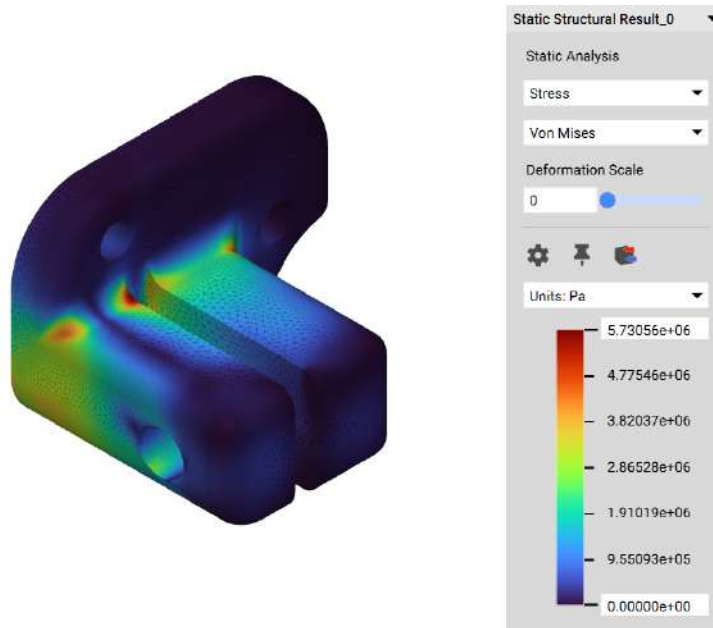


Figure 101: LDM bracket FEA result.

The FEA is meant to represent a worst case scenario where the full landing force is applied to the tip of a single leg. This results in a force of about 140 N on the the leg bracket, half of which is applied to each of the vertical faces. Figure 101 shows the resultant stress concentrations caused by this force and also shows that the part will not break under this loading. This conclusion is also supported by the testing discussed immediately below.

In order to verify both the qualitative data from field tests and the FEA analysis, load testing of the LDM system was conducted. This was done by fixing the LDM bulk plate in a clamp, as shown in Figure 102 below, and incrementally adding weight to the end of the leg.



Figure 102: Single LDM bulk plate clamped for testing.



Figure 103: Full view of test setup while loaded.

The weight was added perpendicular to the axis of the leg, which should represent the highest bending moment possible. The loading was increased statically up to 8 lb, which represents the mass of the payload. At this loading, the leg deflected a total of 3.25 in but does not permanently deform. This deformation can be seen in Figure 104 below.



Figure 104: Leg deforming under loading from applied weight. Total deformation distance can be seen.

When the 8 lb was dropped a distance of 27 in to mimic the expected landing velocity of 12.4 fps, the leg deformed slightly but no component were permanently damaged. This situation simulates a scenario where the full weight of the payload impacted one leg, perfectly perpendicular to the axis of the leg, and in which the payload is rigidly mounted. This, clearly, cannot happen in a practical setting. Therefore, this testing significantly overloads the leg in comparison to the forces that it will actually experience. This lends confidence to the team, because there was no breakage in the bulk plate, leg bracket, or leg, even in this extreme loading scenario. Based on this load testing, field testing, and FEA analysis, the team has a high level of confidence that the LDM will not break under any loading experienced in the field and will perform nominally throughout the mission.

The LDM stores the legs within the airframe throughout rocket flight and releases them once the payload separates from the main rocket. This is achieved by placing the rocket side of the payload within the coupler to the main body of the rocket, which allows the coupler to retain the legs. Once the command is sent and the latches release the lander, the legs are free of this restraint and can swing out under the force from the attached torsional springs. Images of the stowed and deployed landing legs are shown below.

The largest change to the legs since PDR is the increase in both the length of the legs and their angle relative to the body of the lander. These two parameters have allowed the legs to provide a clearance of 1.7 in with the ground after landing, whereas they provided no clearance previously. In order to achieve this, the mounting point of the leg brackets was shifted inwards from where it was at PDR. This shift allowed the angle to be increased without the bracket or hardware striking the outer shell of the lander. The purpose of this increase, as discussed in Section 4.1.3, is to provide some shock absorption upon landing and reduce the bouncing of the payload. This should reduce the landing Gs experienced by the payload, thereby increasing the survivability of the STEMnauts. The functionality of the lander is the same with this redesign, but needs to be verified through drop testing.

In the event that bouncing is still an issue after this design change, a suspension concept has also been explored as a solution. To achieve this, the leg bracket is mounted by a single bolt which acts as a pivot point. A stiff torsional spring is placed about this mounting bolt, and is housed within the lower portion of the leg bracket. A damper is placed centrally on the bulk plate and is attached to each of the two leg brackets either by string or a thin, stiff rod. This creates a combined spring/damper system that provides a more formal suspension to the lander. A prototype of this suspension has already been created, and can be implemented fairly quickly if necessary. The primary reason this concept is not currently being pursued is because it complicates packaging within the payload a fair amount, and also because it is unclear if this extensive of a suspension system will be necessary. The team believes that it is most prudent to pursue the simpler design change, discussed in the paragraph immediately above, prior to pursuing this more complex design change. However, the team is prepared to pursue either of these designs depending on the results of testing which will be completed prior to full scale flight.

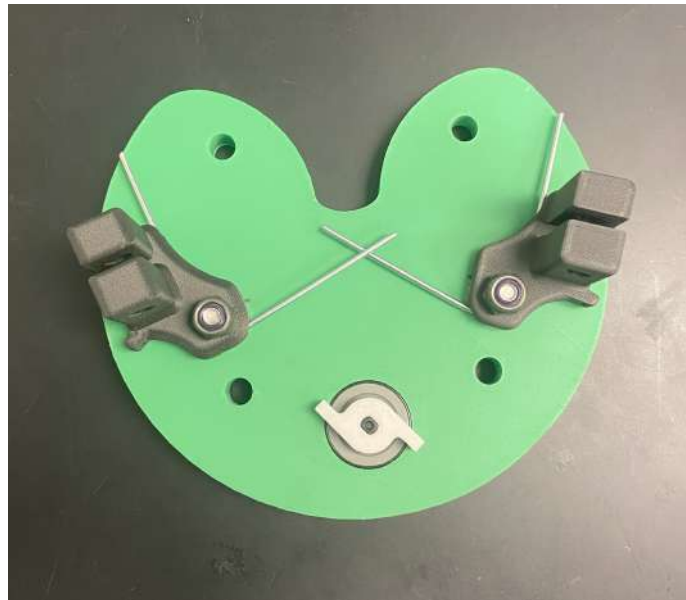


Figure 105: Suspension prototype.

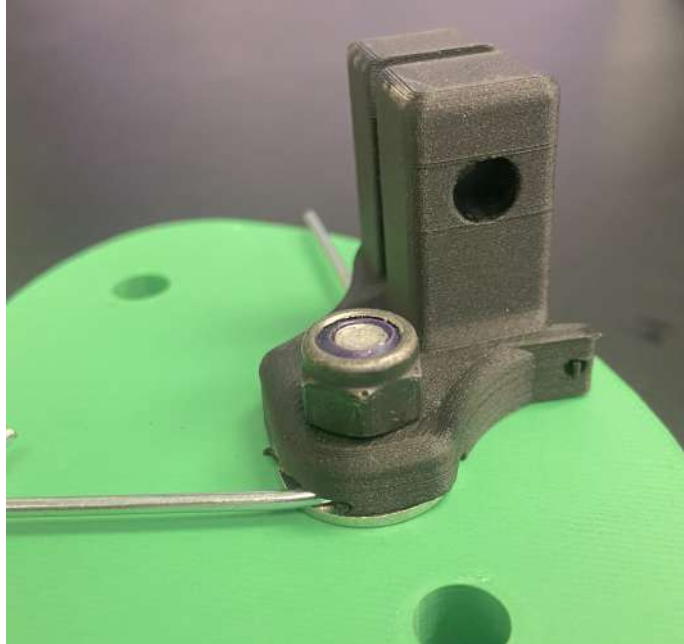


Figure 106: Closeup of leg bracket for suspension prototype.

The functionality of the legs as a ground plane for the antenna is achieved through electrical connection to a common antenna plate. The antenna is located in the forward portion of the payload, near the nose cone, and grounding wires run from this point to each of the four legs. These wires are attached to the arm of the torsional spring, which allows them to move with the legs as they swing out, providing a reliable connection throughout operation. This design is optimal because it removes the risk of pulling out the grounding wires upon leg deployment, or of leg deployment being restricted by attachment to the grounding wires. Another benefit of the aluminum legs is that they serve as the ground plane without the addition of wires along the length of the leg. This greatly simplifies the manufacturing process, allowing the system to be more modular and quicker to assemble.

4.2.3 Parachute Detachment Mechanism

The purpose of the PDM is to prevent the parachute from dragging the payload around after landing. This poses the risk of toppling the lander so that it is no longer in an upright orientation, and potentially damages the antenna during dragging. As mentioned in Section 4.1, the payload parachute size has been increased from a diameter of 4 to 5 ft. Additionally, the shock cord has been decreased in length from 50 to 30 ft to maintain the same parachute opening force on the PDM latch. These 2 changes have the effect of increasing the likelihood that the parachute will drag the payload.

During all previous full-scale drop tests, although the PDM was properly actuated whenever the payload flight computer detected landing, the PDM failed to properly actuate. This can be seen in Figure 107 below.



Figure 107: Full-scale lander prototype post drop. The PDM successfully actuated, however the parachute shock cord caught on the latch and did not detach properly. This can be seen on the right where the shock cord loops are stuck on the sides of the latch, even when open.

The cause of this failure was from the shock cord loops catching on the sides of the PDM latch. The team found a way to fix this by adding a cover to the top of the PDM latch to cover the sides of the latch and thus prevent the shock cord from catching.

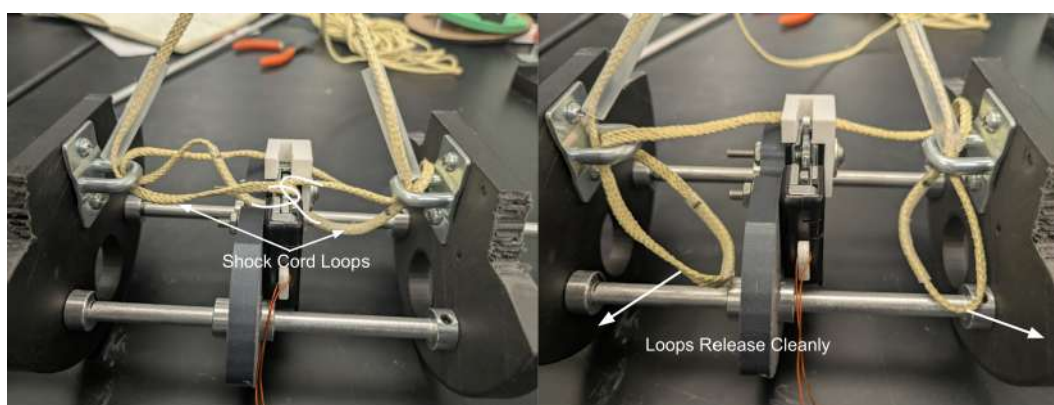


Figure 108: PDM with new latch cover to prevent shock cord loops from catching on latch during detachment. The latch cover is made of PLA and adds negligible mass (6 grams). The hook bulk plates have not yet been re-manufactured. The previous iteration is visualized.

Beside this addition to the PDM latch, the bulk plates holding the hooks (the bulk plates to the left and right of the central bulk plate holding the PDM latch in Figure 108) were modified to accommodate changes made to the LDM. These changes include the addition of deeper slots cut into the bulk plate to allow the legs to recess further in, and an extension of the top of the bulk plate to connect with all 4 rods of the skeleton as opposed to just the bottom 2 in the PDR design. The drawing of the overall assembly can be seen below in Figure 109.

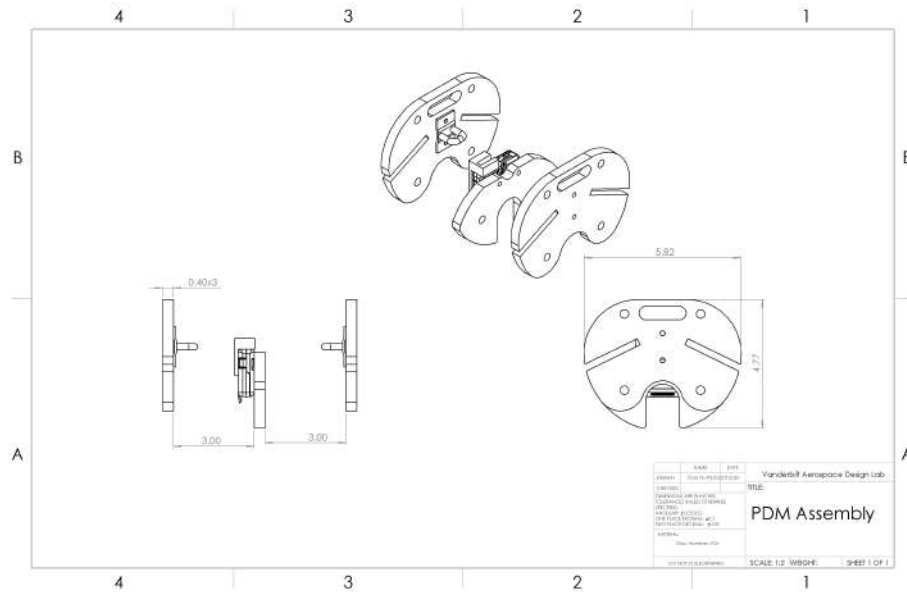


Figure 109: PDM assembly drawing. The smooth rods making up the lander skeleton are not visualized for clarity.

The team conducted FEA on the bulk plates to ensure that they met the safety factor requirements imposed by the team in VADL requirement 2.2.2. The maximum force experienced by the PDM is during payload parachute opening. This induces an acceleration of 16 G onto the entire payload. The mass of the payload is 8.05 lbs, thus the PDM experiences a force of 129 lbf during parachute opening. The FEA boundary conditions were set assuming each bulk plate holds all of the 129 lbf load.

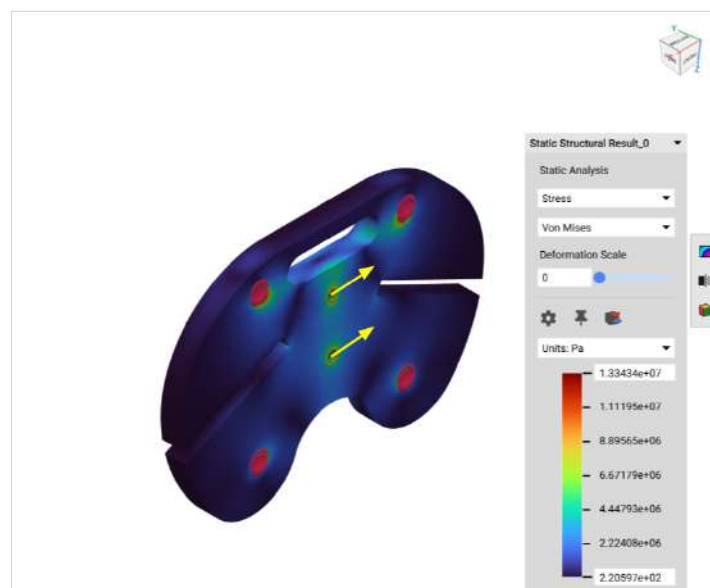


Figure 110: PDM hook bulk plate FEA results. Red faces represent fixed regions, while yellow arrows indicate force of magnitude 129 lb (574 N) pointing at a 45 degree angle in the YZ plane. The minimum safety factor on this part was calculated to be 3.

The boundary and loading conditions are produced in the simulation to most accurately represent reality. It is determined from calculations that the maximum tensile force acting on the bolts is 129 lbf (574 N) from the maximum acceleration of main parachute deployment (16 Gs) multiplied by the weight (in newtons) of the entire payload. Because the shock cord is pulling on the parachute hook at approximately a 45 degree angle, it is also pulling on the bulk plate at a 45 degree angle. Also, since the 4 corner screw-holes are attached to the body of the rocket, they will be deemed as the fixed sections.

Using maximum deflection energy formula, while taking the ultimate tensile strength of Onyx filament to be 40 MPa , the factor of safety can be calculated as the following:

$$F.O.S. = \frac{SUT}{\sigma_{study}} = \frac{40 * 10^6 Pa}{1.33 * 10^7 Pa} = 3 \quad (37)$$

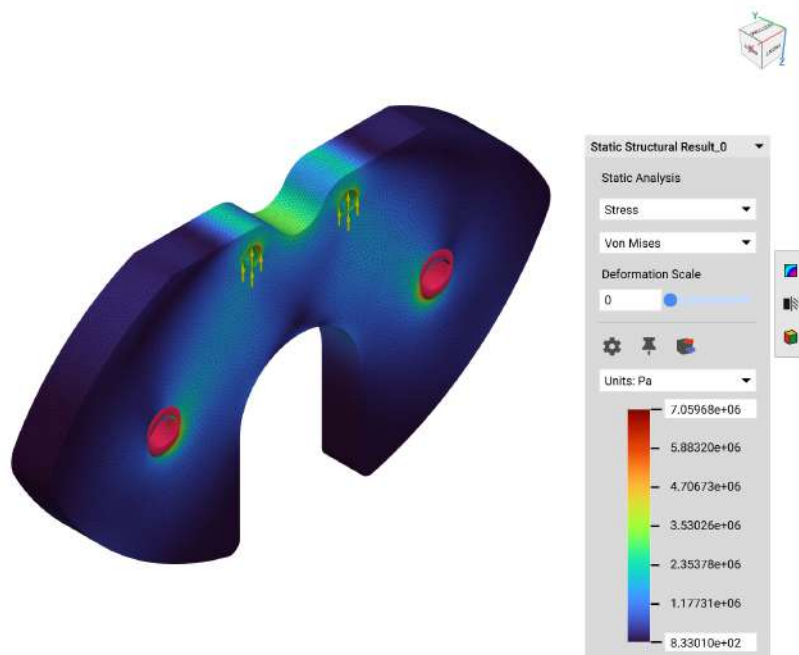


Figure 111: PDM latch bulk plate FEA results. Red faces represent fixed regions, while yellow arrows indicate force of magnitude 129 lb (574 N) pointing perpendicular to the YX plane. The minimum safety factor on this part was calculated to be 5.7.

The two bottom holes were constrained as fixed boundary conditions, and all load was transferred through the 2 latch mounting holes. The factor of safety can be calculated as the following:

$$F.O.S. = \frac{SUT}{\sigma_{study}} = \frac{40 * 10^6 Pa}{7.06 * 10^6 Pa} = 5.7 \quad (38)$$

4.2.4 Antenna Extension Mechanism

The purpose of the AEM is to ensure the line of sight of the antenna with the NASA ground receiver. Each AEM contains a single antenna/tape spring that self extends when released. There are 3 AEMs attached to a bulk plate located in the fore section of the lander. Each of these antennas points about 120 degrees from each other to ensure that at least one antenna is pointed away from the ground, regardless of landing configuration.

The AEM structure can be seen in Figures 112 and 115. The individual AEM assembly structure can be seen in Figures 113, 114, and 116

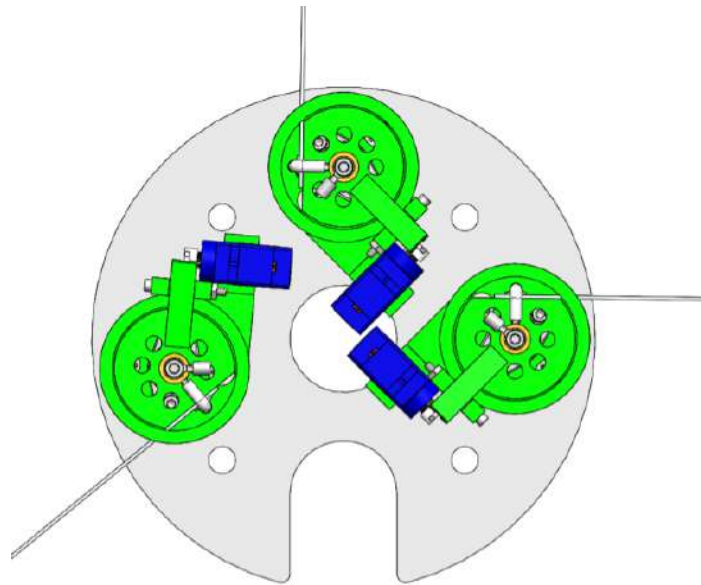


Figure 112: CAD rendering of AEM assemblies on lander bulk plate. Antenna/tape springs are shortened for visual clarity.

Each AEM contains two 18 in long, $\frac{1}{2}$ in wide tape springs that are screwed into a damped rotating cylinder. These tape springs also act as the antennas themselves. The antennas are attached to a screw using ring terminals. This screw is then wired to the RF board using a ring terminal. To reduce friction between the wire and the cylinder, a brass bushing was placed between the 2 terminals. To ensure electrical connectivity, a steel spring was inserted between the 2 terminals.

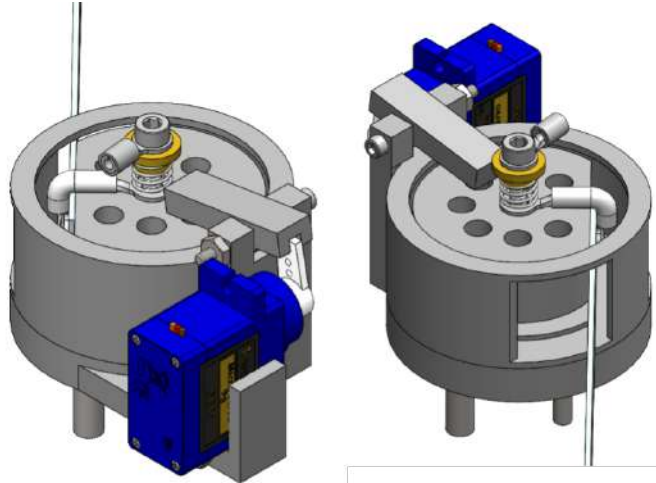


Figure 113: CAD rendering of individual AEM sub-assembly. Antenna/tape spring is shortened for visual clarity.

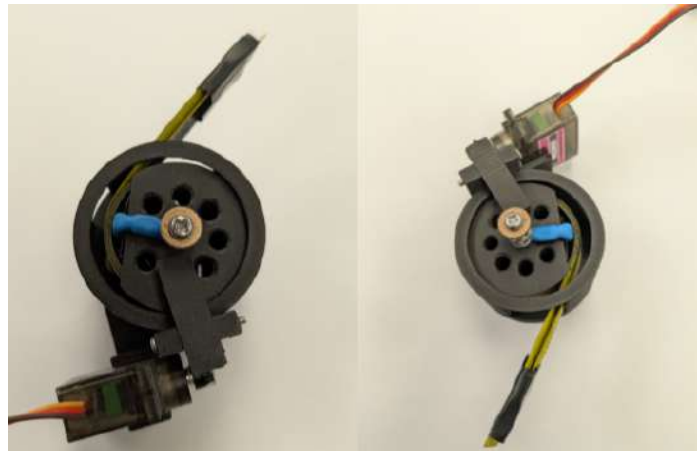


Figure 114: Individual AEM sub-assembly

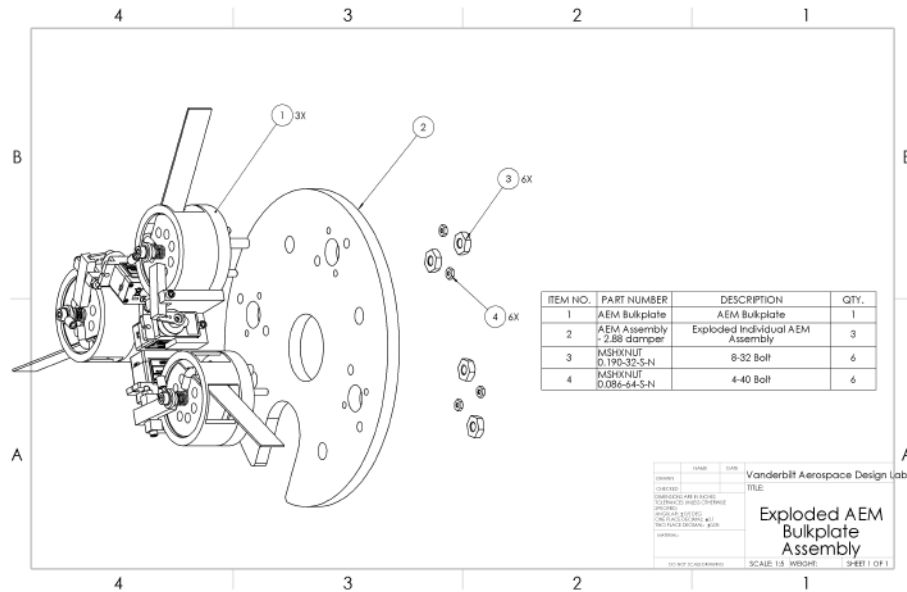


Figure 115: Exploded view of high-level AEM assembly drawing. All dimensions are in inches. Antenna/tape spring is shortened for visual clarity.

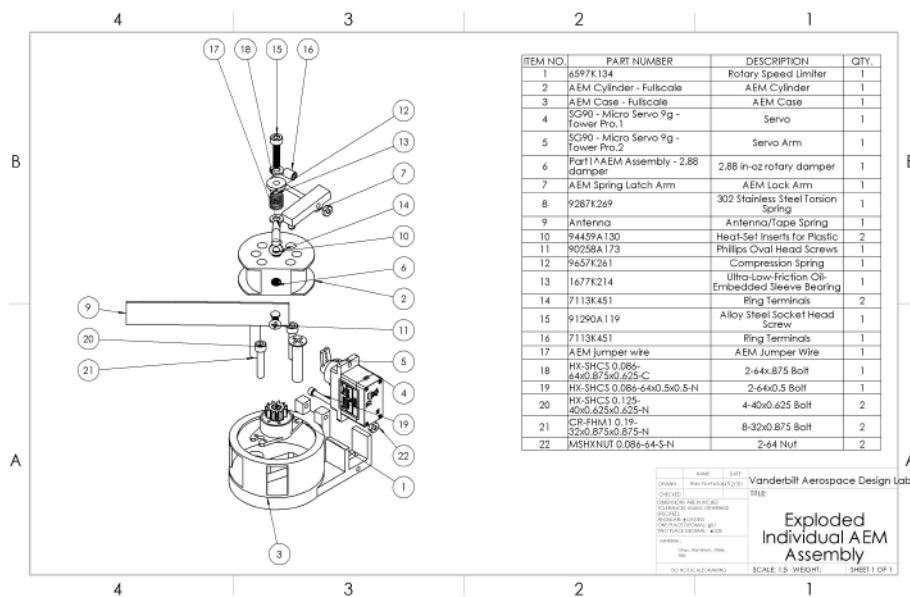


Figure 116: Exploded individual AEM drawing with call outs.

To prevent the antenna from binding within the AEM case during extension, a 2.88 in-oz rotary damper retains the cylinder and slows extension. Testing has revealed that if the tape extends too quickly it binds internally. The design of the tape spring was inspired by the use of self-deployable composite booms that are often used in the space industry. These are often used on satellites to deploy solar panels and even solar sails. Because the antennas extend from within the payload bay, there needs to be a cutout to allow the antenna to pass through. These cutouts are covered with an aerodynamic cover to prevent these holes from increasing drag. As was discussed in Section 3.2, these cutouts did not have an effect on flight, and were not simulated.

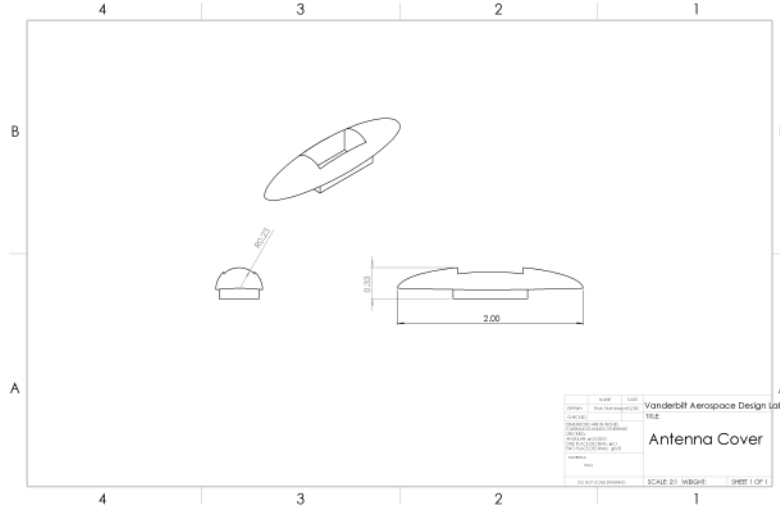


Figure 117: Antenna cover drawing. All dimensions are in inches. This cover was designed to extend less than 0.25 in above the airframe surface in accordance with NASA requirement 4.2.2.

The combined mass of the AEM assembly is 1 lbm. This includes the coaxial cables and connectors to the Light APRS board, and cables to the ground plane/legs of the lander. Each individual AEM has a mass of 0.15 lbm. There are 4 bolts fixing each AEM to the main bulk plate - 2x 4-40 bolts, and 2x 10-32 bolts. The maximum loading upon this bulk plate occurs at main parachute opening, and is about 32 G of acceleration. During this, the AEM bolts shall experience a tensile load of about 4.8 lbf, which is well within the safety margin for the PLA bulk plate and bolts. The bolts are made of alloy steel with a tensile strength of about 170 kpsi according to the manufacturer's specifications. A single 4-64 bolt with a minor diameter of 0.0849 in is capable of holding more than 962 lbf in tension. The factor of safety in tension can be calculated as such:

$$F.O.S._{Tension} = \frac{\frac{\pi}{4} * 170kpsi * 0.0849^2 in^2}{4.8lbf} = 200 \quad (39)$$

Assuming the shear strength is half of the tensile strength, the bolt can hold more than 297 lbf in shear. If it is assumed that the 32 G of acceleration occurs while the lander is horizontal, 297 lbf is still well above 4.8 lbf. It is also unlikely that the bulk plate will deform, allowing the AEM to come free. The factor of safety in shear can be calculated as such:

$$F.O.S._{Shear} = \frac{\frac{\pi}{4} * 85kpsi * 0.0668^2 in^2}{4.8lbf} = 100 \quad (40)$$

Although there is little risk of physical damage to the AEMs and their structure during flight, if the antenna begins to uncoil from the cylinder due to in-flight forces, they will get bound inside the AEM case and not extend properly. Subsequently, the RF transmission will be put in jeopardy. To prevent this, numerous drop tests were conducted to ensure that impacts would not jostle the tapes and cause them to unwind prematurely. Additionally, the subscale flight would verify that the tapes would extend properly even after withstanding a rocket flight.

Previous iterations of the AEM were flown on prototype landers without RF transmission to verify the extension mechanism itself. During the CDR cycle the team verified antenna extension and transmission on a full-scale lander dropped from 300 ft AGL. It fulfilled all mission requirements by extending upon landing, and transmitting telemetry over RF at a range of around 150 ft. An AEM assembly was also flown on the subscale rocket flight. During this, a cold solder caused the payload computer to disconnect upon launch, and therefore the landing detection and transmission did not occur. However, the AEMs themselves survived the flight completely intact, and did not unlock due to the high accelerations experienced during parachute deployment and opening. Additionally, once the tapes were manually unlocked, the tapes extended to a proper 18 in length. The second subscale launch confirmed the same results, as the AEMs remained in a flight-worthy state afterwards.

Table 41: Summarized AEM testing results.

Mechanism Aspect	Goal	Verifying Test	Test Results
AEM Cylinder Locking	AEM must not unlock until commanded to by the payload computer	Drop testing	AEM did not unlock after repeated drop tests
		Subscale launch and landing	AEM did not unlock after launch and landing
Antenna Extension	Antennas must extend a full 18 in when the cylinder is released	Full-scale lander drop test from 300 ft AGL	Top antenna successfully extended, two side antennas failed to extend
RF Transmission	Antennas must transmit a distance of >3000 ft over uneven terrain	Standalone transmission test outside lander	Successful transmission to a distance of 1300 ft
		Full-scale lander drop test from 300 ft AGL	Successful transmission to a distance of 150 ft after landing
Structural integrity	The AEM and bulk plate must be re-flyable	Full-scale lander drop test from 300 ft AGL	Entire AEM assembly was intact after drop, multiple drop tests were successfully conducted Subscale AEM assembly - which shares common parts with the full scale payload - remained intact after 2 launches
		Subscale launch and landing	



Figure 118: Full antenna extension in full-scale drop test from 300 ft AGL. During this test, the lander successfully transmitted telemetry via RF to the team ground station at a distance of 150 ft.

4.2.5 STEMnaut Crew Module (SCM)

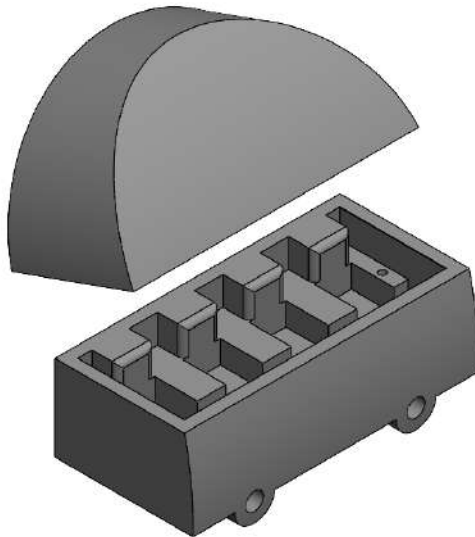


Figure 119: STEMnaut crew module 3D rendering and exploded view showing the seats.

In the figure above, the STEMnaut Crew Module, or the SCM, will be secured in the nose cone section of the rocket. As mentioned in the PDR milestone, this housing was inspired by the SpaceX Dragon capsule where the astronauts are stationed in the nose cone section, see Figure 120. With this configuration, the astronauts, in this case the Lego figures, will be laying on their back which allows a more equal distribution of the g-forces faced upon launch. This will allow a low-strain flight which further ensures their safety.

As per NASA requirements, the SCM will house four astronauts tied down with a long seat-belt that runs across their laps and through the holes on the arm rest on either side of the module. This also ensures that no blood flow is lost and the secures them to their seats.

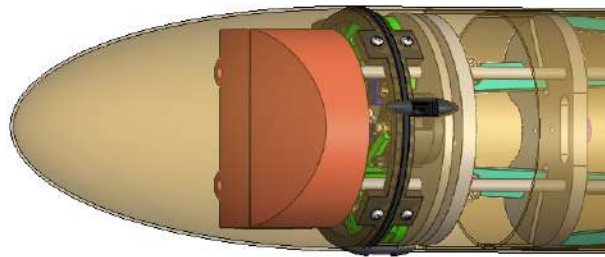


Figure 120: Assembly of the nose cone in an isometric view showing the location of the STEMnaut Crew Module.

While the team has focused extensively on collecting qualitative engineering data for the CDR milestone, the team will continue performing structural health monitoring on the payload legs. As declared in the PDR milestone, this will be done through the breakwire system. The system will undergo rigorous testing to ensure the collection of authentic data.

4.3 Electrical Subsystems

The objective of the electrical systems is to ensure reliable data collection and real-time analysis of flight performance and survivability metrics. Key tasks include managing parachute detachment, antenna deployment, and landing detection to ensure proper sequencing during descent and landing. The system packages data using the APRS protocol and transmits it via RF to ground stations.

Previously, a system based on a Teensy microcontroller with limited functionalities was presented. This time, the system's Raspberry Pi version, as proposed in the PDR report, is presented.

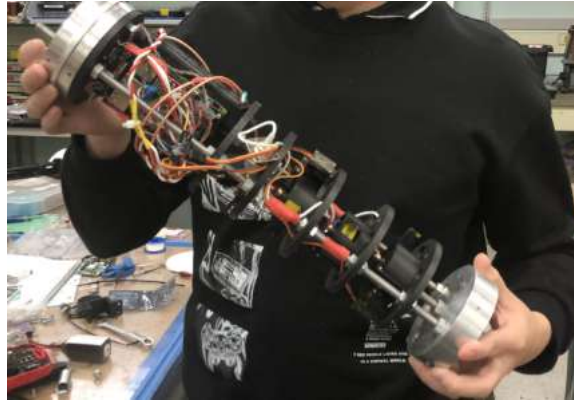


Figure 121: Assembled subscale payload. subscale and full-scale payloads share the same electronics.

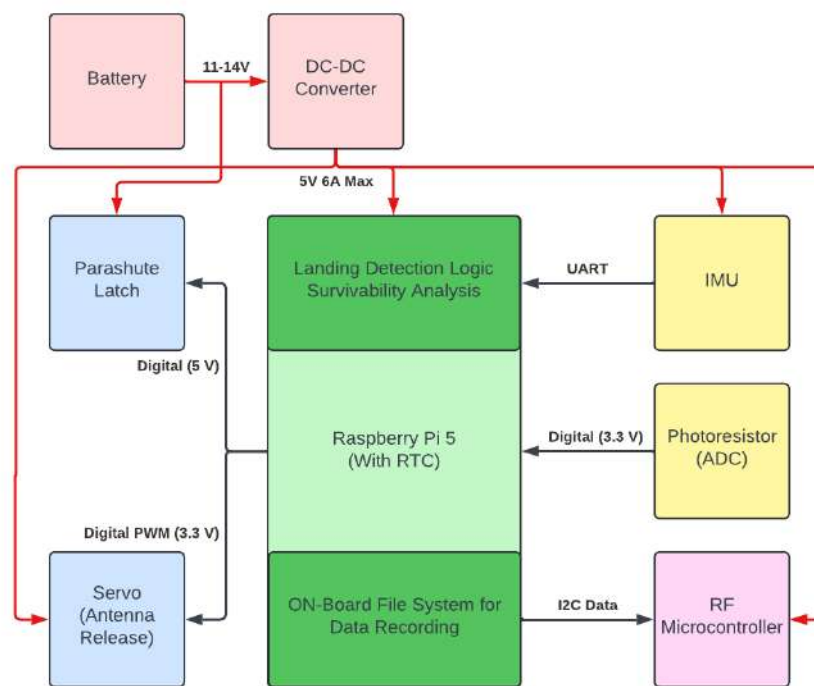


Figure 122: System Block Diagram.

4.3.1 System Overview

The system-level diagram is identical to the one proposed in the PDR, and it has proven to be an effective way to modularize a complicated project. To reiterate, the payload's electrical system, with the Raspberry Pi 5 at its core, is responsible for data recording, landing detection, and survivability analysis. It operates on an 11-14V power supply, regulated to 5V. The Raspberry Pi connects to an IMU through UART, measures light using an ADC-linked photoresistor (to be implemented in the final design iteration), and communicates with an RF microcontroller via I2C. It also controls the parachute latch and antenna release servos.

The following sections provide a detailed overview of the latest design iteration for specific subsystems, along with a failure case analysis from the subscale launch that informed the subsequent improvements. It is important to note that for the sake of vehicle verification, flying with payload electronics is optional; however, this experience has been instrumental in developing a deeper understanding of designing safe, redundant, and reliable systems.

4.3.2 Software Architecture

Previously, a multitasking scheme based on network sockets to enable real-time communication between tasks was proposed. However, this approach was observed to introduce significant delays, hindering the effectiveness of landing detection.

To ensure rapid post-landing actuation while retaining multitasking modularity, a new software architecture using Python's multiprocessing library was adopted. The multiprocessing library enables parallel execution of tasks by creating separate processes, each with its own memory space, allowing true parallelism on multi-core systems. This approach avoids the limitations of Python's threading library, which is constrained by the Global Interpreter Lock (GIL). The threading library uses mechanisms like mutexes for shared memory management but does not achieve true parallel execution due to shared memory space. By utilizing multiprocessing, bottlenecks are eliminated, ensuring faster, more reliable task execution critical for landing detection and subsequent actuation.

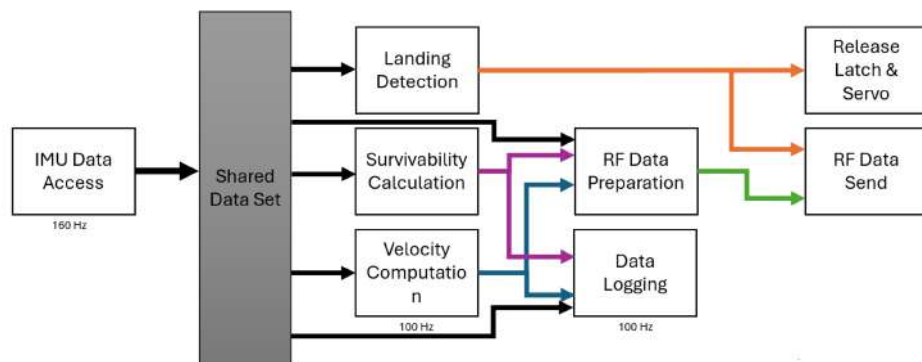


Figure 123: Software High-level Data Flow Diagram

The current software architecture modularizes each subtask as an individual process. The IMU Data Access process reads binary data via UART from the Vector VN100 IMU at 160 Hz, parses it, and updates the globally shared IMU data array. The Landing Detection process reads data from the shared IMU array safely and performs detection, as detailed in 4.3.3. Upon detecting the landing, it sets a globally shared landing detected boolean to true.

The Survivability Calculation process continuously estimates survivability based on the Dynamic Response Index (DRI) and updates the shared RF data array. The Velocity Computation process calculates velocity based on changes in altitude (discussed in 4.3.4) and writes the results to the shared RF data variable.

The RF Data Preparation process aggregates all necessary global data into the shared RF data array for RF transmission and transfers it to the RF microcontroller via I2C. The Data Logging process reads global variables and writes to files consistently, running in a timed loop at 100 Hz to ensure steady data processing.

When the global landing detected boolean is set to true, the Release Latch and Servo process toggles the GPIO pins to actuate the parachute latch and antenna servo, while the RF Data Send process activates the RF board by toggling its enable signal line. This design ensures efficient modular execution, quick response for landing events, and reliable data transmission.

4.3.3 Redundancy in Landing Detection Strategy

Previously, the landing detection strategy relied solely on altitude and acceleration spikes. This approach was heavily dependent on acceleration data, which is highly sensitive to landing conditions. For instance, if the lander were to land on a tree or in bushes, the expected acceleration spike might not occur, rendering the detection unreliable. Furthermore, both altitude and acceleration data were sourced from a single sensor. If this sensor became disconnected due to high-G acceleration, the system would fail immediately without any redundancy. Such vulnerabilities raise significant safety concerns, as a failure in landing detection could compromise the recovery of critical mission data and hardware. In scenarios where a false or missed detection occurs, the lander might also violate regulatory compliance.

In the new iteration, a new landing detection strategy is presented that incorporates additional redundancy by integrating a photoresistor and two timeout timers.

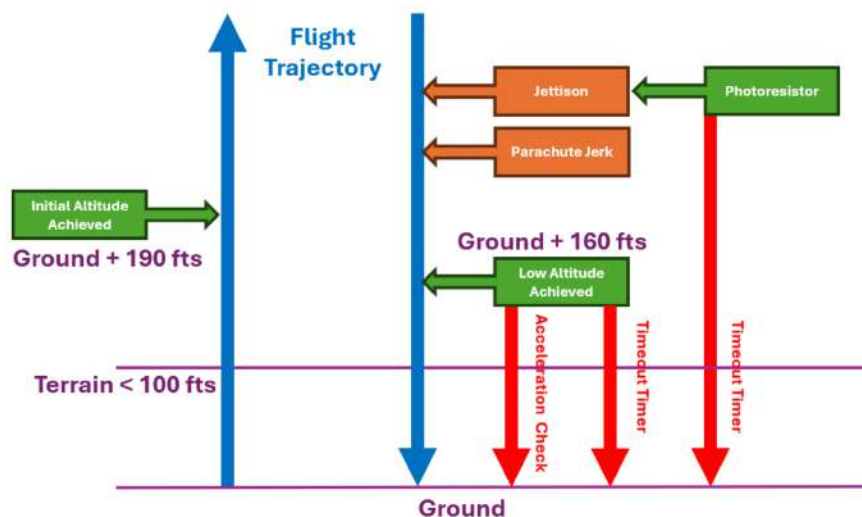


Figure 124: Landing Detection Algorithm with Built-in Redundancy. Height values are specific to the drone drop test; terrain height is exaggerated.

The system's altitude-based logic follows a dual-stage process: it first detects launch by surpassing a predefined high-altitude threshold and then confirms descent when the altitude falls below a low-altitude threshold (the number in the figure is arbitrary and requires tuning). This design prevents false triggers before launch, ensuring the system only activates during actual flight events.

To handle different landing scenarios, the system incorporates three distinct detection paths. The Primary Path uses acceleration magnitude to detect landing when it exceeds a defined threshold, providing a reliable solution for standard, high-impact landings.

The Secondary Path introduces a timeout mechanism to account for softer landings where acceleration may not meet the detection threshold. Once the altitude drops below the low-altitude threshold, a timer starts. If no acceleration spike is detected within the calibrated timeout period, the system still confirms landing. The timer duration requires more tuning and will require analysis from the simulation side.

The Jettison-Based Path adds further robustness through sensor diversity. This path uses a photoresistor to detect payload deployment by sensing changes in light levels. Although this method has not yet been implemented for the subscale rocket, it introduces redundancy by utilizing an independent sensor and a simulation-calibrated timeout to enhance reliability.

By combining these layered mechanisms, the new design mitigates the risks of missed detections, adapts to a variety of landing conditions, and ensures system reliability through redundancy, sensor diversity, and simulation-backed calibration.

4.3.4 Velocity Estimation

Another significant advancement in the development is the implementation of velocity estimation. Initial attempts to integrate acceleration data revealed challenges. The data was not only noisy, causing drift, but also highly unstable due to abrupt events such as parachute deployment and landing. This led to the decision to estimate velocity using the derivative of altitude, as the motion is primarily vertical.

To address the noise in the altitude data, a simple average filter was implemented. This filter utilizes a queue data structure of size 100, where each sampling iteration enqueues the newest data and dequeues the oldest, calculating the average for the sampling period. The filtered data appears smoother while still reflecting the actual altitude changes.

The derivative operation, modeled as a practical derivative in the Laplace domain, was converted to the z-domain using the Tustin transformation. This was then reformulated into a difference equation suitable for embedded system applications, enabling real-time velocity estimation.

$$\frac{V(s)}{A(s)} = \frac{s}{\tau s + 1}$$

$$(T + 2\tau)V(z) + (T - 2\tau)z^{-1}V(z) = 2A(z) - 2z^{-1}A(z)$$

$$(T + 2\tau)V(k) + (T - 2\tau)V(k - 1) = 2A(k) - 2A(k - 1)$$

$$V(k) = \frac{2}{T + 2\tau}A(k) - \frac{2}{T + 2\tau}A(k - 1) - \frac{T - 2\tau}{T + 2\tau}V(k - 1)$$

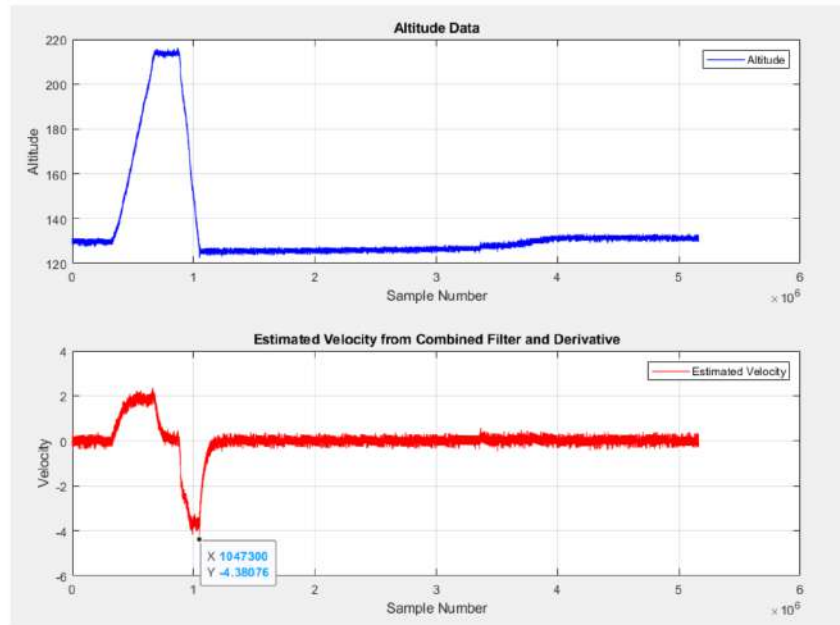


Figure 125: Initial test with MATLAB, using simulation data.

The difference equation was validated using previous drop test data in MATLAB, where its functionality was thoroughly tested and verified to ensure accurate velocity estimation under real-world conditions. The resulting landing velocity is within 5% of the simulation expectation (altitude is measured in meters (*m*) and velocity in meters per second (*m/s*)).

When migrating the difference equation to the Raspberry Pi, additional tuning was required. Through testing, the filter constant was optimized and found to perform best at a value of 0.75. This ensures accurate and stable velocity estimation in the embedded system environment.

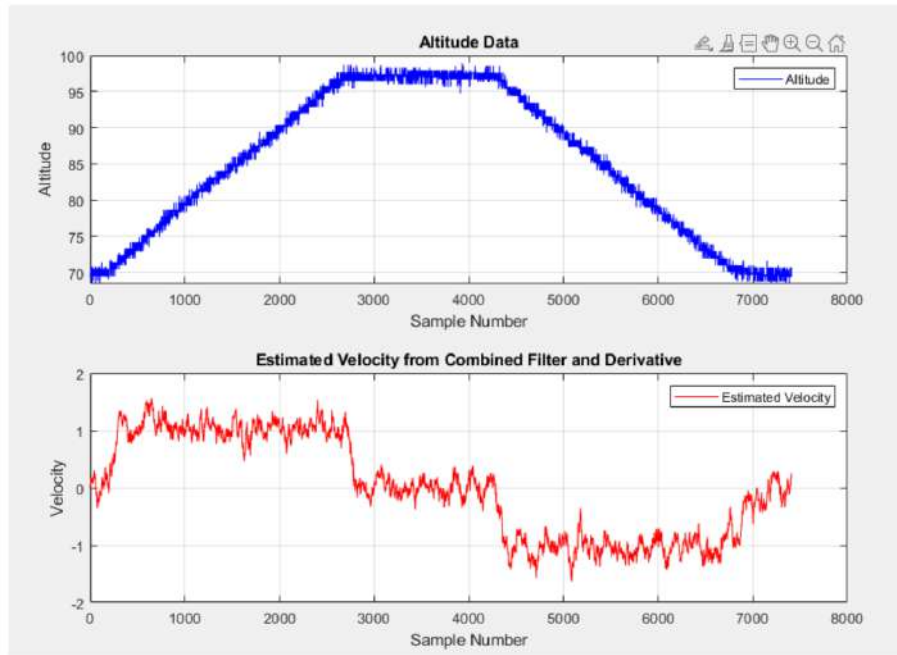


Figure 126: Test performed on the elevator, with the safety office. Real-time velocity estimation.

A test was conducted on the Python implementation of the difference equation by moving the payload up and down in an elevator. The velocity shown in the figure represents real-time data calculated and collected on the Raspberry Pi (altitude is measured in meters (m) and velocity in meters per second (m/s)). A linear regression on the altitude data revealed slopes of 1.06 m/s for ascent and -1.07 m/s for descent, demonstrating the program's accuracy in estimating velocity. However, additional real-time testing, including drop tests and launches, is necessary to fully validate the estimation's performance.

This strategy, if proven successful, will also be applied to survivability estimation, which is currently in the preliminary stages of development.

4.3.5 RF Transmission

The RF system is designed to transmit flight and processed data through APRS packages at a designated frequency upon landing. The LightAPRS-1.0 board was selected for this purpose due to its Atmega1284P-AU architecture, which provides programmability and supports both SPI and I2C interfaces. I2C was chosen for communication with the Raspberry Pi because it minimizes pin usage and is suitable for non-time-sensitive tasks. The unused SPI pin has been repurposed as an ENABLE pin, allowing the system to initiate transmission only after landing is detected.

In previous testing, the RF system underwent rigorous evaluation during drop tests, resulting in several updates and improvements. The APRS board (Top of Figure 129) now incorporates two input mechanisms: an enabling signal, set to HIGH by the Raspberry Pi upon landing detection, and updated RF data arrays sent from the Raspberry Pi via the I2C protocol. These enhancements ensure reliable and accurate data transmission immediately after landing.

On the software side, the LibAPRS.h library was used for managing APRS communications. Data processing is handled through I2C, where received data is stored in a global array, processed, and parsed. To ensure reliable transmission, data is sent at 10-second intervals, alternating between two groups: a short message for better reliability and a more detailed message. This alternating pattern provides a balance between redundancy and data fidelity.

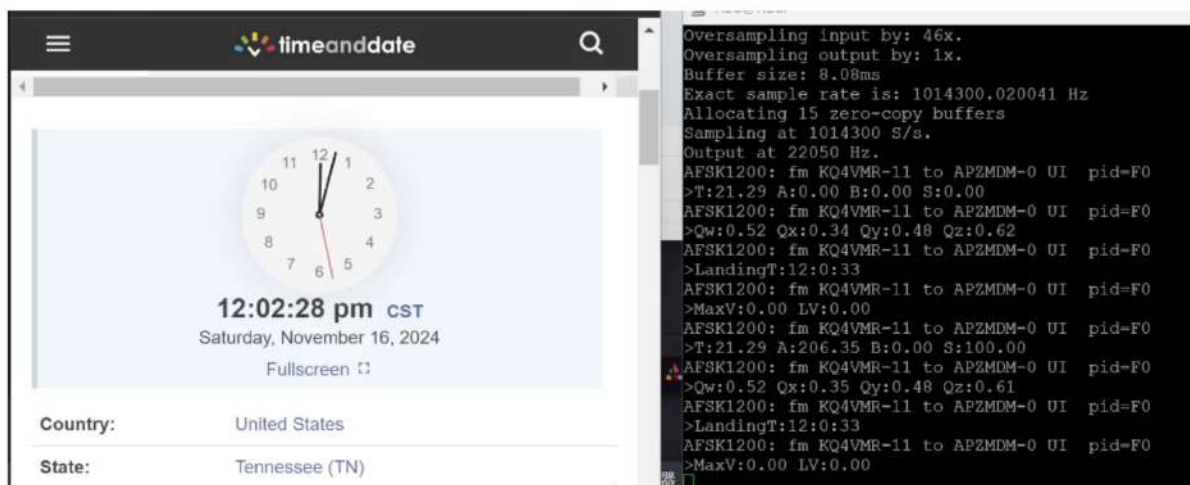


Figure 127: Screenshots from the receiver test in one of the drop tests.

Currently, an RTL-SDR dongle connected to a separate Raspberry Pi is used for receiving and decoding messages. A sample of the data collected during one of the drop tests shows that all transmitted data was correctly received. Notably, the landing time remains constant after landing, while quaternion data continues to update. However, velocity and battery monitoring were not implemented during the drop test and will be added in future iterations.

A challenge identified during testing involves competition between the I2C bus and the enable signal on the APRS board, causing the first few transmitted data points to be junk. To address this, a solution has been planned to delay the APRS transmission or give priority to I2C communication, using one-time non-blocking delays, which will be implemented in future updates. This test was performed before velocity estimation was developed hence on the receiver side there is no velocity data.

The transmission was successfully verified at a distance of over 1,300 feet. However, further testing with actual quantifiable data is required. The next steps include utilizing an affordable Baofeng handheld radio, connecting its audio output to a computer via a cable, and assessing RSSI signal strength using professional software like SDRUno. These steps will help improve the reliability and performance of the RF system.

4.3.6 Prototype Board and Power System

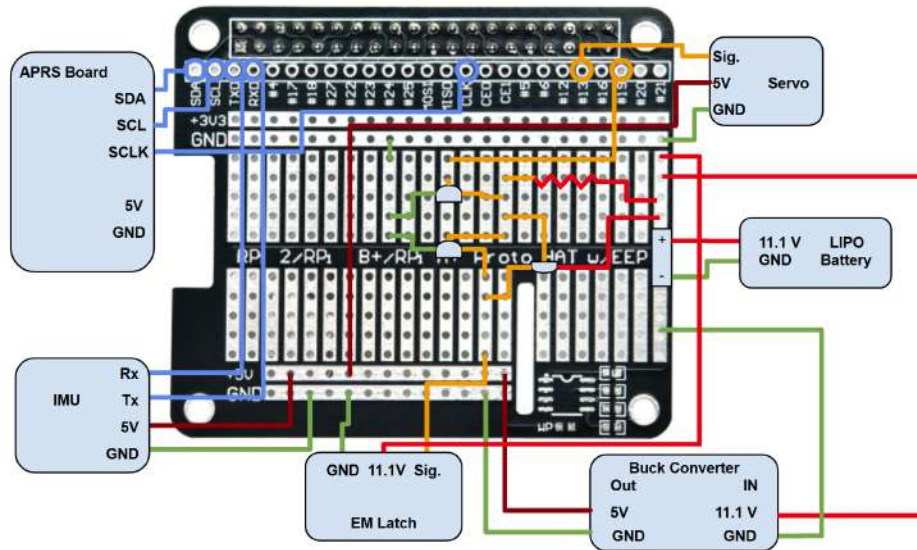


Figure 128: Assembled prototype board.

The prototype board shown in the Figure 128 above was designed by first organizing the placement of various voltage levels. It features dedicated power rails for GND, 3.3 V, and 5 V, corresponding to the output voltages provided by the Raspberry Pi. Additionally, an 11.1 V power supply was included specifically to power the electromagnetic (EM) latch. This 11.1 V power rail was connected to the board through a screw terminal block along the rightmost vertical rails. A pin from the 11.1 V rail was routed to a DROK Buck Converter, which stepped down the voltage to 5 V at a maximum of 5 A, sufficient to power the Raspberry Pi 5. The +5 V output from the Buck Converter was also used to power the servos, the IMU, and the APRS board. Connections to peripheral devices were made directly from the prototype board, with SDA, SCL, and SCK wired to the APRS board; Tx and Rx to the IMU; and GPIO13 to the servos used in the Adjustable Equilibrium Mechanism (AEM).

The other rails on the prototype board were used to mount the resistor and transistors used to amplify the unlatch signal from the Raspberry Pi GPIO19. When the IO pin sends a high output upon detecting landing, the NMOS transistor turns on, pulling the drain of that transistor to 0 V. This gets sent through an inverter made from two other transistors, which pulls the signal up to the 11.1 V, triggering the latch and releasing the parachute.

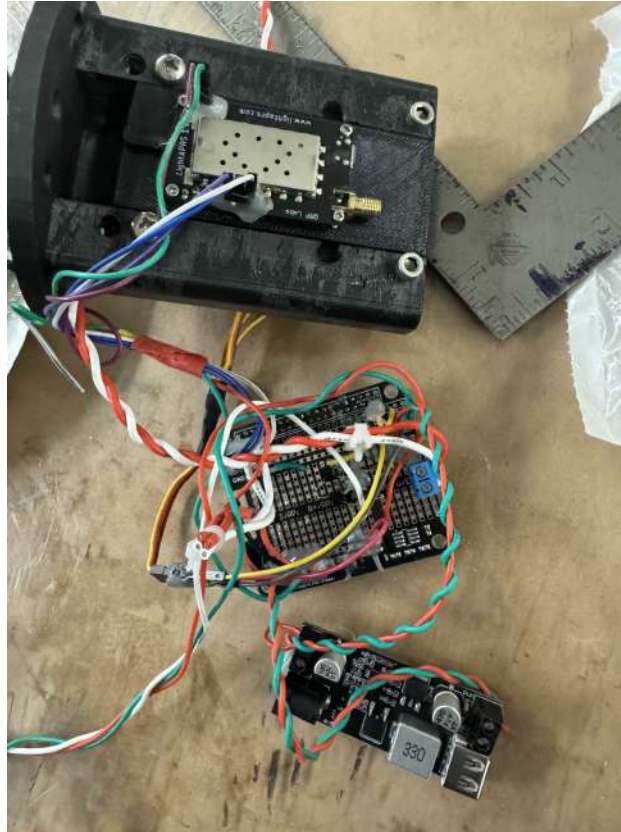


Figure 129: Assembled prototype board.

This prototype board worked as intended during drop tests, however, in the subscale launch, the soldered connections weren't able to withstand the forces of the launch, and subsequently, the Raspberry Pi lost power. Improvements to these poor solder connections are detailed in Section 4.3.8.

Table 42: Payload Power Consumption

Component	Voltage (V)	Amps (A)	Power Draw (W)	Operating Time (s)	Power Consumption (Wh)	Necessary Supply (Wh)
Computer (on-pad)	5	0.8	4	10800	12	15
Computer (flight and transmission)	5	5	25	400	2.7	3.5
EM Latch (Idle)	12	0.02	0.24	11200	0.75	0.93
EM Latch (Trigger)	12	1	12	1	0.0033	0.004
Servo (Idle) x3	5	0.0027	0.0135	11200	0.042	0.053
Servo (Turning) x3	5	0.4	2	2	0.0011	0.0014
IMU	5	0.04	0.2	11200	0.62	0.78
APRS (Sleep)	5	0.005	0.025	10800	0.075	0.094
APRS (Tx)	5	0.76	3.8	400	0.42	0.53

The power calculations from the PDR report have changed slightly as the team determined that the Raspberry Pi 5 operates best when it draws 5 A. When testing the capabilities of the Raspberry Pi in the lab, it was discovered that the performance of the Pi was significantly slower when powered

with the previous 3 A supply. The buck converter chosen is capable of producing a 5 A current, which allows the Pi to run as expected. Another fine-tuning to the calculations was made for the components where the power draw is affected by what mode of operation the devices are in. With a power efficiency factor of 1.25 times the total power consumption, the necessary power supply of the payload electronics is 21.0 Wh.

4.3.7 GPS Selection

As described in Section 3.3.6, the GPS module has been updated to the BigRedBee 70 cm 100 mW GPS/APRS Transmitter. One unit will be placed in the payload and another in the rocket body to meet NASA's requirements. Switching to the 70 cm band for data transmission reduces interference with other teams operating on the 33 cm band.

For future full-scale launches, the payload GPS will be isolated from other payload electronics and powered by an 850 mAh, 3.7 V LiPo battery. Once the system's performance is validated in isolation, it will be integrated into the payload's main electronics. According to the user guide, the module can tolerate up to 6V without damage. Eventually, it will be powered by the 5V output from the buck converter on the PCB. This GPS unit will also operate under Time Slotting.

4.3.8 Failure Case Study and Plans

Though not required by NASA to include payload electronics for subscale flight, the team integrated the payload into the subscale launch vehicle to test the system under realistic conditions, with the expectation of gaining maximum field experience to enhance the reliability of the electronics.

Power Loss During Launch Acceleration

The payload experienced a loss of power during launch acceleration. Diagnostic testing revealed a cold solder joint on the power line, which failed during launch. This issue disrupted electrical continuity, leading to system-level failure during the critical launch phase.

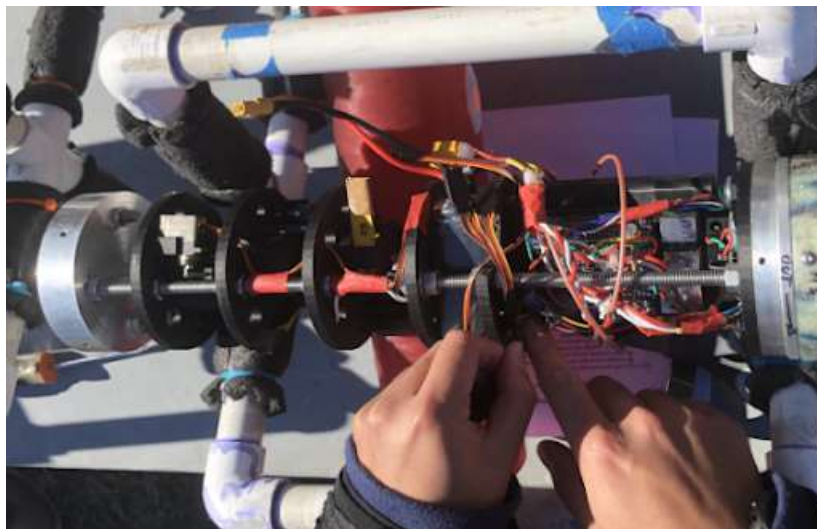


Figure 130: Disorganized wiring made it difficult to check for poor solder joints.

Wire management further exacerbated the problem. Excessive wire lengths and disorganized routing increased the likelihood of accidental shorts and electrical interference. Loose wiring also amplified mechanical stress during launch, adding strain to already weak solder joints.

The failure highlights poor soldering quality, which compromised the reliability of the power line under mechanical stress. Cold solder joints are a result of insufficient heat application during assembly and prevent proper bonding between components and traces.

The combination of poor soldering and inadequate wire management reveals significant gaps in hardware assembly practices. The team will implement proper manufacturing quality control, including visual inspection and rework of suspect joints, to prevent such issues from occurring in the future.

Damage During Ground Troubleshooting

While probing the board with the battery connected, an unintended short circuit occurred when the probe made contact with unintended points on the board. This short caused irreversible damage to several components, including the power regulator and Raspberry Pi, causing the custom-made board inoperable.

This failure underscores the lack of adherence to standard board probing protocols. Probing live circuits without proper isolation introduces risks of accidental shorts, especially in densely packed boards.

Testing Protocols

Testing conducted on the subscale payload and structure was focused on mechanical structure integrity, and electronics were not integrated into the launch vehicle until very close to the launch. This prevented the team from properly verifying the payload's entire functionality under realistic conditions. No vibration, thermal cycling, or mechanical stress tests were performed on the electronics after integration to simulate launch conditions. Similarly, electrical testing gaps, such as continuity and load testing, left critical power lines invalidated.

The lack of environmental and electrical testing meant that electronic hardware weaknesses remained undetected until flight and troubleshooting conditions exposed them. Stress testing replicates real-world dynamic forces, helping identify failure points before critical missions. Detailed continuity and load testing of power lines would have ensured the system's robustness. Moving forward, strict testing protocols shall be followed including acceleration and drop testing with all electronic components integrated.

Future Work

To improve reliability, modularity, and system organization, the electronics will transition to a fully custom-printed PCB design. The architecture will be divided into three distinct layers to streamline communication and reduce interference. Layer 1 will house the main controller and bus manager, centralizing control and ensuring efficient system-wide communication. Layer 2 will include the IMU and photo-resistor circuit, dedicated to sensor data acquisition. Layer 3 will isolate the RF

board for communication to minimize potential interference with other electronics. This layered approach enhances modularity while simplifying integration and troubleshooting.

A well-managed, server-room-style wiring approach will be implemented to maintain reliability while reducing system complexity. Wires will be systematically routed and clearly labeled to create a neat and maintainable infrastructure. This organized approach ensures reduced risk of errors, easy identification of connections, and long-term reliability without over complicating the design.

To address the issues with hard-soldered wires and screw terminals, the new design will incorporate MOLEX connectors for all critical connections. These connectors are robust, durable, and keyed to ensure error-free assembly and prevent incorrect connections. The upgrade will simplify maintenance, enable quick and secure connections, and improve the overall reliability of the system under both operational and testing conditions.

4.3.9 Custom PCB Design

In response to the lessons learned from the subscale payload failure, an industry-standard design of a 4-layer PCB is presented, integrating a custom Buck Converter, Photo-resistor Analog-to-Digital Conversion, venting, and wire organization to a MOLEX Connector. This PCB is designed to mount directly on top of the Raspberry Pi, providing organized cable management, as discussed in the previous section.

At this moment, the PCB is still being manufactured. Tests will be performed in isolation to ensure individual modules are functional before connecting them to the Raspberry Pi. The performance and test results will be presented in the FRR Report. Detailed design considerations and functional explanations will also be provided in detail once verified in the next report.

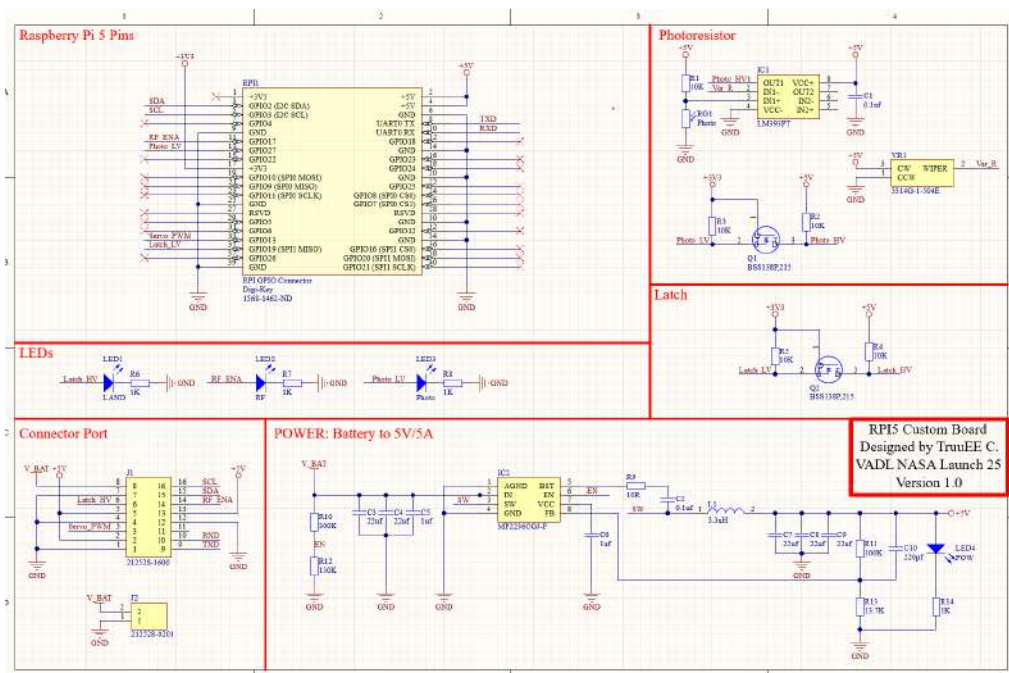


Figure 131: Schematics of Custom PCB.

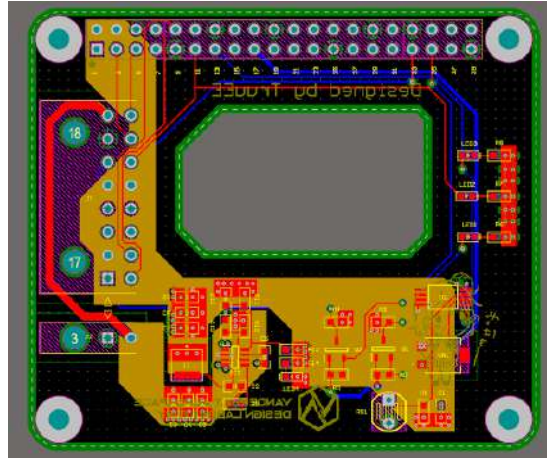


Figure 132: 4-Layer PCB Layout.

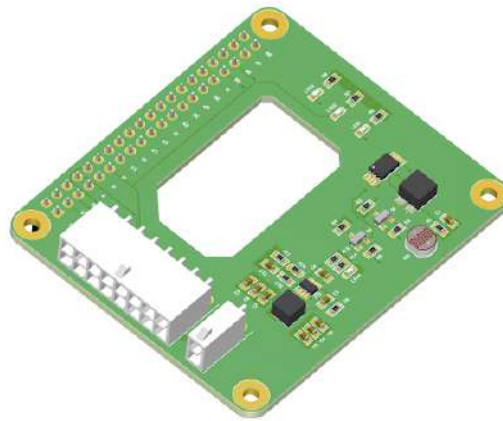


Figure 133: 3D Model of PCB

The cutout in the middle is intended for improved ventilation. In the next iteration, an active cooling fan/battery status ADC is planned to be added above this region.

5 Safety

5.1 Safety Overview

The Vanderbilt Aerospace Design Lab adheres to a strict safety protocol for all activities conducted during the USLI competition. Safety is a top priority for the team and is central to all manufacturing, testing, and launch operations. The day-to-day safety of the team is overseen by the Student Safety Officer, whose responsibilities will be discussed at greater length in Section 5.1.2. Lab safety is guided by a collection of documents, including the VUSE Shop Safety Guidelines, the Material Safety Data Sheets (MSDS), and the VADL Cardinal Safety Rules. All three of these guides are stored in the safety binder - which is easily accessible in the lab - and the Cardinal Rules are additionally posted around the lab for easy reference. Launches are governed by the above listed documents, as well as the NAR and FAA guidelines. Extensive mitigation strategies for potentially dangerous activities have been prepared by the team, all of which can be found in Section 5.2 through Section 5.5. These strategies and their organization is heavily guided by the MIL-STD-882, which is the standard system safety document for the Department of Defense. These sections are also influenced by the NASA System Safety Handbook, which helped to supplement the guidelines of the MIL-STD-882.

5.1.1 VADL Safety Structure

VADL follows a straightforward approach to safety in the lab, which is guided by the documents mentioned in Section 5.1 and by the VADL safety personnel covered in Section 5.1.2. This structure is detailed in the flowchart below.

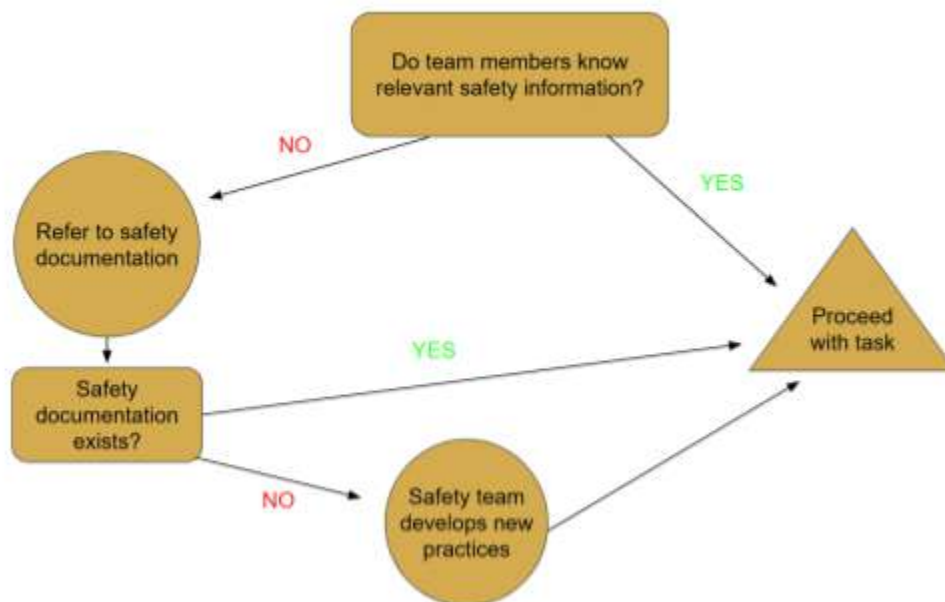


Figure 134: Safety flowchart.

The primary purpose of this process is to take a preemptive approach to lab safety, with an emphasis

on stopping accidents before they happen. By referring to safety protocols for each step in the manufacturing, testing, and launch process, the team can prevent most accidents from occurring in the first place. In the event that an accident does occur, the safety guidelines will be carefully reviewed by the safety officer, and adjustments to lab practices will be made to avoid such accidents in the future. These amended practices are then added to the safety binder.

5.1.2 Safety Personnel

While all team members are responsible for maintaining a safe environment, three individuals hold the ultimate responsibility for directing and ensuring this safety. These are the Student Safety Officer, Assistant Safety Officer, and Safety Mentor. The Assistant Safety Officer is Mark Thelen. He is responsible for overseeing all of the mechanical engineering lab spaces at Vanderbilt, which includes the Aerospace Design Lab. Mark works closely with the team throughout the project cycle and is regularly consulted about lab equipment and practices. Chris Dondanville is the Safety Mentor for the team. He is responsible for preparing the rocket motor in the field prior to each launch, and is also responsible for approving the construction of the rocket. Chris makes himself available to the team during the design process and serves as an invaluable resource for information pertaining to recovery systems, motor selection, and general rocket construction. The Student Safety Officer (SSO) this year is Zachary Stutsman. His primary role is to maintain a safe lab, testing, and launch environment throughout each phase of the project. This is accomplished through a variety of means, which are discussed below.

Each Monday, the SSO gives a weekly safety briefing to the team detailing potential hazards that could be encountered during the manufacturing and testing activities planned for that week. This serves as an important preventative measure, as it provides immediately relevant safety information to each team member and thus greatly reduces the likelihood of accidents occurring. In addition to these weekly briefings, the SSO gives an additional briefing prior to each launch or drop test, as these scenarios involve numerous potential safety hazards. These briefings are an important component of the mitigation strategies which will be discussed in more detail in Section 5.2 through Section 5.5. The SSO is responsible for keeping the Material Safety Data Sheets, NAR, FAA, and Vanderbilt regulations up to date in the Safety Binder, and for ensuring that all team members are aware of its location and contents. Another responsibility of the SSO is to regularly check the first aid kits, PPE, and fire extinguishers in the lab to ensure that they are in their proper place and are well-stocked. As an extension of these tasks, the SSO should be well versed in all safety related practices, and should be able to address any safety related questions posed by the team members or mentors. This can involve either directly addressing the question or knowing which resources to utilize to find an answer.

5.2 Safety Matrix Design

Safety matrices are used as a means to succinctly yet methodically list, categorize, and address hazards and failure points that could be encountered at any point throughout the design, building, and testing cycle. The matrices are split into three main sections: personnel hazards, failure modes, and environmental hazards.

The structure of these matrices draws heavily on the standards set by MIL-STD-882, especially

with regard to severity classification, probability levels, and risk assignment. The descriptions of each severity and probability level have been adjusted to fit within the context of the USLI competition, but the structure of this system remains largely consistent with the MIL document. This is done in an effort to acknowledge and implement industry best-practices, thus ensuring a robust and proven safety framework for the project.

5.2.1 Severity Classifications

An important part of the categorization of hazards is assigning severities to them. This allows for hazards and failure points to be systematically sorted based on the criteria that they meet. Table 43 below lists the severity levels associated with different criteria.

Table 43: Severity classifications.

Severity Classifications		
Severity	Classification	Result of Mishap
Catastrophic	1	Could result in loss of life or severe injury, extreme environmental impact, or complete loss of rocket or payload.
Critical	2	Could result in injury to personnel, significant impact to environment, or substantial damage to the payload or rocket.
Marginal	3	Could result in minor injury to personnel or reversible moderate damage to the rocket, payload, or environment.
Negligible	4	No injury risk to personnel, and chance of minor reversible damage to rocket, payload, or environment.

This classification system encompasses all possible hazards or failure modes that could be encountered in the project, allowing for a thorough evaluation of each potential risk.

5.2.2 Probability Levels

Probability levels are the second core component of meaningful risk assessment. This allows for events to be sorted by the likelihood that they will occur. These range from frequent to improbable, with a separate eliminated category that addresses a risk which has been removed through mitigation measures. This classification is described in Table 44 below.

Table 44: Probability levels.

Probability Levels		
Probability	Classification	Description
Frequent	A	Likely to occur often
Probable	B	Will likely occur several times
Occasional	C	Will likely occur at some point
Remote	D	Unlikely but could possibly occur
Improbable	E	Unlikely to occur
Eliminated	F	Incapable of occurrence

With respect to probability, the goal of mitigation strategies is to lower the chance of occurrence as much as possible. By minimizing the likelihood of an event, it is possible to offset the impact of more severe hazards, even if it is not possible to lower the severity of these hazards.

5.2.3 Risk Assessment Matrix

The risk assessment matrix assigns a risk level to each hazard or failure mode based on its probability and its severity. These levels are color coded for ease of reading and are listed in Table 45 below.

Table 45: Color key.

Color Key	
	High
	Substantial
	Moderate
	Low
	Eliminated

As can be seen, there are five risk levels, ranging from high to eliminated. A high risk event is one that has both high severity and probability, while a low risk event is one that has both low severity and probability of occurrence. There is also a section for eliminated risks, which accounts for hazards that are removed through mitigation strategies. Ideally, all hazards would be completely eliminated, but this is not a realistic expectation. Therefore, the aim of the mitigation strategies conducted by the team is to reduce the risk level as much as possible by lowering the severity and/or probability of an event occurring. The risk assessment matrix is shown in Table 46 below.

Table 46: Risk assessment matrix.

Risk Assessment Matrix				
	Catastrophic (1)	Critical (2)	Marginal (3)	Negligible (4)
Frequent (A)	1A	2A	3A	4A
Probable (B)	1B	2B	3B	4B
Occasional (C)	1C	2C	3C	4C
Remote (D)	1D	2D	3D	4D
Improbable (E)	1E	2E	3E	4E
Eliminated (F)	Eliminated			

As mentioned above, the goal of mitigation measures is to shift hazards and failure modes as close to the bottom right corner of this table as possible. This risk level classification will be used throughout the safety matrices in Sections 5.3 through 5.5 to denote the risk level of each hazard or failure mode.

Each hazard will be assigned two risk levels. One is the risk level prior to mitigation and the second is the risk level after mitigation. The risk level before mitigation represents the risk associated with that hazard, assuming that all of its potential causes represent the actual conditions in the lab or field. Many of these will be higher level risks than desired. The post-mitigation risk assessment, however, represents the risk after all mitigation measures have been applied. This represents the actual risk associated with these activities as they are conducted by the team, and these should all be within the team-defined acceptable risk range. An acceptable risk is defined as a hazard having a post-mitigation risk level of moderate, low, or eliminated. Hazards will be continually evaluated based on experience gained and/or safety incidents. Once re-evaluated, the hazard mitigations will be revised as necessary to ensure the post-mitigation risk level remains at moderate or lower.

5.2.4 Hazard and Failure Mode Key

The safety matrices cover personnel hazards, environmental hazards, and failure modes. Each of these high level areas comprise specific risks which can occur across many phases of the project. For that reason, each safety matrix is broken into subsections pertaining to the project phase in which they can occur. These subsections are described in Table 47 below.

Table 47: Safety matrix subsections.

Safety Matrix Subsections		
Personnel and Environmental Hazards		
Designation	Meaning	Description
M	Manufacturing	Safety hazards that could arise during the manufacturing process.
L (S)	Launches	Safety hazards associated with the launch and recovery of the rocket. All hazards apply to the full-scale launch vehicle, and the subset that also apply to the subscale launch vehicle are denoted by an (S).
DT	Drop Testing	Safety hazards that could arise during drop testing. This designation specifically refers to testing conducted on the hexacopter drone owned by the team.
LT	Lab Testing	Safety hazards associated with testing beyond the hexacopter drop tests.
Failure Modes		
P	Payload	Failure modes associated specifically with the payload portion of the mission.
V	Vehicle	Failure modes associated specifically with the vehicle portion of the mission.
T	Team (team functioning)	Failure modes associated with team or organizational hazards.

The personnel and environmental hazard matrices will have the same subsections, as they both are concerned with hazards that can arise during the project. The failure modes table will have its own set of subsections, as it is concerned with project failure modes instead of hazards. These subsections make the safety matrices more readable and provide a framework to more thoroughly define risks.

Hazards or failure modes that apply to one or more of these subsections will have all relevant subsections listed in the ‘Designation’ column of the matrix. For ease of reading, all hazards which have multiple designations will be listed at the bottom of the matrix, following the items which have only a single designation.

5.3 Personnel Hazard Analysis

The matrix below details the risks to any personnel associated with the project throughout the duration of its cycle. This primarily means VADL team members but also includes team mentors, spectators, and occupants of the areas surrounding both laboratory and field sites. Each hazard is assigned a pre and post mitigation risk level, as detailed in Section 5.2.3. Higher risk levels, in this table, indicate the potential for more severe injuries to personnel, with death being a possibility in the highest risk cases.

Section 5.2.4 can be referred to for information about the designation of each hazard, which helps to organize the matrix based on what phase of the project a hazard could occur. All hazards with more than one designation can be found at the bottom of the table.

Several of the verification columns in this matrix mention that only approved team members are allowed to use certain equipment. This refers to the Equipment Operation Approval Sheet, which was discussed in depth in PDR. Basically, this sheet lists which team members are approved to use each piece of power equipment in the lab. In order to be approved, individuals must be trained by a team member who is already approved on the equipment and then be checked off by the SSO.

It is the ultimate responsibility of the SSO to regularly check that mitigation strategies are being followed. The verification measures listed below and in the subsequent matrices have been established by the SSO to help ensure that each mitigation measure is being met every day. This forces the entire team to take personal responsibility for safety, which greatly increases the overall effectiveness of the preventative measures.

Personnel Hazard Analysis							
Designation	Hazard Description	Causes of Hazard	Hazard Effect	Risk Level	Mitigation Strategy	Verification of Mitigation	Post-Mitigation Risk Level
M	Tripping and falls	- Untidy lab spaces - Power cords strung across walkways	- Cuts or bruises - Potential head injury	3B	- Power cords will be kept clear of walkways - Lab will be regularly cleaned and walkways will be kept clear - Team members will be alert while moving about lab spaces	- Rais is assigned to check the lab each day for cords or cluttered walkways - Weekly safety briefings will include reminders about this and other pertinent details	3E
M	Power drill related injuries	- Improper fixturing of work piece - PPE not worn - Hands or other body parts too close to work area	- Cuts or puncture wounds - Eye injury or loss of vision from debris	2B	- Eye protection and work gloves will be worn when operating drills - Respirator will be worn as well when drilling into fiberglass - Work pieces will be firmly restrained while drilling - Other team members will stand clear of the work area	- Eye protection, gloves, and respirators are readily available in the lab at all times and their location is known by every team member - Clamps are also available and their location is known by all team members - Weekly safety briefings will include this information when drilling is expected during that week - Only approved team members will use drills	4D
M	Chop saw related injuries	- Improper fixturing of work piece - PPE not worn - Hands or other body parts too close to work area	- Loss of fingers or limbs - Severe cuts - Eye damage or loss of vision from debris - Inhalation of damaging particulates - Hearing damage	1C	- Gloves, safety glasses, ear protection, and respirator will be worn throughout saw operation - Hands will be kept well clear of blade whenever saw is plugged in - Work piece will be firmly clamped in place	- Gloves, eye protection, ear protection, and respirators are available in the lab and their location is known by all team members - Clamps are easily accessible - Only approved members will use the chop saw - Two or more members will be present during saw usage - Weekly safety briefings will include mitigation information when saw usage is expected	3D
M	Dremel related injuries	- Improper fixturing of work piece - PPE not worn - Hands or other body parts too close to work area	- Cuts or abrasions - Eye damage or loss of vision from debris - Inhalation of damaging particulates	2C	- Eye protection, gloves, and respirator will be worn while using dremel - Hands will be kept clear of work area and piece will be firmly clamped - Other team members will be kept clear of work area	- Eye protection, gloves, and respirators are available in the lab - Only approved team members will use the dremel - Weekly safety briefings will include this mitigation information when dremel usage is expected	4D
M	Table sander related injuries	- Loose clothing or hair near sanding belt - Inattention while sanding - PPE not worn	- Severe abrasions - Inhalation of harmful particulates - Damage to hearing	2C	- Eye, ear, and hand protection will be worn while operating sander - Operator will be fully focused on sanding - Hands will be kept clear of sanding belt	- Gloves, ear, and eye protection are easily accessible in the lab - Only approved team members will use the table sander - Weekly safety briefings will include this mitigation information when table sander usage is expected	4D
M	Electrical shock	- Frayed or damaged power cords - Cutting of power cord during tool operation - Improper wiring of electronics	- Electric burns - Muscle spasms - Nerve damage - Irregular heartbeat - Cardiac arrest	1C	- Power cords will be regularly checked to ensure they are insulated properly - Power cords will be kept out of harms way during tool operation - Electronics layouts will be checked by at least two team members and only trained members will wire electrical components - Power sources will be kept to the minimum viable strength	- Rais is responsible for checking power cords for damage each week - Patrick is responsible for overseeing electronics checks - Cord management is included as part of power tool training	3E
M	Inhalation of damaging fumes or particulates	- Cutting, drilling, or sanding of fiberglass without a respirator - Use of epoxy, polyurethane, or other chemical without proper ventilation or mask - Soldering with Lead Solder	- Brain or lung damage	2B	- No chemicals are to be used without adequate ventilation - All members near fiberglass manufacturing will wear respirators and other appropriate PPE	- MSDS are easily accessible in the lab and are consulted prior to use of chemicals - Respirators and other PPE are easily accessible - Fiberglass manufacturing best practices have been discussed at length with full team	3D
M	Mill or lathe related injuries	- Inattention or improper PPE while working on mill or lathe - Loose clothing; dangling jewelry or hair - Hands or other body part too close to work piece - Improper fixturing of tools or work piece	- Severe injury or death if pulled into machine - Eye or body wounds from shrapnel or debris	1C	- Loose clothing will not be worn in the shop - Dangling jewelry will be removed and hair will be put up - Eye protection will be worn at all times in the shop - Only team members who have completed the machining course or have approval from the shop supervisor can use the mill or lathe - Any and all additional requirements of the shop supervisor will be followed	- Special approval by shop supervisor is required to use machining equipment - Shop supervisor will directly supervise any team members working in the machine shop and will be available for any questions	3D
M	Injury during blast charge preparation	- Preparation of charges near heat or electrical source - Ematch wires not wrapped together - Preparer is not grounded prior to assembling charges	- Ignition of charge leading to burns on hands, body, or face of preparer - Potential damage to eyes	2C	- Charges will be prepared far from any source of heat or electricity - Ematch wires will be wrapped together prior to placement in charge to ensure electricity cannot flow through match - Preparer will ground themselves prior to charge preparation	- SSO will personally prepare blast charges to ensure safety measures are met - Blast charge preparation procedure will be carefully followed to ensure all mitigation strategies are adhered to	F
M	Injuries associated with electrical system preparation	- Soldering iron left on or held too close to skin - Heat gun left on or held too close to skin - Molten solder not allowed to cool before handling - Poor labeling of system leading to unknown live wires	- Potential burns, especially on hands, from soldering iron, heat gun, or molten solder - Electrical shock from live wires	3B	- Soldering irons and heat guns will be kept off and unplugged when not in use - Electronics will not be handled until solder has had sufficient time to cool - Electrical systems will be constructed in accordance with documented diagrams, and will only be handled by those familiar with the layout	- Only approved team members will use soldering equipment - All electrical engineers must complete solder training prior to participation in manufacturing activities - Patrick is responsible for approving electrical diagrams prior to circuit construction	3E

M	Fire in lab space	- Exposed electrical wiring - Overheating of power tools - Hot box left on - Heat gun or soldering iron left on - Exposure of energetic material to heat	- Minor to major burns on team members - Damage to or destruction of lab and possibly of Olin Hall	1C	- Hot box, heat gun, and soldering irons will always be turned off and unplugged before leaving - Extension cords and other wires will be regularly checked to ensure proper insulation - Energetic material will be kept in MAGloc box whenever not in use; heat sources will be kept away from these boxes and away from energetics during usage - Power tools will be allowed to cool after use prior to storage - Fire extinguishers are easily accessible and regularly inspected	- Fire extinguishers will be inspected by the SSO each month - Team members are reminded of extinguisher locations each week in the safety briefing - Rais is responsible for checking extension cords for damage each week - MAGloc boxes are locked at all times - Cord management is included in all power tool training	3E
M	Explosions in lab	- Pressure cooker left on or allowed to exceed safe operating pressure - Black powder or fuel grains exposed to heat - Black powder or fuel grains stored outside of proper containers	- Shrapnel strikes team members or damages other lab equipment - Shockwave damages hearing or vision - Fire is started by explosion of energetics	1C	- Pressure cooker will be kept under operating limit by at least a safety factor of 1.5 - Energetics will be stored in MAGloc boxes when not in use and will be kept clear of heat sources when being used - MAGloc boxes will be kept locked at all times	- Vehicle lead will oversee operation of pressure cooker - MAGloc boxes will be locked at all times - Removal of energetic material from MAGloc boxes will be overseen by either the SSO or the team president, both of whom are well versed in the risks associated with energetic material handling	F
L (S)	Burns during launch or launch preparation	- Blast charge or motor exposed to heat source during preparation - Multimeter used to check electrical connections on prepared altimeter/blast charge system - Improper wiring or coding of avionics leading to premature ignition of blast charge - Motor ignition charge exposed to electricity or heat during installation - Motor ignition charge installed before rocket placed on launch rail	- Minor to severe burns on face or body of team members - Loss of fingers - Chance of death	1C	- Blast charges will be prepared in a clear area away from heat or electrical sources - Only essential personnel will be near blast charge/motor preparation and arming - Avionics will be thoroughly tested and calibrated prior to wiring of blast charges - Multimeters will not be used to check electronic connections of energetics at any point - Motor ignition charge will not be installed until rocket is on rail and properly oriented and will only be installed by the rocketry mentor, Chris Dondanville	- SSO will oversee avionics preparation and motor assembly - Avionics will be assembled by the same team member each time, who has been specially trained on safe avionics assembly - Chris Dondanville will be the only person who handles the assembled motor - SSO will ensure area is clear prior to the placement of the motor ignition charge by Chris Dondanville - The avionics checklist, launch procedure, and safety checklist will be carefully followed on launch day	2E
L (S)	Tripping and falls	- Ground tarp not staked down - Stakes left sticking up above ground level - Inattention of team members while moving about rocket preparation area - Toolboxes or other items left in walkways - Rocket preparation area near large rocks or potholes - Wet conditions leading to slick surfaces	- Scrapes, bruises, cuts - Potential broken bones or head injury	3B	- Rocket preparation area will be selected away from ground hazards such as rocks or pot holes - Tables, chairs, and toolboxes will be kept in designated areas and won't be left in walkways - Team members will be attentive while moving about rocket preparation area - Stakes will be fully hammered into ground and tarp will be fully staked down - Launch day procedures will be thoroughly rehearsed, reducing chance of mishaps	- Launch rehearsal will be attended by all team members - Notes will be taken on any hazards noticed during rehearsal - Each team member will be reminded of their personal responsibility for safety during the launch day safety briefing	4C
L (S)	Back injury	- Using improper form while lifting heavy objects - Lifting more than physically capable of - Risk objects include: launch rail, tool chests, tent, tables, and MAGloc boxes	- Acute or sustained pain in back - Muscular or nervous system injury	2C	- Heavy objects will be moved by two or more team members, as necessary - Team members will not try to lift more than they are capable of - Heavy objects will be lifted using legs and while maintaining a straight back	- Dangers of heavy lifting were discussed by full team at beginning of year - Weekly safety briefings will include a reminder of this information prior to launch week, as launch week generally contains most of the heavy lifting done during the project	3D
L (S)	Team members or spectators struck by rocket	- Launch rail angled incorrectly with respect to wind - Team members located too close to launch rail during launch - Strong wind gusts or gusts of changing direction - Failed parachute deployment - Improper rocket assembly leading to falling debris or rocket sections - Improper sizing of rocket fins	- Minor to severe injury to team members or spectators - Possible death	1C	- Launch rail angle will be checked and approved by the SSO and rocketry mentor prior to launch; launch rail will be properly secured - Team members will be located at least 300 feet away from the launch rail, as designated by the NAR codes. - Launch will not proceed in winds higher than 20 mph, or in unpredictably gusty scenarios - Rocket fins will be sized to provide an appropriate stability margin of approximately 2.2 - Rocket will be carefully assembled following a strict procedure - Blast charges will be carefully prepared and sized according to data from both calculations and deployment tests - All team members and spectators will keep eyes on the rocket from launch until landing	- Launch must be approved by the SSO, team president, and Safety Mentor - Wind speed will be measured in field with an anemometer to ensure compliance - All relevant checklists and procedures leading up to launch will be carefully followed - Rocket construction must be approved by Chris Dondanville prior to launch	3E
L (S)	CATO	- Improper spacing or preparation of motor grains - Use of cracked or otherwise damaged grains - Introduction of foreign material into motor chamber - Improper fastening of nozzle - Improper fastening of end cap - Improper insertion of ignition charge	- Team members struck by flying debris	3C	- Motor will be prepared exactly as shown in the instruction booklet - Motor assembly will be done by the rocketry mentor and overseen by the SSO - Grains will be carefully checked for cracks or other damage prior to assembly - Nozzle and end cap will be securely fastened - Motor casing will be checked for damage prior to each launch - Metal blast deflector will be placed beneath motor exhaust prior to each launch - Team members will maintain at least the NAR minimum distance from the launch rail and will maintain visual contact with the rocket during launch operations	- SSO will review motor assembly instructions prior to launch day - SSO will provide rocketry mentor with motor components and will remain with him until motor is fully assembled; this ensures that oversight is provided at each step - Grain inspection will be conducted by the SSO and the rocketry mentor	4D

DT	Loss of control of hexacopter drone	- High winds render drone control unreliable - Incompetence of drone pilot - Drone runs out of battery during flight	- Significant injury from impact of drone with team member - Potential death	1C	- Hexacopter drone will ONLY be flown by trained and designated drone pilot - Drone will not be flown if winds are too high - Drone batteries will be checked prior to each flight to ensure sufficient charge - Team members will stand clear of drop point	- SSO will consult drone pilot prior to each flight to ensure wind speeds are manageable and batteries are charged - Drone pilot has flown the hexacopter drone extensively and is very capable of controlling the drone	2E
DT	Injury from impact with drone blades	- Team member standing too close to drone during takeoff or landing - Inattention from team members or drone operator	- Minor to severe cuts or lacerations	2C	- Team members will stand well clear of drone during takeoff and landing - All parties will remain aware of their surroundings at all times - Drone operator will issue verbal notice prior to turning drone on	- SSO and drone pilot must visually scan the area surrounding the drone and issue a verbal warning prior to startup of drone	F
LT	Load frame related injuries	- Debris thrown when airframe pieces fail - Team members standing too close to load frame - Team members handling test piece or other portions of load frame during use	- Eye injury or vision loss from debris - Crushing of fingers, hands, or arms	1C	- Plexiglass shield will be placed in front of load frame prior to use - All team members will wear eye protection and stand clear of load frame - Piece will be securely fastened to frame - Frame will not be touched at any point during operation	- Load frame operation procedures will be carefully followed - Load frame will only be operated when Mark Thelen is present to oversee operation - Eye protection is readily available in the load frame facility	4E
LT	Leg loading tests	- Team members standing too close to test area - Weights fastened to legs insecurely - Leg breaks during testing	- Bruising - Chance of minor cuts - Chance of broken fingers or toes	3C	- Team members will keep feet clear while weight is being applied to legs - Weight will be securely fastened to legs and will be steadily and carefully applied - Foam will be placed beneath leg to soften any falling weight	- SSO approved testing setup and plan prior to tests beginning	4D
LT	- Debris or falling components during small scale drop tests - Debris or falling components during parking garage drop testing	- Team or passerby standing too near to drop point - Improper sizing of shock cord - Improper attachment of shock cord to payload or to fixed point - Poor construction of payload leading to breakage upon shock cord extension - Improper length of shock cord used leading to higher than expected forces	- Head or eye injury to team members or passerby - Bruising - Minor cuts	2B	- Team members will stand at least 30 feet from point beneath falling payload (SMALL SCALE DROPS) - Team members will stand at least 75 feet from point beneath falling payload (GARAGE DROPS) - Sufficient number of team members will be present to keep lookout and ensure area is clear of passerby - Drop tests will be conducted outside of peak foot traffic hours - Calculations will be carefully conducted and payload will be constructed to proper specifications - Team members will maintain visual contact with payload throughout drops	- SSO must be present to oversee garage drops - At least two members are required to be present for small scale drops; more are required if the area is moderately populated by passerby - At least two team members are required at the drop point and an additional minimum of two members must be on the ground for garage testing. This ensures that eyes are on the payload at all times and no steps are overlooked - Prior to any drop testing, a procedure will be written out, walked through, and then carefully followed for testing	3E
M, L (S), DT, LT	Exhaustion/fatigue	- Long work hours - Insufficient sleep - High temperatures at launch or tests - Lack of food/water	- Heat exhaustion - Inattention or forgetfulness leading to a variety of unsafe conditions - Dehydration	2C	- Milestones will be reasonably set and work will steadily be done to reach them - No one team member will be responsible for work exceeding their capacity - Food and water will be brought to every launch or field test - Team members will be intentional about their sleep schedule	- Sufficient food and water was brought to subscale launch and the same will be done for full scale - The importance of sleep will be regularly mentioned by the SSO during weekly safety briefings	3D
L (S), DT	Hypothermia or frostbite	- Cold weather during outdoor activities - Lack of proper clothing - Extended exposure to the elements - Wet or windy conditions	- Team members experience hypothermia, frostbite, or both - Discomfort and distraction, potentially leading to safety oversights in other areas	3C	- Launches will not be held if weather is excessively cold, especially if also rainy and/or windy - Team members will dress properly for weather conditions - Time outside will be limited to that necessary for the testing or launch - Hand warmers and emergency blankets will be on hand when cold weather is expected	- Safety briefing will cover cold weather hazards if expected for launch or drop test - Hand warmers and emergency blankets are included on packing list for launches and drop tests	4D
L (S), DT	LiPO battery fire	- Battery overheats from temperature or sun exposure - Battery is punctured during flight or landing	- Potential burns incurred by team members - Potential of falling debris if fire occurs in air	2D	- Battery will be checked for bloating or puncture after each flight - Safe clearance will be maintained throughout drone operation - Team members on ground will remain attentive throughout drone operation	- SSO will inspect all batteries used after each launch or drop test - Payload and vehicle leads are responsible for ensuring their respective LiPO batteries are transported in the proper LiPO container - Battered or otherwise damaged LiPOs will be discarded immediately as specified in the MSDS	3E
L, DT	Payload failure	- Payload unable to withstand parachute deployment forces - Falling debris from broken lander legs or other broken components	- Bruising and potential head injury from falling components	2C	- Team members will maintain a safe clearance from drop point at all times - Payload will be thoroughly tested prior to field drop test - Visual contact with payload will be maintained throughout drop	- SSO and team president will both individually verify that area is cleared prior to launches or drops - Verbal warnings will be issued before both drops and launches by the team president - Payload lead will ensure that proper testing has been conducted on the payload prior to launch and that the construction of the payload is adequate	F

L, DT	Team members struck by payload	- Parachute packed incorrectly leading to rapid descent of payload - Quick link not secured properly - Team located too close to drop point - Excessive or unpredictable wind	- Bodily injury, ranging from minor to significant - Slim chance of death	2C	- Parachute packing will be done carefully and overseen by at least two team members - Quick link will be fully tightened and checked for attachment prior to payload flight - Team will be located sufficiently far from drop point. This location will be determined based on wind speed and drop height - Team members will maintain visual contact with payload throughout drop	- Payload lead will be in charge of parachute packing, with oversight from one of several team members who are familiar with the parachute packing procedure - Dry runs of parachute packing will be conducted before each launch - Payload lead is responsible for checking quick links for damage prior to each flight; checks must be approved by SSO before flight - Wind speed will be measured with an anemometer prior to drop or launch - Team president is responsible for providing a verbal warning prior to drop or launch	3E
M, L (S), DT, LT	Team members experience damage to mental health	- Excessive working without breaks - Insufficient communication of needs among members	- Exacerbation of mental health issues - Loss of motivation - Decrease in team productivity	2B	- Mental health will be prioritized by all team members - Honest and respectful communication will be practiced by all team members - Breaks will be taken as needed by each individual	- The team lead will promote a culture of mental health awareness - The team lead will regularly check in with each team member to ensure they are in a healthy mental state	2D

5.4 Analysis of Failure Modes

The matrix below details potential failure modes for the vehicle, payload, and team as a whole. The vehicle and payload portions are focused primarily on failures that could occur during the competition rocket flight, whereas the team failure modes are more focused on team or organizational level issues that prevent the production of a viable final product. For the purposes of this matrix, failure is defined as an event or occurrence that leads to full or partial incompleteness of the mission. The risk level assigned to each failure mode reflects whether that particular failure represents a full or partial incompleteness of the mission, with full incompleteness yielding higher risk classifications.

Analysis of Failure Modes							
Designation	Failure Mode	Causes of Failure Mode	Effect of Failure Mode	Risk Level	Mitigation Strategy	Verification of Mitigation	Post-Mitigation Risk Level
P	Antennas fail to extend	- Improper winding of AEM - Blockage of AEM opening - Loss of electrical connection to servo - Failure to detect landing - Loss of power prior to landing - Servo strength insufficient to release AEM - Misalignment of payload within rocket body	- No data transmission - Mission failure	2B	- Landing detection software will be thoroughly tested through lab tests, drop tests, and launches - Electrical components will be thoroughly ruggedized and tested - Battery will be sized to provide at least 3 hours of power to full system - Rocket will be carefully constructed - Assembly will be practiced many times prior to launch and a careful assembly procedure will be laid out - Servo will be tested in both lab and field settings - Three AEMs will be included in the payload to render blockage of all improbable	- Antennas have been tested under a variety of operating conditions - Antennas have proven capable of extending from within an airframe - Antennas have been successfully reset in the field multiple times	2D
P	Parachute does not deploy	- Tape is not removed before packing parachute - Parachute is packed in a manner that does not expose it to aerodynamic forces - Parachute is improperly folded - Shock cord is wrapped around parachute	- Payload is not slowed in its descent - Major damage or complete destruction of payload - Mission failure	1C	- Parachute will be folded carefully along each gore to ensure swift opening - Shock cord will not be wrapped around chute - Folding and packing will be practiced prior to launch and a thorough procedure will be followed each time	- Payload and vehicle leads will fold and pack their respective parachutes - Parachute folding and packing will be overseen by at least one other team member familiar with the process - Trial runs have been conducted prior to the subscale launch and drop tests and will continue to be conducted, especially with the larger chutes which will be used for the full scale vehicle	1E
P	Legs fail to deploy	- Misalignment of payload within airframe - Insufficient strength of torsional springs to push legs out - Excessive friction with airframe or with other payload components	- Failure to land upright - Partial mission failure	3B	- Legs will be carefully constructed - Assembly will be practiced prior to launch and all components will be tested to ensure proper fit - Leg deployment will be thoroughly tested in both lab and field settings	- Payload lead is responsible for ensuring that proper deployment testing was conducted on the legs prior to launch - Payload assembly will be carefully practiced multiple days before launch, both inside and outside. Walking through assembly outdoors more closely emulates field conditions, which prepares the team better for field testing	3E
P	LSM bulk plate failure	- Excessive loading forces during main parachute deployment - Insufficient thickness of LSM attachment bulk plate - Shaft collars are attached poorly or are of insufficient strength to withstand main parachute deployment	- Premature release of payload from rocket - Partial destruction of LDM - Possible partial or full destruction of electronics - Potential drifting outside of transmission range - Possible mission failure	2C	- LSM bulk plate will be constructed from aluminum - FEA analysis will be performed on LSM bulk plate and hooks to ensure that it can withstand loading - Sufficient safety factors will be used in bulk plate construction - Shaft collars will be tested to ensure they can withstand forces greater than the expected loading\	- Payload lead is responsible for ensuring that proper strength testing has been conducted on the LSM bulk plate - FEA analysis shows a factor of safety greater than 3 for the LSM bulk plate	2E
P	Improper Handling of Electronics	- Wet conditions introduce moisture to electronics - Excessive or unpredicted launch or landing forces	- Short circuiting or breakage of electronics - Failure to detach parachute, collect data, extend antennas, and transmit data	1C	- Electronics will be prepared under a tent to protect from rain/weather - Launch will not be held in excessively rainy conditions - Electronics will be extensively ruggedized and tested in field conditions - Battery will be tested to ensure sufficient power is provided	- Electronics preparation has been practised in the lab in order to streamline the field preparation - Electronics have been ruggedized a sufficient amount to withstand drop testing - Electronics will continue to be ruggedized to ensure they will withstand the launch and deployment forces experienced during full scale flight	1E
P	Software failure	- Improper setting of altitude thresholds - Logic error in code - Insufficient testing of software under operating conditions - Improper integration of sensors into central logic	- Failure to detect landing - Failure to collect data - Failure to extend antennas and/or transmit data	1C	- Software will be thoroughly tested in the lab and in the field while integrated with the electronics system - Full sensor suite will be integrated and tested prior to launch - Software logic will be checked by multiple team members	- Software has been tested in both lab and field settings and has performed nominally - Altitude setting procedure has been practised in the field multiple times with success - Remaining sensors will be integrated into software for full scale flights	1E
V	CATO	- Improper spacing or preparation of motor grains - Use of cracked or otherwise damaged grains - Introduction of foreign material into motor chamber - Improper fastening of nozzle - Improper fastening of end cap - Improper insertion of ignition charge	- Significant damage to rocket and possibly payload - Possible complete destruction of rocket and possibly payload - Likely removal of ability to relaunch - Likely complete mission failure	1C	- Motor will be prepared exactly as shown in the instruction booklet - Motor assembly will be done by the rocketry mentor and overseen by the SSO - Grains will be carefully checked for cracks or other damage prior to assembly - Nozzle and end cap will be securely fastened - Motor casing will be checked for damage prior to each launch	- SSO will review motor assembly instructions prior to launch day - SSO will provide rocketry mentor with motor components and will remain with him until motor is fully assembled; this ensures that oversight is provided at each step - Grain inspection will be conducted by the SSO and the rocketry mentor	1E

V	Improper stability margin	- Location of CP is not calculated correctly - Location of CG is too close to or far from CP	- Unstable flight - Possible destruction of rocket	1C	- CP will be carefully calculated, checked by multiple team members, and approved by the rocketry mentor prior to flight - CG will be carefully measured - Stability margin of approximately 2.2 will be achieved through sizing of fins and manipulation of CG	- Rocketry Mentor will check and approve rocket CP and CG prior to flight - CFD and flight simulations have been conducted and indicate stable flight of the launch vehicle	2E
V	Fin flutter	- Fins are poorly constructed - Improper sizing of fins - Poor choice of fin material	- Breakage of fins - Potential crashing of rocket	1C	- Fins will be constructed from carbon fiber and will be firmly attached to airframe - Fin flutter will be calculated to ensure that critical values are not met	- Fins performed nominally during subscale flight. This lends confidence in the continuation of carbon fiber as the fin material for the full scale vehicle	2D
V	Airframe failure	- Airframe material too weak - Bulk plate bolts placed too close to edge of airframe - Greater launch or separation forces than expected - Blast charges blow apart airframe	- Loss of rocket and/or payload - Possible mission failure	1D	- Airframe will be constructed from fiberglass - Bolts will be at least 1.5 to 2 bolt diameters in distance from the edge of the airframe to minimize risk of tearout - Load testing will be performed on airframe and bulk plates to ensure that it can withstand all flight forces with a substantial safety factor - Blast charges will be sized based on both calculations and deployment testing	- The team has conducted load frame testing and verified the safety factor for airframe hoop stress, bolt tear out stress, and axial compressive stress are greater than 2.5 - Blast charges have been successfully sized, tested, and implemented for subscale. The same procedure will be followed for full scale preparation	2E
V	Deployment failure	- Blast charges too small to separate sections - Parachute bay not sufficiently sealed from rest of rocket - Excessive friction between overlapping airframe pieces - Altimeter fails to register apogee	- Rocket sections do not separate - Drogue and/or main parachute does not deploy - Probable destruction of rocket and payload - Probable mission failure	1C	- Rocket sections will be sized so that they slide easily into each other - Blast charges will be sized based on both calculations and deployment tests which will verify that sufficient black powder is being used - Openings out of the parachute bay will be blocked with putty prior to each launch - Altimeters will be calibrated prior to launch and will be tested in the subscale and full scale flights	- Blast charges have been successfully sized, tested, and implemented for subscale. The same procedure will be followed for full scale preparation - Putty was successfully used to seal off parachute compartment for subscale and will continue to be used for future flights - Altimeters were successfully calibrated for subscale. The same process will be used for the full scale flight - Rocket construction was successfully completed for subscale. Similar manufacturing techniques will be applied to full scale production	1E
V	Sticking on launch rail	- Buttons not in line with each other - Rocket fins in contact with launch rail - Excessive friction between rail buttons and rail - Ice buildup on launch rail or rocket	- Rocket does not reach target apogee - Rocket exits launch rail at angle that is not conducive to safe flight	2C	- Launch rail will be greased with WD-40 prior to each launch - Rail buttons will be carefully placed and will be sized to fit launch rail - Rocket will be dry fitted to rail before launch day to ensure good fit - Launch will be postponed in excessively icy conditions	- WD-40 was successfully used during subscale flight - Rail buttons were successfully placed for subscale flight - Rocket behaved as expected on the launch rail. These methodologies will be applied to full scale development and launch	2E
T	Planning failures	- Project timelines not created - Time not set aside for payload testing - Team unaware of goals and milestones - Manufacturing gets behind schedule - Poor communication between team and mentors	- Necessary testing is not completed - Project gets behind schedule - Manufacturing is poor or incomplete - Potential mission failure	2B	- Team will hold meetings with team mentor twice per week - Team will meet nightly for short update meetings - Timelines and goals will be discussed in depth by team and will be adhered to	- Team meets twice weekly with the team mentor - Team has consistently held nightly meetings to update on progress and organize - Timelines have been kept thus far and goals have been clearly communicated - These practices will be continued for the duration of the project	2D
T	Report writing failures	- Poor team participation - Lack of communication on section responsibilities and deadlines - Poor proofreading of sections - Lack of full project knowledge from team members - Not leaving enough time to write	- Poor scores on report sections - Significant negative impact to team standing in competition	2B	- Full team will be involved in report writing process - Team will develop an in depth document where section responsibilities, proofreading responsibilities, and report completion will be tracked - All sections will be thoroughly proofread by multiple team members - Report will be started at least one week in advance of due date - Testing and manufacturing will be well documented, which will help strengthen reports	- The team has successfully produced two reports thus far - Report writing has been streamlined for CDR to ensure efficient and thorough completion - PDR feedback was taken into account to guide adjustments to writing styles and practices	2D
P, V	Loss of power	- Battery does not supply enough power - Battery dies before or during launch - Launch forces cause battery connection to separate - Electrical components disconnect from battery due to launch or landing forces	- No or incomplete collection of data - No or incomplete transmission of data - Failure to disconnect parachute - Failure to trigger blast charges - Failure to complete mission	2B	- Careful power calculations and extensive testing will be done on the chosen batteries - Batteries will be fully charge prior to launch and spares will be brought to the field - All electrical connections will be firmly fastened and thoroughly ruggedized	- Power calculations have been done - Spare batteries have been brought to each field test and launch and will continue to be brought - Electrical connection issues have been identified and fixed - Batteries have proven to be of sufficient power during testing thus far	1E

P, V	Parachute fails during deployment	- Damaged parachute is used - Parachute strings or material is unable to withstand shock loading experienced - Fire blanket is improperly wrapped, leading to burning of chute - Blast charges are too large, leading to excessive deployment forces - Sections of the rocket or payload strike or tangle with parachute during deployment	- Rapid descent of rocket and/or payload - Complete or partial destruction of rocket and/or payload - Mission failure	1C	- Parachutes will be carefully inspected for damage prior to each launch - Deployment loadings will be carefully calculated and parachutes will be chosen to withstand these loads with an added safety factor - Fire blanket will be wrapped fully around blast-deployed chutes - Blast charges will be carefully prepared and sized based on both calculation and deployment testing	- Vehicle and payload leads are responsible for checking their respective parachutes prior to launch or drop - SSO and Rocketry Mentor will oversee packing of vehicle parachutes, including the wrapping of the fire blanket, the attachment of the shock cord, the placement of the fire ball, and the addition of fire retardant	1E
P, V	Shock cord fails during deployment	- Damaged shock cord is used - Loading is higher than predicted - Shock cords do not have enough safety margin	- Complete or partial destruction of rocket and/or payload - Mission failure	1C	- Shock cords will be visually inspected prior to each launch or drop test - Loading will be carefully calculated and verified through testing - Shock cords will be selected with a safety margin of at least four	- Rocketry Mentor must approve shock cords prior to launch - Vehicle and payload leads will check shock cords both in the lab and in the field before launch - Spare shock cord will be brought if available	1E
P, V	Shock cord bulk plate failure	- Excessive loading during parachute deployment - Poor material choice for bulk plate - Insufficient bulk plate thickness	- Partial or complete destruction of payload - Probable mission failure	1C	- FEA analysis will be conducted on plates - Extensive testing of bulk plates and deployment events will be performed - Bulk plates will be sufficiently strong with an added safety factor	- FEA analysis has been conducted on all shock cord bulk plates. Safety factors greater than 2.5 were observed on each, which meets the VADL minimum for recovery components - Completed testing of bulk plates indicates that airframe should fail before bulk plates do	1E
P, V	Landing in unintended areas	- Excessively high launch day winds - Failure to correctly choose launch angle - Improper SSM - Poor choice of launch site	- Vehicle and payload potentially sustain water damage - Vehicle and payload are potentially struck by motor vehicle - Vehicle and payload are potentially stuck in trees	2C	- Launch site will meet NAR guidelines and will be well clear of potential hazards such as water, roadways, and trees - Launch site must be approved by the rocketry mentor prior to launch - Launch angle and rail setup approved by rocketry mentor prior to launch - Launch will not be carried out if winds exceed 20 mph	- Anemometer was used to check wind speeds before launch and will continue to be used - Relevant approvals from Chris Dondanville are and will be received prior to launch - Launches are carried out only at NAR approved sites	3D

5.5 Environmental Hazard Analysis

The following matrix details potential environmental hazards and their associated risks. Higher risk levels on this table indicate more extreme or far reaching impacts to the environment. This matrix follows the same type of designations as the Personnel Hazard Analysis matrix and has some overlap in hazard causes. However, the result of hazard, risk levels, and mitigation strategies differ significantly. The goal of mitigation strategies in this table is to reduce the risk to the environment as much as possible. In the event that mitigation strategies for environmental factors would result in unsafe conditions for personnel, personnel safety will always be prioritized. However, no facet of the project will be undertaken if its personnel, failure, or environmental risk level exceeds the VADL defined upper limit of moderate.

Environmental Hazard Analysis							
Designation	Hazard Description	Causes of Hazard	Hazard Effect	Risk Level	Mitigation Strategy	Verification of Mitigation	Post-Mitigation Risk Level
M	Fire in lab	- Exposed electrical wiring - Overheating of power tools - Hot box left on - Heat gun or soldering iron left on - Exposure of energetic material to heat	- Expulsion of toxic smoke into atmosphere - Spillage of chemicals - Significant physical damage to area surrounding Olin Hall	1C	- Hot box, heat gun, and soldering irons will always be turned off and unplugged before leaving - Extension cords and other wires will be regularly checked to ensure proper insulation - Energetic material will be kept in MAGloc box whenever not in use; heat sources will be kept away from these boxes and away from energetics during usage - Power tools will be allowed to cool after use prior to storage - Fire extinguishers are easily accessible and regularly inspected	- Operating procedures for hot items have been successfully followed so far and will continue to be adhered to - Rais has regularly checked the extension cords and will continue to do so - Energetic material is locked at all times. SSO oversees any handling of energetic material - Power tools continue to be discussed in safety briefings when relevant	2E
M	Explosions in lab	- Pressure cooker left on or allowed to exceed safe operating pressure - Black powder or fuel grains exposed to heat - Black powder or fuel grains stored outside of proper containers	- Damage to building and natural areas immediately surrounding it in extreme cases	3C	- Pressure cooker will be kept under operating limit by at least a safety factor of 1.5 - Energetics will be stored in MAGloc boxes when not in use and will be kept clear of heat sources when being used - MAGloc boxes will be kept locked at all times	- Vehicle lead oversees any operation of pressure cooker - MAGloc boxes are locked at all times. SSO oversees any handling of energetic material	F
L (S)	Fires/explosions during launch or launch preparation	- Blast charge or motor exposed to heat source during preparation - Multimeter used to check electrical connections on prepared altimeter/blast charge system - Improper wiring or coding of avionics leading to premature ignition of blast charge - Motor ignition charge exposed to electricity or heat during installation - Motor ignition charge installed before rocket placed on launch rail	- Scorched ground or destroyed plantlife surrounding burn/blast area - Cratering in the event of a blast	2C	- Blast charges will be prepared in a clear area away from heat or electrical sources - Avionics will be thoroughly tested and calibrated prior to wiring of blast charges - Multimeters will not be used to check electronic connections of energetics at any point - Motor ignition charge will not be installed until rocket is on rail and properly oriented and will only be installed by the rocketry mentor, Chris Dondanville	- SSO will oversee avionics preparation and motor assembly - Avionics will be assembled by the same team member each time, who has been specially trained on safe avionics assembly - Chris Dondanville will be the only person who handles the assembled motor - SSO will ensure area is clear prior to the placement of the motor ignition charge by Chris Dondanville - The avionics checklist, launch procedure, and safety checklist will be carefully followed on launch day	3E
L (S)	Rocket impact with landscape, plants, or animals	- Launch rail angled incorrectly with respect to wind - Strong wind gusts or gusts of changing direction - Failed parachute deployment - Improper rocket assembly leading to falling debris or rocket sections - Improper sizing of rocket fins	- Damage to trees or other plant life near launch site - Damage to landscape around launch site, primarily through impact craters - Pollution of land immediately surrounding launch site in the event that debris is scattered to the point that team is unable to fully clean it	3C	- Launch rail angle will be checked and approved by the SSO and rocketry mentor prior to launch; launch rail will be properly secured - Launch will not proceed in winds higher than 20 mph, or in unpredictably gusty scenarios - Rocket fins will be sized to provide an appropriate stability margin of approximately 2.2 - Rocket will be carefully assembled following a strict procedure - Blast charges will be carefully prepared and sized according to data from both calculations and deployment tests	- Chris Dondanville approves the launch rail assembly and angle prior to each launch - Wind speed is checked with an anemometer prior to each launch to ensure compliance - Rocket construction and SSM was adequate for subscale launch. Similar construction and design procedures will be followed for full scale - Blast charges have been successfully prepared, tested, and implemented. The same procedure will be followed for the full scale blast charges	3E
L (S)	CATO	- Improper spacing or preparation of motor grains - Use of cracked or otherwise damaged grains - Introduction of foreign material into motor chamber - Improper fastening of nozzle - Improper fastening of end cap - Improper insertion of ignition charge	- Scorching of ground surrounding launch rail - Start of fire near launch rail - Release of toxic fumes from burning plastic components - Spreading of debris about launch site - Potential cratering immediately beneath rocket	2C	- Motor will be prepared exactly as shown in the instruction booklet - Motor assembly will be done by the rocketry mentor and overseen by the SSO - Grains will be carefully checked for cracks or other damage prior to assembly - Nozzle and end cap will be securely fastened - Motor casing will be checked for damage prior to each launch - Metal blast deflector will be placed beneath motor exhaust prior to each launch	- SSO will review motor assembly instructions prior to launch day - SSO will provide rocketry mentor with motor components and will remain with him until motor is fully assembled; this ensures that oversight is provided at each step - Grain inspection will be conducted by the SSO and the rocketry mentor	3D
DT	Loss of control of hexacopter drone	- High winds render drone control unreliable - Incompetence of drone pilot - Drone runs out of battery during flight	- Harm or death of wildlife - Damage to plant life or test field - Puncturing of LiPO (discussed in detail below)	2C	- Hexacopter drone will ONLY be flown by trained and designated drone pilot - Drone will not be flown if winds are too high - Drone batteries will be checked prior to each flight to ensure sufficient charge - Area will be visually scanned for wildlife prior to each flight	- SSO will consult drone pilot prior to each flight to ensure wind speeds are manageable and batteries are charged - Drone pilot has flown the hexacopter drone extensively and is very capable of controlling the drone	2E

LT	- Debris or falling components during small scale drop tests - Debris or falling components during parking garage drop testing	- Animal life too near to drop point (GARAGE DROPS) - Improper sizing of shock cord - Improper attachment of shock cord to payload or to fixed point - Poor construction of payload leading to breakage upon shock cord extension - Improper length of shock cord used leading to higher than expected forces	- Damage to landscape from payload impact - Harm to animals from impact of whole or parts of payload	3D	- Payload will be thoroughly tested prior to field drop test - Drop area will be visually checked for animal presence prior to launch	- SSO must be present to oversee garage drops - At least two members are required to be present for small scale drops - At least two team members are required at the drop point and an additional minimum of two members must be on the ground for garage testing. This ensures that eyes are on the payload at all times and no steps are overlooked - Prior to any drop testing, a procedure will be written out, walked through, and then carefully followed for testing	F
M, L (S), DT, LT	Improper disposal of waste	- No trash bags available in field - Windy conditions blow trash away from prep area - Negligence from team members - Chemicals or hazardous material not disposed of through proper channels	- Potential harm or death to wildlife - Pollution of land or bodies of water near test sites or near garbage dump - Potential contamination of drinking water supply	2B	- Trash bags will be on hand for every launch or field test - Team members will be reminded about trash disposal prior to each field test as a part of the safety briefing - Any team members handling chemicals or hazardous materials will refer to the appropriate MSDS for disposal instructions	- Trash has been properly discarded at every field test thus far and will continue to be taken care of - MSDS have been successfully followed so far. They will continue to be consulted when chemicals are being handled - Safety briefings have covered relevant trash and chemical handling and will continue to do so	F
L (S), DT	LiPO battery fire	- Battery overheats from temperature or sun exposure - Battery is punctured during flight or landing - Drone battery is punctured during landing or in loss of control event	- Fire starts prior to take off or after landing - Plants and animals damaged or destroyed as a result - Spillage of harmful chemicals or fumes	2C	- Battery will be checked for bloating or puncture before and after each flight - Sun exposure will be limited as much as possible - Launch/drop will only be conducted when area is free from animal life - Launch/drop site will be chosen in area as free from plant life as possible	- SSO will inspect all batteries used after each launch or drop test - Payload and vehicle leads are responsible for ensuring their respective LiPO batteries are transported in the proper LiPO container - Bloating or otherwise damaged LiPOs will be discarded immediately as specified in the MSDS	2E
L, DT	Payload failure	- Payload unable to withstand parachute deployment forces	- Pollution of launch site from falling debris - Potential impact with animals	3C	- Payload will be thoroughly tested prior to field drop test - Launch or drop area will be visually checked for animal presence prior to launch	- Payload lead is responsible for ensuring that payload is well constructed - Payload lead is responsible for ensuring that necessary strength testing has been completed - SSO and team president are responsible for scanning the drop or launch area for wildlife prior to each launch or drop	4E
L, DT	Payload impact with landscape, plants, or animals	- Parachute packed incorrectly leading to rapid descent of payload - Quick link not secured properly - Drop area not checked for wildlife before drop - Excessive or unpredictable wind	- Injury or death of animals - Damage to trees or other plants - Damage to field in which drop is conducted	3C	- Parachute packing will be done carefully and overseen by at least two team members - Quick link will be fully tightened and checked for attachment prior to payload flight - Area will be visually checked for wildlife prior to each drop or launch	- Payload lead will be in charge of parachute packing, with oversight from one of several team members who are familiar with the parachute packing procedure - Dry runs of parachute packing will be conducted before each launch - Payload lead is responsible for checking quick links for damage prior to each flight; checks must be approved by SSO before flight - Wind speed will be measured with an anemometer prior to drop or launch - SSO and team president are individually responsible for scanning the area prior to launch or drop	3E

5.6 Launch Concerns and Operations Procedures

It is imperative to have detailed documentation of procedures and safety considerations on launch day. As such, the Student Safety Officer has compiled checklists covering the following categories: avionics, payload, recovery, motor, vehicle, launch rail, launch, post-flight, and general safety considerations. Each checklist details the required materials, personnel, and assembly procedure. Further, the responsible personnel must sign off on each sheet pending the procedure's completion. The following sheets show the checklists used for the subscale flight, which will be adapted for the full-scale flights.



VADL Subscale Flight

12/7/2024

1

Avionics Checklist

Required Materials

- ☐ Avionics assembly
 - ☐ Avionics housing
 - ☐ Entacore AIM altimeter (x2)
 - ☐ Screw terminal (x2)
 - ☐ 9v battery (x3)
- ☐ Avionics bulk plate
- ☐ Maglok
 - ☐ Main blast charge (x3)
 - ☐ Secondary blast charge (x3)
- ☐ Blast putty
- ☐ Blast caps
- ☐ Safety glasses
- ☐ Avionics Box
- ☐ Hardware
 - ☐ M3x20 screws x4
 - ☐ M3x16 screws x4
 - ☐ M3 nuts x8
 - ☐ 5/16-18 eyebolt
 - ☐ 1/4-20 threaded rods x2
 - ☐ 1/4-20 washers+nuts x4
 - ☐ 10-32 screws x4
 - ☐ 10-32 nuts x4
 - ☐ Zip Ties
- ☐ Tools
 - ☐ M3 hex key
 - ☐ 1/4-20 nut driver
 - ☐ 10-32 screw driver
 - ☐ Altimeter screwdriver
 - ☐ Screw terminal screwdriver

Required Personnel

Name	Role
Santino Clemente	Avionics Engineer
Ian Morgan	President
Zachary Stutsman	Student Safety Officer
Chris Dondanville	Rocketry Mentor

Pre-Launch Day Procedure

PPE: Face shield and gloves.

- ☐ Verify supply of charged 9V batteries, Entacore AIM altimeters, and screw terminals.
- ☐ Wire altimeters to screw terminals and batteries.

1 of 4



VADL Subscale Flight 12/7/2024

1

Avionics Checklist

- ☐ Configure altimeters for deployment at apogee and apogee + 1s for primary and backup, respectively.
- ☐ Verify altimeter configuration through pressure testing.
- ☐ Assemble 9v batteries into avionics housing with zip ties.

WARNING: Neglecting to complete previous steps may cause failed deployment and destruction of rocket and payload.

- ☐ Assemble altimeters and screw terminals into avionics housing with M3 screws and nuts.
- ☐ Measure and store calculated blast charge amounts for primary and backup deployment.

WARNING: Failure to store blast charge quantities will result in failed deployment.

Launch Day Procedure

PPE: Face shield and gloves.

- ☐ Arm altimeters and verify deployment settings with rocketry mentor.
- ☐ Disarm altimeters.
- ☐ Place blast charges into blast caps.

WARNING: Ensure static discharge by touching ground wire, else risk detonation of blast charge.

- ☐ Wire charges to altimeter.

WARNING: Ensure the altimeter is not powered and is not facing into wind, as doing so may cause inadvertent apogee detection.

- ☐ Verify connections of primary and backup wires to their respective blast charges.
- ☐ Assemble avionics bulk plate and housing with 1/4-20 rods and nuts.
- ☐ Screw 5/16-18 eye bolt into bulk plate.
- ☐ Place blast putty over remaining openings in the bulk plate.

Inter-Launch Procedure

PPE: Face shield and gloves.

- ☐ Disarm altimeters.

WARNING: Failure to complete this step risks inadvertent apogee detection.

- ☐ Verify deployment of both blast charges.

2 of 4



VADL Subscale Flight 12/7/2024

1

Avionics Checklist

- ☐ Unscrew avionics bulk plate and remove assembly from rocket.
- ☐ Repeat Launch Day Procedure above

Post-Launch Procedure

- ☐ Repeat steps 1-3 of Inter-Launch Procedure
- ☐ Safely store all avionics equipment

WARNING: Failure to complete this step may result in damage or destruction of avionics equipment.

3 of 4



VADL Subscale Flight
12/7/2024

1

Avionics Checklist

Signatures

Name	Signature
Santino Clemente	_____
Ian Morgan	_____
Zachary Stutsman	_____
Chris Dondanville	_____

4 of 4



VADL Subscale Flight

12/07/2024

2

Payload Checklist

Required Materials

- ☐ Ground Station Electronics
 - ☐ 2x 2s and 3s Lipo
 - ☐ 4x charged Lithium Ion batteries
 - ☐ Computer + Charger
 - ☐ USB-C adapter
 - ☐ SD Card reader
 - ☐ USB-MiniUSB connector
- ☐ Assembled Electronics Housing
 - ☐ SD card
 - ☐ VectorNav IMU
 - ☐ Battery (7.4 LiPO)
 - ☐ 3.3V regulator
 - ☐ 5V regulator
 - ☐ Light APRS board
 - ☐ Coax cable with male SO239 connector
 - ☐ 3 way servo cable splitter
- ☐ AEM Assembly
 - ☐ 18" tape spring x3
 - ☐ M3x10 bolts x3
 - ☐ Jumper wires (w/ crimped ring connectors) x3
 - ☐ Female SO239 connector with crimped and soldered ring jumper
 - ☐ AEM bulk plates x3
 - ☐ 3x 9g servo
- ☐ Power
 - ☐ Jackery battery (wall voltage supply)
 - ☐ LiPo charger (including wall connector/adaptor and lipo cable)
- ☐ Hardware
 - ☐ Bulk plate aft and fore
 - ☐ 5/16-18 threaded rods x2
 - ☐ 5/16-18 carriage bolts x2
 - ☐ 5/16-18 nuts and washers x25
 - ☐ 5/16-18 eyebolt
 - ☐ 10-32 radial bolts x4 (aft bulkplate)
 - ☐ 6-32 radial bolts + washers x3 (fore bulkplate)
 - ☐ M2x16 screws x4 (PCB)
 - ☐ M3x14 screws + washers x5 (IMU + nosecone)
 - ☐ M3 nuts x2 (IMU)
 - ☐ Electrical tape
 - ☐ Multimeter
 - ☐ Zip ties
- ☐ Tools
 - ☐ 3mm phillips head screwdriver (Teensy/SD Mount)
 - ☐ 1.8mm allen key (Vector Nav)
 - ☐ 1.4mm allen key (Pi Mount)
 - ☐ 12mm wrench
 - ☐ 1/2 in wrench (5/16 - 18 nuts)
 - ☐ Driver set

1 of 4



VADL Subscale Flight

12/07/2024

2

Payload Checklist

Required Personnel

Name	Role
Rais Nurhidajat	Vice President
Zachary Stutsman	Payload Engineer
Patrick Chen	Electronics Engineer

Pre-Launch Day Procedure

- ☐ Validate LiPo Battery and backup are charged to 12.65 V
- ☐ Assemble Vector Nav, Light APRS board, Battery, and PCB into electronics housing
- ☐ Individually test AEM deployment
- ☐ Stake connectors (VN, Latches), verify software and electromechanical actuation
- ☐ Change testing parameters in code to flight values
 - ☐ Launch Detection threshold
 - ☐ Timing for apogee + landing
 - ☐ Buffer size
- ☐ Unplug all electronics until morning
- ☐ Attach electronics housing to fore bulk plate
- ☐ Attach payload to electronics to fore and aft bulk plate
- ☐ Verify all holes (radial and AEM slots) line up properly
 - ☐ Mark out nut placement with marker and tape
- ☐ Verify electrical connection between APRS board and antennas

Launch Day Procedure

- ☐ Verify all battery charges with multimeter

WARNING: Uncharged batteries will result in payload failure.
- ☐ Dry run of software initialization for launch
 - ☐ Verify threshold parameters
 - ☐ Verify IMU connectivity
 - ☐ Verify servo actuation using spare servo
 - ☐ Power off electronics

WARNING: Software malfunction will ensure data corruption and payload failure.

2 of 4



VADL Subscale Flight

12/07/2024

2

Payload Checklist

- ☐ Stake remaining connectors and wires
 - ☐ Latch connectors on board
 - ☐ SD card
 - ☐ Rx antennas
 - ☐ Zip tie long latch wires for stress removal
 - ☐ Proceed with mechanical connections + relatching
 - ☐ Confirm vector nav data stream with code
 - ☐ Attach housing to skeleton and bulk plate
 - ☐ Attach skeleton to payload bay and nose cone. Tighten top radial bolts
 - ☐ Tighten lower radial bolts
 - ☐ **Re-connect battery and power on electronics when final assembly is ready**
 - ☐ Pull antennas into slots
 - ☐ Connect Recovery Components
 - ☐ Connect to main vehicle body
- WARNING: Failure to complete previous steps will result in payload section separation and destruction

Re-Launch Procedure

- ☐ Remove recovery components
 - ☐ Push antennas back in/ spin skeleton to fold them down
 - ☐ Remove skeleton from payload bay
 - ☐ Download telemetry via usb or wifi
 - ☐ Power off electronics
 - ☐ Verify all fasteners and structure
 - ☐ Change battery and verify charge using multimeter
 - ☐ **Power on electronics when ready for re-launch**
 - ☐ Attach skeleton to fore and aft bulk plates
 - ☐ Pull antennas into slots
 - ☐ Connect recovery components
 - ☐ Connect to main vehicle body
- WARNING: Failure to complete previous steps will result in payload performance failure and destruction

3 of 4



VADL Subscale Flight
12/07/2024

2

Payload Checklist

Name

Signature

Rais Nurhidajat

Zachary Stutsman

Patrick Chen

4 of 4



VADL Subscale Flight

12/07/2024

3

Recovery Checklist

Required Materials

- | | |
|---|--|
| ☐ 48" Iris Ultra parachute | ☐ Hardware |
| ☐ Fore shock cord (10 ft) | ☐ 2-56 shear pins x4 |
| ☐ Aft shock cord (20 ft) | ☐ 1200 lbs quick links x3 |
| ☐ Fireballs x2 | ☐ Swivel link |
| ☐ Dog Barf | ☐ Zip Ties |
| ☐ Fire Blanket | ☐ Painters tape |
| ☐ Baby powder | ☐ Box cutter knife |

Required Personnel

Name	Role
Ian Morgan	Lead Recovery Engineer
Zachary Stutsman	SSO
Chris Dondanville	Rocketry Mentor

Pre-Launch Day Procedure

- ☐ Examine the parachute, shock cords, fireballs, fire blanket, quick links, and swivel link for any damage.
- ☐ Z-fold the shock cords and secure them with a single layer of painter's tape, ensuring not to crease or damage the tape.
- ☐ Practice assembling and storing recovery equipment in the recovery bay of the vehicle, ensuring sufficient packing volume.

Launch Day Procedure

1 of 3



VADL Subscale Flight 12/07/2024

3

Recovery Checklist

- ☐ Lay out and examine all recovery equipment for any damage.
- ☐ Attach quick links to outer ends of the shock cords.
- ☐ Proceed with parachute packing procedure:
 - ☐ Untangle shroud lines and separate into respective three groups
 - ☐ Unfurl parachute and lay out in its natural form, with center tucked in slightly to conform with toroidal shape.
 - ☐ Beginning with the labeled gore, fold each gore on the left side of the parachute toward the center.
 - ☐ Once complete, place weights atop folded half and repeat the process for the right side.
 - ☐ Fold halves lengthwise.
 - ☐ Lay shroud lines atop folded parachute in a snaking pattern.
 - ☐ Z fold the parachute from top to bottom and roll closed.
- ☐ Attach inner loops of the shock cords to the parachute quick link.
- ☐ Attach the fire blanket to the parachute quick link.
- ☐ Wrap the fire blanket around the parachute and tape closed until final assembly.
- ☐ Score the taped shock cords.
- ☐ Attach the 10 ft fore shock cord to the upper bulk plate.
- ☐ Attach the 20 ft aft shock cord to the avionics bulk plate.
- ☐ Coat the fire blanket with baby powder.
 - WARNING: Failure to complete previous steps will result in parachute deployment malfunction and subsequent destruction of vehicle and payload.**
- ☐ Add dog barf to the recovery bay.
- ☐ **Remove red tape from the fire blanket.**
- ☐ Load shock cords, parachute, and fireballs in the recovery bay, ensuring that the parachute quick link is facing away from the blast (toward the fore section).
 - WARNING: quick link placed towards aft section may cause parachute tangling during deployment and prevent full unfurling.**
- ☐ Adjust fireball location on shock cords such that they reside within the airframe and zip tie in place.
- ☐ Assemble vehicle sections together and fasten with shear pins.

2 of 3



VADL Subscale Flight
12/07/2024

3

Recovery Checklist

Signatures

Name	Signature
Ian Morgan	_____
Zachary Stutsman	_____
Chris Dondanville	_____

3 of 3



Motor Checklist

Required Materials

- | | |
|---|--|
| ☐ Aerotech I366R reload kit | ☐ Ignition system |
| ☐ Liner tube propellant grains | ☐ Lithium Grease |
| ☐ Primary O-rings | ☐ RMS-38/600 Hardware |
| ☐ Red Cap with black powder | ☐ Motor Casing |
| ☐ Smoke Grain | ☐ Forward and Aft Closure |
| ☐ PPE | ☐ Spacers |
| ☐ Latex gloves | ☐ Delay/eject module |
| ☐ Safety glasses/face shield | ☐ Igniter x3 |
| | ☐ Backup Retaining Ring Cap |

Required Personnel

Name	Role
Ian Morgan	Vehicle Engineer
Zachary Stutsman	SSO
Chris Dondanville	Rocketry Mentor

Pre-Launch Day Procedure

- ☐ Inspect motor hardware and reload kit for defects.
WARNING: Defects in motor hardware will cause rocket flight irregularities and vehicle damage.
- ☐ Store reload kit in MAGLock for transport.

Launch Day Procedure

Reference attached RMS-38/600 assembly instructions for procedure. Motor assembly should be overseen by the Rocketry Mentor, the SSO, and one other team member to



VADL Subscale Flight
12/07/2024

4

Motor Checklist

ensure that all procedures are carefully followed. These three members are designated above, and their approval of motor assembly quality is located below.

Signatures

Name	Signature
Ian Morgan	
Zachary Stutsman	
Chris Dondanville	

2 of 2



General Vehicle Checklist

Required Materials

- | | |
|--|--|
| ☐ Nose cone | ☐ Hardware |
| ☐ Payload section | ☐ Pro-38 |
| ☐ Tail section | ☐ 4x M3 bolts |
| ☐ Recovery System | ☐ 6x 10-32 bolts |
| ☐ Aerotech I366R motor (x3) | ☐ 6x 6-32 bolts |
| ☐ RMS-38/600 hardware | ☐ Tools |
| ☐ Hanging Scale | ☐ Wrench |
| ☐ Silver and black sharpie | ☐ Screw driver |
| ☐ Boat tail | ☐ Mallet |
| | ☐ Gloves |
| | ☐ Allen Key 10-32 |
| | ☐ Allen Key 6-32 |
| | ☐ Allen Key M3 |

Required Personnel

Name	Role
Ian Morgan	President
Zachary Stutsman	SSO
Saksham Sharma	Vehicle Engineer
Rais Nurhidajat	Payload Engineer

Pre-Launch Day Procedure

- ☐ Fully assemble vehicle and payload as closely to launch day conditions as possible.
- ☐ Find the dry mass of the vehicle via hanging scale.
- ☐ Find and mark CG of vehicle without ballast.
- ☐ Determine optimal ballast and place ballast at appropriate location.
- ☐ Find and mark CG of vehicle with ballast.

PPE: Facial Protection, gloves



General Vehicle Checklist

- ☐ Ensure rail button alignment by mounting to launch rail.

Dry mass (ballasted)	CG	CP	Dry SSM

Wet mass (ballasted)	CG	CP	Wet SSM

Ensure SSM is between 2 and 3 for both wet and dry configurations.

Launch Day Procedure

- ☐ Attach boat tail to tail section via shear pins
- ☐ Secure avionics bulk plate to tail section via 10-32 bolts
- ☐ Ensure payload is fully assembled, including raised coupler
- ☐ Attach payload section and nose cone to raised coupler using M3 and 6-32 bolts
- ☐ Secure aft payload bulk plate to payload body tube via 10-32 bolts
- ☐ Secure shock cords to corresponding bulk plates via quick links
- ☐ Couple payload and tail sections
- ☐ Insert four 2-56 shear pins into parachute bay shear pin holes with small hammer.
- ☐ To be completed only when ready to move to launch rail.
 - ☐ Insert fully assembled motor into motor tube.
 - ☐ Find wet CG and confirm adequate SSM.

WARNING: Failure to complete any previous steps will result in improper vehicle separation and destruction of vehicle



VADL Subscale Flight
12/07/2024

5

General Vehicle Checklist

Signatures

Name	Signature
Ian Morgan	
Zachary Stutsman	
Saksham Sharma	
Rais Nurhidajat	

3 of 3



VADL Subscale Flight 12/07/2024

6

Launch Rail Checklist

Required Materials

- | | |
|---|--|
| ☐ Launch rail base | ☐ Tools |
| ☐ Launch rail extension | ☐ Wrench |
| ☐ WD-40 | ☐ Hex key |
| ☐ Stakes | ☐ Anemometer |
| ☐ Sand Bags | ☐ Level |
| ☐ Launch Rail Hardware Box (stakes, bolts) | ☐ Hex Wrench |
| ☐ Blast deflection plate | ☐ Crescent Wrench |
| | ☐ Mallet |
| | ☐ Angle Finder |

Required Personnel

Name	Role
Zachary Stutsman	Payload Engineer
Saksham Sharma	Vehicle Engineer
Benjamin Racape	Payload Engineer

Launch Day Procedure

- ☐ A level area must be chosen that is a minimum of 500 feet from any structures which could interfere with the flight.
Warning: Failure to follow risks rocket collision with obstacles, and injury to launch spectators.
- ☐ Unfold launch rail legs and turn the apparatus such that it is facing into the wind. Wind direction should be determined using the anemometer.
- ☐ Ensure launch rail base is level within 2 degrees using level.
Warning: Failure to follow risks improper rocket launch conditions and potentially unstable flight
- ☐ Secure legs using stakes or sand bags.
- ☐ Slide launch rail extension onto main launch rail, ensuring that all bolt heads are guided into the slot of the X-frame. Ensure that the rail surface is flush, then tighten all bolts to secure rail.

1 of 3



Launch Rail Checklist

Warning: Non-flush rail surface risks improper launch conditions, and damage to personnel and rocket equipment.

- ☐ Adjust the launch rail stop so that it is a maximum of 18 inches from the ground. Ensure that stop is tightened.
- ☐ Place blast plate on the ground beneath launch rail stop. Hold in place with a sandbag or with clamps attached to launch frame legs
- ☐ Apply WD-40 to launch rail.
- ☐ Inspect the assembled launch rail, double check tightness of all bolts, and ensure that rail is sufficiently lubricated with WD-40.
- ☐ Decide on launch rail angle based on wind conditions. Launch rail angle must be approved by the Rocketry Mentor, Chris Dondaville, the Student Safety Officer, Zachary Stutsman, and Lead Simulations Engineer, Sitan Huang. The launch rail angle is checked using the angle finder.

Warning: Improper angle and launch direction may cause windcocking and increase drift.

Signatures

Selected Rail Angle:	_____
Chris Dondanville	_____
Zachary Stutsman	_____
Sitan Huang	_____
Name	Signature
Zachary Stutsman	_____
Saksham Sharma	_____
Benjamin Racape	_____



VADL Subscale Flight 12/7/2024

7

Launch Procedure

Required Materials

- ☐ Fully assembled vehicle
- ☐ Ignition system
- ☐ Assembled launch rail

Required Personnel

Name	Role
Ian Morgan	President and Launcher
Zachary Stutsman	Student Safety Officer
Santino Clemente	Avionics Engineer
Chris Dondanville	Rocketry Mentor

Pre-Launch Day Procedure

- ☐ Launch requirements from **Subscale Safety Checklist** are to be committed to memory by the predesignated launcher and SSO.

Launch Day Procedure

- ☐ Confirm the vehicle is fully assembled and ready for launch.
WARNING: Improper vehicle assembly risks unstable flight and/or loss of vehicle.
 - ☐ Carry vehicle to launch rail location.
 - ☐ Lower launch rail as needed and slide on the vehicle via launch rail buttons.
 - ☐ Raise launch rail to predetermined launch rail angle.
WARNING: Incorrect launch rail angle risks wincocking and increased total drift.
 - ☐ Ensure a minimum ground clearance of 18 inches from the bottom of the vehicle.
- PPE: Facial covering, gloves
- ☐ Arm altimeters.

1 of 3



VADL Subscale Flight 12/7/2024

7

Launch Procedure

- ☐ Confirm altimeters are armed and configured by listening to series of beeps (two sets of three).
 - ☐ Confirm continuity between avionics and e-matches by listening to subsequent series of beeps (two sets of three).
 - ☐ Clear all non-essential personnel within a minimum radius of 100 ft from the launch rail.
 - ☐ Listen for continuous altimeter beeps to confirm launch has not been prematurely detected.
 - ☐ Perform launch countdown procedure:
 - ☐ Confirm all personnel are clear of the launch pad within a minimum radius of 100 ft.
 - ☐ Confirm airspace is clear.
 - ☐ Confirm all other launch minimums are met.
 - ☐ Launcher yells "CLEAR... 5... 4... 3... 2... 1...".
 - ☐ Launch vehicle.
- WARNING: Improper clearance of area and launch callout may injure personnel and risk mission success.
- ☐ In the case of failure to launch:
 - ☐ Attempt launch countdown procedure again as necessary.
 - ☐ Vehicle may not launch if any criteria are not met, per **Subscale Safety Checklist**.

Signatures

2 of 3



VADL Subscale Flight
12/7/2024

7

Launch Procedure

Name	Signature
Ian Morgan	
Zachary Stutsman	
Chris Dondanville	



VADL Subscale Flight 12/07/2024

8

Safety Checklist

Identification of Safety Team

Name	Role
Ian Morgan	President
Zack Stutsman	Student Safety Officer
Chris Dondanville	Rocketry Mentor

Launch Criteria

If not all criteria are met, launch must be delayed or cancelled.

Airspace

- ☐ Flight must comply with any and all applicable Federal Aviation Authority Regulations.
- ☐ Airspace must be clear of all traffic.
- ☐ The predicted apogee must be below the flight ceiling for the launch (2000 ft AGL).

WARNING: Unclear airspace will result in hazards to rocket and air traffic.

Environment

- ☐ No precipitation may be falling at the time of launch.
- ☐ Ground level wind must be below 20 mph.
- ☐ All personnel must be at a distance of a least 100 ft from the launch rail at the time of launch, per NAR regulations.

WARNING: Improper clearance of personnel beyond listed minimum safety distance may result in personnel injury and vehicle damage.



VADL Subscale Flight

12/07/2024

8

Safety Checklist

Payload

- ☐ Antenna Extension Mechanism (AEM) must be wound within the outer confines of the airframe prior to launch.
- ☐ No payload activity may be allowed to affect the safe flight and recovery of the rocket.

Vehicle Readiness

- ☐ The vehicle must be deemed fit to fly by rocketry mentor Chris Dondaville or another applicable, certified rocket mentor.
- ☐ The recovery system and motor must likewise be deemed suitable for the launch vehicle.
- ☐ Student Safety Officer and team president must also approve all subsystems and safety measures prior to launch

The safety team and/or rocket mentor may cancel or postpone the flight for any reason, regardless of its presence on the checklist above.

Delay Procedure

- ☐ If the launch criteria have not been met, but it is expected that they may be met within three hours (maximum launch rail time), the launch may be delayed until the criteria are met.
 - ☐ If the delay-causing-criteria has now been met, all criteria must be reevaluated before launch attempt.
 - ☐ If not, then refer to cancellation procedures.

WARNING: Failure to meet launch criteria may jeopardize mission success and launch spectator safety.

Cancellation Procedures

- ☐ Disconnect ignition
- ☐ Disarm avionics and other electronics
- ☐ Remove vehicle from the launch rail
- ☐ Remove motor

PPE: Facial covering, gloves



VADL Subscale Flight 12/07/2024

8

Safety Checklist

WARNING: Failure to follow previous steps may result in accidental discharge of ignition or activation of deployment mechanisms, damaging vehicle and personnel.

- ☐ Complete post flight procedures

Packing List

- ☐ Safety Binder
- ☐ Appropriate PPE (specific items listed in subsystem checklists)
- ☐ Fire extinguisher
- ☐ First aid kits (2-3)
- ☐ Water and food
- ☐ Appropriate clothing (cold weather and wind expected)

WARNING: Missing items jeopardizes team and vehicle safety.



VADL Subscale Flight
12/07/2024

8

Safety Checklist

Flight on:

By signing below, I acknowledge that the launch procedure was properly followed, and to the best of my knowledge, the vehicle is safe to fly.

Name

Signature

Ian Morgan

Zach Stutsman

Chris Dondanville

Notes for improvement on safety procedures:



VADL Subscale Flight 12/7/2024

9

Post-Flight Checklist

Required Personnel

Name	Role
Santino Clemente	Avionics Engineer
Ian Morgan	President
Zachary Stutsman	Student Safety Officer
Chris Dondanville	Rocketry Mentor

Procedure

PPE: Face shield, gloves, close-toed shoes.

- ☐ Disarm altimeters.

WARNING: Failure to complete this step risks inadvertent blast charge detonation.

- ☐ Verify deployment of both blast charges.

WARNING: Some or all blast charges may not have detonated. Therefore, no personnel can enter a 100 ft radius of the vehicle within 60 seconds of landing.

- ☐ If any blast charges have not detonated, remove carefully with proper PPE.

- ☐ Properly dispose of black powder charges

WARNING: ANY improper handling risks personnel and vehicle damage.

- ☐ Recover all sections of rocket from landing site and return to tent.
- ☐ Disconnect parachute from rocket.
- ☐ Unscrew avionics bulk plate and remove assembly from rocket.
- ☐ Inspect lander and vehicle for structural damage.
- ☐ Record apogee data from altimeters.
- ☐ Safely store all equipment for transportation.

WARNING: Failure to properly store equipment may result in structural or electronic damage.

1 of 2



VADL Subscale Flight
12/7/2024

9

Post-Flight Checklist

Signatures

Name	Signature
Santino Clemente	_____
Ian Morgan	_____
Zachary Stutsman	_____
Chris Dondanville	_____

6 Project Plan

6.1 Testing

VADL ensures that design and fabrication meet NASA and team imposed standards through the use of rigorous testing. This will ensure that the launch vehicle and payload are robust enough to withstand the rigors of rocket flight, recovery, and landing. It is not enough for the launch vehicle and payload to merely perform the required mission once, but it must be capable of re-flight within 2 hours of recovery. (NASA Requirement 2.5) Testing is a cyclical process in which requirements are derived and verified based on previous test results and design adjustments. Key tests - such as integrated lander function tests - shall be repeated to prove to the team that the design is ready for the USLI launch.

6.1.1 Required Testing

The team has defined measurable and strict testing goals to ensure that each test serves to verify NASA or the team-imposed requirements. Certain tests are only conducted as required because of their destructive nature. Such tests include airframe structural load tests, in which specimens are loaded to failure. Key tests are repeated even if marked complete to ensure that no aspect of launch is overlooked in the event that a material has failed in an undetectable way that jeopardizes mission success.

VADL has refined the testing tables originally presented at the PDR milestone. Tests are broadly grouped into vehicle and payload tests, with additional subcategories for further classification. Each test has a clear and specific objective, with an actionable plan for verification. The tested variable and pertinent requirement(s) are outlined, and a definitive success criteria is described. The status of each test is then listed; as of the CDR milestone, all tests are either complete or in progress. The vehicle and payload testing tables are provided on the following pages.

LAUNCH VEHICLE		Objective	Verification Plan	Variable Tested	Pertinent Technical Requirement(s)	Success Criteria	Status
Test Code	Verification Method						
1.1 Structural Testing							
1.1.1	Drop Test	Verify that the airframe can withstand any forces sustained during a landing impact	The team will drop the each rocket section from a height that simulates the terminal velocity of the rocket body after main parachute deployment; the team will also add masses to each rocket section to reach a representative landing energy	Strength of G-12 fiberglass	NASA 2.3 VADL 2.2.1.3, 2.2.1.4	The airframe does not sustain any damage in any of the simulated impacts	In-Progress
1.1.2	g-Load Test	Verify the structural integrity of the internal scaffolding of the payload section by itself without the integrated payload	The team will place the internal scaffolding in the g-load frame and subject it to accelerations expected during flight	Structural integrity of payload section	NASA 2.3 VADL 2.2.1.3	Visual and physical inspection determines that the structural integrity was not compromised and no component came loose or out of place	In-Progress
1.1.3	Load Test	Determine the modulus of elasticity of the quasi-isotropic carbon fiber and epoxy "Dragon Plate" material	The team will find the stress-strain curve of this material by suspending weights from the end of a piece of material held in a vice (a calculated bending stress) and applying strain gages to determine the strain for each given stress	Modulus of elasticity of the quasi-isotropic carbon fiber and epoxy material	NASA 2.3, VADL 2.2.2	Find the material properties of the quasi-isotropic carbon fiber and epoxy material material to determine whether this material is fit for use in vehicle fins	In-Progress
1.1.4	Load Test	Determine the ultimate tensile stress of the quasi-isotropic carbon fiber and epoxy "Dragon Plate" material	The team will place a piece of this material in a load frame and apply stress until it fails in tension	Ultimate tensile strength of quasi-isotropic carbon fiber and epoxy material	NASA 2.3, VADL 2.2.2		In-Progress
1.1.5	Load Test	Determine the critical hoop stress of the fiberglass for a geometry that is representative of the rocket body	The team will place a representative geometry of the rocket body into the load frame and apply compressive stress until it fails	Critical hoop stress of fiberglass	NASA 2.3, VADL 2.2.2	The critical hoop stress is greater than the maximum loads expected to be sustained by the rocket body throughout the mission sequence by a safety factor of 2.5	Complete
1.1.6	Load Test	Determine the critical bolt tear out stress of the fiberglass for a geometry that is representative of the rocket body	The team will place a section of the rocket body into the load frame and apply tensile stress to two bulkplates attached to airframe with bolts until failure	Critical bolt bearing stress of fiberglass	NASA 2.3, VADL 2.2.2	The critical tear out stress is greater than the maximum expected load sustained by the rocket body throughout the mission sequence by a safety factor of 2.5.	Complete
1.1.7	Load Test	Determine the modulus of elasticity of fiberglass airframe material	The team will place a section of the rocket body into the load frame and apply axial compressive loading until failure.	Modulus of elasticity of fiberglass	NASA 2.3, VADL 2.2.2	Ensure that the loads on the vehicle during flight are within the elastic range of fiberglass deformation, and ensure the material is fit for use as the vehicle airframe.	Complete
1.1.8	Impact Test	Ensure the fiberglass material can withstand the expected impact loads during landing.	The team will place a section of the rocket body into the drop tower and drop a weight of 25 lbm from a height of 36 in to simulate a 75 lb-ft impact	Impact strength of fiberglass	NASA 2.3	Ensure the vehicle section can withstand a 75 lb-ft impact without permanent deformation.	Complete
1.1.9	Load Test	Determine the failure load of recovery quick-link and swivel-link hardware	The team will place a quick-link and swivel-link into the load frame and apply tensile loading until failure.	Ultimate tensile strength of quick-link and swivel-link	NASA 2.3, VADL 2.2.2	Ensure that the failure load on the quick-link and swivel-link are at least 2.5 times greater than the maximum expected force during recovery deployment.	In-Progress
1.2 Recovery Testing							
1.2.1	Battery Test	Confirm that the battery life for recovery electronics is sufficient to remain on the launch pad for three hours before launch	Team members will connect the recovery electronics to their batteries and leave them on for three hours	Battery Life	NASA 2.6.3.5 VADL 3.12	The recovery electronics still functioning after being turned on for three hours	Complete
1.2.2	Altimeter Test	Verify that the altimeter can detect apogee and trigger the parachute blast charges	The team will place the altimeter with its blast charge igniters in a vacuum chamber. The vacuum will lower the pressure in the chamber until the pressure corresponding to projected apogee is reached. Team members will listen for the sound of the ignition trigger which would deploy the drogue parachute. NOTE: There will be no gun powder present during this test.	Proper altitude detection	NASA 3.1.2	The team hears the ignition trigger at the correct pressure readings corresponding to parachute deployment	Complete
1.2.3	Deployment Test	Verify that the calculated sizes of the blast charges can separate the launch vehicle for parachute deployments	The team will detonate the blast charges inside the rocket body and see if they effectively separate the rocket sections. This test will be performed under the supervision of the SSO and the team mentor to ensure safety	Blast charge mass	NASA 3.2	The rocket body sections are separated after detonation	Complete; to be repeated with full-scale vehicle

LAUNCH VEHICLE		Objective	Verification Plan	Variable Tested	Pertinent Technical Requirement(s)	Success Criteria	Status
Test Code	Verification Method						
1.2.4	Load Test	Ensure that the bulk plates can withstand the anticipated stresses sustained during flight, especially during parachute deployments	Team members will put the bulk plates in the load frame and subject them to tensile stress until they fail	Ultimate Tensile Strength of Bulk Plates	NASA 2.3	The ultimate tensile strength of the bulk plates is beyond the maximum expected stresses on the bulk plates by a safety factor of 2.5	In-Progress
1.2.5	Radio Interference Test	Verify that the avionics bay will not be subject to radio interference during flight	Team members will put an RF receiver in the avionics bay and send increasingly stronger signals until a signal is received	Radio noise cancellation of avionics bay	NASA 3.13, VADL 3.5	The team determines that the signal strength at which the receiver received a signal is higher than the expected level of any radio interference	In-Progress

PAYLOAD		Objective	Verification Plan	Variable Tested	Pertinent Technical Requirement(s)	Success Criteria	Status
Test Code	Verification Method						
2.1 Lander Separation Mechanism							
2.1.1	Command Response Time Test	Ensure that the LSM disengages from the rocket within 1 second of command	Perform ground-based timed tests, activating the LSM through command and measuring the separation time	Separation time	VADL 4.2.1	The LSM disengages within 1 second of the command with no significant delay	Complete
2.1.2	RF Transmission Test	Verify that the LSM is capable of receiving RF transmissions	Team members shall perform RF transmission tests to ensure that the LSM successfully receives commands without interference from surrounding materials	Signal strength and reliability	VADL 4.2.2	The LSM reliably receives RF transmissions without any significant signal interference	Complete
2.1.3	Altitude Operation Test	Ensure that the LSM functions reliably at altitudes between 200-400 ft AGL	A team member shall raise the LSM to an altitude between 200-400 ft AGL, and verify the separation event occurs as expected	Altitude at activation	VADL 4.2.3	The LSM successfully disengages during the specified altitude range in multiple tests	Complete
2.1.4	Vibration and Shock Load Test	Ensure the LSM withstands vibration and shock loads experienced during rocket ascent and separation	A tem member shall connect the LSM to a shock cord and drop off the side of a stairwell to simulate the jerk of parachute opening forces.	Vibration and shock resistance	VADL 4.2.4	The LSM remains fully operational and intact after exposure to vibration and shock loads simulating rocket flight	Complete
2.2 Leg Deployment Mechanism							
2.2.1	Buckling Load Test	Ensure that the legs do not buckle due to the force caused by impact in case there is a direct impact on a leg where buckling can occur	Using a load frame, team members will apply a load upon the vertical axis of the leg to determine the critical buckling load	Critical Buckling Load, Strength of aluminum legs	VADL 4.3.1, VADL 2.2.2	The critical buckling load is above the expected load the legs are expected to experience by a safety factor of at least 2.5	In-Progress
2.2.2	Dynamic Loading/ Drop Tests	Ensure that the legs do not break under dynamic loading similar to that experienced during landing. This includes moments and multidirectional forces	The primary verification for this will be from multiple drop tests. The initial drop tests will be from 4 ft AGL, without parachute, to simulate the forces exerted during an actual landing. These will include a quiescent drop from 80 ft AGL, as well as angled drops and drops with initial horizontal velocity. This will prove out both the strength of the legs and the righting ability of the legs. Final step is drop from drone at 300 ft AGL under parachute to precisely emulate rocket deployment conditions.	Righting ability, strength of leg/linkage system, strength of spring/damper necessary	VADL 4.3.1, VADL 2.2.2	The legs can survive a drop, in multiple orientations, with landing speeds that are equal or greater than those expected during competition. The legs should also enable the lander to land upright under these various conditions.	Complete
2.2.3	LDM Deployment Test	Confirm that the legs will successfully deploy upon release from rocket body	Test leg opening ability in lab conditions and under simulated drop conditions	Hardware necessary for deployment (springs needed or is gravity sufficient?)	VADL 4.3.2	Legs are able to deploy and lock under forces greater than what is expected during rocket flight.	Complete
2.3 Parachute Detachment Mechanism							
2.3.1	Release Time Test	Verify that parachute detachment occurs in the proper time frame, not too early such that the landing energy increases drastically nor too late such that the parachute destabilizes the payload	The team will run PDM in the lab as if it were during flight and make sure it functions without IMU input within a proper time frame; PDM will run based on input from microcontroller timer	PDM run time/proper microcontroller algorithm	VADL 4.4.1	PDM functions within 10s of input from microcontroller, so the parachute detaches in a timely manner	Complete
2.3.2	FEA analysis	Confirm that the PDM latch survives parachute opening forces.	The team will create a finite element analysis model of the load the mechanism will have to withstand due to the main parachute	PDM structural integrity	VADL 4.4.2	PDM can endure the load from the parachute with a safety factor of 5 applied	Complete
2.3.3	PDM Landing Detection/Deployment Test	Ensure that PDM safely detects landing prior to its deployment in order to prevent unacceptable landing energies	The team will place the IMU with its connections to a microcontroller and PDM inside a pressure chamber and decrease and increase the pressure inside to simulate the mission sequence. (This IMU contains a barometer.) The team then will watch to see that PDM only deploys once the pressure chamber increases back to ground pressure	IMU Performance, Microcontroller to PDM communication	VADL 4.4.1	The team observes that PDM only deploys at ground pressure and does so in a timely manner	Complete
2.3.4	PDM Locking Test	Verify that accelerations or vibrations will not cause PDM to prematurely deploy	A team member will violently shake PDM with its connection to the main parachute and expose it to vibrations	Robustness of PDM against accelerations and vibrations	VADL 4.4.2	PDM does not prematurely deploy under any accelerations or vibrations	Complete
2.4 Antenna Extension Mechanism							
2.4.1	Extension test	Verify that the AEM is able to fully extend to 18 in from within the airframe	A team shall repeated wind up and deploy the AEM to ensure that it does not bind internally. Antenna must also extend through the airframe without getting stuck.	Tape extension, servo unlocking	VADL 4.5.1	The AEM successfully deploys to full length 20 out of 20 times repeatedly	Complete
2.4.2	Locking Mechanism Test	Verify that the cylinder lock will not open unless actuated	A team member shall repeatedly drop the AEM and see if the locking mechanism remains locked	Locking mechanism robustness against vibrations and impact	VADL 4.5.4	The locking mechanism withstands 20 drops from a height of 4 feet AGL.	Complete
2.5 RF Transmission System							

PAYLOAD							
Test Code	Verification Method	Objective	Verification Plan	Variable Tested	Pertinent Technical Requirement(s)	Success Criteria	Status
2.5.1	Reliability Test	Ensure Transmission without issue for long period	Send out APRS continuously for 1 hour.	APRS reliable operation duration	VADL 4.6.1	RF system transmits for at least 1 hour	Complete
2.5.2	Distance Test	Verify Transmission over long range	Send APRS over 2 mile distance with varying environmental conditions	APRS reliable operation distance	VAD: 4.6.3	RF system should be error-free when there is ground clearance, and should also function with minimal degradation in more challenging environment	In-Progress
2.5.3	Transmission with damaged antenna	Verify Transmission when antenna fails to deploy, or is damaged during flight	Have different sets of antenna, some optimal, some broken in different ways, and transmit signals and measure signal strength.	RF system failure resilience	VADL 4.6.2	RF system should be able to broadcast in the range half of the optimal range	In-Progress
2.6 Data Collection Sensors							
2.6.1	Static Bench Testing	Verify that sensors produce reasonable outputs at 160 Hz when stationary	Place IMU on flat, stable surface and run for > 3 hrs	Accelerometer, gyroscope, altimeters calibration	VADL 4.7.1	Check whether accelerometers read $\sim 9.8 \text{ m/s}^2$ on the Z-axis, gyro reads zero, and altimeter reads stable value.	Complete
2.6.2	Drop Test and Rotation	Verify how IMU responds to changes in motion, orientation, and free fall	Drop the IMU from a known height to simulate rapid acceleration	Accelerometer	VADL 4.7.7	Ensure accelerometer detects proper acceleration changes	Complete
2.6.3	Altitude and Pressure Test	validate barometer / altimeter under varying pressure conditions to mimic altitude changes	place the sensor at different height and potentially pressure chamber	Barometer	VADL 4.7.4	Report correct altitude reading as expected	Complete
2.6.4	Apogee Detection	Confirm sensor can properly detect apogee	Test for rapid pressure drop and stabilization	Apogee	VADL 4.7.5	Report correct apogee as expected	Complete
2.6.5	Drop testing	Test complete system to ensure all sensors and detection events trigger as expected	Perform drop test using the entire payload	Landing and jettison detection	VADL 4.7.2, 4.7.3	System detects key flight events accurately	In-Progress
2.7 Payload Structure							
2.7.1	Load Test	Verify that the shaft collars can withstand expected loads during flight, recovery, and landing	A team member shall perform a load test on a shaft and shaft collar to determine the force required to move the collar.	Shaft collar strength	NASA 2.3, VADL 2.2.2	Shaft collars withstand a force at least 2.5 times greater than the maximum expected force	Complete

6.1.2 Completed Testing

The team has developed a methodical approach to testing documentation. For each test the objective, test variables, success criteria, and responsible team members are outlined. The procedure is then described, and key safety considerations are outlined. Photos are included to visualize the experimental setup. Upon completion of the test, key takeaways and design implications are described. The following sheets outline the testing completed thus far. As the design evolves and full-scale vehicle fabrication begins, additional testing will be drafted and done as required to ensure no aspects of vehicle launch, recovery, or landing are overlooked. VADL remains vigilant in ensuring that the USLI product is rigorously tested and analyzed for all possible failure modes.



Figure 135: Status of testing as of CDR cycle. Additional tests shall be done and re-done in order to maximize the chances of mission success.



VADL Testing Protocol

Airframe Hoop Load Test (COMPLETED)

1.1.5

Overview:

Objective / Purpose	To determine the structural integrity of the fiberglass airframe in hoop loading to inform vehicle design decisions and ensure adequate safety factor in flight.
Testing Variable(s)	Critical hoop stress of fiberglass airframe
Success Criteria	Airframe returns to normal shape after a critical hoop stress at least 2.5 times greater than the maximum hoop stress expected during vehicle flight and landing.
Responsible Team Members	Rais Nurhidajat, Zachary Stutsman

Methodology:

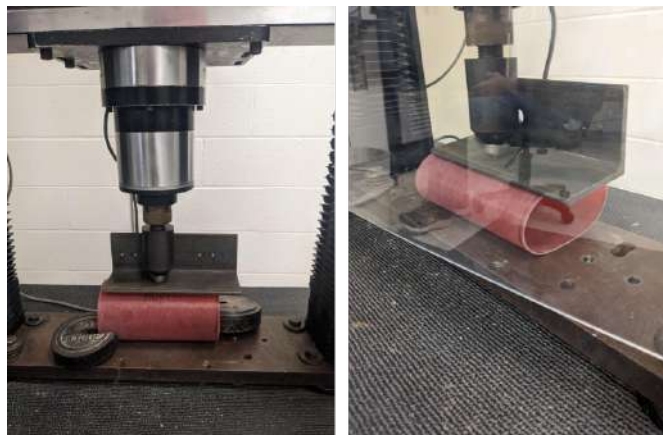
Procedure

1. Position airframe section beneath load transfer plate and base of load frame
2. Place 2 weights on either side of airframe to prevent sliding during load test
3. Close fracture shield an
4. Don safety glasses
5. Manually preload airframe section to 100 lbf
6. Initiate automated loading sequence
7. Load test is complete upon airframe section failure (cracking, delamination) abort load test, or wait for automatic cutoff once limit switches are reached
8. Reset machine and compile data to excel file

Key Safety Considerations

- Wear safety glasses and ensure the fracture shield is in front of the load frame to prevent injury due to flying debris.
- Keep clear of the load frame during operation and tie back loose hair to prevent pinch or crush injuries.
- One person must be in close proximity to the emergency stop switch at all times

Experimental Setup Photos

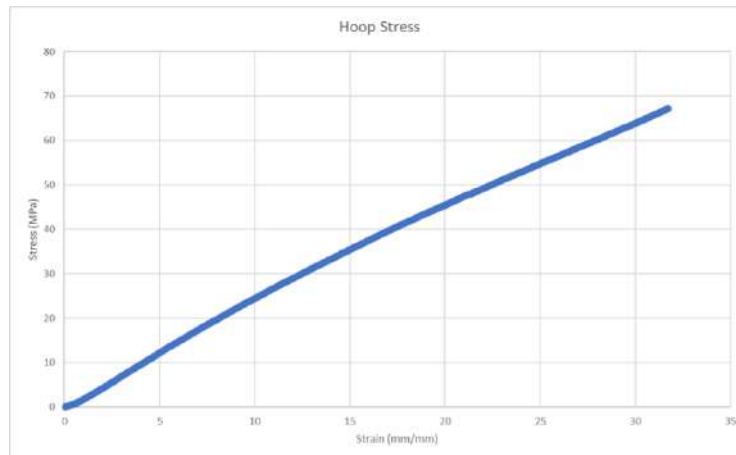




VADL Testing Protocol

Airframe Hoop Load Test (COMPLETED)

1.1.5



Results

Key Takeaways:

- Load test automatically cutoff at 22.2 kN (67.2 MPa)
- Maximum expected hoop load during recovery deployment and landing is 1.3 MPa
- Safety factor of 51 is much greater than the desired 2.5.

Design Implications:

It is unlikely that the airframe will fail in hoop loading. Even so, this will be mitigated through the use of tight-fitting bulk plates and internal skeleton.



VADL Testing Protocol

Airframe Bolt Tear Out Load Test (COMPLETED)

1.1.6

Overview:

Objective / Purpose	To determine the structural integrity of the fiberglass airframe for bolt tear out and inform vehicle design decisions and ensure adequate safety factor in flight.
Testing Variable(s)	Critical bolt tear out stress of fiberglass airframe
Success Criteria	Critical bolt tear out stress must be 2.5 times greater than the expected maximum bolt tear out stress expected during vehicle flight and landing for the avionics bay.
Responsible Team Members	Rais Nurhidajat, Zachary Stutsman

Methodology:

Procedure

1. Position airframe section between the 2 load frame hooks and ensure quick links are secure
2. Close fracture shield and don safety glasses
3. Manually preload airframe section to 100 lbf
4. Initiate automated loading sequence
5. Critical bolt tear out is when bolt tear out of 5mm occurs
6. Load test is complete upon bolt tear out of 1 inch or load frame limit switches are tripped
7. Reset machine and compile data to excel file

Key Safety Considerations

- Wear safety glasses and ensure the fracture shield is in front of the load frame to prevent injury due to flying debris.
- Keep clear of the load frame during operation and tie back loose hair to prevent pinch or crush injuries.
- One person must be in close proximity to the emergency stop switch at all times

Experimental Setup Photos

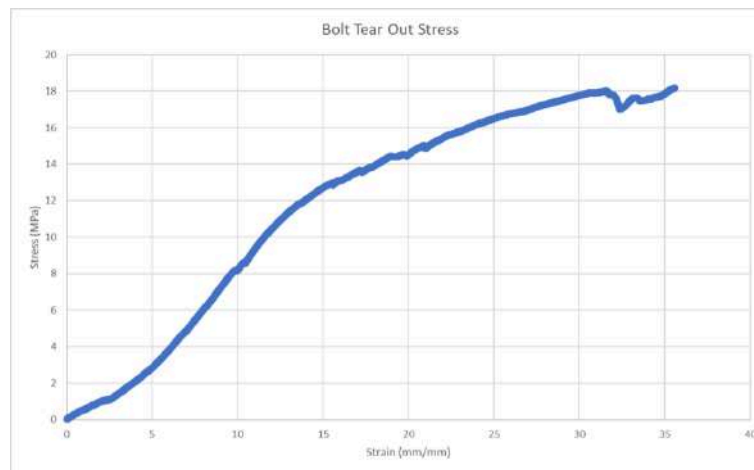




VADL Testing Protocol

Airframe Bolt Tear Out Load Test (COMPLETED)

1.1.6



Results

Key Takeaways:

- Critical bolt tear out stress of 8 MPa
- Calculated maximum bolt tear out stress of 1.8 MPa, which provides a safety factor of 4.4.
- Bolts deformed severely at the end of the load test once the material was taken to complete failure (1 inch of bolt tear out)

Design Implications:

Both bolt tear out and bolt deformation will be mitigated by properly securing the bulk plates to the internal skeleton.



VADL Testing Protocol

Airframe Axial Compression Test (COMPLETED)

1.1.7

Overview:

Objective / Purpose	To determine the structural integrity of the fiberglass airframe in axial compression and inform vehicle design decisions and ensure adequate safety factor in flight.
Testing Variable(s)	Critical axial compressive stress
Success Criteria	Critical axial compressive stress is at least 2.5 times greater than the maximum expected compressive stress experienced during vehicle flight and landing.
Responsible Team Members	Rais Nurhidajat, Zachary Stutsman

Methodology:

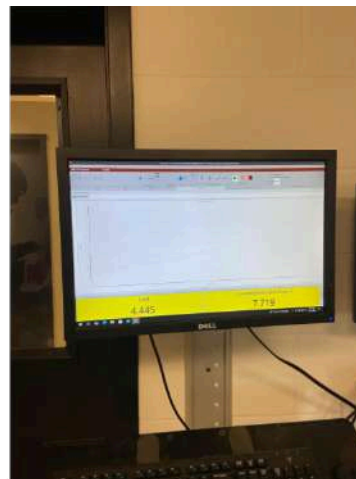
Procedure

1. Position airframe section between load transfer plate and center of load frame bed
2. Close fracture shield and don safety glasses
3. Manually preload airframe section to 100 lbf
4. Initiate automated loading sequence
5. Load test is complete upon airframe section failure (cracking, buckling) or load frame exceeds safety cutoff
6. Reset machine and compile data to excel file

Key Safety Considerations

- Wear safety glasses and ensure the fracture shield is in front of the load frame to prevent injury due to flying debris.
- Keep clear of the load frame during operation and tie back loose hair to prevent pinch or crush injuries.
- One person must be in close proximity to the emergency stop switch at all times

Experimental Setup Photos

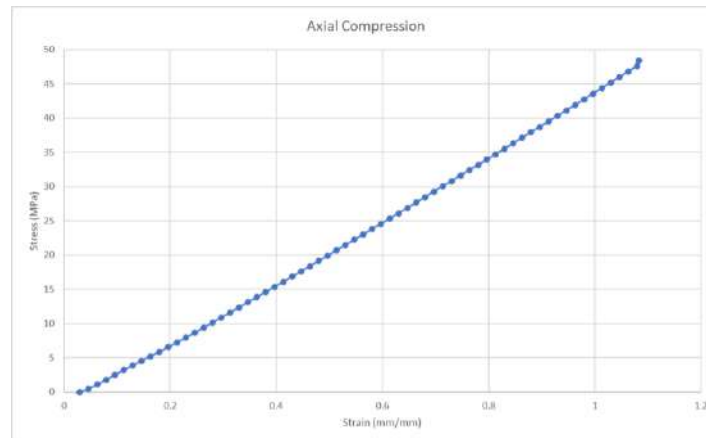




VADL Testing Protocol

Airframe Axial Compression Test (COMPLETED)

1.1.7



Results

Key Takeaways

- Load frame automatically cutoff at 22.6 kN (49.5 MPa)
- Maximum expected compressive stress is 1.89 MPa
- Safety factor of 26 is much greater than the desired 2.5

Design Implications:

It is unlikely that the airframe shall fail in compressive loading during launch. To increase structural stability the motor retention rings also serve to distribute the load at the base of the rocket.



VADL Testing Protocol

Airframe Impact Test (COMPLETED)

1.1.8

Overview:

Objective / Purpose	Ensure the fiberglass material can withstand the expected impact loads during landing.
Testing Variable(s)	Impact strength of fiberglass
Success Criteria	Ensure the vehicle section can withstand a 75 lb-ft impact without permanent deformation.
Responsible Team Members	Rais Nurhidajat, Zachary Stutsman

Methodology:

Procedure

1. Place the airframe section at the bottom of the drop tower and secure it in place.
2. Place weights on the drop plate and secure in place.
3. Raise drop plate to desired height.
4. Ensure that all personnel and appendages are clear of the drop zone and release the drop plate.
5. Compile data to Excel file.

Key Safety Considerations

- Wear safety glasses to prevent injury due to flying debris.
- Keep all appendages clear of the drop tower during operation to prevent crush injuries.

Experimental Setup Photos



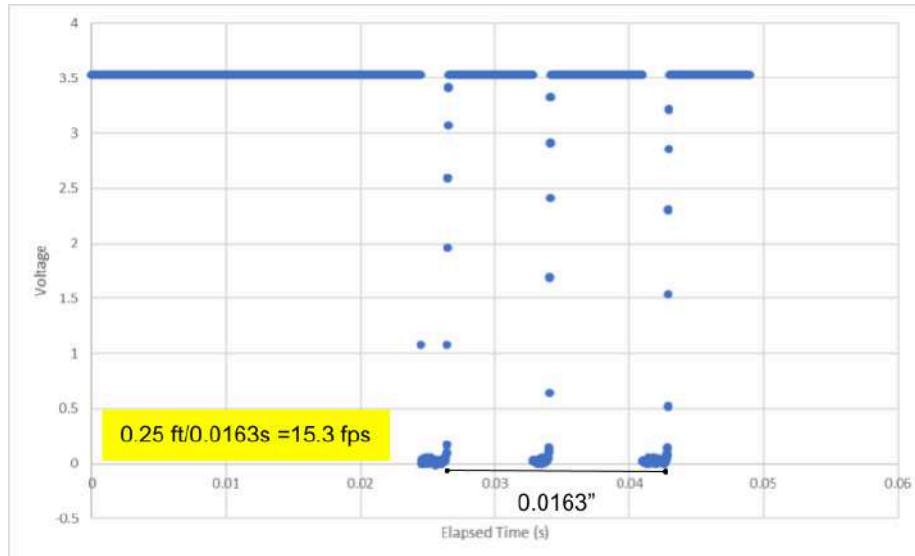


VADL Testing Protocol

Airframe Impact Test (COMPLETED)

1.1.8

Results



Key Takeaways

- Voltage vs. time data confirms the velocity and subsequent L.K.E. of 75 lb-ft was achieved
- Airframe withstood impact with no structural damage incurred

Design Implications:

It is unlikely that fiberglass will experience structural damage upon landing. As such, fiberglass is verified to be a suitable airframe material selection for the full-scale flight.



VADL Testing Protocol

Swivel-Link Load Test (COMPLETED)

1.1.9

Overview:

Objective / Purpose	Determine the failure load of recovery swivel-link hardware
Testing Variable(s)	Ultimate tensile strength of swivel-link
Success Criteria	Ensure that the failure load on the swivel-link is at least 2.5 times greater than the maximum expected force during recovery deployment.
Responsible Team Members	Rais Nurhidajat, Zachary Stutsman

Methodology:

Procedure

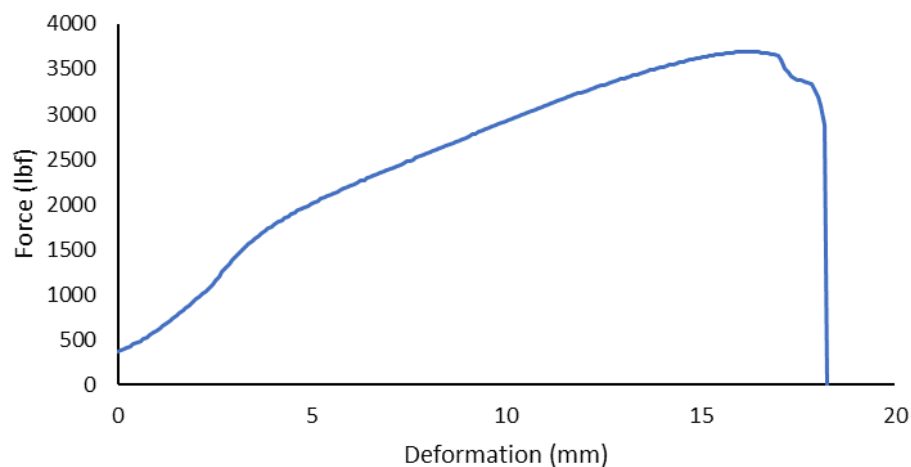
1. Place link between load frame hooks
2. Close fracture shield and don safety glasses
1. Manually preload link to 100 lbf
2. Initiate automated loading sequence
3. Load test is complete upon link section failure (cracking, stretching) or load frame exceeds safety cutoff
4. Reset machine and compile data to excel file

Key Safety Considerations

- Wear safety glasses and ensure the fracture shield is in front of the load frame to prevent injury due to flying debris.
- Keep clear of the load frame during operation and tie back loose hair to prevent pinch or crush injuries.
- One person must be in close proximity to the emergency stop switch at all times

Experimental Setup Photos

Applied Force vs Deformation





VADL Testing Protocol

Swivel-Link Load Test (COMPLETED)

1.1.9



Results

Key Takeaways

- Swivel-Link can withstand 3692 lbf before failing
- The expected loading on the link is 1204 lbf, yielding a safety factor of 3.06

Design Implications

The safety factor provided by the swivel-link is 3.06, which is well above the VADL minimum of 2.5 for recovery components, not including the shock cords. These swivel-links will be kept for full scale flight.



VADL Testing Protocol

Recovery Avionics Battery Test (COMPLETED)

1.2.1

Overview:

Objective / Purpose	Confirm that the battery life for recovery electronics is sufficient to remain on the launch pad for three hours before launch
Testing Variable(s)	Avionics battery life
Success Criteria	Avionics still functional after being turned on for over 3 hours
Responsible Team Members	Saksham Sharma, Santino Clemente

Methodology:

Procedure

1. Ensure battery voltage is above 9V
2. Connect battery to altimeter and arm
3. Every 10 minutes, validate altimeter is still live by listening to altimeter buzzer
4. Test is complete after 3 hours have elapsed

Key Safety Considerations

- Ensure battery polarity is correct before arming altimeter to prevent electrical fires or damage to altimeter electronics

Experimental Setup Photos



Results

Key Takeaways:

- 9V battery lasted well after 3 hours, and only died after 12 hours

Design Implications:

A single 9V battery is sufficient to power one of the avionics boards throughout the rocket sitting on the pad and through launch.



VADL Testing Protocol

Recovery Avionics Altimeter Test (COMPLETED)

1.2.2

Overview:

Objective / Purpose	Verify that the altimeter can detect apogee and trigger the parachute blast charges
Testing Variable(s)	Proper altitude detection
Success Criteria	The team hears the ignition trigger at the correct pressure readings corresponding to parachute deployment
Responsible Team Members	Saksham Sharma, Santino Clemente

Methodology:

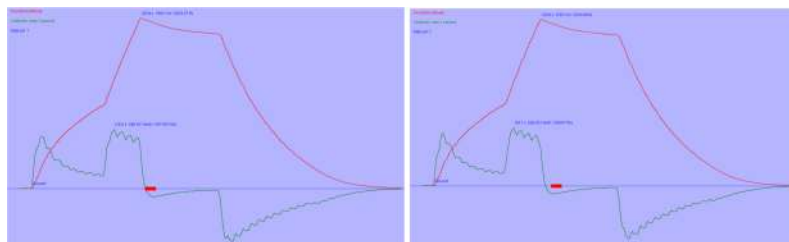
Procedure

1. Set altimeters to deploy at apogee with backup at apogee +1 second
2. Don safety glasses
3. Verify battery terminal orientation on altimeter, and connect 9V battery, and e-match
4. Arm altimeter using screw switch
5. Don safety glasses
6. Turn shock tube electronics on and open vacuum chamber
7. Place altimeters within the vacuum chamber and secure the clamps
8. Lower the pressure within the shock tube to correspond to 5000 ft altitude
9. Open the release valve and allow the pressure to slowly return to ground pressure within the chamber, listening for e-match ignition
10. Connect to altimeter user interface and verify deployments occurred at apogee

Key Safety Considerations

- Ensure the altimeter is properly armed and setup to prevent premature e-match ignition
- Wear safety glasses to protect eyes in the event of a premature e-match ignition

Experimental Setup Photos



Main altimeter shown on left, backup shown on right. The red line represents when deployment occurred

Results

Key Takeaways

- Altimeter results were within 15 feet, with detections matching to the tenth of a second.
- Deployments occurred at apogee and apogee +1 second.

Design Implications

Altimeter requires sufficiently sized port holes in order to function properly. Altimeters have slightly different calibrations, which necessitates the use of a backup altimeter to ensure deployments occur properly.



VADL Testing Protocol

Recovery Deployment Test (**COMPLETE**)

1.2.3

Overview:

Objective / Purpose	Verify that the calculated sizes of the blast charges can separate the launch vehicle for parachute deployments
Testing Variable(s)	Blast charge mass
Success Criteria	The rocket body sections are separated after detonation, the parachute is completely outside of the bay
Responsible Team Members	Saksham Sharma, Santino Clemente

Note:

Test has been conducted using subscale launch vehicle to verify blast charge calculations. **Test shall be repeated** upon completion of full scale launch vehicle.

Methodology:

Procedure

1. Don safety glasses
2. Attach recovery hardware to eyebolts with the fire blanket and fire retardant inside the parachute bay
3. Secure the rocket together with nylon shear pins
4. Attach rocket to x-frame atop tarp and foam to prevent damage to the surroundings and rocket
5. Wire e-matches to deployment box.
6. Trigger e-match and observe deployment. Verify recovery hardware is completely separated from the rocket itself

Key Safety Considerations

- Verify terminal polarity to prevent electrical fires or sparks that may cause premature detonation of the blast charge
- Wear safety glasses to shield eyes from debris
- Stand well clear of the test setup and away from the fore section of the rocket
- Keep a fire extinguisher on standby in the event the deployment causes a fire

Experimental Setup Photos



Results

Key Takeaways

- Blast charge calculations verified
- Recovery hardware properly deployed, parachute experienced no burns

Design Implications

Sufficient fire retardant is required to prevent parachute burn damage. Bulk plates must be properly designed to prevent blast charge gasses from leaking through to the avionics or other sections



VADL Testing Protocol

LSM Command Response Test (COMPLETE)

2.1.1

Overview:

Objective / Purpose	Ensure that the LSM disengages from the rocket within 1 second of command.
Testing Variable(s)	Separation time
Success Criteria	The LSM disengages within 1 second of the command with no significant delay
Responsible Team Members	Rais Nurhidajat, Santino Clemente

Methodology:

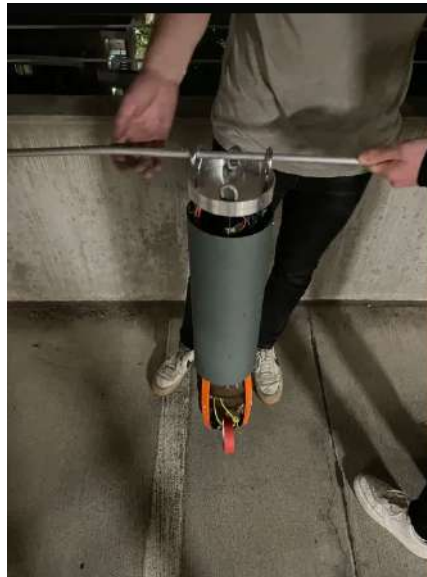
Procedure

1. Prepare LSM system according to setup instructions.
2. Connect LSM to a simulated payload to test release mechanism functionality.
3. Arm the system and perform pre-test checks.
4. Send command signal to initiate LSM separation.
5. Measure time from command to LSM disengagement using a high-speed camera or timer.
6. Repeat the test three times to ensure consistency.

Key Safety Considerations

- Verify electrical connections to prevent unintended activation of the LSM.

Experimental Setup Photos



Results

Key Takeaways

- The LSM successfully disengaged within 1 second of command in all three trials, meeting the success criteria.
- The separation mechanism operated without delay or malfunction, confirming system reliability.

Design Implications

No immediate changes required for LSM design. System is confirmed to be reliable and effective for mission deployment.



VADL Testing Protocol

LSM RF Transmission Test (COMPLETE)

2.1.2

Overview:

Objective / Purpose	Verify RF transmission to LSM electronics from long distances
Testing Variable(s)	RF signal strength and communication reliability over distance.
Success Criteria	Successful transmission and response from the LSM electronics at a distance of approximately 1400 feet without signal loss or interference.
Responsible Team Members	Rais Nurhidajat, Santino Clemente, Patrick Chen

Methodology:

Procedure

1. Prepare LSM electronics with RF receiver powered and armed for testing.
2. Ensure all electrical components are safely connected and insulated.
3. Position the RF transmitter while the receiver is stationed approx. 1400 ft away.
4. Initiate RF transmissions using pre-configured command signals.
5. Monitor LSM receiver for response to each transmission and verify successful commands.
6. Record any delays, signal degradation, or failed transmissions.
7. Repeat test multiple times to ensure consistency and reliability.

Key Safety Considerations

- Verify all electronic connections to prevent RF interference with other devices in the area.
- Maintain clear communication between testing locations for safety and efficiency.

Experimental Setup Photos



Results

Key Takeaways

- RF transmissions successfully reached the LSM receiver at 1400 feet with no signal loss or delays.

Design Implications

The RF system in the LSM is capable of maintaining communication within the required range for mission operations.



VADL Testing Protocol

LSM Altitude Operation Test (COMPLETE)

2.1.3

Overview:

Objective / Purpose	Verify the LSM operates effectively at deployment altitudes between 200 and 400 feet.
Testing Variable(s)	Deployment altitude.
Success Criteria	The LSM successfully disengages at the specified altitude range with consistent performance and no malfunction.
Responsible Team Members	Rais Nurhidajat, Santino Clemente, Patrick Chen

Methodology:

Procedure

1. Attach the LSM to the Aurelia drone.
2. Perform pre-flight safety checks on the drone and LSM, ensuring secure attachment and proper functionality.
3. Fly the drone to the specified altitude between 300 and 400 feet.
4. Issue the release command to the LSM while maintaining altitude.
5. Observe and record the LSM's response time and deployment behavior.
6. Repeat the process at varying altitudes within the range to ensure consistent performance.

Key Safety Considerations

- Ensure the drop area is clear of personnel and obstructions.
- Verify secure attachment of the LSM to the drone to prevent accidental releases.
- Follow all FAA and drone operation guidelines for altitude and area restrictions.

Experimental Setup Photos



Results

Key Takeaways

- The LSM successfully disengaged upon command at altitudes ranging from 200 to 400 feet in all trials.
- No delays or malfunctions were observed during deployment, and the mechanism demonstrated consistent reliability.

Design Implications

The LSM is confirmed to be functional and reliable at mission-required altitudes.



VADL Testing Protocol

LDM Dynamic Loading/ Drop Tests (Complete)

2.2.2

Overview:

Objective / Purpose	Ensure LDM can withstand landing forces without breaking
Testing Variable(s)	Loading capacity of leg bulk plate, leg bracket, and leg
Success Criteria	No components break or deform past functionality
Responsible Team Members	Zachary Stutsman, Rais Nurhidajat

Methodology:

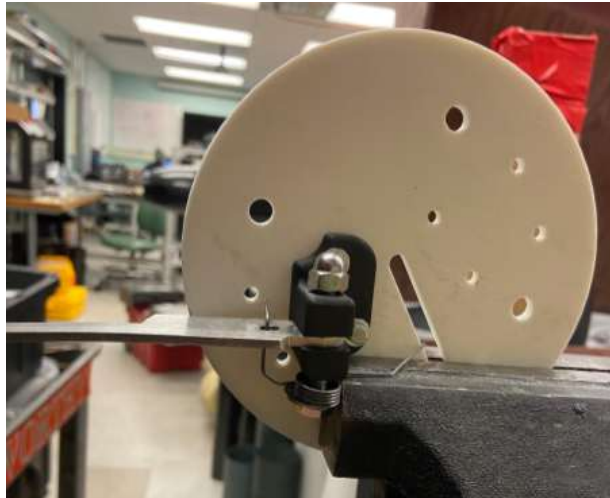
Procedure

1. Assemble one leg and bracket onto leg bulk plate
2. Clamp bulk plate into table vise so that leg is parallel to ground
3. Run string through hole at end of leg
4. Attach 1 lb weight to the end of the string and gently lower. Increase weight until 8 lb is reached
5. Once 8 lb static loading is reached, proceed to dynamic testing
6. With leg still clamped and 8 lb of mass still attached, raise weight 6 in from neutral leg position and drop
7. Continue dropping greater distances until a drop of 27 in is completed

Key Safety Considerations

- Safety glasses and closed toed shoes will be worn at all times during testing
- Team members will stand clear of dropping weights
- Incremental increases are instrumental in avoiding unexpected failures

Experimental Setup Photos

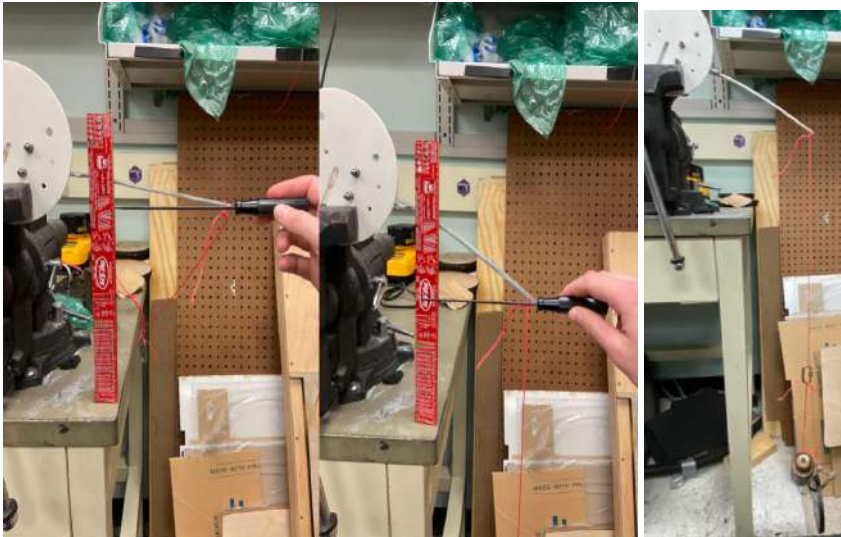




VADL Testing Protocol

LDM Dynamic Loading/ Drop Tests (Complete)

2.2.2



Results

Key Takeaways

- Leg and bracket withstood static testing with bending but no permanent deformation
- Dynamic testing at maximum drop height created some deformation but did not impact functioning

Design Implications

Leg bracket and bulk plate experienced no breakage or permanent deformation. Leg experienced bending without deformation during static loading and some deformation during the 27 in drop. This deformation could be undone relatively easily and was not great enough to prevent the proper functioning of the LDM. Therefore, the leg, bracket, and bulk plate design and material choices are sufficient.



VADL Testing Protocol

LDM Deployment Test (Complete)

2.2.2

Overview:

Objective / Purpose	Prove that legs can be deployed from airframe upon lander separation
Testing Variable(s)	Ability of legs to deploy
Success Criteria	Consistent deployment
Responsible Team Members	Zachary Stutsman

Methodology:

Procedure

1. Fold legs of constructed payload
2. Place payload into airframe
3. Slide airframe into rocket coupler section
4. Release payload so that it can fall freely. Make sure to hold shock cord when dropping
5. Visually check to confirm that all four legs fully deploy
6. Repeat until convinced deployment is consistent

Key Safety Considerations

- Wear closed toe shoes and safety glasses
- Team members who are not dropping payload will stand clear of drop area

Experimental Setup Photos



LDM deployment re-verified in Full-scale drop test from 300 ft AGL

Results

Key Takeaways

- Legs can successfully deploy from within airframe
- Deployment is consistent and happens immediately upon separation and shock cord jerk

Design Implications

The current, torsional spring driven deployment of the legs is sufficient for successful deployment during flight. This conclusion is supported by field payload testing.



VADL Testing Protocol

PDM Release Time Test (COMPLETE)

2.3.1

Overview:

Objective / Purpose	Verify that parachute detachment occurs in the proper time frame, not too early such that the landing energy increases drastically nor too late such that the parachute destabilizes the payload
Testing Variable(s)	PDM run time/proper microcontroller algorithm
Success Criteria	PDM functions within 10s of input from microcontroller, so the parachute detaches in a timely manner
Responsible Team Members	Rais Nurhidajat, Patrick Chen, Timothy Shaheen

Methodology:

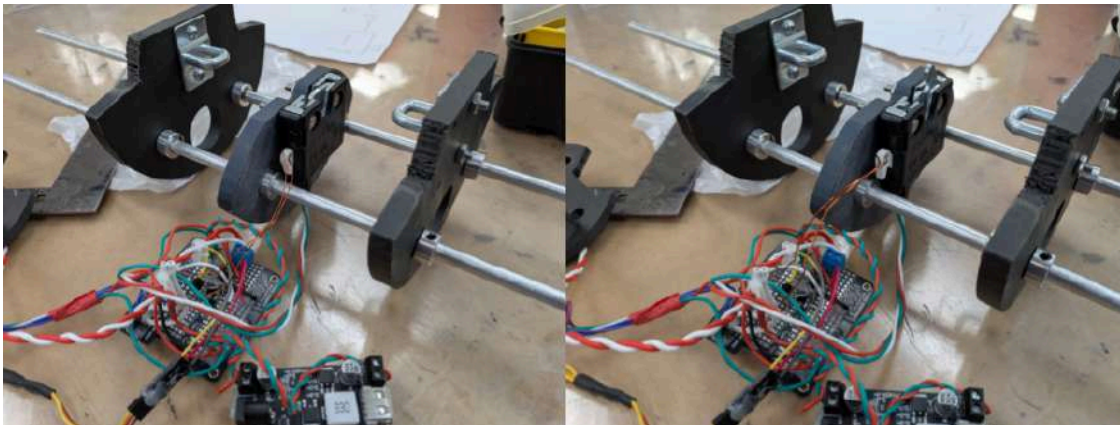
Procedure

1. Connect the PDM latch to the payload electronics and Raspberry Pi
2. Within the main launch code, adjust landing acceleration check such that it can be triggered by tapping the IMU on a hard surface.
3. Run main launch code.
4. Impose some acceleration to the IMU to simulate landing impact and observe PDM release.

Key Safety Considerations

- Tape up bare electrical wires and observe proper polarity to prevent electrical shocks and fires

Experimental Setup Photos



Results

Key Takeaways

- The PDM latch opened within 10 seconds of IMU acceleration spike for all 10 tests conducted.
- The payload electronics properly actuate the rotary latch

Design Implications

Testing and tuning will be required to properly set the acceleration threshold for landing detection.



VADL Testing Protocol

AEM Extension Test (COMPLETE)

2.4.2

Overview:

Objective / Purpose	Verify that the AEM is able to fully extend to 18 in from within the airframe
Testing Variable(s)	Tape extension, servo unlocking
Success Criteria	The AEM successfully deploys to full length 20 out of 20 times repeatedly
Responsible Team Members	Rais Nurhidajat

Methodology:

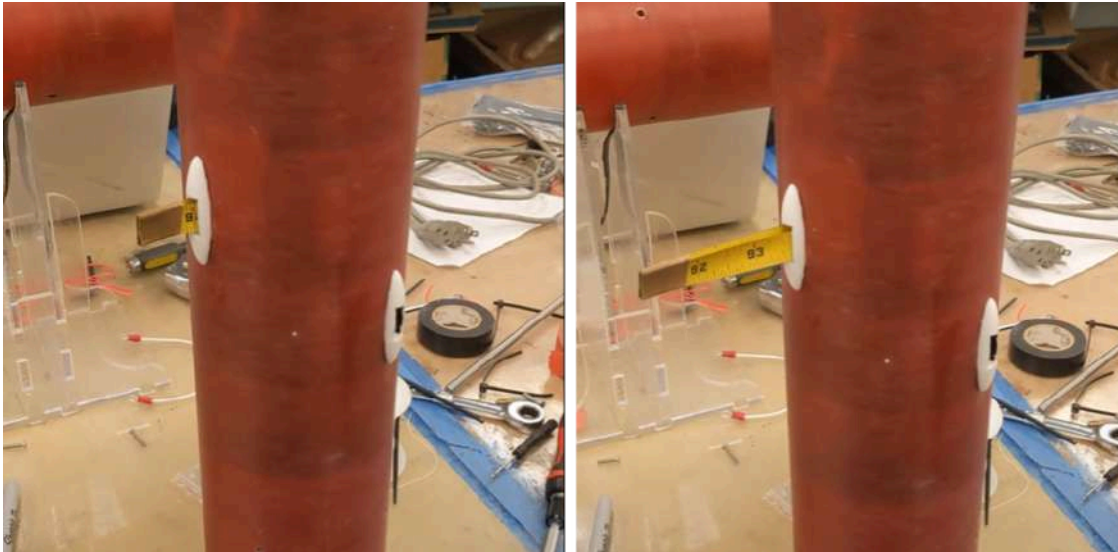
Procedure

1. Wind AEM and set servo lock
2. Turn on payload electronics and SSH into raspberry pi
3. Place AEM and bulkplate within airframe and pull antenna into slot cover
4. Run servo release command
5. Verify full extension length of 18 in

Key Safety Considerations

- Electronics shall be inspected prior to use to prevent shock and/or electrical fires

Experimental Setup Photos



Results

Key Takeaways

- Antenna fully extended to 18 in 17 out of 20 trials
- In 3 trials the antenna partially extended to 12 in. The cause was determined to be improper slot cutout alignment.

Design Implications

Slot cutouts must be carefully cut to ensure the tape lines up as closely to the center of the slot as possible.



VADL Testing Protocol

AEM Locking Mechanism Test (COMPLETE)

2.4.3

Overview:

Objective / Purpose	Verify that the cylinder lock will not open unless actuated
Testing Variable(s)	Locking mechanism robustness against vibrations and impact
Success Criteria	The locking mechanism withstands 20 drops from a height of 4 feet AGL
Responsible Team Members	Rais Nurhidajat

Methodology:

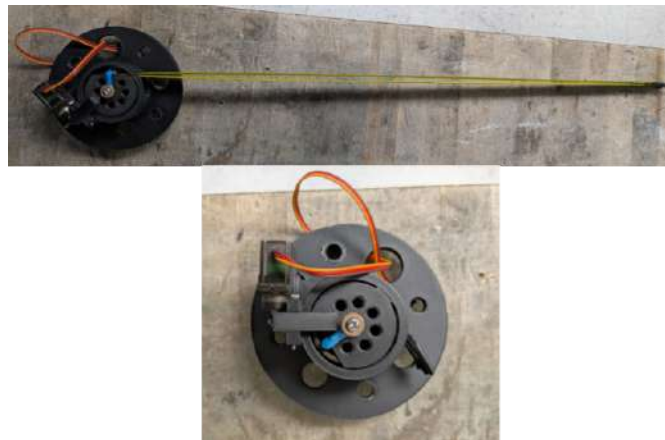
Procedure

1. Set cylinder lock by moving servo arm
2. Drop AEM repeatedly from a height of 4 feet and verify the cylinder does not unlock after repeated drops

Key Safety Considerations

- Wear safety glasses to protect eyes from potential debris breaking off AEM

Experimental Setup Photos



Results

Key Takeaways

- AEM cylinder did not unlock after over 20 drops
- Antenna can be jostled leading to slight expansion within the AEM case. When this was observed, the cylinder was unlocked to ensure tape extension. In these trials the tape did not extend as smoothly.

Design Implications

The antenna must be wrapped as tightly around the cylinder as possible prior to each launch. Teflon tape will be added to the tip of the antenna to lower friction between the antenna and case as much as possible.



VADL Testing Protocol

Sensor Drop Test (**COMPLETE**)

2.6.2

Overview:

Objective / Purpose	Ensure IMU can detect dramatic changes in the state of the payload. Verify its reliability in free-fall conditions.
Testing Variable(s)	Acceleration and altitude data changes during the test.
Success Criteria	Confirm the accelerometer detects proper acceleration changes, including rapid deceleration upon impact.
Responsible Team Members	Patrick Chen

Methodology:

Procedure

1. Prepare the IMU for testing by securely mounting it with the system on a prototype payload with a parachute.
2. Drop the IMU from the selected height.
3. Collect accelerometer data during the drop, ensuring the IMU records acceleration changes in all axes.
4. Analyze the data to ensure it reflects expected behavior.

Key Safety Considerations

- Conduct the test in a clear and designated area free from bystanders, obstacles, and hazardous materials.
- Ensure the landing surface is aligned with the drop point to avoid unexpected payload trajectory.
- Test the parachute system beforehand to confirm proper deployment and functionality.
- Maintain a safe distance from the drop zone during testing.

Experimental Setup Photos



Results

Key Takeaways

- Acceleration spike is correctly recorded and plotted.

Design Implications

The original plan to use the landing acceleration spike for detection has been successfully validated. This confirms the feasibility and effectiveness of the approach.



VADL Testing Protocol

Altimeter Test (COMPLETE)

2.6.3

Overview:

Objective / Purpose	Ensure barometer data is changing with respect to the change of altitude.
Testing Variable(s)	Pressure and altitude readings from the barometer/altimeter.
Success Criteria	The sensor must report correct altitude readings consistent with the expected pressure changes.
Responsible Team Members	Patrick Chen

Methodology:

Procedure

1. Attach the payload system, including the barometer/altimeter, securely to a drone.
2. Verify the drone's flight controls and payload detachment mechanism are functioning properly.
3. Fly the drone to the desired altitude for testing.
4. At the designated height, detach the payload and deploy the parachute to ensure a controlled descent.
5. Retrieve the payload after landing.
6. Plot the data to visualize altitude changes over time and validate sensor performance.

Key Safety Considerations

- Conduct the test in a designated safe area, clear of bystanders, to minimize risks.
- Ensure the payload and parachute system are securely attached and tested beforehand to avoid malfunctions.
- Maintain visual control of the drone throughout the flight and descent.

Experimental Setup Photos





VADL Testing Protocol

Altimeter Test (COMPLETE)

2.6.3

Results



Key Takeaways

- Both acceleration and altimeter data are confirmed to be working.

Design Implications

The core components of the landing detection algorithm have been successfully validated. As a result, the team will proceed with the originally proposed algorithm, which utilizes altitude changes and acceleration spikes to reliably detect landing events.



VADL Testing Protocol

Apogee Detection Test (COMPLETE)

2.6.4

Overview:

Objective / Purpose	Ensure the apogee of the flight can be detected and recorded by the payload system.
Testing Variable(s)	Barometric pressure readings from the altimeter to identify changes indicative of apogee
Success Criteria	The sensor correctly detects apogee by identifying the rapid pressure drop. The result is cross-referenced against simulation data.
Responsible Team Members	Patrick Chen, Sitan Huang

Methodology:

Procedure

1. Attach the payload system to a drone, ensuring the altimeter sensor is calibrated and securely mounted.
2. Fly the drone to a predetermined altitude simulating apogee conditions.
3. Collect barometric pressure readings during the ascent, at apogee (when the drone stabilizes at maximum altitude), and during descent.
4. Analyze the data to detect the pressure drop at apogee and confirm stabilization, comparing the results against simulation data.

Key Safety Considerations

- Conduct the test in a designated flight zone free of bystanders, obstacles, and restricted airspace.
- Ensure the drone's flight controls and payload attachment mechanisms are thoroughly inspected before the test.
- Maintain a safe distance and clear communication among team members during the flight.

Experimental Setup Photos



Results

Key Takeaways

- The apogee was correctly recorded and was within $\pm 5\%$ of the commercial altimeter on the drone (plus the length of the shock cord).
- The apogee data was also successfully transmitted via APRS.

Design Implications

This functionality was verified and requires no changes to the system.



VADL Testing Protocol

Landing Detection Test (COMPLETE)

2.6.5

Overview:

Objective / Purpose	Landing detection must be verified. It triggers all post-landing procedures.
Testing Variable(s)	Accelerometer, Barometer data. Confirm the activation of key detection events.
Success Criteria	The latch needs to detach, the AEM needs to deploy, and RF transmission needs to begin immediately after landing.
Responsible Team Members	Patrick Chen

Methodology:

Procedure

1. Prepare the payload system, ensuring all sensors are calibrated and detection mechanisms are operational.
2. Attach the payload to a controlled drop mechanism (drone) to simulate the descent and landing phases.
3. Drop the payload from a predetermined height to mimic real flight conditions.
4. Validate that the landing and jettison events are detected accurately and match the expected behavior.

Key Safety Considerations

- Perform the test in a secure area away from bystanders and obstacles.
- Use protective materials to minimize potential damage to the payload upon landing.
- Maintain a safe distance for all team members during the drop.

Experimental Setup Photos



Results

Key Takeaways

- Landing was detected in 4 out of 5 tests.
- The only time of not detecting landing was caused by human error in calibrating the threshold data.

Design Implications

The checklist was updated to avoid human error. Now the software system requires three people to cross-check to ensure reliability in calibration processes.

6.2 Requirements Compliance

6.2.1 NASA Requirements

VADL is committed to a requirements-driven approach to project development. The requirements laid out in the 2025 Student Launch handbook drive all design decisions; as such, it is imperative to develop detailed and actionable derived design requirements and verification plans for each NASA requirement. These plans, initially presented at the PDR milestone, have since been updated as the project has matured. These changes are reflected in the following tables. The verification methods, informed by the NASA Systems Engineering Handbook, are consistent with those shown in the PDR report and are described as follows.

- **Analysis (A):** This method refers to the use of computational simulations, mathematical models, or hand calculations to analytically verify the performance of a design and its compliance with the requirement. A-type verification will be implemented when prototyping and/or physical testing is deemed infeasible due to constraints in time, cost, or a lack of necessary manufacturing or testing infrastructure. Kinematic analysis, MATLAB simulations, FEA, flight dynamics, CFD, and recovery system calculations are all examples of analysis methods utilized by the team.
- **Demonstration (D):** This method refers to the operation of a mechanism, electronics system, component, etc. to verify that it functions as intended, thus satisfying the requirement. Examples include demonstrating that a payload system functions in subscale flight, or that the full-scale vehicle can be prepared within the required timeframe.
- **Inspection (I):** This method refers to the examination of a system, subsystem, or component to verify that its physical attributes comply with the respective requirement(s). Inspection is employed when operating the hardware is not necessary to confirm requirement fulfillment. Visually checking the number of independent sections on the launch vehicle and measuring coupler lengths are both instances of I-type verification.
- **Test (T):** This method refers to the experimental operation of a system, subsystem, or component. In particular, said operation must be well-documented with detailed data to quantitatively confirm compliance with the requirement. Examples include deployment testing, payload drop testing, and RF transmission testing.

The verification status options remain the same from PDR: In Progress (IP), Verified (V) and Not Verified (NV). The status of each verification plan has been updated and can be seen in the following tables.

NASA General Requirements		Design Requirements	Method	Verification Details	Team Member(s) - Initials	Status
Req #	Description					
1.1.	Students on the team will do 100% of the project, including design, construction, written reports, presentations, and flight preparation with the exception of assembling the motors and handling black powder or any variant of ejection charges, or preparing and installing electric matches (to be done by the team's mentor). Teams will submit new work. Excessive use of past work will merit penalties.	The team shall complete 100% of USLI project work.	I	The team will maintain accountability through self-inspection to verify only students work on the project, with the exception of motors and energetics handled by the team mentor.	All	IP
1.2.	The team will provide and maintain a project plan to include, but not limited to the following items: project milestones, budget and community support, checklists, personnel assignments, STEM engagement events, and risks and mitigations.	The team shall regularly update key project management items, to be presented in key NASA deliverables.	I	The team will meet weekly outside of design meetings to discuss these aspects of the project; the President, Student Safety Officer, and Outreach Lead will manage these items.	All	IP
1.3.	Team members who will travel to the Huntsville Launch shall have fully completed registration in the NASA Gateway system before the roster deadline. Team members shall include:	The team shall announce all members attending launch week by CDR and ensure one mentor and no more than two educators.	I	The team lead has inspected that all members, mentors, and educators attending launch week were included on the submitted Huntsville roster.	IM	V
1.3.1	Students actively engaged in the project throughout the entire year.					
1.3.2	One mentor (see requirement 1.13).					
1.3.3	No more than two adult educators.					
1.4	Teams shall engage a minimum of 250 participants in Educational Direct Engagement STEM activities. These activities can be conducted in-person or virtually. To satisfy this requirement, all events shall occur between project acceptance and the FRR addendum due date. A template of the STEM Engagement Activity Report can be found on pages 40 – 43.	The team has planned events to remain on track with this requirement and shall continue to engage more participants.	I	The team will inspect completed engagement reports to ensure outreach milestones are met in a timely manner.	All	IP
1.5	The team shall establish and maintain a social media presence to inform the public about team activities.	The team shall designate a social media manager to ensure timely and frequent social media presence.	I	The social media manager will inspect all social media presence and ensure timely, informative postings. The team will inspect these accounts as well.	SS	IP
1.6.	Teams shall email all deliverables to the NASA project management team by the deadline specified in the handbook for each milestone. In the event that a deliverable is too large to attach to an email, inclusion of a link to download the file will be sufficient. Late submissions of PDR, CDR, or FRR milestone documents will NOT be accepted. Teams that fail to submit the PDR, CDR, or FRR milestone documents will be eliminated from the project.	All deliverables shall be sent to the NASA project management team in a timely manner.	I	The team lead, with assistance from other team members, will ensure that all deliverables are submitted by the deadline.	IM	IP
1.7	Teams who do not satisfactorily complete each milestone review (PDR, CDR, FRR) will be provided action items to be completed following their review and will be required to address action items in a delta review session. After the delta session, the NASA management panel will meet to determine the teams' status in the program, and the teams will be notified shortly thereafter.	N/A				
1.8.	All deliverables shall be in PDF format.	The team shall submit deliverables in PDF format.	I	The team lead, with assistance from other team members, will inspect submission documents.	IM	IP
1.9.	In every report, teams will provide a table of contents including major sections and their respective sub-sections.	A table of contents shall be included in every report.	I	The table of contents will be updated as the report is constructed, inspected by the team lead.	All	IP
1.10.	In every report, the team shall include the page number at the bottom of the page.	Page numbers shall be included on every report page.	I	The report outline will automatically include page numbers, inspected by the team lead.	All	IP
1.11.	The team shall provide any computer equipment necessary to perform a video teleconference with the review panel. This includes, but is not limited to: a computer system, video camera, speaker telephone, and a sufficient Internet connection. Cellular phones should be used for speakerphone capability only as a last resort.	Team members shall procure and test video conferencing equipment ahead of review panel meetings.	T	The team will test all equipment prior to presentations and ensure it meets required parameters.	All	IP
1.12.	All teams attending Launch Week shall be required to use the launch pads provided by Student Launch's launch services provider. No custom pads will be permitted at the NASA Launch Complex. At launch, 8-foot 1010 rails and 12-foot 1515 rails will be provided. The launch rails will be canted 5 – 10 degrees away from the crowd on Launch Day. The exact cant will depend on Launch Day wind conditions.	The launch vehicle shall be designed to integrate with provided launch pads.	I	The Vehicle Team Lead will ensure the vehicle is constructed to integrate with the provided launch rails.	SS	IP
1.13.	Each team shall identify a "mentor." A mentor is defined as an adult who is included as a team member, who will be supporting the team (or multiple teams) throughout the project year, and may or may not be affiliated with the school, institution, or organization. The team mentor shall not be a student team member. The mentor shall maintain a current certification, and be in good standing, through the National Association of Rocketry (NAR) or Tripoli Rocketry Association (TRA) for the motor impulse of the launch vehicle and must have flown and successfully recovered (using electronic, staged recovery) a minimum of two flights in this or a higher impulse class, prior to PDR. The mentor is designated as the individual owner of the rocket for liability purposes and must travel with the team to Launch Week. One travel stipend will be provided per mentor regardless of the number of teams he or she supports. The stipend will only be provided if the team passes FRR and the team and mentor attend Launch Week in April.	The team shall identify their rocketry mentor in the General Information section of all reports.	I	The mentor certifications have been inspected by the team lead to ensure they are up-to-date and meet requirements.	IM	V
1.14.	Teams will track and report the number of hours spent working on each milestone.	Team members shall record their weekly hours in a common spreadsheet.	I	The team lead will inspect the hours tracking spreadsheet and ensure all members are tracking hours in a timely and accurate manner.	IM	IP
NASA Vehicle Requirements		Design Requirements	Method	Verification Details	Team Member(s)	Status
Req #	Description					
2.1.	The vehicle shall deliver the payload to an apogee altitude between 4,000 and 6,000 feet above ground level (AGL). Teams flying below 3,500 feet or above 6,500 feet on their competition launch will receive zero altitude points towards their overall project score and will not be eligible for the Altitude Award.	Team simulations will utilize confidence intervals to confirm that the vehicle will deliver the payload to the desired apogee range regardless of environmental conditions.	AD	The vehicle altitude will be analyzed with redundant simulation methods, and demonstrated in a full-scale test flight.	SH	NV
2.2.	Teams shall declare their target altitude goal at the CDR milestone. The declared target altitude shall be used to determine the team's altitude score.	The target apogee shall be specified in VADL derived requirement 1.1.1.	AI	The target altitude has been analyzed with redundant simulation methods and reported in the CDR submission.	SH	V
2.3	The launch vehicle shall be designed to be recoverable and reusable. Reusable is defined as being able to launch again on the same day without repairs or modifications.	The launch vehicle shall be designed to be reusable, and the payload will be entirely reusable with battery recharging.	AD	Calculations, computational models, and flight demonstrations will verify the launch vehicle is recoverable and reusable.	All	IP
2.4.	The launch vehicle shall have a maximum of four (4) independent sections. An independent section is defined as a section that is either tethered to the main vehicle or is recovered separately from the main vehicle using its own parachute.	The launch vehicle shall have four independent sections.	I	The launch vehicle design has been inspected to confirm there are 3 independent sections pre-jettison and 4 post-jettison.	All	V

2.4.1	Coupler/airframe shoulders which are located at in-flight separation points shall be at least two airframe diameters in length. (One body diameter of surface contact with each airframe section.)	The separation point couplers shall all be designed with 2 body diameter length.	I	The launch vehicle design has been inspected to ensure separation point couplers and shoulders are all at least 2 body diameter in length.	All	V
2.4.2	Coupler/airframe shoulders which are located at non-in-flight separation points shall be at least 1.5 airframe diameters in length. (0.75 body diameter of surface contact with each airframe section.)	If non-separation point couplers shall all be designed with 1.5 body diameter length.	I	The launch vehicle design will be inspected to ensure non-separation point couplers and shoulders are all at least 1.5 body diameter in length.	All	IP
2.4.3	Nosecone shoulders which are located at in-flight separation points shall be at least 1/2 body diameter in length.	There is no nosecone shoulder in the full-scale vehicle design.				
2.5.	The launch vehicle shall be capable of being prepared for flight at the launch site within 2 hours of the time the Federal Aviation Administration flight waiver opens.	The launch vehicle shall be constructed in a modular fashion to allow for rapid assembly.	D	The launch vehicle will be prepared on a full-scale test to ensure the vehicle can be prepared within 2 hours.	All	NV
2.6.	The launch vehicle and payload shall be capable of remaining in launch-ready configuration on the pad for a minimum of 3 hours without losing the functionality of any critical on-board components, although the capability to withstand longer delays is highly encouraged.	The payload shall be equipped with sufficient battery power and cooling methods to ensure launch-ready configuration for greater than two hours.	AT	The launch vehicle and payload will be analyzed and tested to ensure its launch-ready configuration can be maintained for at least 3 hours.	All	IP
2.7.	The launch vehicle shall be capable of being launched by a standard 12-volt direct current firing system. The firing system shall be provided by the NASA-designated launch services provider.	The launch vehicle will be designed to integrate with a 12-volt direct current firing system.	D	Flight demonstrations will validate that a launch can occur with a 12-volt direct current firing system.	All	NV
2.8.	The launch vehicle shall require no external circuitry or special ground support equipment to initiate launch (other than what is provided by the launch services provider).	The launch vehicle shall be designed to operate without reliance on special ground support equipment.	ID	All design and integration plans will be inspected to ensure no external support equipment is needed, validated by demonstration flights.	PC, SS	IP
2.9	Each team shall use commercially available ematches or igniters. Hand-dipped igniters shall not be permitted.	The team shall use MJG Firewire electric matches to ignite blast charges.	I	The team lead and student safety officer will inspect the electric match selection to ensure that they are commercially available and not hand-dipped.	IM, ZS	V
2.10.	The launch vehicle shall use a commercially available solid motor propulsion system using ammonium perchlorate composite propellant (APCP) which is approved and certified by the National Association of Rocketry (NAR), Tripoli Rocketry Association (TRA), and/or the Canadian Association of Rocketry (CAR).	The launch vehicle shall use the TRA-certified Cesaroni L1720 motor.	I	Motor selection and vehicle plans have been inspected to ensure the rocket motor meets NASA requirements.	SS, SH	V
2.10.1	Final motor choices shall be declared by the Preliminary Design Review (PDR) milestone.	The team shall verify their final motor selection by PDR.	I	The team lead has inspected that the final motor choice was included in the PDR report.	IM	V
2.10.2	Any motor change after PDR shall be approved by the NASA management team or NASA Range Safety Officer (RSO). Changes for the sole purpose of altitude adjustment shall not be approved. A scoring adjustment against the team's overall score shall be incurred when a motor change is made after the PDR milestone. The only exception is teams switching to their secondary motor choice provided the primary motor choice is unavailable due to a motor shortage.	The team shall present any motor changes after PDR to the NASA RSO for approval.	I	Any motor changes requested will be sent to NASA's RSO for approval.	IM	NV
2.11.	The launch vehicle shall be limited to a single motor propulsion system.	The launch vehicle shall be a single stage.	I	Inspection of vehicle designs confirm the vehicle is a single stage.	All	V
2.12.	The total impulse provided by a College or University launch vehicle shall not exceed 5,120 Newton-seconds (L-class).	The motor selected will not exceed the maximum impulse limits.	I	Inspection of the motor selection confirms the motor does not exceed L-class limitations.	SS, SH	V
2.13.	Pressure vessels on the vehicle must be approved by the RSO and shall meet the following criteria:	The launch vehicle and payload shall not include any pressure vessels.				
2.13.1.	The minimum factor of safety (Burst or Ultimate pressure versus Max Expected Operating Pressure) will be 4:1 with supporting design documentation included in all milestone reviews.					
2.13.2.	Each pressure vessel shall include a pressure relief valve that sees the full pressure of the tank and is capable of withstanding the maximum pressure and flow rate of the tank.					
2.13.3.	The full pedigree of the tank shall be described, including the application for which the tank was designed and the history of the tank. This will include the number of pressure cycles put on the tank, the dates of pressurization/depressurization, and the name of the person or entity administering each pressure event.					
2.14.	The launch vehicle shall have a minimum static stability margin of 2.0 at the point of rail exit. Rail exit is defined at the point where the forward rail button loses contact with the rail.	The design of the rocket shall estimate the SSM to exceed 2.0 at the rail exit, and then verified by two methods of simulation.	A	Simulations confirm that the static stability margin exceeds 2.0 at the launch rail.	SH	V
2.15.	The launch vehicle shall have a minimum thrust to weight ratio of 5.0 : 1.0.	The motor choice shall provide a high enough thrust relative to the vehicle weight to meet the requirement.	A	Simulations confirm that the thrust to weight ratio exceeds NASA requirements.	SH	V
2.16.	Any structural protuberance on the rocket shall be located aft of the burnout center of gravity. Camera housings will be exempted, provided the team can show that the housing(s) causes minimal aerodynamic effect on the rocket's stability.	There are no structural protuberances in the full-scale vehicle design.				
2.17.	The launch vehicle shall accelerate to a minimum velocity of 52 fps at rail exit.	The simulations shall calculate velocity at rail exit.	ADT	Simulations confirm the minimum velocity is exceeded, and flight tests will confirm the flight velocity at the rail exit.	SH	IP
2.18.	All teams shall successfully launch and recover a subscale model of their rocket. Success of the subscale is at the sole discretion of the NASA review panel. The subscale flight may be conducted at any time between proposal award and the CDR submission deadline. Subscale flight data shall be reported in the CDR report and presentation at the CDR milestone. Subscale are required to use a minimum motor impulse class of E (Mid Power motor).	The team shall conduct a subscale launch on Dec. 7, 2024.	ID	The team conducted a subscale flight test on December 7 and ensured it was reported in the CDR report.	All	V
2.18.1.	The subscale model should resemble and perform as similarly as possible to the full-scale model; however, the full-scale will not be used as the subscale model.	The subscale shall be designed and built to represent a scaled version of the full-scale vehicle.	ID	The subscale was designed and built, with its performance validated through a successful flight.	All	V
2.18.2.	The subscale model shall carry an altimeter capable of recording the model's apogee altitude.	The subscale model shall carry an onboard altimeter.	I	Subscale inspection confirmed redundant Entacore AIM altimeters were used to capture altitude data.	SS, SC	V
2.18.3.	The subscale rocket shall be a newly constructed rocket, designed and built specifically for this year's project.	The subscale rocket shall be designed from scratch and constructed entirely separately from the full-scale vehicle.	I	The subscale was verified by the team as a newly constructed vehicle.	All	V
2.18.4.	Proof of a successful flight shall be supplied in the CDR report.	Subscale proof of flight requirements shown below.				

2.18.4.1	Altimeter flight profile graph(s) OR a quality video showing successful launch, recovery events, and landing as deemed by the NASA management panel are acceptable methods of proof. Altimeter flight profile graph(s) that are not complete (liftoff through landing) will not be accepted.	The team shall include altimeter flight profile graphs in the CDR report.	I	The team lead will inspect the CDR report to ensure that altimeter flight profile graphs were included.	IM	V
2.18.4.2	Quality pictures of the as landed configuration of all sections of the launch vehicle shall be included in the CDR report. This includes, but is not limited to: nosecone, recovery system, airframe, and booster.	The team shall include pictures of the rocket and payload sections in their landed configurations in the CDR report.	I	The team lead inspected the CDR report to ensure that the required pictures were included.	IM	V
2.18.5.	The subscale rocket shall not exceed 75% of the dimensions (length and diameter) of your designed full-scale rocket. For example, if your full-scale rocket is a 4" diameter 100" length rocket your subscale shall not exceed 3" diameter and 75" in length.	The subscale rocket shall be designed such that the length and diameter do not exceed 75% of the full-scale rocket.	I	Subscale design inspection confirmed the rocket's dimensions did not exceed 75% of the full-scale vehicle	SS, SC, IM	V
2.19.	All teams shall complete demonstration flights as outlined below.					
	Verification is outlined as follows.					
2.19.1.	Vehicle Demonstration Flight — All teams shall successfully launch and recover their full-scale rocket prior to FRR in its final flight configuration. The rocket flown shall be the same rocket to be flown for their competition launch. The purpose of the Vehicle Demonstration Flight is to validate the launch vehicle's stability, structural integrity, recovery systems, and the team's ability to prepare the launch vehicle for flight. A successful flight is defined as a launch in which all hardware is functioning properly (drogue chute at apogee, main chute at the intended lower altitude, functioning tracking devices, etc.). The following criteria shall be met during the full-scale demonstration flight:	The team shall launch and recover their full-scale rocket prior to FRR.	ID	The team will verify the launch vehicle is the same as the launch day vehicle, and the flight demonstration will validate success.	All	NV
2.19.1.1.	The vehicle and recovery system shall have functioned as designed.	The vehicle and recovery system data shall be analyzed to ensure proper functionality.	ID	The vehicle and recovery performance will be inspected and validated through the demonstration flight to ensure proper functionality.	All	NV
2.19.1.2.	The full-scale rocket shall be a newly constructed rocket, designed and built specifically for this year's project.	The team shall design and construct an entirely new full-scale rocket.	I	The team will inspect the launch vehicle to ensure it is a newly constructed rocket.	All	NV
2.19.1.3.	The payload does not have to be flown during the full-scale Vehicle Demonstration Flight. The following requirements still apply:	The team shall fly its payload during the full-scale demonstration flight.	I	The team will inspect the launch vehicle and payload to ensure it is the same payload being flown during the competition launch.	All	NV
2.19.1.3.1.	If the payload is not flown, mass simulators shall be used to simulate the payload mass.					
2.19.1.3.2.	The mass simulators shall be located in the same approximate location on the rocket as the missing payload mass.					
2.19.1.4.	If the payload changes the external surfaces of the rocket (such as camera housings or external probes) or manages the total energy of the vehicle, those systems will be active during the full-scale Vehicle Demonstration Flight.	The payload will not include external surfaces.				
2.19.1.5.	Teams shall fly the competition launch motor for the Vehicle Demonstration Flight. The team may request a waiver for the use of an alternative motor in advance if the home launch field cannot support the full impulse of the competition launch motor or in other extenuating circumstances.	The team shall fly the L1720 motor for the Vehicle Demonstration Flight.	ID	The vehicle lead will inspect that the launch motor is the L1720 for the Vehicle Demonstration Flight.	SS	NV
2.19.1.6.	The vehicle will be flown in its fully ballasted configuration during the full-scale test flight. Fully ballasted refers to the maximum amount of ballast that will be flown during the competition launch flight. Additional ballast shall not be added without a re-flight of the full-scale launch vehicle.	The team does not intend to fly with ballast for the full-scale vehicle.				
2.19.1.7.	After successfully completing the full-scale demonstration flight, the launch vehicle or any of its components shall not be modified without the concurrence of the NASA management team or Range Safety Officer (RSO).	Any changes made after the full-scale demonstration flight will only be performed with concurrence of the NASA RSO.	I	The team will confirm that no launch vehicle components are modified without NASA RSO approval.	All	NV
2.19.1.8.	Proof of a successful flight shall be supplied in the FRR report.	Proof of successful full-scale flight requirements shown below.				
2.19.1.8.1.	Altimeter flight profile data output with accompanying altitude and velocity versus time plots is required to meet this requirement. Altimeter flight profile graph(s) that are not complete (liftoff through landing) shall not be accepted.	The team shall provide proof of a successful flight in FRR with altimeter flight profile data and required analysis.	I	The team lead will inspect that the FRR flight data is included in the FRR report.	IM	NV
2.19.1.8.2.	Quality pictures of the as landed configuration of all sections of the launch vehicle shall be included in the FRR report. This includes, but is not limited to: nosecone, recovery system, airframe, and booster.	The team shall provide pictures of the landed rocket and payload sections in the FRR report.	I	The team lead will inspect that the required pictures are included in the FRR report.	IM	NV
2.19.1.8.3.	Raw altimeter data shall be submitted in .csv or .xlsx format.	The team shall submit raw altimeter data in .csv or .xlsx format.	I	The team lead will inspect that the raw altimeter data has been submitted in the proper format.	IM	NV
2.19.1.9.	Vehicle Demonstration flights shall be completed by the FRR submission deadline. No exceptions will be made. If the Student Launch office determines that a Vehicle Demonstration Re-flight is necessary, then an extension may be granted. THIS EXTENSION IS ONLY VALID FOR RE-FLIGHTS, NOT FIRST TIME FLIGHTS. Teams completing a required re-flight shall submit an FRR Addendum by the FRR Addendum deadline.	The team shall complete it full-scale flight prior to FRR and request an extension if re-flight is deemed to be necessary.	ID	The team will comply with FRR requirements and ensure that the full-scale flight and potential reflight is successfully completed.	All	NV
2.19.2	Payload Demonstration Flight — All teams shall successfully launch and recover their full-scale rocket containing the completed payload prior to the Payload Demonstration Flight deadline. The rocket flown shall be the same rocket to be flown as their competition launch. The purpose of the Payload Demonstration Flight is to prove the launch vehicle's ability to safely retain the constructed payload during flight and to show that all aspects of the payload perform as designed. A successful flight is defined as a launch in which the rocket experiences stable ascent and the payload is fully retained until it is deployed (if applicable) as designed. The following criteria shall be met during the Payload Demonstration Flight:	The team shall fly a payload demonstration flight in Spring 2025 at a partner launch location.	ID	The completed flight vehicle and payload will be inspected to ensure it is ready for launch, and the launch demonstration will validate that the system meets all requirements.	All	NV
2.19.2.1.	The payload shall be fully retained until the intended point of deployment (if applicable), all retention mechanisms shall function as designed, and the retention mechanism shall not sustain damage requiring repair.	The payload shall be retained via the LSM until 300 ft.	ID	The team will inspect that all components are secured and that the payload is jettisoned at 300 ft, demonstrated through flight.	All	NV
2.19.2.2.	The payload flown shall be the final, active version.	The team shall complete a flight in its final configuration prior to the FRR deadline and follow proper NASA procedures for vehicle re-flights.	I	The team will inspect the payload and ensure it is ready to be flown in its final version.	All	NV
2.19.2.3.	If the above criteria are met during the original Vehicle Demonstration Flight, occurring prior to the FRR deadline and the information is included in the FRR package, the additional flight and FRR Addendum are not required.	The team shall complete a flight in its final configuration prior to the FRR deadline and follow proper NASA procedures for vehicle re-flights.	I	FRR requirements inspection will ensure a successful demonstration flight, and, if required, inspection of an FRR addendum will ensure a successful flight as well.	All	NV
2.19.2.4.	Payload Demonstration Flights shall be completed by the FRR Addendum deadline. NO EXTENSIONS WILL BE GRANTED.					NV

2.20.	An FRR Addendum shall be required for any team completing a Payload Demonstration Flight or NASA-required Vehicle Demonstration Re-flight after the submission of the FRR Report.	The team shall complete a flight in its final configuration prior to the FRR deadline and follow proper NASA procedures for vehicle re-flights.	I	The team will verify that all requirements of the FRR and, if applicable, FRR addendum are validated.	All	NV
2.20.1.	Teams required to complete a Vehicle Demonstration Re-Flight and failing to submit the FRR Addendum by the deadline will not be permitted to fly a final competition launch.					NV
2.20.2.	Teams who complete a Payload Demonstration Flight which is not fully successful may petition the NASA RSO for permission to fly the payload during launch week. Permission will not be granted if the RSO or the Review Panel have any safety concerns.					NV
2.21.	The team's name and Launch Day contact information shall be in or on the rocket airframe as well as in or on any section of the vehicle that separates during flight and is not tethered to the main airframe. This information shall be included in a manner that allows the information to be retrieved without the need to open or separate the vehicle.	The team shall provide the team name and contact information on each section of the airframe.	I	The Vehicle Lead will ensure that the required information is present on the vehicle in the manner requested.	SS	NV
2.22.	All Lithium Polymer batteries shall be sufficiently protected from impact with the ground and will be brightly colored, clearly marked as a fire hazard, and easily distinguishable from other payload hardware.	The team shall protect lithium polymer batteries and provide identifiable colorings and fire hazard markings.	I	The Avionics Lead will ensure that all LiPo batteries are sufficiently protected and marked, inspected by the safety lead.	SC, ZS	IP
2.23.	Vehicle Prohibitions:					
2.23.1.	The launch vehicle shall not utilize forward firing motors.					
2.23.2.	The launch vehicle shall not utilize motors that expel titanium sponges (Sparky, Skidmark, MetalStorm, etc.)	The launch vehicle shall use a single, rear-facing, solid rocket motor.	I	The Vehicle Lead/Safety Lead has confirmed that the motor selection and integration does not violate any vehicle prohibitions.	SS, ZS	V
2.23.3.	The launch vehicle shall not utilize hybrid motors.					
2.23.4.	The launch vehicle shall not utilize a cluster of motors.					
2.23.5.	The launch vehicle shall not utilize friction fitting for motors.					
2.23.6.	The launch vehicle will not exceed Mach 1 at any point during flight.	The launch vehicle shall be designed to ensure all vehicle prohibition requirements are met.	I	The Simulations Lead will verify that the launch meets these stated requirements.	SH	IP
2.23.7.	Vehicle ballast shall not exceed 10% of the total unballasted weight of the rocket as it would sit on the pad (i.e. a rocket with an unballasted weight of 40 lbs. on the pad may contain a maximum of 4 lbs. of ballast).					
2.23.8.	Transmissions from onboard transmitters, which are active at any point prior to landing, shall not exceed 250 mW of power (per transmitter).	The transmitter shall be selected to ensure that it abides by prohibition requirements.	IT	The Payload Lead will inspect and test relevant systems to ensure these requirements are satisfied.	RN	IP
2.23.9.	Transmitters shall not create excessive interference. Teams shall utilize unique frequencies, handshake/passcode systems, or other means to mitigate interference caused to or received from other teams.	The transmitter shall be selected to ensure that it abides by prohibition requirements.	IT	The Payload Lead will inspect and test relevant systems to ensure these requirements are satisfied.	RN	IP
2.23.10.	Excessive and/or dense metal shall not be utilized in the construction of the vehicle. Use of light-weight metal will be permitted but limited to the amount necessary to ensure structural integrity of the airframe under the expected operating stresses.	The launch vehicle shall be designed with a combination of aluminum and rapid prototyping materials.	I	The team has inspected the design of the vehicle to ensure proper material selection, and FEA has been performed to verify the structural integrity of load-bearing components.	All	V
NASA Recovery System Requirements						
Req #	Description	Design Requirements	Method	Verification Details	Team Member(s)	Status
3.1	The full scale launch vehicle shall stage the deployment of its recovery devices, where a drogue parachute is deployed at apogee, and a main parachute is deployed at a lower altitude. Tumble or streamer recovery from apogee to main parachute deployment is also permissible, provided that kinetic energy during drogue stage descent is reasonable, as deemed by the RSO.	The launch vehicle drogue parachute will be designed to deploy within one second of apogee, and the main parachute will be designed to deploy at 600 ft.	AD	The altimeters in the avionics bay will be configured such that the blast charges responsible for parachute deployment are triggered at apogee for drogue and 600 ft for main, which will be demonstrated through test flights.	SC	IP
3.1.1.	The main parachute shall be deployed no lower than 500 feet.	The main parachute deployment shall occur at 600 ft.	AD	The altitude sensing performance of the altimeter will be tested prior to launch, and demonstrated with full-scale flights.	SC	NV
3.1.2.	The apogee event shall contain a delay of no more than 2 seconds.	The apogee event shall occur within one second of apogee.	AD	The apogee event timing will be analyzed by altimeter data of a full-scale launch and the event timing will be verified.	SC	NV
3.1.3.	Motor ejection is not a permissible form of primary or secondary deployment.	The mission sequence shall not involve motor ejection.				
3.2.	Each team shall perform a successful ground ejection test for all electronically initiated recovery events prior to the initial flights of the subscale and full scale vehicles.	A ground test of parachute ejection shall be performed in a safe outdoor location, with events monitored by the rocketry mentor and Student Safety Officer.	TD	The team performed successful ejection tests of the blast charges prior to the subscale flight. The team will perform the required ground ejection tests (drogue and main) and record data to validate performance for the full-scale flights.	All	IP
3.3.	Each independent section of the launch vehicle will have a maximum kinetic energy of 75 ft-lbf at landing. Teams whose heaviest section of their launch vehicle, as verified by vehicle demonstration flight data, stays under 65 ft-lbf will be awarded bonus points.	The maximum kinetic energy shall be calculated in simulations and design parameters shall incorporate the maximum kinetic energy limitations.	AD	Flight simulations have verified that the launch vehicle landing energy does not exceed the limits, and a flight demonstration will further verify.	All	IP
3.4.	The recovery system shall contain redundant, commercially available barometric altimeters that are specifically designed for initiation of rocketry recovery events. The term "altimeters" includes both simple altimeters and more sophisticated flight computers.	The vehicle avionics shall contain two EasyMini altimeters for the recovery system.	I	The team will inspect the design and will inspect the constructed vehicle to ensure the recovery system contains redundant altimeters.	All	IP
3.5.	Each altimeter shall have a dedicated power supply, and all recovery electronics shall be powered by commercially available batteries.	Each altimeter shall be powered by independent, commercially available 9V batteries.	IT	The Avionics Lead has ensured each altimeter has independent power supplies, and will test that they are sufficient to power the altimeters.	SC	IP
3.6.	Each altimeter shall be armed by a dedicated mechanical arming switch that is accessible from the exterior of the rocket airframe when the rocket is in the launch configuration on the launch pad.	The rocket body tube shall contain external entry points for altimeter access on the launch pad.	ID	The vehicle design has been inspected to ensure the altimeters are externally accessible, and demonstration flights will validate successful arming systems.	SC	IP
3.7.	Each arming switch shall be capable of being locked in the ON position for launch (i.e. cannot be disarmed due to flight forces).	The avionics arming switch shall lock into the ON position for launch via screw switches.	ID	Screw switches have been used in the avionics design. The Avionics Lead will ensure the arming switch can be locked in the ON position during demonstration flights.	SC	IP
3.8.	The recovery system, GPS and altimeters, and electrical circuits shall be completely independent of any payload electrical circuits.	The recovery system electronic shall be housed in the avionics bay, separated from payload circuits.	I	Design plan inspections by the Vehicle Lead and Payload Lead have ensured that the electrical circuits are independent of payload circuits.	SS, RN	V
3.9.	Removable shear pins shall be used for both the main parachute compartment and the drogue parachute compartment.	Shear pins shall hold all three rocket sections together prior to deployment.	AD	Mathematical calculations have verified the shear pin requirements, and ejection tests will confirm they meet requirements.	All	IP

3.10.	Bent eyebolts shall not be permitted in the recovery subsystem.	The team shall not include bent eyebolts in the recovery subsystem.	I	The vehicle lead has inspected and confirmed that no bent eyebolts are included in the design of the recovery subsystem.	SS	V
3.11.	The recovery area shall be limited to a 2,500 ft. radius from the launch pads.	The flight vehicle shall be designed to land within a 2,500 ft. radius of the launch pad	AD	Redundant flight simulations have confirmed that the vehicle and payload drift do not exceed 2,500 ft, and flight demonstrations will validate the results.	SH	IP
3.12.	Descent time of the launch vehicle shall be limited to 90 seconds (apogee to touch down). Teams whose launch vehicle descent, as verified by vehicle demonstration flight data, stays under 80 seconds will be awarded bonus points.	The descent time shall be calculated in simulations and design parameters shall change to accommodate this requirement if needed .	AD	Redundant flight simulations have confirmed vehicle and payload descent time do not exceed 90 seconds, and flight demonstrations will validate the results.	SH	IP
3.13.	An electronic GPS tracking device shall be installed in the launch vehicle and will transmit the position of the tethered vehicle or any independent section to a ground receiver.	A GPS device shall be installed in the launch vehicle.	TD	The GPS tracking device will be tested utilizing ground tests, and demonstrations will validate the results.	AD	IP
3.13.1.	Any rocket section or payload component, which lands untethered to the launch vehicle, shall contain an active electronic GPS tracking device.	The payload shall contain a GPS tracking device.	TD	The GPS tracking device will be tested utilizing ground tests, and demonstrations will validate the results.	AD	IP
3.13.2.	The electronic GPS tracking device(s) shall be fully functional during the official competition launch.	The GPS device shall communicate with the base station on launch day.	ID	The GPS device will be inspected to ensure it is active, and validated as functional in demonstration flights.	AD	NV
3.14.	The recovery system electronics shall not be adversely affected by any other on-board electronic devices during flight (from launch until landing).	The Recovery system components shall all be housed in an independent avionics bay and shielded from all other onboard electronics to prevent interference.	IT	The vehicle designs will be inspected to ensure recovery electronics are sufficiently isolated from other electronics, and testing will validate the reliability of the designs.	All	NV
3.14.1.	The recovery system altimeters shall be physically located in a separate compartment within the vehicle from any other radio frequency transmitting device and/or magnetic wave producing device.	The Recovery system components shall all be housed in an independent avionics bay and shielded from all other onboard electronics to prevent interference.	IT	The vehicle designs have been inspected to confirm the altimeters are in a separate compartment, and testing will validate they are shielded from wave producing devices.	SS, SC	IP
3.14.2.	The recovery system electronics will be shielded from all onboard transmitting devices to avoid inadvertent excitation of the recovery system electronics.	The Recovery system components shall all be housed in an independent avionics bay and shielded from all other onboard electronics to prevent interference.	ID	The vehicle designs have been inspected to ensure proper isolation and shielding, and demonstration flights will validate reliability.	All	IP
3.14.3.	The recovery system electronics shall be shielded from all onboard devices which may generate magnetic waves (such as generators, solenoid valves, and Tesla coils) to avoid inadvertent excitation of the recovery system.	The Recovery system components shall all be housed in an independent avionics bay and shielded from all other onboard electronics to prevent interference.				
3.14.4.	The recovery system electronics shall be shielded from any other onboard devices which may adversely affect the proper operation of the recovery system electronics.	The Recovery system components shall all be housed in an independent avionics bay and shielded from all other onboard electronics to prevent interference.				
NASA Payload Experiment Requirements		Design Requirements	Method	Verification Details	Team Member(s)	Status
Req #	Description					
4.1.	Teams are tasked with designing, building, and flying a STEMnaut flight capsule capable of safely retaining four STEMnauts and transmitting, via radio frequency, relevant rocket and STEMnaut landing site data to a NASA-owned receiver located at the launch site. STEMnauts are physical representations of the crew on-board the rocket. The method(s)/design(s) utilized to complete the payload mission shall be at the team's discretion and will be permitted so long as the designs are deemed safe, obey FAA and legal requirements, and adhere to the intent of the challenge. NASA reserves the right to require modifications to a proposed payload.	The team shall design, build, and fly a STEMnaut flight capsule to complete the 2025 NASA Student Launch Competition challenge.	ADIT	Analysis, demonstration, inspection, and testing will all be undertaken in order to satisfy the core NASA mission requirements.	All	IP
4.2.	STEMCRaFT Mission Requirements:					
	Mission requirements are met as follows.					
4.2.1.	Teams shall choose a minimum of 3 pieces of data from the below list to a maximum of 8 to transmit to the NASA receiver. Temperature of landing site • Apogee reached • Battery check/power status • Orientation of on-board STEMnauts • Time of landing • Maximum velocity • Landing velocity, G-forces sustained • Calculated STEMnaut crew survivability	The team shall implement the data collection and transmission systems to collect and transmit a minimum of 3 pieces of data from the NASA-provided list.	ADIT	Analysis, demonstration, inspection, and testing will all be performed to ensure completion of the data collection and RF transmission mission.	All	IP
4.2.2.	The payload shall not have any protrusions from the vehicle prior to apogee that extends beyond a quarter inch exterior to the airframe.	The payload shall be designed such that it can be stowed entirely within the nose cone and payload bay.	I	The payload lead and vehicle lead have inspected the payload design to confirm that no protrusions exist.	RN, SS	V
4.2.3.	Payload shall transmit on the 2-M band. A specific frequency shall be given to the teams later. NASA shall use the FTM-300DR transceiver.	The RF transmission system shall be designed such that it is capable of transmitting on the designated band.	DT	The RF transmission system will be thoroughly tested on the ground and demonstrated on the subscale and full-scale flights.	PC	IP
4.2.4.	All transmissions shall start and stop with team member call sign.	The transmission signals shall be configured to begin and end with the team member call sign.	DT	The transmission signals will be confirmed to contain the call sign through ground testing and flight demonstrations.	PC	IP
4.2.5.	Teams shall submit a list of what data they will attempt to transmit by NASA receiver by March 17.	The team shall finalize the data selections no later than March 17	D	The electrical lead will finalize the data list, and the team lead will confirm the information is sent to NASA prior to March 17.	PC, IM	IP
4.2.6.	Teams shall transmit with a maximum of 5W and transmissions shall not occur prior to landing.	The transmission system shall be designed such that its power usage does not exceed 5W nor begin transmitting prior to landing.	DT	Thorough testing will be conducted to verify power usage stays within the specified limit; flight demonstrations will confirm that transmission does not occur prior to landing.	PC	IP

4.2.6.1.	Teams shall not transmit on the specified NASA frequency on launch day prior to landing.	The team shall refrain from transmitting on the specified NASA frequency on launch day prior to the payload mission occurring.	D	The team will verify this requirement by not transmitting on the NASA frequency at any time on launch day prior to landing.	All	NV
4.3.	General Payload Requirements:	General payload requirements are met as follows.				
4.3.1.	Black Powder and/or similar energetics are only permitted for deployment of in-flight recovery systems. Energetics will not be permitted for any surface operations.	Energetics shall only be used for deployment of launch vehicle recovery systems.	I	The team has inspected system designs to ensure energetics are used in the proper manner.	All	V
4.3.2.	Teams shall abide by all FAA and NAR rules and regulations.	The team shall follow all FAA and NAR rules and regulations, overseen by the Student Safety Officer.	I	The team will continuously self-inspect procedures to confirm they abide by regulations.	All	IP
4.3.3.	Any experiment element that is jettisoned during the recovery phase shall receive real-time RSO permission prior to initiating the jettison event, unless exempted from the requirement at the CDR milestone by NASA.	The lander separation mechanism shall be designed such that it receives a manual release signal following RSO approval.	DT	The team will demonstrate through drop tests and flight demonstrations that the lander separation mechanism executes following manual receipt of the separation signal.	All	IP
4.3.4.	Unmanned aircraft system (UAS) payloads, if designed to be deployed during descent, shall be tethered to the vehicle with a remotely controlled release mechanism until the RSO has given permission to release the UAS.	The payload shall be structurally retained to the vehicle.	I	The team will inspect the payload to ensure it is retained to the vehicle prior to separation and prove through flight demonstrations.	All	IP
4.3.5.	Teams flying UASs will abide by all applicable FAA regulations, including the FAA's Special Rule for Model Aircraft (Public Law 112-95 Section 336; see https://www.faa.gov/uas/faqs).	The team shall follow all FAA and NAR rules and regulations, overseen by the Student Safety Officer.	I	The team will continuously self-inspect procedures to confirm they abide by regulations.	All	IP
4.3.6.	Any UAS weighing more than .55 lbs. shall be registered with the FAA and the registration number marked on the vehicle.	The team shall register the lander with the FAA and mark the registration number on the vehicle.	I	The team will inspect that the payload is registered and the number is marked.	All	IP
NASA Safety Requirements						
Req #	Description	Design Requirements	Method	Verification Details	Team Member(s)	Status
5.1.	Each team will use a launch and safety checklist. The final checklists will be included in the FRR report and used during the Launch Readiness Review (LRR) and any Launch Day operations.	The team shall develop a launch and safety checklist for FRR and launch.	I	The team will inspect all launch and safety procedures and ensure proper checklists are developed for FRR and LRR.	IM	IP
5.2.	Each team shall identify a student safety officer who will be responsible for all items in section 5.3.	The team shall identify a student safety officer.	I	The team has designated Zachary Stutsman as the Student Safety Officer.	IM	V
5.3.	The role and responsibilities of the safety officer will include, but are not limited to:	Verifications are documented for each requirement below.				
5.3.1.	Monitor team activities with an emphasis on safety during:					
5.3.1.1.	Design of vehicle and payload	The Student Safety Officer shall ensure designs comply with NASA, local, state, NAR, and FAA rules and regulations.	I	The team will self-inspect its practices to ensure all rules and regulations are followed.	ZS, IM	IP
5.3.1.2.	Construction of vehicle and payload components	The Student Safety Officer shall ensure construction protocols are written and followed during construction.	I	The team will self-inspect its practices to ensure all construction protocols are followed.	ZS, IM	IP
5.3.1.3.	Assembly of vehicle and payload	The Student Safety Officer shall ensure assembly protocols are written and followed during assembly.	I	The team will self-inspect its practices to ensure all assembly protocols are followed.	ZS, IM	IP
5.3.1.4.	Ground testing of vehicle and payload	The Student Safety Officer shall ensure that ground testing procedures are written and followed for all ground testing.	I	The team will self-inspect its practices to ensure all ground testing protocols are followed.	ZS, IM	IP
5.3.1.5.	Subscale launch test(s)	The Student Safety Officer shall produce subscale launch checklists and ensure procedures are followed.	I	The team has self-inspected its practices to ensure all subscale testing protocols were followed, with checklists verified for launches.	ZS, IM	V
5.3.1.6.	Full-scale launch test(s)	The Student Safety Officer shall produce full-scale launch checklists and ensure procedures are followed.	I	The team will self-inspect its practices to ensure all full-scale testing protocols are followed, with checklists verified for launches.	ZS, IM	NV
5.3.1.7.	Competition Launch	The Student Safety Officer shall produce and compile all launch day checklists and ensure procedures are followed during launch.	I	The team will self-inspect its practices to ensure all competition testing protocols are followed, with checklists verified for launches.	ZS, IM	NV
5.3.1.8.	Recovery activities	The Student Safety Officer shall develop a recovery checklist and ensure procedures are followed during launch.	I	The team self-inspected its practices to ensure all recovery protocols were followed for the subscale flight, with checklists verified for full-scale launches.	ZS, IM	IP
5.3.1.9.	STEM Engagement Activities	The Student Safety Officer shall communicate with educational outreach members to ensure STEM engagement activities follow proper safety protocol.	I	Safety procedures will be inspected prior to all engagement activities.	ZS, IM	IP
5.3.2.	Implement procedures developed by the team for construction, assembly, launch, and recovery activities.	The Student Safety Officer shall develop a comprehensive set of operational protocols to ensure safety in regards to personal protective equipment, operational checklists, and testing overview.	I	The safety procedures will all be followed by the team and enforced and updated by the Student Safety Officer.	ZS, IM	IP
5.3.3.	Manage and maintain current revisions of the team's hazard analyses, failure modes analyses, procedures, and MSDS/chemical inventory data.	The Student Safety Officer shall maintain a safety binder in the lab and digitally containing all required safety documentation.	I	The safety binder shall be inspected regularly to ensure all information is present and relevant.	ZS, IM	IP
5.3.4.	Assist in the writing and development of the team's hazard analyses, failure modes analyses, and procedures.	The Student Safety Officer shall lead the development of the team's hazard analyses, failure modes analyses, and procedures.	I	Hazard and failure analyses and procedures will be inspected by the team and lead by the Student Safety Officer.	ZS, IM	IP

5.4.	During test flights, teams will abide by the rules and guidance of the local rocketry club's RSO. The allowance of certain vehicle configurations and/or payloads at the NASA Student Launch does not give explicit or implicit authority for teams to fly those vehicle configurations and/or payloads at other club launches. Teams should communicate their intentions to the local club's President or Prefect and RSO before attending any NAR or TRA launch.	The Student Safety Officer shall communicate with the local NAR organization and rocketry mentor to ensure compliance with pertinent rules and regulations.	I	Procedural and system-wide inspections will take place to ensure the team abides by the rules and regulations of NAR and TRA launches.	ZS, IM	IP
5.5.	Teams will abide by all rules set forth by the FAA.	The Student Safety Officer shall ensure compliance with FAA rules are regulations.	I	Inspections will take place to confirm that procedures and mission profiles abide by FAA rules and regulations.	ZS, IM	IP
NASA Final Flight Requirements		Design Requirements	Method	Verification Details	Team Member(s)	Status
Req #	Description					
6.1.	NASA Launch Complex					
6.1.1.	Teams are not permitted to show up at the NASA Launch Complex outside of launch day without permission from the NASA management team.	The team shall abide by all NASA Launch Complex rules and regulations.	I	The team lead will continuously inspect activities throughout the Launch Week to verify compliance with rules and regulations.	IM	NV
6.1.2.	Teams shall complete and pass the Launch Readiness Review conducted during Launch Week.					
6.1.3.	The team mentor shall be present and oversee rocket preparation and launch activities.					
6.1.4.	The scoring altimeter shall be presented to the NASA scoring official upon recovery. The scoring altimeter shall be one of the altimeters used for recovery events.					
6.1.5.	Teams may launch only once. Any launch attempt resulting in the rocket exiting the launch pad, regardless of the success of the flight, will be considered a launch. Additional flights beyond the initial launch, will not be scored and will not be considered for awards.	The team shall abide by all NASA Launch Complex rules and regulations.		The team lead will continuously inspect activities throughout the Launch Week to verify compliance with rules and regulations.		
6.2.	Commercial Spaceport Launch Site					
6.2.1.	The launch shall occur at a NAR or TRA sanctioned and insured club launch. Exceptions may be approved for launch clubs who are not affiliated with NAR or TRA but provide their own insurance, such as the Friends of Amateur Rocketry. Approval for such exceptions shall be granted by NASA prior to the launch.	The team shall comply with all Commercial Spaceport Launch Site guidelines.	I	The team lead will actively monitor team activities during Launch Week to ensure adherence to all guidelines.	IM	NV
6.2.2.	Teams shall submit their rocket and payload to the launch site Range Safety Officer (RSO) prior to flying the rocket. The RSO shall inspect the rocket and payload for flight worthiness and determine if the project is approved for flight. The local RSO shall have final authority on whether the team's rocket and payload may be flown.					
6.2.3.	The team mentor shall be present and oversee rocket preparation and launch activities.					
6.2.4.	BOTH the team mentor and the Launch Control Officer shall observe the flight and report any off- nominal events during ascent or recovery on the Launch Certification and Observations Report.					
6.2.5.	The scoring altimeter shall be presented to BOTH the team's mentor and the Range Safety Officer. The scoring altimeter shall be one of the altimeters used for recovery events.	The team shall comply with all Commercial Spaceport Launch Site guidelines.	I	The team lead will actively monitor team activities during Launch Week to ensure adherence to all guidelines.	IM	NV
6.2.6.	The mentor, the Range Safety Officer, and the Launch Control Officer must be three separate individuals who must ALL complete the applicable sections of the Launch Certification and Observations Report. The Launch Certification and Observations Report document will be provided by NASA upon completion of the FRR milestone and shall be returned to NASA by the team mentor upon completion of the launch.					
6.2.7.	The Range Safety Officer and Launch Control Officer certifying the team's flight shall be impartial observers and shall not be affiliated with the team, individual team members, or the team's academic institution.					
6.2.8.	Teams may launch only once. Any launch attempt resulting in the rocket exiting the launch pad, regardless of the success of the flight, will be considered a launch. Additional flights beyond the initial launch will not be scored and will not be considered for awards.					

6.2.2 VADL Derived Requirements

Derivation

VADL has refined its list of derived requirements, initially presented at the PDR milestone, to reflect the project's maturation. Further design iterations and testing have informed the need for additional requirements, and these changes are reflected in the following tables. VADL is guided by the following criteria when deriving requirements.

- **Mission Purpose:** The mission purpose outlines the project's scope, with NASA's requirements shaping the design and objectives of the payload and vehicle. The team has aligned their requirements with NASA's directives.
- **Mission Constraints:** NASA's requirements impose constraints that must be addressed during the design and development of the product. For instance, NASA Requirement 2.13 mandates that the SSM of the vehicle be a minimum of 2.0 at rail exit. These constraints must be carefully evaluated to ensure that the derived requirements align with them.
- **Safety Considerations:** Safety is a top priority for VADL. The team is committed to proceeding with the launch only after thorough safety evaluations confirm that both the vehicle and payload are secure. This underscores the importance of incorporating safety measures into the technical requirements development process.

Along with these driving factors, an added emphasis has been placed on the requirements hierarchy. Since the NASA mission objective primarily dictates the payload design, the payload's requirements are especially important. As such, the requirements of the vehicle must be driven, at least in part, by those of the payload. For instance, the requirement of the payload to transmit data via RF informed the selection of fiberglass, an RF-friendly material, as the vehicle's airframe.

The categories used by the team to group requirements remains the same: General, Vehicle, Recovery, Payload, and Safety. The improved list of VADL derived requirements and associated justifications are provided in the tables below.

1. VADL General Requirements		
Req #	Description	Justification
1.1 Budgetary Requirements		
1.1.1	The team shall maintain an itemized list of all expenditures	Having this updated list of expenses will keep the team within budget
1.1.2	The president and treasurer shall review the budget and all expenses	Review by multiple people will prevent any budget mishaps
1.1.3	The team shall keep 10% of the budget in reserve for emergency expenses	Maintaining a healthy amount of contingency funding is necessary for unforeseen expenditures, such as an expensive component failing unexpectedly.
1.1.4	All purchases shall be reviewed and approved by team leadership	Allows for time to review lab inventory before purchases, minimize the risk of unnecessary purchases
1.2 Social Media Requirements		
1.2.1	The team shall post to the VADL instagram at least once a week	A social media presence is necessary for proper engagement with the greater outreach community
1.2.2	Key news and major updates shall be cross-posted to the VADL twitter account	Twitter is a key resource used by the University, NASA, and Academia in general, so important updates must be viewable by those audiences
1.3 STEM Engagement Requirements		
1.3.1	VADL shall conduct STEM outreach and engagement events both in the Nashville community and across the country	Engaging with the community, both in Nashville and elsewhere, is a key tenet of VADL's outreach program
1.3.2	VADL shall reach a minimum of 500 participants	Outreach is of the utmost importance to VADL, so the team wishes to engage at least twice the total designated by NASA requirement 1.4
1.3.3	VADL shall have at least 8 STEM engagement events over the lifetime of this project cycle	Consistent STEM engagement over the project lifecycle is key to any outreach program
1.3.4	Every team member shall participate in at least one STEM outreach and engagement event	STEM engagement is a responsibility for all team members, so that outreach participants will learn about all aspects of the team's project
1.3.5	All team members shall lead one engagement event in their hometown over Thanksgiving or Winter break	VADL would like to impact not just the surrounding TN communities, but give back to the teachers and educators who helped the team along their educational career
1.4 Quality Control Requirements		
1.4.1	Any technical writing shall be proofread by at least two other team members before submission	Proofreading will maintain quality of any written material
1.4.2	All requirement verification that demands testing shall be done according to a rigorous testing protocol that clearly defines success parameters	Rigorous testing is the only way to ensure the verification of the requirements that VADL has imposed on itself
2. VADL Vehicle Requirements		
Req #	Description	Justification
2.1 System-Level Requirements		
2.1.1	The vehicle shall reach an apogee within 5% of the predicted apogee of 4200 ft.	The NASA-specified apogee range is between 4000 and 6000 ft, and flight simulations indicate a projected apogee of 4200 ft.
2.1.2	The launch vehicle shall have a static stability margin between 2.0 and 2.5 calibers on the launch pad.	The launch vehicle static stability margin must be large enough to ensure a stable launch, but not too large so that it causes overstable flight and excessive pitch of the vehicle into the wind.
2.1.3	Both the full scale and subscale launch vehicles shall have an average Thrust-to-Weight ratio of 5.5 minimum.	A higher thrust to weight ratio ensures that the rocket leaves the launch rail with an aerodynamically stable velocity.
2.1.4	The launch vehicle shall be designed and fabricated to be recoverable and reusable, meaning that it can be re-launched within 2 hours after recovery.	The team wants to take sustainability into account; this would aid especially for large scale applications where a reusable vehicle would save costs and materials.
2.2 Structural Requirements		
2.2.1.1	The vehicle shall be manufactured with a lightweight and durable material.	The vehicle must withstand launch and landing loads and in order to do so must be manufactured with durable material.
2.2.1.2	The nose cone shall withstand the maximum landing kinetic energy expected for the lander.	The nose cone must protect the STEMnauts and the rest of the payload components during landing.
2.2.1.3	The fins shall avoid fluttering.	Fin flutter must be prevented, as it can cause structural failures.
2.2.1.4	LiPo batteries shall be properly stored and protected from excessive damage and heat.	LiPo batteries can catch fire if they are damaged/punctured, so it's important to properly protect them from potential hazards.
2.2.1.5	The motor shall provide sufficient thrust to achieve the required apogee designated in Req. 2.1.1	Achieving the projected apogee is an important element of mission success.
2.2.1.6	All structural components shall withstand expected temperatures on the launch pad.	Structural components must be designed to withstand launch pad temperatures so that they do not deform and cause structural damage.
2.2.2	All structural components shall have a safety factor of at least 2.5.	Structural components must be designed with large safety factors in order to mitigate any unpredicted errors that may occur.
2.3 Flight Analysis Requirements		
2.3.1	Flight Simulations shall be conducted that shall predict the flight trajectory at winds from omph to 20mph and launch angles from 0 to 10 degrees	Full flight simulations verify that all flight requirements will be met regardless of envorinomental condions
2.3.2	Simulations shall benchmark the performance of various motors	These motor simulations will allow the team to choose the best motor for flight that also adheres to NASA's launch and flight requirements
2.3.3	Analysis from commercial software shall verify the vehicle's SSM	Redundancy with regards to the SSM will ensure flight safety. Several softwares such as RockSim and OpenRocket are valuable alternatives to MATLAB program
2.3.4	Descent time and drift calculations will be calculated for varying wind speeds and deployment heights	Necessary to determine feasible range of parachute sizes and deployment times
2.3.5	Flight simulations in the case of denied jettison shall be conducted.	Jettison is not guaranteed; simulations are necessary to ensure compliance against NASA requirements in this context.
2.3.6	Analysis performed in MATLAB shall track and simulate pitch angle throughout the duration of flight	Pitch angle is a difficult flight parameter to predict, and a data-informed prediction will improve accuracy of the secondary payload algorithm
2.3.7	The launch thrust-to-weight ratio shall be estimated with simulation data and verified after construction of the vehicle	Determining a value for the thrust-to-weight ratio will confirm the team has selected the right motor and that the vehicle will achieve the necessary velocity for launch rail exit and the necessary acceleration to reach the estimated apogee
2.3.8	Massing estimates shall be continually updated as designs progress, and verified once component masses are known	Accurate massing is essential to achieve accurate SSM and flight trajectory analysis
2.4 Aerodynamic Analysis Requirements		
2.4.1	CFD shall be conducted to verify the drag coefficient values yielded by OpenRocket	The drag coefficient dictates the descent time and landing energy, and it should be well understood
2.4.2	Thorough CFD analysis shall be performed at various flight speeds and at different angles of attack	The axial and normal coefficients of drag play a major role in determining flight dynamics and altitude performance
2.4.3	OpenRocket analysis shall be used to optimize design of the boat tail.	Proper boat tail design is essential for the aerodynamic performance of the rocket.
2.4.4	OpenRocket analysis shall be used to optimize design of the fins.	Proper fin design is necessary for ensuring the stability of the vehicle.
3. VADL Recovery Requirements		
Req #	Description	Justification
3.1	There shall exist a primary and backup charge for both main and drogue deployment	A larger backup charge will be used to ensure that recovery hardware deploys to avoid catastrophic recovery failure
3.2	In the event of failed jettison, the descent velocity of the vehicle shall not exceed 20 fps upon landing.	Parachute selections should be designed to ensure soft landing even in case of jettison failure.
3.3	Shock cords shall be sown, as opposed to being tied in knots	Knots create unnecessary stress concentration and additional points of failure
3.4	Parachutes shall be wrapped in fireproof fabric such that they can easily be dislodged and inflated	Fire and heat can create holes or melt parts together given the synthetic construction material. Either poses a massive hazard to recovery system performance
3.5	All recovery electronics shall be shielded from RF interference in an independent section	RF interference can impede the the proper functionality of altimeters and cause premature / delayed deployment. Extension of NASA 3.8
3.6	The shock cords shall be configured such that no independent sections have to roll over and reconfigure during main deployment	Unnecessary reorienting and maneuvering increases the chance of collision between sections and dimensions

3.7	Independent altimeters shall be responsible for the primary and backup charges	In case of failure, a redundant altimeter be able to activate charges in both the main and drogue bays
3.8	A recoverable, electrically independent 'black box' shall be included that is intended to record all flight data, as well as transmit GPS data	An independent, redundant system is useful to guarantee that flight data is recovered regardless of the performance of the primary payload
3.9	Recovery system shall utilize firebails to prevent zippering of the rocket body due to tensioning of the shock cord	To comply with NASA requirement 2.3, the vehicle should not incur severe / irreparable damage during deployment
3.10	Batteries used for recovery electronics shall be tested prior to launch day, and backups brought to the field	Mission success is required at all launches, therefore thorough power supply testing and options will be necessary prior to any rocket ignition
3.11	The terminal descent velocity under drogue parachute of the rocket shall not exceed 115 fps.	Substantially high drogue terminal velocity places dangerous stress to rocket structures and shock cords during main parachute deployment.
4. VADL Payload Requirements		
Req #	Description	Justification
4.1 General Payload Requirements		
4.1.1	The payload shall provide a retention method for four STEMnauts.	To comply with NASA requirement 4.1, it is necessary to provide a retention method for the STEMnauts
4.1.2	The payload shall land with a velocity no greater than 24 fps.	To meet survivability requirements for the STEMnauts, it is necessary to limit the landing velocity
4.1.3	The payload shall have an antenna capable of transmitting flight data via RF	To transmit data via RF, as imposed by NASA requirement 4.1, an antenna is needed
4.1.4	The payload shall be capable of being prepared in the field in less than 45 min	Flight preparation within a two hour window necessitates an efficient payload preparation in order to ensure enough time is available for full vehicle assembly
4.1.5	The team shall develop software unit tests for all software integrated systems	Integration of software into the payload is both mission critical and involves many potential failure points, so rigorous testing is required
4.1.6	The Dynamic Response Index (DRI) for the STEMnauts shall be no greater than 17.7 in the vertical axis.	To comply with NASA requirement 4.1, it is necessary to provide a metric for determining whether or not the STEMnauts survive flight.
4.1.7	The payload shall land in an upright orientation.	To maximize the chances for RF transmission, orientation of the transmission antenna requires a known payload landing orientation
4.1.8	The payload electronics shall be able to withstand launch acceleration and landing impact	Electronics must be able to withstand these forces for data collection and transmission to be achieved
4.2 Lander Separation Mechanism		
4.2.1	The LSM shall disengage the lander from the main rocket body within 1 second of reaching the target altitude and receiving command signal.	Timely separation is critical to ensure that the lander can safely deploy its parachute and legs
4.2.2	The LSM shall be able to receive RF transmission	Approval for payload release requires the ability to communicate with the mechanism in flight, necessitating RF capability.
4.2.3	The mechanism shall be capable of operating at altitudes between 200 and 400 ft AGL	Variable altitude capabilities are necessary to determine the optimal release point of the payload
4.2.4	The mechanism shall be capable of withstanding maximum expected shock loads of 32G.	The mechanism must survive launch and flight conditions to ensure successful operation.
4.3 Leg Deployment Mechanism		
4.3.1	The legs shall withstand maximum anticipated landing loads.	Legs must be able to withstand the maximum possible landing force in order to ensure mission success.
4.3.2	The legs shall deploy passively upon lander separation.	Passive deployment is preferable to an active mechanism to limit mechanical complexity and remove potential failure points.
4.3.3	The legs shall be packaged within the body of the payload.	Packaging the legs within the payload body decreases risk of entanglement with parachute.
4.4 Parachute Detachment Mechanism		
4.4.1	The parachute shall detach within 1 second of detected landing	Parachute must be able to detach quickly in order to prevent dragging or toppling of payload.
4.4.2	The detachment latch shall withstand the maximum acceleration experienced during flight.	The latch must stay locked throughout the entire flight to prevent premature deployment
4.5 Antenna Extension Mechanism		
4.5.1	The AEM shall extend the antenna 18"	The antenna requires a 0.5m/18" transmission length
4.5.2	The AEM shall be capable of extending even if partially obstructed	The antenna needs to achieve the required clearance regardless of obstructions in its path
4.5.3	The AEM shall not interfere with the RF transmission	The antenna must transmit collected data across the field to NASA ground station. Any interference in the RF system from the use of non-transmitting metal within the RF system must be minimized.
4.5.4	The AEM locking mechanism shall withstand maximum expected loading during flight	The AEM must not extend until the payload has landed
4.5.5	The AEM shall be capable of extending regardless of payload jettison, and have at least 1 antenna extending away from the ground regardless of landing orientation	RF transmission is required regardless of whether or not the payload is jettisoned from the launch vehicle. Without jettison and subsequent leg deployment, orientation is not guaranteed.
4.6 RF Transmission System		
4.6.1	Transmission shall occur given any landing orientation.	Data transmission is the primary mission objective and must occur regardless of landing orientation
4.6.2	The antenna system shall be able to reliably transmit up to a distance of 1 mile.	The maximum predicted drift distance is 2500ft. Applying a safety factor of 2 yields approximately 1 mile as the maximum transmission distance
4.7 Data Collection Sensors		
4.7.1	Data sampling frequency shall be 160 Hz (TBR) throughout data collection.	Data sampling frequency must be high enough to collect all relevant data and is required for calculations
4.7.2	The sensors shall be able to detect landing	Provides data to be transmitted and prevents transmission from occurring before landing
4.7.3	The sensors shall be able to detect jettison	Transmission sequence will be affected depending on whether payload is jettisoned or not
4.7.4	The sensors shall be able to determine altitude	Altitude is needed for transmission of apogee value, as well as for the landing detection sequence
4.7.5	The sensors shall be able to detect apogee	Required to send data selected by VADL
4.7.6	The sensors shall be able to detect orientation	
4.7.7	The sensors shall be able to detect acceleration	
4.7.8	The sensors shall be able to detect battery voltage	
4.7.9	The software shall be able to estimate velocity	Software velocity calculation is preferable to sensor reading because it reduces the number of instruments required on the payload
4.7.10	The software shall be able to estimate survivability	Verifying STEMnaut survival is required for successful mission completion
5. VADL Safety Requirements		
Req #	Description	Justification
5.1 General Safety Requirements		
5.1.1	Team members shall wear proper PPE as indicated in VADL safety protocols.	Team safety depends on the proper use of PPE
5.1.2	Team members shall clean up after themselves and engage in collective lab cleanings when deemed necessary by the Student Safety Officer.	A clean lab promotes the safety and good organization bolsters team efficiency.
5.1.3	Team members shall participate in mandatory weekly safety briefings delivered by the Student Safety Officer.	Safety briefings help maintain safe practices throughout the year and are an important part of the VADL commitment to safety.
5.1.4	Energetic material shall be stored in appropriate explosives magazines and shall be kept locked at all times.	The safety of team members and general community is reliant on proper storage of energetic material.
5.1.5	Team members shall only use equipment that they are trained to operate.	Proper equipment training helps to reduce preventable accidents.

Verification

As with the requirements derivation sheet shown above, the accompanying verification plans have been further developed. Proper tracking of verification progress is critical to ensuring that requirements are being fulfilled in a methodical manner. The same verification methods and statuses described in Section 6.2.1 are applied here, and the refined plans are shown in the tables below.

VADL General Requirements		Design Requirements	Section	Verification Method	Verification Details	Team Member(s)	Status
Req #	Description						
1.1 Budgetary Requirements							
1.1.1	The team shall maintain an itemized list of all expenditures	The president and treasurer shall track all expenses incurred by the team in a spreadsheet	6.3.3	I	The expense spreadsheet and its proper maintenance shall keep the team on budget	IM, TS	IP
1.1.2	The president and treasurer shall review the budget and all expenses	The president and treasurer shall discuss the budget and the expenses that must be made at various points during the project	6.3	I	These discussions shall maintain a clear picture of the budget for the team's leadership	IM, TS	IP
1.1.3	The team shall keep 10% of the budget in reserve for emergency expenses	The team shall not spend all of its budget and shall maintain an emergency reserve of 10% of the total budget	6.3	I	Team leadership shall review the budget to determine how the reserve will be maintained	IM, TS	IP
1.1.4	All purchases shall be reviewed and approved by team leadership	All team purchases shall go through the president and vice presidents	6.3.3	I	All purchases shall be made by team leadership, so that the team stays on budget	IM, TS	IP
1.2 Social Media Requirements							
1.2.1	The team shall post to the VADL instagram at least once a week	The outreach lead shall post to the VADL instagram at least three times a month	VADL Instagram	I	Team members with instagram accounts shall follow the VADL instagram to ensure compliance with this requirement	SS	IP
1.2.2	Key news and major updates shall be cross-posted to the VADL twitter account	The outreach lead shall post noteworthy news, updates, and accomplishments on the VADL twitter account	VADL Twitter	I	Team members with twitter accounts shall follow the VADL twitter account to ensure compliance with this requirement	SS	IP
1.3 STEM Engagement Requirements							
1.3.1	VADL shall conduct STEM outreach and engagement events both in the Nashville community and across the country	All team members shall participate in STEM outreach events in Nashville and in their hometowns. VADL as a whole shall reach at least 500 participants	N/A	I	The outreach lead shall contact organizations and maintain a regular schedule of events through a shared events spreadsheet	SS	IP
1.3.2	VADL shall reach a minimum of 500 participants	The outreach lead shall track participation at all events to ensure that a minimum of 500 participants have been engaged	N/A	I	The outreach lead will continuously inspect the number of participants reached to ensure that the minimum requirement has been achieved	SS	IP
1.3.3	VADL shall have at least 8 STEM engagement events over the lifetime of this project cycle	The outreach lead with the help of graduate student mentors shall schedule at least 8 STEM engagement events over the length of the project cycle	N/A	I	Team members shall keep track of STEM engagement events to ensure VADL has at least 8 STEM engagement events	All	IP
1.3.4	Every team member shall participate in at least one STEM outreach and engagement event	Team members shall sign up for events via the event spreadsheet to ensure they have met the minimum requirement for participation	N/A	I	The outreach lead shall track who has gone to each event so that everyone attends at least one event	SS	IP
1.3.5	All team members shall lead one engagement event in their hometown over Thanksgiving or Winter break	Team members shall make the necessary arrangements to organize and lead the event	N/A	I	The outreach lead will inspect that all team members have conducted outreach events in their hometowns	SS	IP
1.4 Quality Control Requirements							
1.4.1	Any technical writing shall be proofread by at least two other team members before submission	Team members shall proofread material that they did not write such that all writing is proofread at least twice	N/A	I	A chart shall track which sections have been proofread and by whom it was proofread	All	IP
1.4.2	All requirement verification that demands testing shall be done according to a rigorous testing protocol that clearly defines success parameters	The team shall create a testing plan that defines each test and its success criteria. Tests and verification shall be performed according to NASA Technical Standard 5001-B	6.1	I	The testing plan shall be updated and proper documentation of the tests shall be kept to ensure compliance with this requirement	All	IP
VADL Vehicle Requirements							
Req #	Description	Design Requirements	Section	Verification Method	Verification Details	Team Member(s)	Status
2.1 System-Level Requirements							
2.1.1	The vehicle shall reach an apogee within 5% of the predicted apogee of 4200 ft.	Team members shall model the vehicle and its flight in MATLAB, OpenRocket, and RockSim simulations to verify the apogee	3.4	AD	As the design of the rocket changes, new simulations shall be performed to verify its flight performance. Compliance with this requirement will be verified through full-scale flight demonstrations.	SS, SH	IP
2.1.2	The launch vehicle shall have a static stability margin between 2.0 and 2.5 calibers on the launch pad.	Vehicle mass distribution and fin design shall be completed with the focus of maintaining a SSM in this range.	3.1	AD	Simulations shall be performed under various constraints to optimize the design and achieve a reasonable SSM within the expected range. Compliance with this requirement will be verified through full-scale flight demonstrations.	SS, SH	IP
2.1.3	Both the full scale and subscale launch vehicles shall have an average Thrust-to-Weight ratio of 5.5 minimum.	Team members shall use OpenRocket simulations to conduct trade studies for selected motors.	3.1	AD	Using different simulation platforms and motor trade studies, determine best thrust to weight ratio. Compliance with this requirement will be verified through full-scale flight demonstrations.	SS, SH	IP
2.1.4	The launch vehicle shall be designed and fabricated to be recoverable and reusable, meaning that it can be re-launched within 2 hours after recovery.	Team members shall ensure the material durability and reusability and analyze them in datasheets and simulations.	3.3	AD	Simulations shall be performed under different wind conditions, so that the vehicle shall comply with this requirement under any flyable conditions (winds up to 20 MPH). Compliance with this requirement will be verified through full-scale flight demonstrations.	SS, SH	IP
2.2 Structural Component Requirements							
2.2.1.1	The vehicle shall be manufactured with a lightweight and durable material.	The vehicle airframe shall be manufactured using fiberglass.	3.1.2	AT	Massing analysis has been performed to ensure the vehicle is sufficiently lightweight, and load testing has been completed to ensure the airframe can withstand expected loading.	SS, SC	V
2.2.1.2	The nose cone shall withstand the maximum landing kinetic energy expected for the lander.	The nose cone shall be manufactured using fiberglass cloth laid up on a 3D printed mold.	3.1.3	T	After fabrication, drop tests will be conducted to ensure durability with added mass to mimic the landing energy.	SS, SC	IP
2.2.1.3	The fins shall avoid fluttering.	The fins shall be sized such that fin flutter is avoided.	3.1.9	AD	Calculations will be performed to ensure fin flutter is avoided. After fabrication, the team will conduct flights to test the reliability and structural integrity of fins and make changes as required.	SS, SC	IP
2.2.1.4	LiPo batteries shall be properly stored and protected from excessive damage and heat.	Secure, protective structures for LiPo batteries shall be designed.	4.1.2	IT	Inspection of casings and hardware as well as tests will be conducted to ensure durability.	SS, PC	IP
2.2.1.5	The motor shall provide sufficient thrust to achieve the required apogee designated in Req. 2.1.1	The Cesaroni L1720 shall be used as the competition motor.	3.4	AD	MATLAB simulations and test flights shall verify motor specifications along with flight reliability. Compliance with this requirement will be verified through full-scale flight demonstrations.	SS, SH	IP
2.2.1.6	All structural components shall withstand expected temperatures on the launch pad.	Structural components shall be made with aluminum or 3D printed materials that can withstand temperatures up to at least 100F.	3.2	AT	Material thermal properties will be analyzed, and components will be tested in a thermal environment at temperatures exceeding that expected on launch day.	SS, SC	IP
2.2.2	All structural components shall have a safety factor of at least 2.5.	Structural components shall be designed to fail at least 2.5x the maximum load experienced during main parachute deployment.	3.2	AD	FEA will be conducted to analytically verify the safety factor of structural components. Performance will be demonstrated through full-scale vehicle flights.	SH	IP

2.3 Flight Analysis Requirements							
2.3.1	Flight Simulations shall be conducted that shall predict the flight trajectory at winds from 0mph to 20mph and launch angles from 0 to 10 degrees	Members shall simulate flight under various conditions using MATLAB	3.4	A	MATLAB simulations have verified the flight trajectory will meet all flight requirements under any flyable condition	SH	V
2.3.2	Simulations shall benchmark the performance of various motors	Members shall conduct a trade study and simulate the performance of different motors with the full-scale rocket vehicle	3.4	A	MATLAB simulations have verified that the motor choice will satisfy flight requirements	SH	V
2.3.3	Analysis from commercial software shall verify the vehicle's SSM	The team shall analyze the vehicle with software to determine its SSM	3.4	A	Outside software and MATLAB simulations have verified the calculated SSM to ensure flight stability	SH	V
2.3.4	Descent time and drift calculations will be calculated for varying wind speeds and deployment heights	The team shall analyze the flight trajectory under different flight scenarios for different mass distributions	3.4	A	MATLAB simulations have verified maximum drift requirement of 1750 ft is met	SH	V
2.3.5	Flight simulations in the case of denied jettison shall be conducted.	Members shall simulate the flight scenario where payload jettison is denied	3.4	A	MATLAB simulations have determined flight performance in the case that payload jettison is denied	SH	V
2.3.6	Analysis performed in MATLAB shall track and simulate pitch angle throughout the duration of flight	The team shall estimate pitch angles to inform apogee prediction	3.4	A	MATLAB simulations of various pitch angles have verified the reliability of the team's predicted apogee	SH	V
2.3.7	The launch thrust-to-weight ratio shall be estimated with simulation data and verified after construction of the vehicle	MATLAB simulations shall predict thrust-to-weight ratio before verification during construction	3.6	AI	MATLAB simulations shall use vehicle mass and motor data to determine thrust-to-weight ratio; proper massing during construction shall validate results	SH	IP
2.3.8	Massing estimates shall be continually updated as designs progress, and verified once component masses are known	The team shall keep up-to-date massing estimates that shall be used in all simulations	N/A	I	Proper documentations shall ensure the most accurate data is used during any calculations	SS, SC	IP
2.4 Aerodynamic Analysis Requirements							
2.4.1	CFD shall be conducted to verify the drag coefficient values yielded by OpenRocket	The team shall create a CFD model of the entire vehicle	3.4.7	A	A CFD analysis has been performed and results tabulated and compared with expected values from OpenRocket	JL	V
2.4.2	Thorough CFD analysis shall be performed at various flight speeds and at different angles of attack	Members shall perform CFD analysis for various flight scenarios	3.4.7	A	A thorough CFD model has been analyzed to validate flight performance under any flyable condition	JL	V
2.4.3	OpenRocket analysis shall be used to optimize design of the boat tail.	The team shall utilize the OpenRocket optimizer tool to facilitate design of the boat tail.	3.1.9	A	The OpenRocket analysis has been thoroughly examined to ensure adequate boat tail design	SS, SC	V
2.4.4	OpenRocket analysis shall be used to optimize design of the fins.	The team shall utilize the OpenRocket optimizer tool to facilitate design of the fins.	3.1.9	A	The OpenRocket analysis has been thoroughly examined to ensure adequate fin design	SS, SC	V
VADL Recovery Requirements							
Req #	Description	Design Requirements	Section	Verification Method	Verification Details	Team Member(s)	Status
3.1	There shall exist a primary and backup charge for both main and drogue deployment	The team shall place independent blast charges inside the vehicle for separate parachute deployment events	3.3.3	IA	Inspection during vehicle construction shall ensure that separate charges have been placed in their respective chambers; analysis prior to launch shall determine the size of the charges	SC	IP
3.2	In the event of failed jettison, the descent velocity of the vehicle shall not exceed 20 fps upon landing.	The team shall select a drogue parachute such that the descent velocity is greater than 70 fps	3.4	AD	Simulations in MATLAB have determined descent velocity for various drogue parachute sizes; the subscale launch provided altimeter data to verify the simulation	SS, SH	V
3.3	Shock cords shall be sown, as opposed to being tied in knots	Team members shall sew all shock cords used in the vehicle	N/A	I	Inspection during vehicle construction shall verify that all shock cords are sown rather than tied in knots	SS	IP
3.4	Parachutes shall be wrapped in fireproof fabric such that they can easily be dislodged and inflated	Team members shall use fireproof fabric when wrapping the parachutes during vehicle construction	N/A	I	Inspection of the recovery hardware will ensure that fireproof fabric is used to wrap the parachutes	SS	IP
3.5	All recovery electronics shall be shielded from RF interference in an independent section	The team shall shield the avionics bay with RF-opaque aluminum bulk plates	3.1.7	I	Inspection of the fabricated avionics bay will confirm the use of aluminum bulk plates to shield the avionics bay from the RF transmitter.	SS, SC	IP
3.6	The shock cords shall be configured such that no independent sections have to roll over and reconfigure during main deployment	The team shall determine proper shock cord lengths and configurations such that main parachute deployment runs smoothly	N/A	I	Inspection of the assembled recovery hardware will confirm the proper selection of cord lengths.	SS	IP
3.7	Independent altimeters shall be responsible for the primary and backup charges	The vehicle shall contain two independent altimeters for primary and backup blast charges	3.3.3	I	Inspection of the assembled avionics will ensure the use of two sets of independent altimeters	SC	IP
3.8	A recoverable, electrically independent 'black box' shall be included that is intended to record all flight data, as well as transmit GPS data	The team shall create an independent electrical system that shall record all flight data and transmit GPS data	3.3.4	I	Inspection of the assembled vehicle will ensure that the GPS is electrically isolated from other systems	PC	IP
3.9	Recovery system shall utilize fireballs to prevent zippering of the rocket body due to tensioning of the shock cord	Team members shall include fireballs on the shockcords	N/A	I	The assembled recovery hardware will be inspected prior to flight to ensure the use and proper placement of fireballs	SS	IP
3.10	Batteries used for recovery electronics shall be tested prior to launch day, and backups brought to the field	The team shall thoroughly test the batteries for any electronic equipment	4.3.6	T	Battery life testing will verify the lifetimes of the batteries used on launch day	PC	IP
3.11	The terminal descent velocity under drogue parachute of the rocket shall not exceed 115 fps.	The drogue parachute shall be selected such that the rocket descent velocity does not exceed 115 fps	3.4	AD	Flight simulations have been conducted to verify the descent velocity, and data from flight demonstrations will be reviewed to ensure compliance with this requirement	SH	IP
VADL Payload Requirements							
Req #	Description	Design Requirements	Section	Verification Method	Verification Details	Team Member(s)	Status
4.1 General Payload Requirements							
4.1.1	The payload shall provide a retention method for four STEMnauts.	The payload shall contain a STEMnaut crew module to house the four STEMnauts	4.2.5	D	The requirement has been verified by demonstrating that the crew module has been built and integrated into the lander	BR	V
4.1.2	The payload shall land with a velocity no greater than 24 fps.	The payload shall descend under parachute, with sizing informed by analysis	4.1.3	AT	Descent simulations have been run to verify that the landing velocity does not exceed the imposed limit, and drop tests have been performed to empirically confirm landing velocity	SH	V
4.1.3	The payload shall have an antenna capable of transmitting flight data via RF	A quarter-wave ground plane antenna shall be designed and integrated into the lander to perform RF transmission	4.3.5	D	The requirement has been verified by demonstrating the payload's ability to transmit RF data through drop tests	PC, AK	V
4.1.4	The payload shall be capable of being prepared in the field in less than 45 min	In order to complete assembly of the rocket within two hours, the team shall be able to assemble the payload in the field in less than 45 min	N/A	D	The team shall practice preparing the payload and time how long it takes to prepare in the field during full-scale payload demonstration	All	IP
4.1.5	The team shall develop software unit tests for all software integrated systems	The team shall create robust software by developing software unit tests	4.3.2	D	Software unit tests will allow the team to verify that all electronics and software are integrated properly and are functioning nominally	PC	IP

4.1.6	The Dynamic Response Index (DRI) for the STEMnauts shall be no greater than 17.7 in the vertical axis.	The payload recovery system and leg damping shall be designed such that the DRI value is below the specified threshold	4.1.3	DA	Simulation data and launch data will be analyzed to ensure that the STEMnauts do not experience a DRI greater than 17.7	SH, PC	IP
4.1.7	The payload shall land in an upright orientation.	The lander shall be designed with a low CG and wide legs.	4.1.2, 4.2.2	D	Drop tests under varying conditions have been conducted to demonstrate that the payload repeatedly and reliably lands upright	ZS, RN	V
4.1.8	The payload electronics shall be able to withstand launch acceleration and landing impact	The electronics system shall be thoroughly ruggedized to ensure survival throughout flight	4.3.8	DT	Electronic connections will be checked in the lab as well as in the field post-launch. Payload operation and transmission of data will indicate successful ruggedization was implemented	PC	IP
4.2 Lander Separation Mechanism							
4.2.1	The LSM shall disengage the lander from the main rocket body within 1 second of reaching the target altitude and receiving command signal.	Proper sensor selection and friction reduction shall be implemented for the LSM.	4.2.1	T	The requirement has been verified by conducting timed separation tests during ground-based trials, ensuring that the lander consistently separates from the main rocket body within 1 second of command. Cameras and timing devices were used to measure separation time.	SC	V
4.2.2	The LSM shall be able to receive RF transmission	No materials that potentially interfere with RF signals shall be used around the LSM	4.2.1	T	RF transmission tests have been conducted by transmitting signals from ground to the LSM to ensure there is no signal degradation. The team verified that the chosen materials do not obstruct RF signals.	SC	V
4.2.3	The mechanism shall be capable of operating at altitudes between 200 and 400 ft AGL	The team shall test the capability of the LSM to reliably function at these altitudes	4.2.1	T	The requirement was verified by flight testing the LSM at different altitudes between 200 and 400 ft AGL.	SC	V
4.2.4	The mechanism shall be capable of withstanding maximum expected shock loads of 32G.	The team shall use durable materials and perform shock load tests on the LSM simulating launch conditions	4.2.1	DT	The requirement shall be verified through vibration and shock load testing in a controlled environment. Full scale launch will further show that the mechanism can withstand the necessary loading	SC	IP
4.3 Leg Deployment Mechanism							
4.3.1	The legs shall withstand maximum anticipated landing loads.	Legs shall be constructed out of material which will not break under the forces experienced at landing	4.2.2	DT	Extensive testing under different loading conditions, dynamic load testing with integrated leg system, and drop testing from drone have been performed to verify leg strength.	ZS	V
4.3.2	The legs shall deploy passively upon lander separation.	Legs shall be spring loaded and released by separation from main body	4.2.2	DT	Extensive testing under different initial conditions, static testing of opening mechanism in lab, testing when being released under tether, and testing when being dropped under parachute have been performed to ensure full deployment under any scenario.	ZS	V
4.3.3	The legs shall be packaged within the body of the payload.	The lander separation bay airframe shall retain the legs within the payload bay airframe until jettison	4.2.2	ID	Inspection of the integrated vehicle and payload, along with flight demonstrations, will ensure that the legs are packaged within the vehicle until jettison.	ZS	IP
4.4 Parachute Detachment Mechanism							
4.4.1	The parachute shall detach within 1 second of detected landing	The PDM shall utilize an EM latch	4.2.3	T	The PDM will be thoroughly tested through drop tests, and the data will be reviewed to confirm that detachment occurs within 1 second of landing detection.	TS	IP
4.4.2	The detachment latch shall withstand the maximum acceleration experienced during flight.	The latch shall be secured in place using sufficiently robust materials. The latch will be capable of withstanding the maximum expected loadings	4.2.3	T	The PDM has been thoroughly tested through drop tests and demonstrated to withstand parachute jerk	TS	V
4.5 Antenna Extension Mechanism							
4.5.1	The AEM shall extend the antenna 18"	The AEM spring tape shall be 18" long	4.2.4	T	Testing has ensured that the AEM is capable of extending even after being subjected to high G loads and impacts	RN	V
4.5.2	The AEM shall be capable of extending even if partially obstructed	Back to back spring tapes shall be rolled together to strengthen their extension force	4.2.4	T	The extension force of the AEM shall be measured and verified to ensure that light debris does not prevent the AEM from extending to full length	RN	IP
4.5.3	The AEM shall not interfere with the RF transmission	The AEM spring tape shall be electrically isolated from the antenna with a nylon coating.	4.2.4	T	Testing shall be conducted to ensure that transmission is not degraded by the use of a metal spring tape in the AEM	RN	IP
4.5.4	The AEM locking mechanism shall withstand maximum expected loading during flight	The AEM cylinder lock shall be designed to be robust and able to withstand high loads and accelerations	4.2.4	T	The AEM has been dropped from a height that simulates landing velocity to ensure that the mechanism remains locked	RN	V
4.5.5	The AEM shall be capable of extending regardless of payload jettison, and have at least 1 antenna extending away from the ground regardless of landing orientation	There shall be three AEMs equally spaced radially to ensure one is always facing upward	4.2.4	T	The payload bay shall be dropped repeatedly to ensure that one AEM is facing upward in all landed configurations	RN	IP
4.6 RF Transmission System							
4.6.1	Transmission shall occur given any landing orientation.	The signal lost relative to optimal orientation should be less than 10 percent.	4.3.5	T	Signal strength in dBm will be measured at different landing orientations.	NB	IP
4.6.2	The antenna system shall be able to reliably transmit up to a distance of 1 mile.	Need to have reliable transmission within 1 mi in clear air space.	4.3.5	T	Testing in different environment will be performed to ensure reliable transmission over long range.	NB, PC, AAJ	IP
4.7 Data Collection Sensors							
4.7.1	Data sampling frequency shall be 160 Hz (TBR) throughout data collection.	Electronics shall be configured such that they sample at 160 Hz	4.3.2	T	The sampling frequency will be tested by leaving the system running for > 1hr and ensuring that data points are consistent with almost no noise or other nonmeaningful values.	AF, PC, JH	IP
4.7.2	The sensors shall be able to detect landing	The thresholds for landing values shall be calibrated to avoid premature parachute detachment. Backup plan is a timer.	4.3.3	T	Landing detection logic has been thoroughly tested with debugging techniques such as blinky and different scenarios, such as at different locations.	AF, PC, JH	V
4.7.3	The sensors shall be able to detect jettison	The thresholds for jettison light level changes shall be tuned	4.3.1	T	Jettison detection will be tested in a controlled environment. Different lighting environment will be tested to ensure reliability.	AF, PC, JH	IP
4.7.4	The sensors shall be able to determine altitude	Pressure readings shall be used to calculate altitude	4.3.1	D	Drone drop test data has been used to verify the functionality of the altimeter.	AF, PC, JH	V
4.7.5	The sensors shall be able to detect apogee	Changes in acceleration shall be utilized to determine when apogee is reached	4.3.1	D	Drone drop test data will be used to verify the functionality of the accelerometer.	AF, PC, JH	V
4.7.6	The sensors shall be able to detect orientation	The selected IMU shall be capable of determining orientation	4.3.1	DT	Orientation values collected during drop testing will be reviewed to ensure they match the actual orientation of the payload. Additional testing of orientation readings will be conducted in the lab	AF, PC, JH	IP

4.7.7	The sensors shall be able to detect acceleration	The selected IMU shall be capable of determining acceleration	4.3.1	DT	IMU has been tested in the field to ensure that it can read acceleration in this environment	AF, PC, JH	V
4.7.8	The sensors shall be able to detect battery voltage	Battery voltage shall be detected through Raspberry Pi.	4.3.1	T	Battery will be monitored during field testing to verify that this capability functions as expected	AF, PC, JH	IP
4.7.9	The software shall be able to estimate velocity	Altitude data shall be used to estimate velocity via a first order filter.	4.3.4	T	Velocity readings will be compared to velocity that can be calculated from drop videos in order to verify that it is accurate	AF, PC, JH	IP
4.7.10	The software shall be able to estimate survivability	Acceleration data shall enable the calculation of DRI, which allows survivability to be estimated.	4.3.1	DT	Survivability will be calculated during drop testing to ensure that code logic and sensors are working properly	AF, PC, JH	IP
VADL Safety Requirements							
Req #	Description	Design Requirements	Section	Verification Method	Verification Details	Team Member(s)	Status
5.1 Safety Requirements							
5.1.1	Team members shall wear proper PPE as indicated in VADL safety protocols.	All team members shall follow VADL safety protocols with respect to PPE.	5.5	I	The Student Safety Officer shall inspect lab activities to ensure that team members are properly wearing the required PPE	ZS	IP
5.1.2	Team members shall clean up after themselves and engage in collective lab cleanings when deemed necessary by the Student Safety Officer.	Team leadership shall enforce guidelines regarding the cleanliness of the lab.	5.3	I	The Student Safety Officer shall inspect the lab to ensure the cleanliness of the lab is up to lab standards	ZS	IP
5.1.3	Team members shall participate in mandatory weekly safety briefings delivered by the Student Safety Officer.	The Student Safety Officer and Team Lead shall enforce attendance at weekly safety briefings.	5	I	The Student Officer and Team Lead will record attendance and inspect that all team members are present.	ZS, IM	IP
5.1.4	Energetic material shall be stored in appropriate explosives magazines and shall be kept locked at all times.	The Student Safety Officer shall store energetic material in a locked MAGloc case.	5	I	The Student Safety Officer will regularly inspect that energetic material is still stored in the MAGloc case and that it remains locked.	ZS	IP
5.1.5	Team members shall only use equipment that they are trained to operate.	The Student Safety Officer shall maintain a list of personnel trained on each piece of equipment.	5	I	All team members will inspect the training certification list to ensure that nobody uses equipment they are not certified to operate.	ZS	IP

6.3 Budgeting

VADL appreciates the importance of detailed budgeting to a well-run project. The following sections detail the team's funding, project budget, spending thus far, and projected remaining expenses. In particular, the on-the-pad costs for the full-scale vehicle and payload are compiled, thus informing spending projections for the rest of the competition year. Thorough documentation of the financial aspects of the project enables the team to see their rocket and payload solutions through to fruition.

6.3.1 Funding Plan

Shown in Figure 136 is the funding plan for VADL for the 2024-2025 competition year, which remains unchanged from the PDR milestone. Mark Dalton, a benefactor of the program, is contributing \$7,000. A grant of \$5,000 was awarded by the Vanderbilt School of Engineering, and the remaining \$4,000 comes from a combination of alumni donations and remaining funds from the previous year.

Total Funding: \$16,000

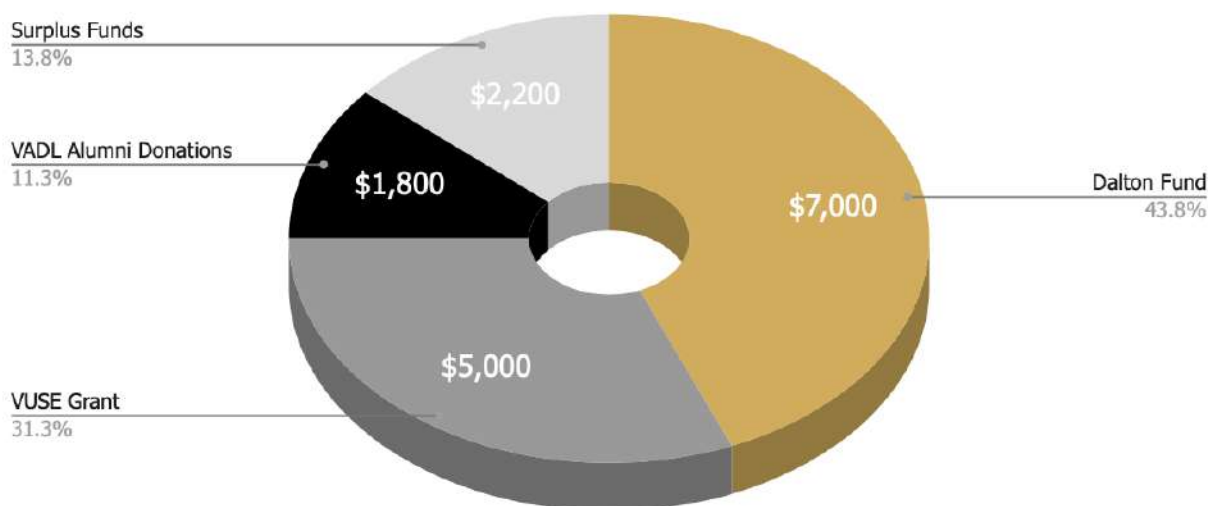


Figure 136: VADL funding plan for the 2024-2025 competition year.

6.3.2 Budget Summary

The VADL budget serves to provide a financial framework with which to guide design, manufacturing, and procurement decision-making. Since the PDR milestone, the budget has been refined to under a more mature design and a better understanding of the project lifecycle. The budget is split into categories of vehicle, payload, travel, and other, each of which is broken down into subcategories indicating the specific project element. The budget summary is visualized in Figure 137, and an itemized list is provided in Table 66. A contingency margin of 10% is added atop the pre-contingency sum to account for unexpected expenses.

Budget: \$14,836

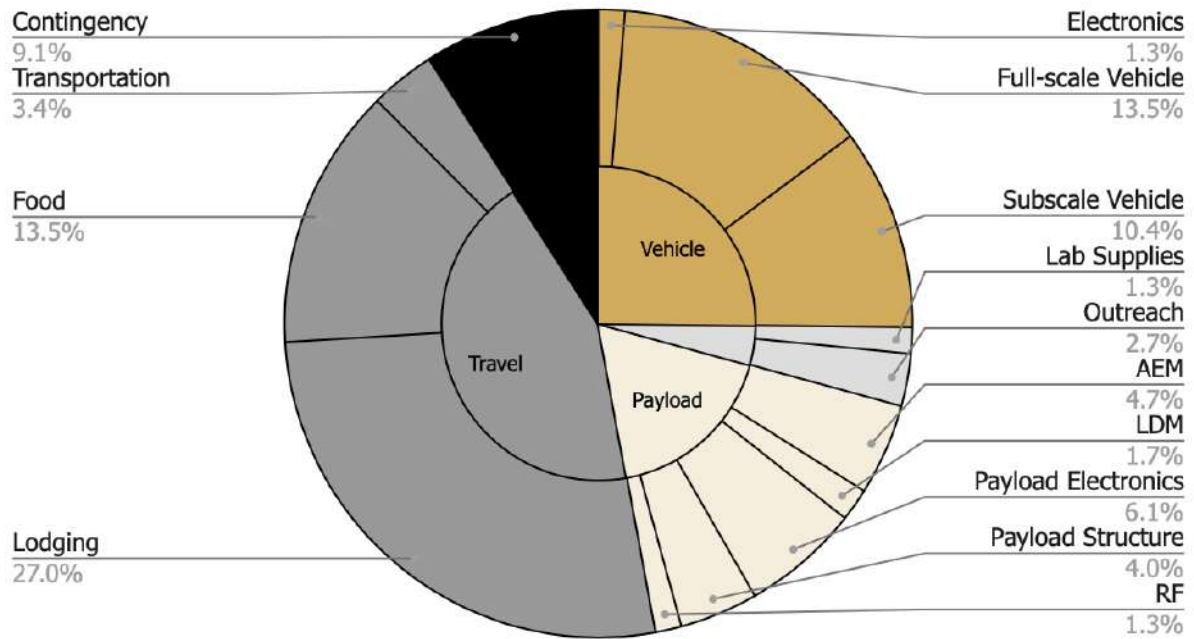


Figure 137: Summary of VADL's budget for the 2024-2025 competition year.

Table 66: Itemized summary of the VADL budget.

Category	Subcategory	Cost
Vehicle		\$3,737
	Full-scale vehicle	\$2,000
	Subscale vehicle	\$1,537
Payload		\$2,650
	AEM	\$700
	LDM	\$250
	Payload Electronics	\$900
	Payload Structure	\$600
	RF	\$200
Travel		\$6,500
	Lodging	\$4,000
	Food	\$2,000
	Transportation	\$500
Contingency		\$1,349
Total		\$14,836

6.3.3 To-Date Spending

Figure 138 shows a summary of the team's spending as of the CDR milestone, with specific project elements grouped into payload, vehicle, and other. Since PDR, the team has spent roughly \$2250, with the majority of recent expenses coming from the subscale vehicle and payload materials for the November drop tests. Additionally, the spending categories have been refined for more clarity. Many of the items previously considered as lab supplies, such as those specifically used for vehicle fabrication, have been shifted to the subscale and full-scale vehicle categories. This modification has resulted in a significantly lower portion of the spending falling in the lab supplies category, with higher shares going to the subscale and full-scale vehicles. An itemized list of all expenses is also provided in the following tables.

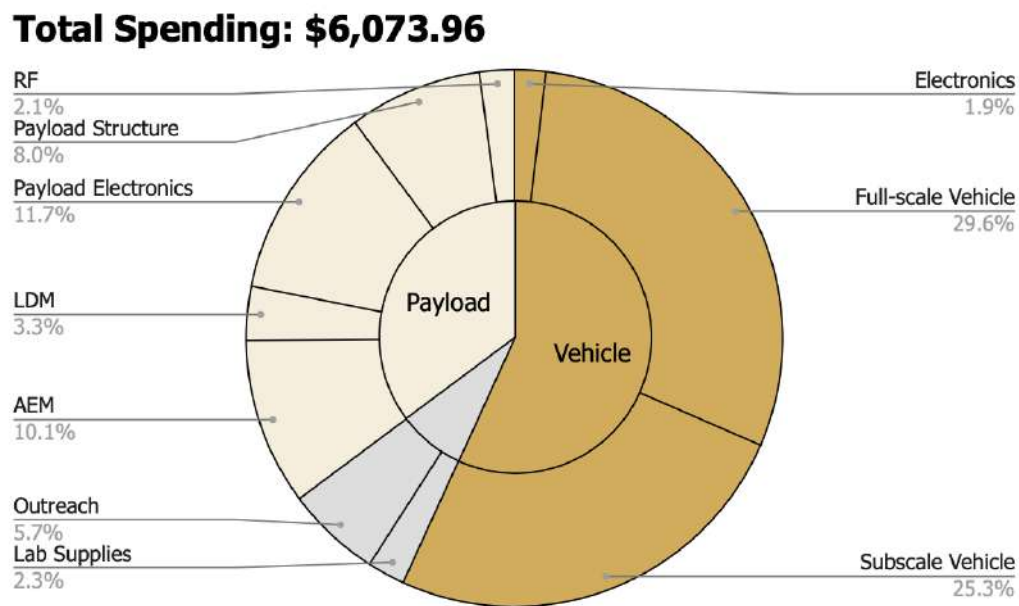


Figure 138: Total to-date spending, as of the CDR milestone.

Project Segment	Date Ordered	Vendor	Quantity	Cost	Total Cost	Description
Category: AEM					\$610.58	
PDR	9/12/24	Amazon	1	\$13.67	\$13.67	2 count 1/4" x 10' tape measure
PDR	9/12/24	Amazon	1	\$7.59	\$7.59	1/2" x 12' tape measure
PDR	9/12/24	Amazon	1	\$14.99	\$14.99	2 count mini solenoid
PDR	9/14/24	McMaster-Carr	2	\$11.40	\$22.80	1-3/8" ID needle roller bearing, 1/2" wide
PDR	9/14/24	McMaster-Carr	1	\$13.05	\$13.05	1/8" diameter flush bearing, 50 ct
PDR	9/24/24	McMaster-Carr	5	\$11.40	\$57.00	1-3/8" ID needle roller bearing, 1/2" wide
PDR	9/24/24	Amazon	3	\$7.59	\$22.77	1/2" x 12' tape measure
PDR	9/26/24	McMaster-Carr	5	\$11.97	\$59.85	Compression Spring 0.375" Long, 0.180" OD, 0.132" ID
PDR	9/26/24	McMaster-Carr	2	\$33.23	\$66.46	Rotary Speed Limiter Gearless, 360 Degree Dual Direction, 21 in.-oz. Torque
PDR	9/26/24	McMaster-Carr	1	\$18.86	\$18.86	Powder Lubricant 1 lbs. Can
PDR	9/26/24	McMaster-Carr	1	\$16.43	\$16.43	Captive Pins 18-8 Stainless Steel, 1/8" Diameter, 1/2" Long, Tps-125-8v
PDR	9/26/24	McMaster-Carr	1	\$12.73	\$12.73	Dowel Pin Aluminum, 1/8" Diameter, 3/4" Long
PDR	9/26/24	McMaster-Carr	10	\$1.51	\$15.10	Dry-Running MDS-Filled Nylon Sleeve Bearing Flanged, for 1/8" Shaft Diameter and 1/4" Housing ID, 1/4" Long
PDR	10/17/24	Amazon	1	\$15.99	\$15.99	Seymour 16-o83 Good Grip Slip Resistant Coating Spray, Black 12 Ounce (Pack of 1)
PDR	10/17/24	Amazon	1	\$14.75	\$14.75	Teflon Glass Cloth Tape, (2-inch x 33 Feet)
PDR	10/17/24	Amazon	3	\$19.09	\$57.27	Rotary Speed Limiter Geared, 360 Degree Clockwise, 2.88 in.-oz. Torque
CDR	11/6/24		2	\$7.59	\$15.18	1/2" x 12' tape measure
CDR	11/6/24	Amazon	1	\$7.99	\$7.99	4 Way Servo Wire Extension Cord
CDR	11/6/24	Amazon	1	\$27.99	\$27.99	10 ct 9g servo
CDR	11/6/24	McMaster-Carr	3	\$6.31	\$18.93	Torsion Spring 90 Degree Left-Hand Wound, 0.16" OD
CDR	11/6/24	Amazon	1	\$12.99	\$12.99	Dip And Grip (Black) Rubberized Plastic Coating Cables Resists Moisture Chemical Abrasion And Corrosion Slip Resistance Air Dry (Black) 8 Fl. Oz
CDR	11/19/24	McMaster-Carr	3	\$19.09	\$57.27	Rotary Speed Limiter Geared, 360 Degree Clockwise, 2.88 in.-oz. Torque
CDR	11/19/24	McMaster-Carr	6	\$3.42	\$20.52	Ultra-Low-Friction Oil-Embedded Sleeve Bearing for 3 mm Shaft Diameter, for 6 mm Housing ID, 4 mm Long
CDR	11/19/24	McMaster-Carr	6	\$3.40	\$20.40	302 Stainless Steel Torsion Spring 180 Degree Right-Hand Wound, 0.217" OD
CDR	11/19/24	McMaster-Carr	1	\$6.72	\$6.72	Compression Spring 0.25" Long, 0.25" OD, 0.21" ID
Category: Electronics					\$114.23	
PDR	9/14/24	Amazon	2	\$6.50	\$13.00	SMA female to UHF male PL-259
PDR	9/14/24	Amazon	2	\$8.88	\$17.76	UHF PL259 Cable RG58 3.3ft(1M) PL259 SO239 Male to Female Extension Cable Low
PDR	9/14/24	Amazon	2	\$6.87	\$13.74	SMA Coax Connector Cable Gender Changers,SMA Male/Female

PDR	9/24/24	Digikey	3	\$12.95	\$38.85	MPU6050 IMU
CDR	11/6/24	Amazon	1	\$30.88	\$30.88	Router
Category: Full-scale Vehicle					\$1,799.94	
PDR	9/12/24	McMaster-Carr	1	\$352.47	\$352.47	6" Diameter Al Mandrel 1/2" thickness. 3 Foot length
PDR	9/24/24	Other	2	\$141.27	\$282.54	6" G-12 Fiberglass airframe
PDR	9/12/24	HomeDepot	1	\$74.66	\$74.66	100 ft. Highly Reflective Mylar
PDR	9/12/24	Fiberglast	1	\$27.95	\$27.95	PVA Release Film 13-B (Gallon)
PDR	9/12/24	Amazon	1	\$14.51	\$14.51	Clay tape / Colorimetrics Gray Putty Tape/Butyl Tape
PDR	10/3/24	Fiberglast	3	\$32.95	\$98.85	OLFA® Rotary Cutter
PDR	9/12/24	Amazon	1	\$17.99	\$17.99	PLA Filament - BLACK
PDR	9/12/24	Amazon	1	\$18.99	\$18.99	PLA Filament - WHITE
PDR	9/12/24	Amazon	1	\$23.99	\$23.99	PLA Filament - GOLD
PDR	9/24/24	Other	2	\$190.00	\$380.00	800cc onyx filament
CDR	11/12/24	Amazon	1	\$11.99	\$11.99	Squeegees for epoxy application
CDR	12/11/24	Other	4	\$124.00	\$496.00	6" G-12 Fiberglass coupler
Category: Lab Supplies					\$137.88	
PDR	9/12/24	Amazon	1	\$20.13	\$20.13	73-Piece Wrench and Screwdriver Set
PDR	10/3/24	Amazon	2	\$7.28	\$14.56	Super glue
PDR	10/3/24	Amazon	1	\$8.11	\$8.11	Band-aids
CDR	11/6/24	Amazon	2	\$8.97	\$17.94	Left-handed scissors
CDR	11/6/24	Amazon	3	\$6.48	\$19.44	Regular use scissors
CDR	11/7/24	Amazon	1	\$12.99	\$12.99	24 Gauge Wire
CDR	11/12/24	Amazon	1	\$7.75	\$7.75	Isopropyl alcohol
CDR	11/12/24	Amazon	1	\$7.99	\$7.99	Cleaning rags
CDR	11/12/24	Amazon	2	\$8.99	\$17.98	Combination Lock
CDR	11/19/24	Amazon	1	\$10.99	\$10.99	1" by 30" sanding belt 24 pack
Category: LDM					\$197.72	
PDR	9/24/24	Amazon	1	\$7.59	\$7.59	8mm ID Linear Bearing
PDR	9/24/24	Amazon	1	\$11.99	\$11.99	8mm Linear Shaft
PDR	10/12/24	McMaster-Carr	5	\$15.55	\$77.75	Spring Pins for Leg Locking System
PDR	10/12/24	McMaster-Carr	10	\$2.25	\$22.50	8mm Set Screw Shaft Collar
PDR	10/12/24	Amazon	1	\$7.59	\$7.59	8mm ID Linear Bearing
PDR	10/12/24	Amazon	2	\$11.99	\$23.98	8mm Linear Shaft
PDR	10/12/24	McMaster-Carr	8	\$5.79	\$46.32	8mm Clamping Shaft Collar
Category: Outreach					\$347.79	
PDR	9/14/24	Amazon	10	\$16.99	\$169.90	Carboard space shuttles
PDR	10/3/24	Amazon	1	\$7.99	\$7.99	Counters for outreach events
PDR	10/17/24	Amazon	10	\$16.99	\$169.90	Space Shuttles for Outreach
Category: Payload Electronics					\$713.20	
PDR	9/12/24	Amazon	4	\$6.00	\$24.00	SO-239 Chassis Connector
PDR	9/12/24	Amazon	1	\$7.59	\$7.59	16-14AWG Wire Crimp Connectors
PDR	9/24/24	Other	1	\$115.00	\$115.00	Light APRS tracker
PDR	9/26/24	Digikey	100	\$13.48	\$13.48	EM Latch 6-Pin Connectors
PDR	9/26/24	Digikey	200	\$16.04	\$16.04	Crimp
PDR	9/27/24	Spark Fun	5	\$1.05	\$5.25	Screw Terminals
PDR	9/27/24	Spark Fun	5	\$1.05	\$5.25	Screw Terminals
PDR	9/27/24	Amazon	2	\$7.99	\$15.98	Desoldering Wick
PDR	9/27/24	Amazon	1	\$16.92	\$16.92	Lipo Battery

PDR	9/27/24	Amazon	1	\$22.69	\$22.69	LiPo Battery Charger
PDR	9/27/24	Amazon	1	\$9.99	\$9.99	DC 5V Converter
PDR	9/27/24	Amazon	1	\$13.99	\$13.99	DC 5V Converter
PDR	10/10/24	Microcenter	1	\$79.99	\$79.99	Raspberry Pi 5
PDR	10/12/24	Microcenter	1	\$79.99	\$79.99	Raspberry Pi 5
PDR	10/12/24	Amazon	3	\$8.79	\$26.37	Micro HDMI to HDMI Display Cable
CDR	11/6/24	Amazon	1	\$35.95	\$35.95	Soldering Iron
CDR	11/6/24	Amazon	3	\$6.11	\$18.33	Photoresistor
CDR	11/6/24	Other	3	\$8.66	\$25.98	Prototyping Board for Raspberry Pi
CDR	11/6/24	Amazon	1	\$39.99	\$39.99	Circuit Board Holder
CDR	11/12/24	Amazon	1	\$15.99	\$15.99	DROK Buck Converter 12v to 5v Pack of 2
CDR	11/19/24	Amazon	3	\$14.99	\$44.97	Electronics-Salon Pi Screw Terminal Block Breakout Module, for Raspberry Pi.
CDR	11/19/24	Amazon	3	\$18.99	\$56.97	GeeekPi GPIO Screw Terminal Block Breakout Board HAT
CDR	11/19/24	Amazon	1	\$22.49	\$22.49	BTECH APRS-K1 Multi-Function Universal Audio Interface Cable
Category: Payload Structure					\$488.43	
PDR	9/26/24	McMaster-Carr	4	\$7.78	\$31.12	1/4"-20 Eye Nuts
PDR	10/10/24	Other	2	\$128.50	\$128.50	1/8" Kevlar shock cord
PDR	10/17/24	McMaster-Carr	1	\$10.19	\$10.19	Copper wire (.016")
PDR	10/17/24	McMaster-Carr	1	\$9.16	\$9.16	Copper Wire (.025")
PDR	10/17/24	McMaster-Carr	1	\$8.83	\$8.83	Copper Wire (.051")
PDR	10/17/24	McMaster-Carr	1	\$12.08	\$12.08	Aluminum wire (.061")
PDR	10/17/24	McMaster-Carr	1	\$12.48	\$12.48	Aluminum wire (.032")
PDR	10/17/24	Amazon	1	\$9.99	\$9.99	Solder wire (2 mm)
CDR	11/6/24	Metal Supermarkets	5	\$9.40	\$46.98	0.313" Diameter by 26" long 6061-T6511 Aluminum round bar
CDR	11/6/24	Amazon	1	\$88.00	\$88.00	Raspberry Pi 5 8GB
CDR	11/6/24	McMaster-Carr	30	\$4.37	\$131.10	5/16" Aluminum Shaft Collar
Category: RF					\$127.50	
PDR	10/10/24	Spark Fun	5	\$1.25	\$6.25	SMA Connector
PDR	10/10/24	Other	1	\$115.00	\$115.00	Light APRS tracker
PDR	10/12/24	Spark Fun	5	\$1.25	\$6.25	SMA Connector
Category: Subscale Vehicle					\$1,536.69	
PDR	9/12/24	McMaster-Carr	1	\$266.88	\$266.88	4" Diameter Al Mandrel 1/2" thickness. 3 Foot length
PDR	9/26/24	Other	2	\$141.27	\$282.54	4" G-12 Fiberglass airframe
PDR	10/3/24	Fiberglast	1	\$37.50	\$37.50	4 oz Fiberglass Cloth x 50" Wide
PDR	10/3/24	Other	1	\$25.75	\$25.75	High temperature mold release
CDR	11/6/24	Madcow Rocketry	2	\$46.00	\$92.00	4" G-12 Coupler
CDR	11/6/24	DragonPlate	1	\$222.84	\$222.84	Quasi-isotropic Solid Carbon Fiber Sheet ~ 1/8" x 12" x 24"
CDR	11/6/24	Wildman Rocketry	1	\$71.39	\$71.39	G-12 Fiberglass air frame
CDR	11/6/24	Other	3	\$82.99	\$248.97	Aerotech I366R motor reload kits for subscale
CDR	11/6/24	Other	1	\$6.48	\$6.48	Regular use scissors
CDR	11/7/24	Apogee Rockets	1	\$108.97	\$108.97	RMS 38/600 Casing w/ forward seal disk
CDR	11/7/24	Apogee Rockets	1	\$51.10	\$51.10	38mm Aft Closure
CDR	11/7/24	Apogee Rockets	1	\$62.22	\$62.22	38mm Standard Forward Closure
CDR	11/7/24	Other	1	\$25.29	\$25.29	18-38mm Delay Tool
CDR	11/7/24	Apogee Rockets	2	\$121.15	\$242.30	Entacore AIM Dual Deployment Altimeters
CDR	11/7/24	Amazon	1	\$11.44	\$11.44	9V Batteries

CDR	11/19/24	McMaster-Carr	4	\$6.27	\$25.08	5/16"-18 steel threaded rods
CDR	11/19/24	McMaster-Carr	2	\$2.33	\$4.66	5/16"-18 aluminum threaded rods
CDR	11/20/24	Amazon	4	\$4.54	\$18.16	5/16"-18 aluminum threaded rods
Total:					\$6,073.96	

6.3.4 Vehicle and Payload Cost Analysis

Alongside the expense tracking provided in Sections 6.3.2 and 6.3.3, and the projection shown in Section 6.3.5, VADL has compiled an estimate of the total on-the-pad cost of the full-scale vehicle and payload. While the team's budgeting considers the supplies already available in the lab, the on-the-pad cost represents the true value of all vehicle and payload components and fabrication tools. It is important to acknowledge the full expenses associated with competing in the NASA Student Launch Challenge, and the following table serves to accomplish this task. Expenses are split between vehicle and payload, each of which are further divided into subcategories. Total prices are shown for each category and subcategory, along with their respective percentage of the category or total cost.

Category	Item	Count	Unit Price	Net Price	
Rocket				\$3,819.24	
Structural				\$1,933.23	50.6%
	6" Fiberglass Airframe, per inch	90	\$2.35		\$211.91
	6" Fiberglass Coupler, per inch	56	\$7.75		\$434.00
	75 mm Bluetube Motortube, per inch	16	\$0.72		\$11.57
	DragonPlate Fin Board	1	\$135.00		\$135.00
	DragonPlate Water Jet Cutting	1	\$60.00		\$60.00
	6" Aluminum 6061 Round Bar for Bulk Plates, per inch	4	\$25.37		\$101.49
	Avionics Sled, 3D Printed	1	\$3.08		\$3.08
	Cesaroni L1720 Motor	1	\$301.74		\$301.74
	Rocket Motor Shipping and Insurance	1	\$138.65		\$138.65
	Motor Retention Ring, 3D Printed	2	\$1.82		\$3.64
	Fiberglass Cloth, per square inch	5400	\$0.01		\$37.50
	Miter Saw	1	\$229.00		\$229.00
	2-ton Epoxy Hardener, per mL	180	\$0.08		\$14.32
	High Temperature Mold Release, per oz	6	\$1.84		\$11.04
	Vacuum Bag	6	\$18.95		\$113.70
	Vacuum Bag Bleeder/Breather	6	\$11.95		\$71.70
	Half Mask Respirator Kit	2	\$18.94		\$37.88
	#2-56 Shear Pin	8	\$0.06		\$0.44
	Launch Rail Button	2	\$2.23		\$4.46
	3/8" Eye Bolt and Nuts	2	\$4.01		\$8.02
	3D Printed Nose Cone Mold	1	\$2.16		\$2.16
	3D Printed Boat Tail	1	\$1.94		\$1.94
Avionics				\$498.08	13.0%
	EasyMini Altimeter	4	\$101.78		\$407.12
	9V Battery	4	\$4.49		\$17.94
	9V Battery Connector	4	\$5.99		\$23.96
	3/8" Eye Bolt and Nuts	4	\$4.01		\$16.04
	Screw Switch	4	\$4.00		\$16.00
	Ejection Canister Cap Kit	4	\$1.88		\$7.52
	4F Black Powder, per g	12	\$0.07		\$0.87
	M2.5 Screw	16	\$0.05		\$0.80
	E-match	4	\$1.18		\$4.71
	18 gauge wire, per ft	4	\$0.30		\$1.20
	1/4"-20 bolts	4	\$0.48		\$1.92
Recovery				\$1,387.92	36.3%
	Fruity Chutes 5' Lander Parachute	1	\$212.85		\$212.85
	Fruity Chutes 18" Drogue Parachute	1	\$70.95		\$70.95
	Fruity Chutes 8' Main Parachute	1	\$382.97		\$382.97
	Quick Link	5	\$1.25		\$6.25
	40', 1/2" Kevlar Shock Cord	1	\$76.15		\$76.15
	20', 1/2" Kevlar Shock Cord	2	\$53.00		\$106.00
	10', 1/2" Kevlar Shock Cord	1	\$33.20		\$33.20
	25', 1/4" Kevlar Shock Cord	1	\$29.49		\$29.49
	Swivel Link	3	\$9.90		\$29.70
	Nomex Fire Blanket	2	\$29.70		\$59.40
	Fireball	4	\$23.99		\$95.96
	70cm 100mw GPS/APRS Transmitter	1	\$285.00		\$285.00
Payload				\$2,338.31	
Structural				\$769.01	32.9%
	3D Printed SCM	1	\$2.93		\$2.93
	3D Printed AEM	3	\$1.47		\$4.41
	8mm Aluminum Rod	4	\$11.99		\$47.96
	8mm Shaft Collar	24	\$5.79		\$138.96
	1/2" x 12' Tape Measure	3	\$7.59		\$22.77
	9g Servo	3	\$2.79		\$8.37
	Torsion Spring	7	\$6.31		\$44.17
	Rotary Speed Limiter	3	\$19.09		\$57.27
	6" 6061 Aluminum Round Bar for Bulk Plates, per inch	1	\$25.37		\$25.37
	Southco EM Latch	3	\$124.97		\$374.91
	3D Printed Battery Housing	1	\$1.78		\$1.78
	3D Printed Electronics Housing	1	\$2.21		\$2.21
	75mm Blue Tube Parachute Tube, per inch	12	\$0.72		\$8.64
	3/8"-16 Eyebolt and Nuts	2	\$4.01		\$8.02
	Latch Hook	4	\$5.31		\$21.24
	3D Printed Leg Mounts				
	3D Printed bulk plates	5	\$7.82		\$39.10
Electronics				\$1,569.30	67.1%
	Raspberry Pi 5 8 GB RAM	1	\$80.00		\$80.00
	HOOVO 3s 11.1 V 6500mAh Lipo Battery	1	\$62.99		\$62.99
	Vector NAV VN100 Rugged	1	\$950.00		\$950.00
	Light APRS Board	1	\$115.00		\$115.00
	Custom PCB & On-board Components, Including Custom Buck Converter and Custom Photoresistor Circuit	1	\$34.08		\$34.08
	SO 239 UHF Female RF Coax Connector w/ 4 hole flange panel mount	1	\$9.99		\$9.99
	70cm 100mw GPS/APRS Transmitter	1	\$285.00		\$285.00
	Wires, Connectors, Miscellaneous	1	\$32.24		\$32.24
Total:				\$6,157.55	

6.3.5 Expense Summary

The to-date spending and vehicle, payload, and travel cost projections were combined to form a chart of cumulative project spending over time, shown in Figure 139. Expenses are categorized into four designations: vehicle, payload, miscellaneous, and travel. The result is a clear picture of the project lifecycle as it pertains to expenses. Minimal costs are incurred at the beginning of the project, followed by a steep uptick as the project is defined and rapid design, build, and test cycles commence. Significant vehicle expenses are also seen in the first half of the project, as the team prepares for the subscale flight and also makes an effort to procure items for the full-scale vehicle well in advance to avoid schedule setbacks due to long lead times. Expenses are projected to plateau in the spring, albeit with a sharp increase at the end of the project due to the steep costs associated with travel to the competition.

Project Spending Over Time

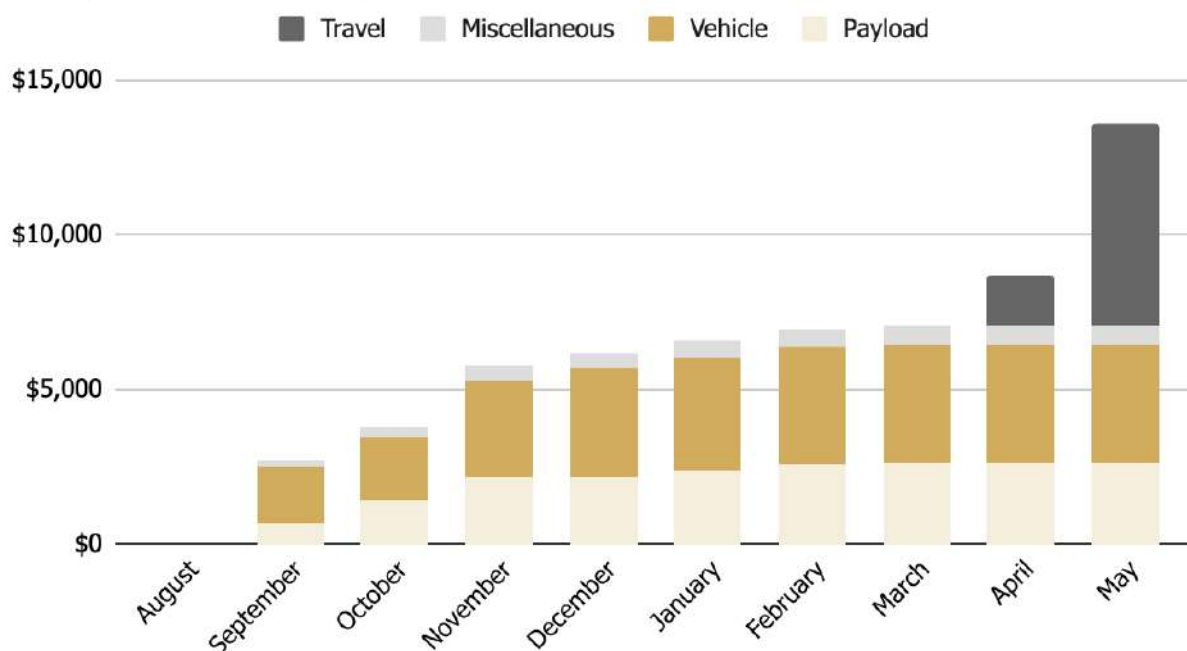


Figure 139: Spending over time for the 2024-2025 competition year, projected for remaining months.

VADL expects spending to level off in the spring semester as the team shifts to full-scale vehicle fabrication with the components and manufacturing materials already procured. Nevertheless, the team's contingency margin, and additional surplus in funds on top of that, will ensure that VADL does not exceed its allocated funding.

6.4 Timeline

VADL adheres to the project lifecycle phases as outlined in the NASA Systems Engineering Handbook. Pre-phase A was dedicated to concept studies, ideation, and divergent thinking, and occurred during the early proposal phase of USLI. Phase A built on those concept studies in an attempt to define a top-level mission architecture and design by methodically researching and narrowing down the available options until one clear choice flew above the rest. Once a leading design was chosen and proposed (in the USLI Proposal), the team entered Phase B and began designing and prototyping systems that would realize the design chosen in Phase A. Upon the conclusion of Phase B, leading designs for each subsystem and system were chosen based on extensive prototyping and analysis. Phase C involves the selection and fabrication of the final design, and Phase D involves comprehensive testing including test launches at the subscale and full scale level. Phase E encompasses the completion of the mission, and Phase F is the closeout phase in which the mission is evaluated and next year's design team is integrated into the laboratory.

The team has updated the timeline presented at the PDR milestone to reflect new learnings and project maturity. In particular, the full-scale vehicle fabrication and testing schedule has been revised, as informed by the team's experience preparing for the subscale flight. Key dates and breakdowns of each phase are visualized in the Gantt chart progression in the following figures.

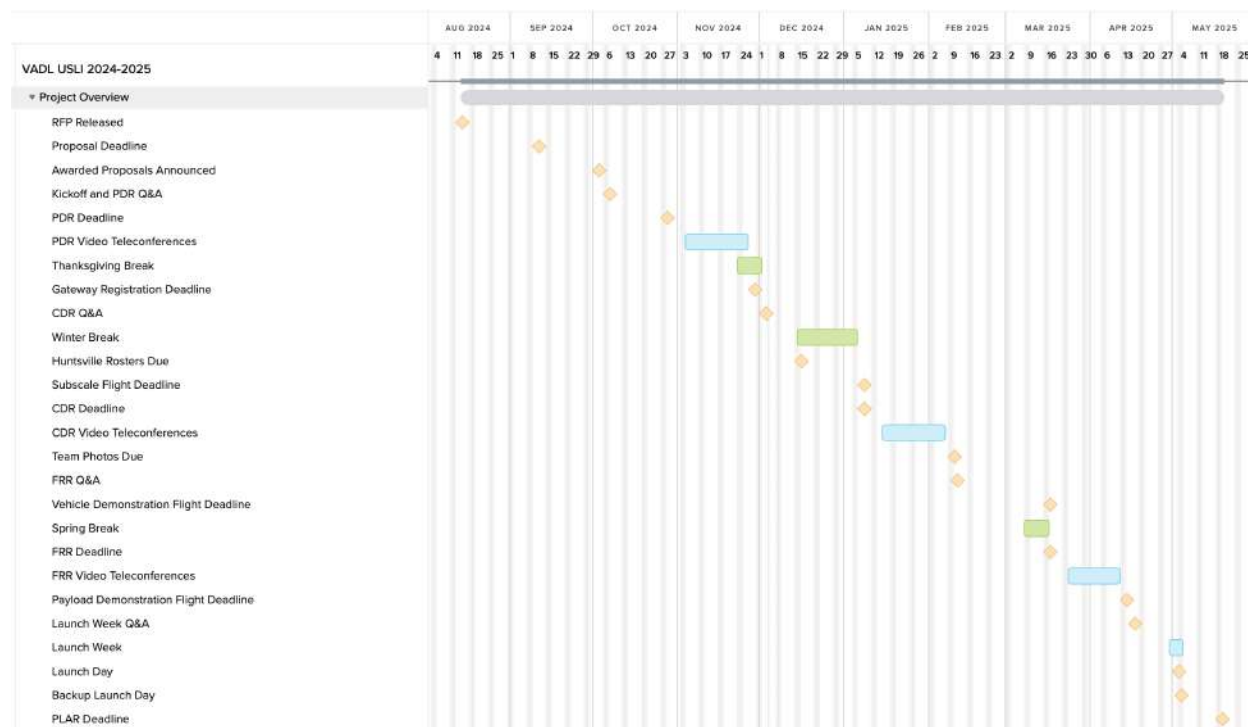


Figure 140: Key project dates and milestones.

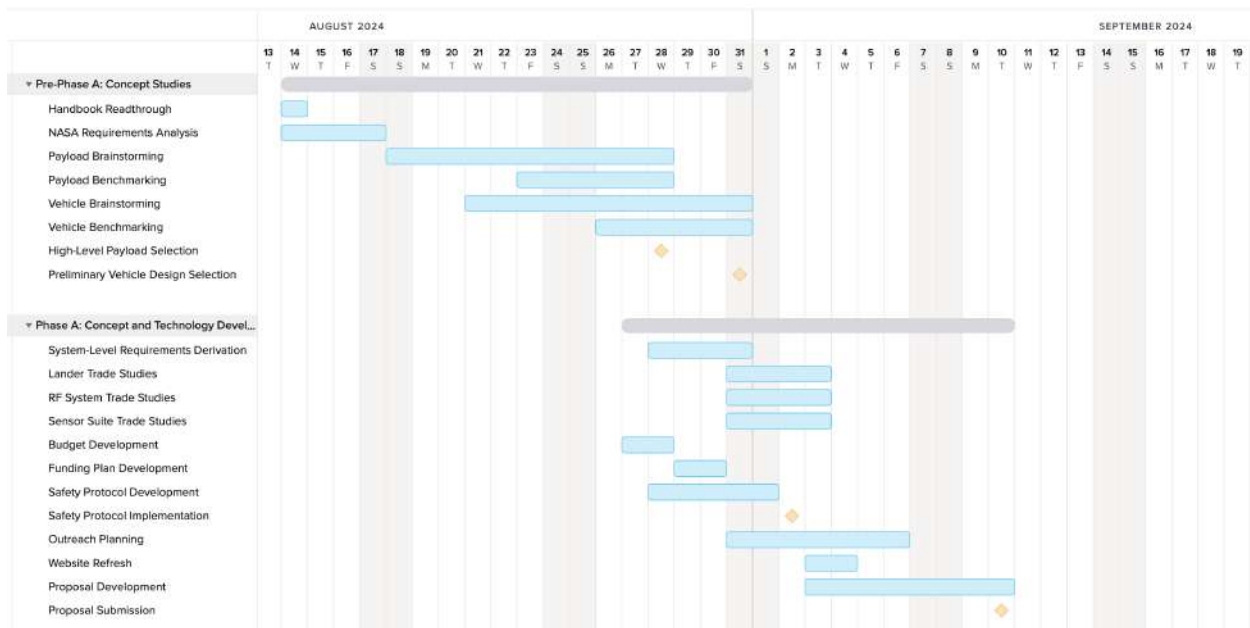


Figure 141: Pre-phase A and Phase A of the project.

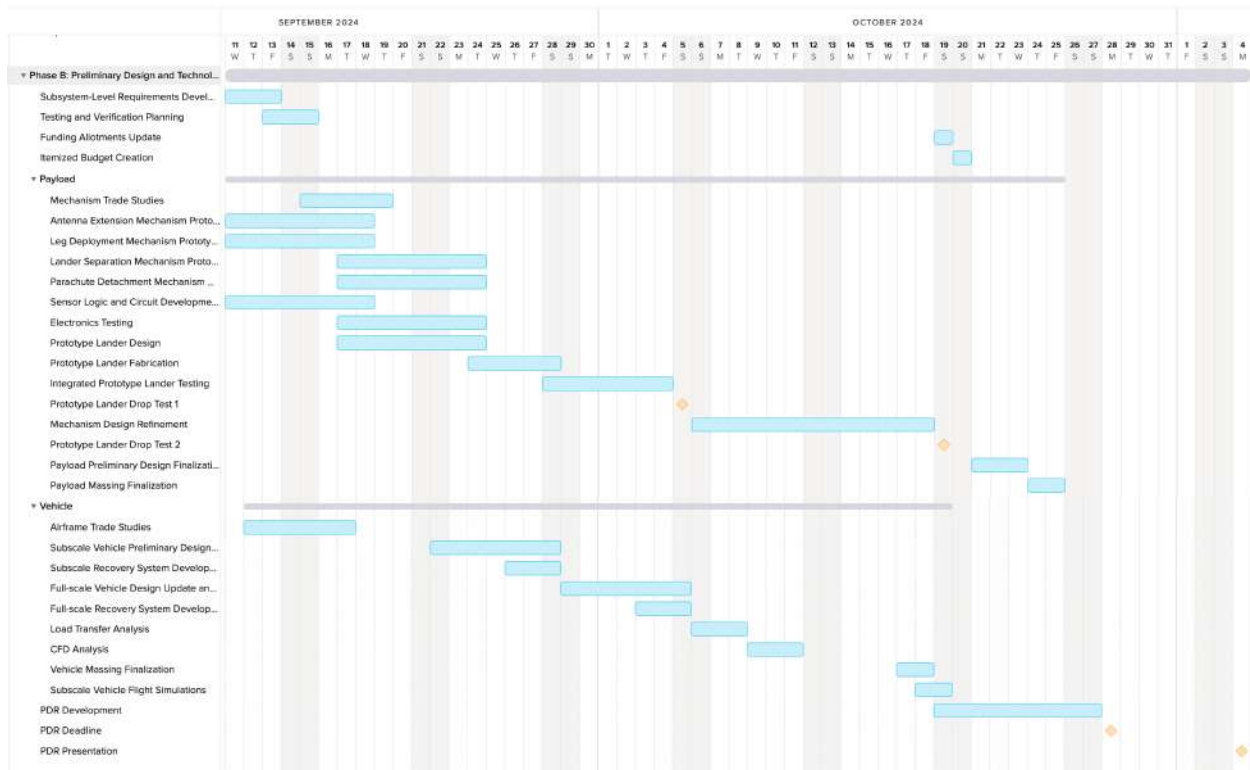


Figure 142: Phase B of the project.

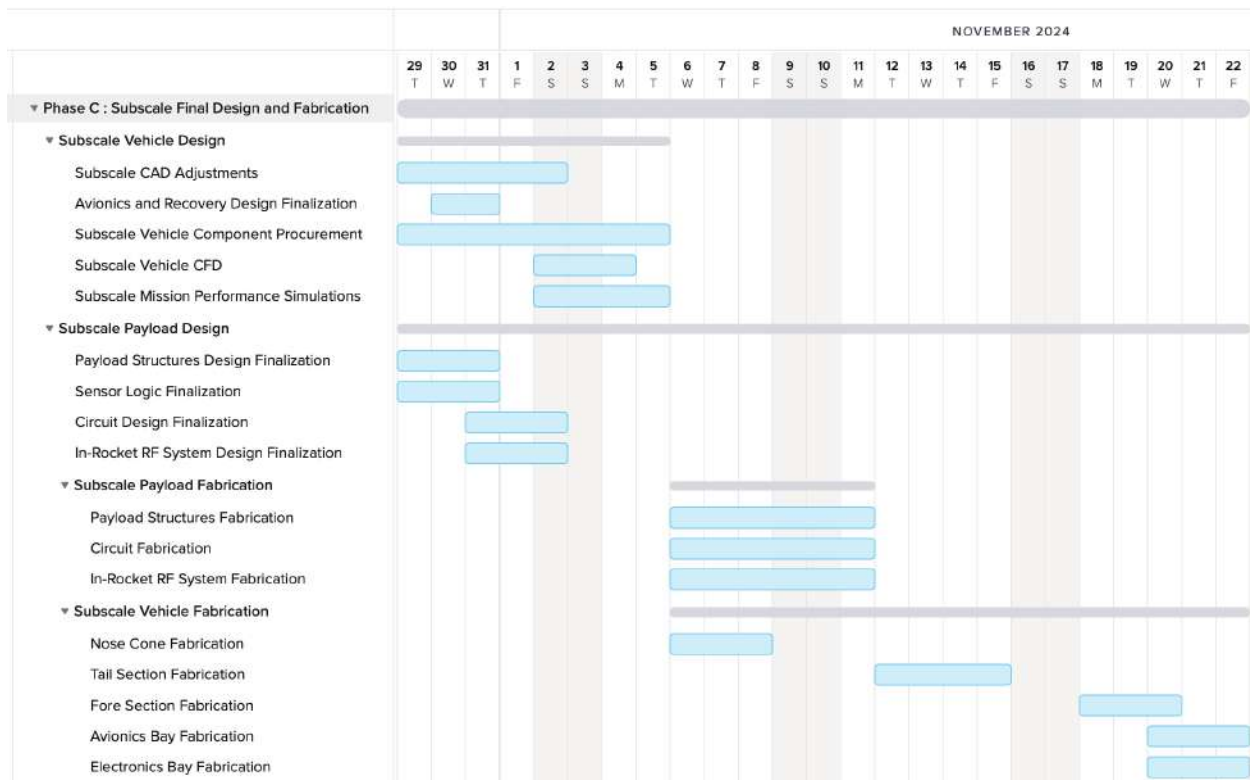


Figure 143: Subscale portion of Phase C of the project.

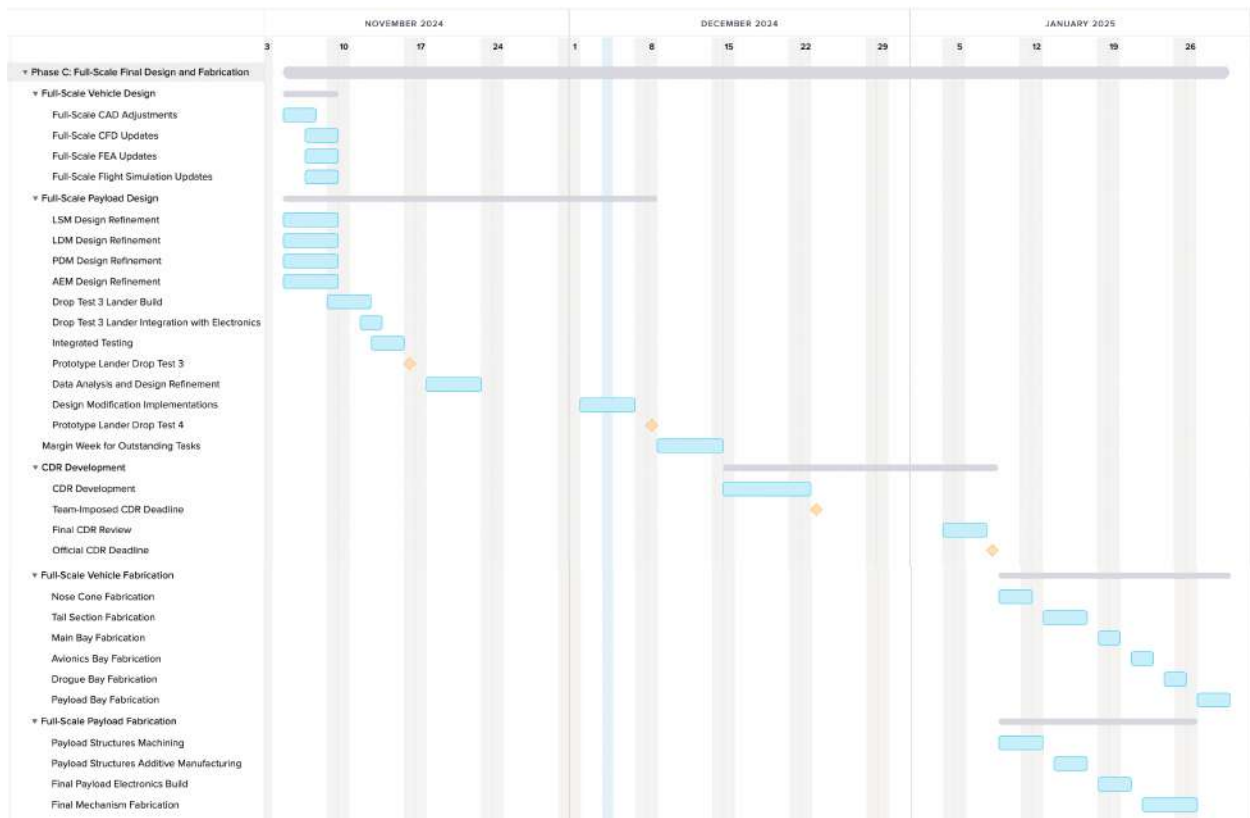


Figure 144: Full-scale portion of Phase C of the project.

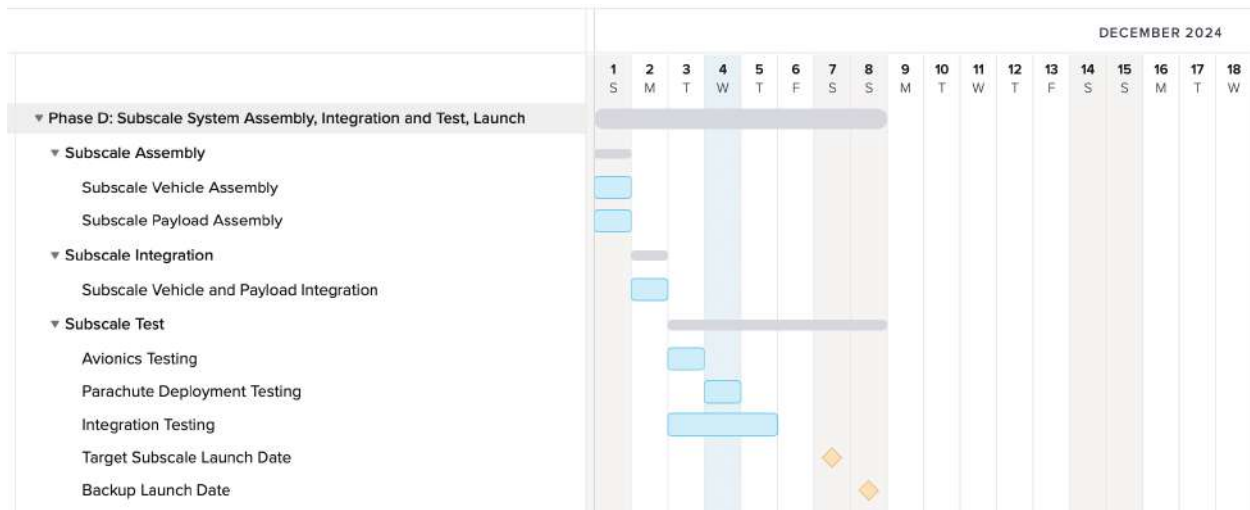


Figure 145: Subscale portion of Phase D of the project.

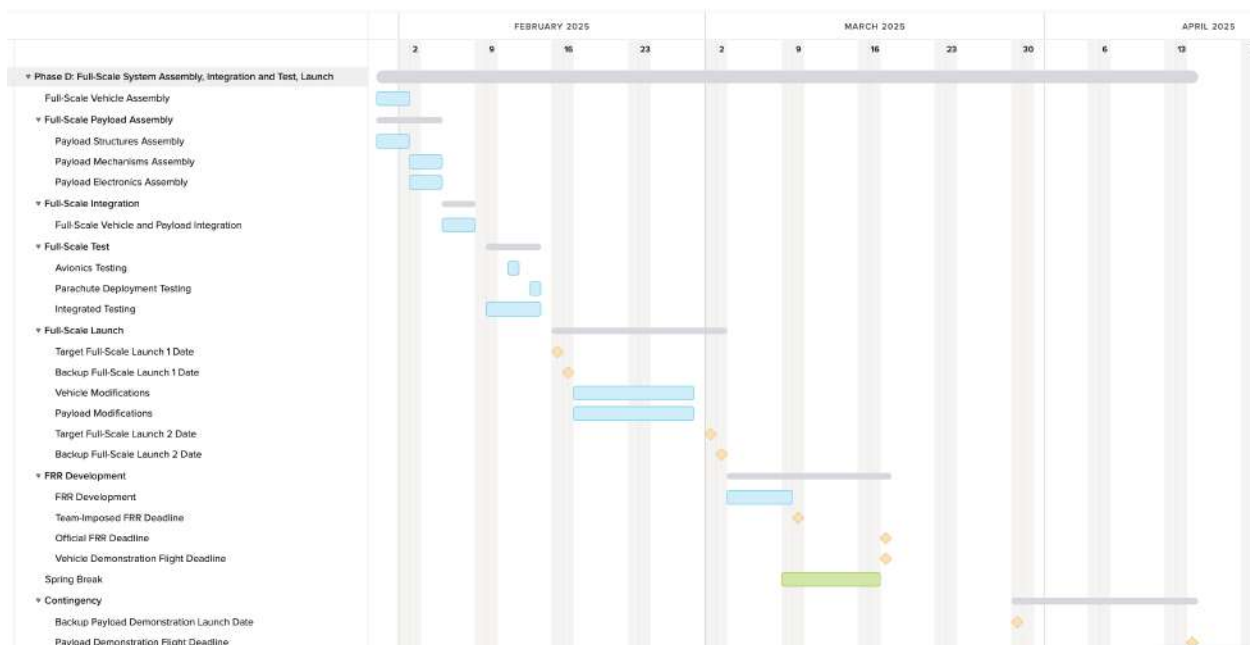


Figure 146: Full-scale portion of Phase D of the project.

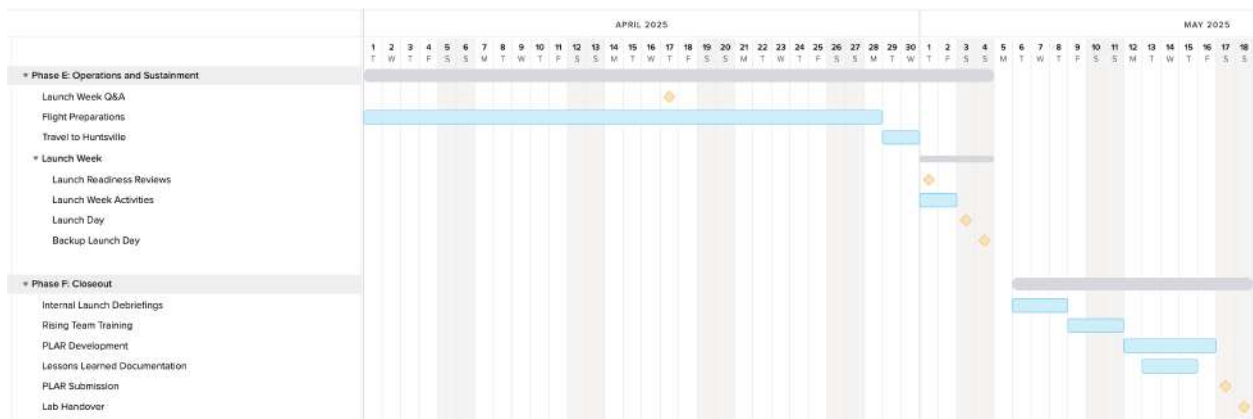


Figure 147: Phases E and F of the project.

A References

- [1] *1L10.50 LOCAL GRAVITY CHART*. URL: https://physics.montana.edu/demonstrations/video/1_mechanics/demos/pics/localgravitychart.jpg.
- [2] *Aerodynamic Coefficients in Cambridge Rocketry*. URL: <https://cambridgerocket.sourceforge.net/AerodynamicCoefficients.pdf>.
- [3] *Cesaroni 3660L1720-P Motor Data*. URL: <https://www.thrustcurve.org/motors/Cesaroni/3660L1720-P/>.
- [4] C. Lawrence et al. “Orion Crew Member Injury Predictions during Land and Water Landings”. In: *Earth Space 2008: Engineering, Science, Construction, and Operations in Challenging Environments* (2008). DOI: <https://ntrs.nasa.gov/api/citations/20080013357/downloads/20080013357.pdf>.
- [5] *Quasi-isotropic Solid Carbon Fiber Sheet 1/8 x 12 x 24*. URL: https://dragonplate.com/Quasi-isotropic-Solid-Carbon-Fiber-Sheet-1_8-x-12-x-24.
- [6] *Simulation and Testing of Rail Launchers*. URL: <https://nar.org/member-only-reports/rd/NARAM-61/N61-Simulation-Testing-Rail-Launchers-Flanigan.pdf>.

B VADL Safety Agreement

Written Safety Agreement

By signing below, I agree to abide by all laws, standards, regulations and guidelines set forth by the National Association of Rocketry (NAR), the Federal Aviation Administration (FAA), the state of Tennessee, and pertinent Material Safety Data Sheets (MSDS). I acknowledge that our rocket must undergo a range safety inspection prior to launch, and I will adhere to the results of that inspection. The ultimate authority on rocket safety lies with the Range Safety Officer (RSO), who has the right to deny our rocket's launch if any safety concerns arise. As the team mentor bears the final responsibility for ensuring the safe flight and retrieval of our rocket, we will not proceed with the launch unless the mentor has reviewed the design, evaluated the construction, and confirmed that it complies with established amateur rocketry design and safety standards. My signature confirms that I have read, understand, and will adhere to all relevant safety procedures as outlined by the Safety Officer.

Print Name

Ian Morgan	Signature: <u>[Signature]</u>	Date: <u>9/6/24</u>
Rais Nurhidajat	Signature: <u>[Signature]</u>	Date: <u>9/6/24</u>
Saksham Sharma	Signature: <u>[Signature]</u>	Date: <u>9/7/24</u>
Sitan Huang	Signature: <u>[Signature]</u>	Date: <u>9/7/24</u>
Zachary Stutsman	Signature: <u>[Signature]</u>	Date: <u>9/6/24</u>
Benjamin Racape	Signature: <u>[Signature]</u>	Date: <u>9/8/24</u>
Santino Clemente	Signature: <u>[Signature]</u>	Date: <u>9/7/24</u>
Samantha Civil	Signature: <u>[Signature]</u>	Date: <u>9/1/24</u>
Timothy Shaheen	Signature: <u>[Signature]</u>	Date: <u>09/07/24</u>
Jonathan Zak	Signature: <u>[Signature]</u>	Date: <u>09/07/24</u>
Jae-Hoon Smith	Signature: <u>[Signature]</u>	Date: <u>9/8/24</u>
Jack Liedel	Signature: <u>[Signature]</u>	Date: <u>9/10/24</u>
Alend Khoshnaw	Signature: <u>[Signature]</u>	Date: <u>9/8/24</u>
Jillian Hornby	Signature: <u>[Signature]</u>	Date: <u>09/07/24</u>
Patrick Chen	Signature: <u>[Signature]</u>	Date: <u>9/6/24</u>
Ashley Fu	Signature: <u>[Signature]</u>	Date: <u>9/8/24</u>
Neal Bagai	Signature: <u>[Signature]</u>	Date: <u>09/07/24</u>

C VUSE Shop Safety Guidelines

Vanderbilt University School of Engineering

Environmental Health and Safety Policy

1.0 Purpose

The School of Engineering is strongly committed to protecting the environment and human health in all of its operations. To fulfill this commitment, the School recognizes that pro-active efforts must be taken to ensure that sound environmental and safety planning is integrated into every level of School decision-making. The School expects that all faculty, staff, and students will share this responsibility for safety and security of themselves, fellow faculty and staff, students, and guests, and maintain reasonable care when using School and University property. It is important that all members of the School of Engineering comply with University written policies and guidelines set forth by offices including but not limited to Vanderbilt Environmental Health and Safety (VEHS), <http://www.safety.vanderbilt.edu>, as well as Vanderbilt University Police Department (VUPD), <http://police.vanderbilt.edu/>, and all Federal, State and local regulatory bodies, including fire and police agencies.

Effective environmental, health and safety performance is important to the School of Engineering in its relationships with students, faculty, staff, regulatory agencies and the general public. All employees are expected to participate in formulation of and to comply with safety practices relating to their work and University guidelines. To assist managers in the School of Engineering in meeting their environmental and safety responsibilities, the following policy is established.

2.0 Policy

It is the policy of the School of Engineering to:

- 2.1 ensure that all operations are conducted safely and in compliance with all Federal, State and local applicable environmental, health, and safety laws and regulations. In the absence of specific laws or regulations, good management practices shall be followed;
- 2.2 design, construct, and operate all facilities in a manner that protects human health, safety of the occupants, and the environment;
- 2.3 ensure that environmental health and safety considerations are an integral part of the School's master planning efforts;
- 2.4 work with Vanderbilt Environmental Health and Safety, <http://www.safety.vanderbilt.edu>, to develop, follow, and continually improve environmental, health, and safety procedures and practices for all facilities, projects, and operations;
- 2.5 perform periodic internal environmental, health, and safety program reviews to correct deficiencies, establish goals, and identify priorities;

- 2.6 promote environmental health, and safety awareness among all faculty, staff and students and ensure that Principal Investigators (PI's) and supervisors proper training using, as appropriate, training courses and other resources provided by Vanderbilt Environmental Health and Safety, <http://www.safety.vanderbilt.edu>;
- 2.7 ensure adherence to University laboratory and Departmental safety procedures;
- 2.8 ensure, as far as is reasonably practicable, maintenance of adequate first aid facilities throughout School premises;
- 2.9 ensure, as far as is reasonably practicable, provision of adequate and correct safety equipment throughout School premises;
- 2.10 ensure, as far as is reasonably practicable, provision of protective clothing and equipment as is necessary to maintain the health and safety of the School's staff, faculty, students and visitors;
- 2.11 encourage the reporting of violations of safe practices to Vanderbilt Environmental Health and Safety, <http://www.safety.vanderbilt.edu>;
- 2.12 ensure the reporting of on-the-job accidents in a timely manner in accordance with OSHA and State Department of Health guidelines to the Risk Management Office (Supervisors are responsible for filing out a First Report of Work injury form as described in HR policy #HR-011, <http://www.vanderbilt.edu/HRS/policies/workaccident.html> . The form itself is available at <http://polproc.mc.vanderbilt.edu/policy/pol9-23.html>.);
- 2.13 ensure the seeking of diagnosis and possible treatment from the Vanderbilt Occupational Health Clinic, <http://www.vanderbilt.edu/HRS/wellness/occhealth.htm>, in the event of personal injury from an on-campus accident requiring immediate attention;
- 2.14 encourage personal accountability and emphasize compliance with standards and conformance with University and School policies and best practices during employee training and in performance reviews;
- 2.15 communicate our desire to continuously improve our performance and foster the expectation that every employee, student, and contractor on Engineering School premises will follow this policy and report any environmental, health, or safety concern to Engineering School management.
- 2.16 Cooperate with any environmental, health, and safety program reviews by Vanderbilt Environmental Health and Safety, <http://www.safety.vanderbilt.edu>, or external regulators and ensure prompt improvement of deficiencies.

**Vanderbilt University
School of Engineering**

Environmental Health and Safety: Roles and Responsibilities

<u>Personnel</u>	<u>Roles and Responsibilities</u>
Dean's Office or Designee	• Supports University policies and training programs, as administered by Vanderbilt Environmental Health and Safety, http://www.safety.vanderbilt.edu, related to the use of appropriate environmental health and safety practices and facilities in research and teaching activities. • Supports safety programs to implement environmental health and safety policies and procedures for safety and compliance in specific departments, institutes, centers, and units. • Reviews annual reports of activities, successes, and problems in environmental health and safety in the Engineering School, and supports departmental efforts to address improvements. • Appoints Chair and members of School of Engineering Safety Committee and ensures that the committee is active.
Department Chair/Center Director	• Supports University and School policies and training programs, as administered by Vanderbilt Environmental Health and Safety, http://www.safety.vanderbilt.edu, related to the use of appropriate environmental health and safety practices and facilities in research and teaching activities in their department/center. • Establishes departmental-specific safety programs and procedures to implement environmental health and safety policies and procedures for safety and compliance in their department/center. • Assumes responsibility for ensuring that good safety practices and environmental compliance are enforced within their department/center. • Supports and establishes the authority of the departmental Safety and Environmental Compliance Officer (SECO) to carry out their duties and to function effectively within the department. • Reports non-compliance and unsafe working

**Principal Investigator
(P.I.)/Faculty**

conditions to the Vanderbilt Compliance Office,
<http://www.vanderbilt.edu/compliance/html/office.htm>

- Ensures departmental personnel receive required training that is appropriate to their work.
- Supports activities of departmental faculty that are assigned to safety committees.
-
- Has knowledge of institutional, local, state, and federal environmental health and safety policies applicable to the research being performed. P.I.'s are responsible for safety, health and compliance issues in their labs or for those experiments being done under their supervision.
- Has knowledge of potential hazardous and/or regulated materials used in research and is trained in methods to minimize the risks of the hazards and to meet institutional compliance requirements.
- Seeks appropriate approvals/guidance through the departmental SECO, School Safety Committee, and VEHS, <http://www.safety.vanderbilt.edu>, for the use of potentially hazardous or regulated materials. Installation requirements of equipment, exhaust systems, etc. involving potentially hazardous or regulated materials should be coordinated through the Vanderbilt Campus Planning and Construction Department, <http://www.cpc.vanderbilt.edu>, prior to requests for construction estimates. Depending on the requirements, engineering analysis and assessment may be required.
- At the initiation of each laboratory intensive project and at least annually during the course of the project, identifies the safety and environmental compliance requirements of that project and discusses them with the departmental SECO.
- Implements appropriate safety policies as they apply to the research being performed.
- Ensures the integrity of equipment and facilities designed to contain potential hazards by providing recommended maintenance and calibration.
- Facilitates internal and external inspections of research facilities and equipment.
- Facilitates or provides environmental health and safety training and education on a continuing basis for research staff and students.
- Takes prompt action when unsafe acts or conditions are reported or noted.

Departmental Safety and Environmental Compliance Officer (SECO)

- Follows written emergency procedures.
- Reports all on-the-job accidents, illnesses, and/or potential exposures to the Vanderbilt Occupational Health Clinic, <http://www.vanderbilt.edu/HRS/wellness/occhealth.htm> . (The P.I. is responsible for filing out a First Report of Work injury form as described in HR policy #HR-011, <http://www.vanderbilt.edu/HRS/policies/workaccident.html> . The form itself is available at <http://polproc.mc.vanderbilt.edu/policy/pol9-23.html> .);
- Requests medical assistance as required and provides information regarding hazards involved to medical personnel.
- Develops and maintains a written Chemical Hygiene Plan (for laboratories that make use of chemicals and hazardous materials) that includes a) Standard Operating Procedures for laboratory processes, b) current inventory of hazardous materials, c) Material Safety Data Sheet (MSDS) for all chemicals, d) Personal Protective Equipment (PPE) for lab personnel, e) proper storage and disposal methods for all hazardous wastes, and f) written emergency procedures.
-
- Maintains a thorough working knowledge of the Vanderbilt Environmental Health and Safety, <http://www.safety.vanderbilt.edu> , web site and the resources provided by VEHS.
- Liaison with Vanderbilt Environmental Health and Safety (VEHS), <http://www.safety.vanderbilt.edu>, Lab Compliance Assistance Program.
- Is current in all required training appropriate for the hazards present in their department.
- At the initiation of each laboratory intensive project and at least annually during the course of the project, identify the safety and environmental compliance requirements of that project and discuss them with the principal investigator.
- Maintains documentation necessary to demonstrate compliance with applicable safety and environmental compliance requirements.
- Notify the Department Chair and Principal Investigator (P.I) when a project or laboratory is out of compliance and communicate the actions required to be

in compliance.

- Periodically monitor instructional and research labs to be sure that health, safety and environmental compliance measures are being properly observed.
- Provides assistance in directing questions from faculty, staff, or students to appropriate resources at VEHS, <http://www.safety.vanderbilt.edu> , or elsewhere.
- Provides assistance in ensuring that lab workers have participated in appropriate courses offered through VEHS, <http://www.safety.vanderbilt.edu> .
- Provides assistance in obtaining appropriate signage through VEHS, <http://www.safety.vanderbilt.edu> , for safety warning, chemical disposal, etc.
- Assists P.I.'s with Chemical Hygiene Plan development.
- Coordinates with the Department Chair and, through the department representative, with the School of Engineering Safety Committee.

**Laser Safety Officer
(LSO)**

- Monitors and enforces the control of laser hazards and effects the knowledgeable evaluation and control of laser hazards for all classes of lasers used in the School. The American National Standard for Safe Use of Lasers (ANSI Z136.1) is adopted as the standard for the safe use of lasers and laser systems. Users of lasers and their supervisors must adhere to the practices and procedures described in the ANSI Standard.
- Classifies laser systems and establishes the Nominal Hazard Zone (NHZ) for all lasers used in the School. Class 3b and 4 unclosed lasers require a well-designated NHZ around the laser within which a person can be exposed to radiation in excess of the Maximum Permissible Exposure (MPE).
- Approves Standard Operating Procedures (SOP's) for all lasers used in the School. Written SOP's are required of all class 3b and 4 lasers and are enforced by the LSO.
- Approves signs and labels.
- Audits, recommends, or approves controls used in ensuring safety.
- Prior to use of new laser systems, approves the laser installation facilities and equipment.
- Periodically audits safety features in each of the laser installations in the School.
- Assists in ensuring that adequate education and

training is obtained by all personnel. As a minimum this requires taking the Vanderbilt Environmental Health and Safety, <http://www.safety.vanderbilt.edu>, training course on Laser Safety. For class 3b and 4 lasers, only authorized, trained personnel may be present in the vicinity of the laser during its operation.

- Determine personnel categories.
- Ensure that appropriate medical surveillance through the Vanderbilt Occupational Health Clinic (OHC), <http://www.vanderbilt.edu/HRS/wellness/occhealth.htm>, is obtained for all workers with class 3b or 4 lasers.
- Maintains records of laser systems in use in the School.
- Investigates accidents and ensures corrective action.

Shop Machinery Safety Officer (SMSO)

- Ensures that safe working conditions are maintained in all areas of the School where machine tools and equipment are in use.
- Provide frequent inspections of work methods and materials/equipment used. Any unsafe equipment/material shall be tagged and rendered inoperative or physically removed from its place of operation.
- Assist in ensuring that all shop users are thoroughly trained in the use of protective equipment, guards, and the safe operation of equipment, machines, and tools they use or operate. Only users who have been trained shall be allowed to use shop equipment, machines, and tools.
- Assist in ensuring that tools and equipment are properly maintained and used.
- Assist in ensuring that required guards and protective equipment are provided, used, and properly maintained.
- Take immediate action to report and correct any violation of safety rules observed or reported to the SMSO.
- Ensure that shop doors are closed after hours or when the supervisor is not present. Unauthorized or untrained people are not allowed into the shop. All shop users must be accompanied by an authorized person at all times.

School Safety Committee

- The School Safety Committee is charged with the continuing development of the overall School Safety and Compliance Plan.

- Serves as a consultative body to the departmental Safety and Environmental Compliance Officers, the School Laser Safety Officer, the Shop Machinery Safety Officer, and the department Chairs.
- Serves as a “translator” of University policy to school safety and environmental compliance needs.
- Serves to evaluate specific safety issues within the School as they arise.
- Performs periodic internal laboratory auditing.

Department of Vanderbilt Environmental Health and Safety (VEHS)

- The Vanderbilt Environmental Health and Safety Department, <http://www.safety.vanderbilt.edu>, serves as the primary technical resource for the Vanderbilt teaching and research community regarding environmental health and safety policies, procedures, and regulatory issues.
- Appoints members for and coordinates the activities of regulatory-required university environmental health and safety committees (e.g., Radiation Safety Committee, Institutional Biosafety Committee, Chemical Safety Committee, Environmental Affairs Committee, Fire Protection Committee).
- Executes enforcement actions as prescribed by institutional policy or committee decision.
- Assists P.I.'s or departments in creating and maintaining inventories of potentially hazardous and regulated materials used in research at Vanderbilt.
- Conducts routine as well as special risk assessments and provide recommendations on appropriate practices and facilities to conduct research and laboratory instruction safely and in compliance with regulatory requirements.
- Conducts periodic review of research and laboratory instructional facilities and/or procedures to ensure compliance.
- Facilitates regulatory inspections by external regulatory entities.
- Provides operational and/or technical support for research and or instructional laboratory activities that require monitoring, tracking, decontamination, or disposal of potentially hazardous and/or regulated materials.
- Provides emergency response as appropriate in handling to spills or other critical events involving potentially hazardous and/or regulated materials.
- Identifies and/or develops training resources for the

	- Vanderbilt teaching and research community. - Identifies, monitors, and interprets local, state, and federal agency actions related to environmental health and safety for implementation within the Vanderbilt community. - Develops draft policies/procedures for oversight committee review. - Provides an annual report of activities to the Environmental Health and Safety Oversight Committee.
Vanderbilt Environmental Health and Safety Committees (Radiation Safety Committee, Institutional Biosafety Committee, Chemical Safety Committee, Environmental Affairs Committee, Fire Protection Committee)	- Have knowledge of institutional, local, state, and federal environmental health and safety policies applicable to research and laboratory instruction. - Establish institutional policies related to content area of committee (e.g., radiation safety, biosafety, etc.). - Assist in notifying the Vanderbilt research and teaching community of responsibilities to environmental health and safety in the conduct of research. - Review research for compliance through assessment of facilities, procedures, practices, and the training/expertise of those conducting the research. - Conduct periodic review of research to ensure compliance. - Take action on non-compliance and unsafe working conditions, as needed. - Report problems, violations, etc. to institutional officials and/or specific regulatory agencies, as appropriate. - Provide an annual report of activities to the Environmental Health and Safety Oversight Committee. - Refer research issues to other research committees as needed.
Division of Sponsored Research (University Central), http://www.vanderbilt.edu/SponsoredResearch/homepg.html	Communicates and coordinates with Environmental Health and Safety, http://www.safety.vanderbilt.edu, regarding appropriate approvals for studies conducted with radioactive, chemical, or biological materials prior to proposal submission (or prior to approval of budget for funded project).
Office of Contract and Grant Accounting (University Central),	Communicates and coordinates with Environmental Health and Safety, http://www.safety.vanderbilt.edu, regarding appropriate approvals for studies conducted

<http://www.vanderbilt.edu/ocga>

with radioactive, chemical, or biological materials prior to approval of budget for funded project if the proper approvals are not apparent.

**Students and others
working in the lab**

- Undergo necessary training as identified by the department SECO.
- Report to the laboratory in appropriate attire.
- Follow safe practices in the laboratory.
- Report safety concerns and/or problems to the department SECO.
- Obtain approval from P.I. or SECO for any deviations from established experimental procedures.

Vanderbilt University School of Engineering

Environmental Health and Safety: General Lab Policies and Procedures

Training: Appropriate training shall be provided for all new personnel (faculty, staff, students) using Research Labs, Teaching Labs, and other Technical Facilities, including Machine Shops, in the School of Engineering. Particular attention should be given to safety and procedures for handling Regulated Hazardous Materials (RHM). Lab Directors and Managers (e.g., PI's, Shop Foreman, Teaching Lab Manager) are responsible for ensuring that this training is delivered in a timely manner and that it is appropriate to the particulars of the Lab, Shop, or Facility. At a minimum, this training should include a review of relevant University Policy and training courses available at Vanderbilt Environmental Health and Safety (VEHS), <http://www.safety.vanderbilt.edu>.

Annual Retraining: All personnel who use Regulated Hazardous Materials must receive annual retraining. That retraining is delivered by Vanderbilt Environmental Health and Safety (VEHS), <http://www.safety.vanderbilt.edu>. Training courses are provided by VEHS in several ways – online through VandySAFE, in regularly scheduled classes, or by the request of the department.

New Projects: At the initiation of each laboratory intensive project and at least annually during the course of the project, the PI shall identify the safety and environmental compliance requirements of that project and discuss them with the departmental Safety and Environmental Compliance Officer (SECO).

Installation Requirements Involving Hazardous or Regulated Materials: PI's shall seek appropriate approvals/guidance through the departmental Safety and Environmental Compliance Officer (SECO), School Safety Committee, and VEHS, <http://www.safety.vanderbilt.edu>, for the use of potentially hazardous or regulated materials. Installation requirements of equipment, exhaust systems, etc., involving potentially hazardous or regulated materials should be coordinated through the Vanderbilt Campus Planning and Construction Department, <http://www.cpc.vanderbilt.edu>, prior to requests for construction estimates. Depending on the requirements, engineering analysis and assessment may be required. Direct requests for installations/construction requirements of this type to the Vanderbilt Plant Operations – Estimating And Design Center, <http://www.vanderbilt.edu/PlantOps/request.html>, will only result in a delay.

Laboratory Closeout Guidelines: Principal Investigators at Vanderbilt are responsible for the safe operation of their laboratory or suite of laboratories. This includes leaving any and all of these facilities in a safe condition when, and if, the premises are vacated. Guidelines outlining the Principal Investigator's responsibilities in the Laboratory Close Out Process may be found on the VEHS web site, <http://www.safety.vanderbilt.edu>.

Student RHM Disposition: RHM should be labeled at all times. As a requirement for graduation, all students must properly dispose of any Regulated Hazardous Materials (RHM) that they have generated and their advisor shall provide a written affirmation to that effect and submit it to the departmental Safety and Environmental Compliance Officer (SECO) for filing.

Flammables Storage Lockers: Storage lockers containing flammable materials must be clearly labeled at all times.

Emergency Contact List: All Research Labs, Teaching Labs, Machine Shops, or other Technical Facilities shall have the Vanderbilt emergency number, 1-1911, posted in a clearly visible location. All P.I.'s must provide one emergency contact with a non-work phone number that emergency responders can contact to determine status and contents of lab. This contact information will be posted on a hazard sign on the door of each lab.

Machine Shop Training: The Shop Machinery Safety Officer (SMSO) shall provide training for all personnel who wish to use the machine shop located on the first floor of Featheringill Hall. The Shop Supervisor for the Model Shop in Olin Hall or PI's, who use machine tools in their laboratory, are responsible for ensuring that users of their equipment are properly trained. They may avail their students of the instruction offered by the Shop Machinery Safety Officer. In all cases, the Shop Machinery Safety Officer is responsible for ensuring that safe working conditions are maintained in all areas of the School where machine tools and equipment are used, and that users have been properly trained in the safe operation of equipment, machines, and tools they use or operate. The Shop Machinery Safety Officer has the final say in who may use any of the School's Machine Shop Facilities and may revoke that privilege at any time for any unsafe or inappropriate use of the facility.

Shop Self-Inspection: Shop Supervisors and/or PI's shall perform and document self-inspections at least annually as part of OSHA's requirement for an effective Injury and Illness Prevention Program (IIPP). The School of Engineering's Shop Self-Inspection Form shall be used for this purpose. This self-inspection form will help document shop safety inspections and will assist shop personnel in identifying and correcting many common unsafe practices and conditions. The results of the self-inspection shall be shared with the Shop Machinery Safety Officer (SMSO), who will maintain a file copy.

Shop Safety Handout: The School of Engineering Shop Safety Handout shall be provided to all personnel using machine tools and related equipment in the School.

"Buddy System": No one shall work with chemicals or perform other hazardous lab work, such as energizing power supplies, alone in the lab.

Laboratory Attire: The School of Engineering operates a number of different laboratories with different potential safety hazards. All personnel, working in any laboratory, shall wear clothing and safety equipment that is appropriate to the laboratory and protective with respect to the hazards that may be present. All personnel in chemical-

using labs should wear a lab coat, closed-toe shoes, and safety glasses at a minimum. More specific attire or protective equipment might be required dependent on the specific hazards in that lab. Standards for appropriate dress, required safety equipment and good laboratory practices shall be developed for each laboratory by the SECO, in concert with VEHS and the department chair, and communicated to all laboratory personnel.

Eating and Drinking in Labs: Eating and drinking are not permitted in School of Engineering Labs unless in an approved clear area.

Vanderbilt University School of Engineering

Environmental Health and Safety: Laser Safety

1. Policy

Lasers can present health and safety hazards to personnel. This policy and procedures manual on laser safety in the School of Engineering outlines methods for assessing and controlling the hazards associated with lasers for the protection of potentially exposed personnel. Installation and operation of lasers for research purposes must be in compliance with the current ANSI Z136.1 standard and the ANSI Z136.5 standard for Safe Use of Lasers in Educational Institutions. A copy of this standard is on file with the School Laser Safety Officer. All laser installations in the School of Engineering must be coordinated with and approved by the School Laser Safety Officer. All Class 3b and 4 lasers must be registered with the Vanderbilt Environmental Health and Safety (VEHS), <http://www.safety.vanderbilt.edu>. A comprehensive Vanderbilt University Laser Safety Program Guide is available on the VEHS web site. A number of additional sources of information on lasers and laser safety can be found on the Vanderbilt Environmental Health and Safety (VEHS), <http://www.safety.vanderbilt.edu>, web site at Safety Links/Laser Safety. The links include a link to the OSHA Technical Manual – Laser Safety, which is an excellent source of additional information. A training course on Laser Safety can be found at Vanderbilt Environmental Health and Safety (VEHS), <http://www.safety.vanderbilt.edu>. All personnel using Class 3a, 3b, or 4 lasers in their research or teaching should avail themselves of this Laser Safety training course.

2. Responsibilities

2.1 Principal Investigators are responsible for:

- The immediate supervision of lasers in the laboratory.
- Providing, implementing and enforcing the safety recommendations and requirements prescribed in this policy.
- Assuring that all laser operators are properly trained and proficient in safety procedures before operating a laser.
- Classifying and labeling all of their lasers.
- Completing a Laser Registry Form and providing it to VEHS.
- Notifying VEHS immediately in the event of any incident involving lasers.

2.2 Laser Operator s are responsible for:

- Following laboratory safety procedures and laser standard operating procedures.
- Keeping the Principal Investigator fully informed of any departure from established safety procedures. This includes notification of an exposure incident.
- Receive Laser Safety Training before working with lasers.

2.3 Vanderbilt Environmental Health & Safety and/or School of Engineering Laser Safety Officer is responsible for:

- Conducting lab inspections to ensure that safety requirements are followed.
- Providing assistance in evaluating and controlling hazards.
- Updating the Laser Safety Policy.
- Maintaining all records of lasers and laser operators.
- Coordinating laser safety training for personnel who are assigned to an area where lasers are operated.
- Participating in accident investigations involving lasers.
- Advising laser operators of the requirements of the Medical Surveillance program.

3. Laser Hazard Classification and Primary Precautions

Lasers are identified by type, e.g., Helium-Neon, wavelength(s), or Laser Hazard Classification. The Laser Hazard Classification is determined by the wavelength(s), maximum duration of exposure, and the average power output of the laser. Manufacturers of lasers are required to label all lasers with the appropriate Laser Hazard Classification “Caution” or “Danger” warnings. If an unlabeled laser is found, contact the School of Engineering Laser Safety Officer. Class 1, 2, and 3a lasers should have a yellow and black “Caution” label, while Class 3b, and 4 lasers should have a red, black, and white “Danger” label. The universal symbol for a laser hazard is a sunburst pattern of short spokes with a single long spoke radiating from the center of the sunburst. Details of the laser classes can be found in the Vanderbilt University Laser Safety Program Guide available at VEHS, <http://www.safety.vanderbilt.edu>.

4. Laser Safety Control Measures

Class 1 Lasers:

- Prior to operation, the laser shall be classified as to its hazard class, and equipped with the appropriate warning label. As previously stated, Manufacturers of lasers are required to label all lasers with the appropriate Laser Classification warning.
- Prior to maintenance, repair, or modifications, a safety plan should be developed to ensure personnel will not be exposed to health or safety hazards.
- A protective housing shall encase the laser.
- The housing shall be interlocked to prevent operation of the laser while the housing is removed.

Class 2 Lasers:

- The laser safety control measures for Class 1 lasers shall be complied with in addition to the following.
- Do not stare into the beam.

- Control all unnecessary shiny reflecting surfaces from the work area and yourself.
- A non-reflective surface should be used as a beam stop. The area along all parts of the path of the beam should be cleared of personnel and reflective material.
- Do not use at eye height.
- A protective housing shall encase the laser.
- The housing shall be interlocked to prevent operation of the laser while the housing is removed.

Class 3a Lasers:

- The laser safety control measures for Class 2 lasers shall be complied with in addition to the following.
- Do not stare into the beam.
- Control all unnecessary shiny reflecting surfaces from the work area and yourself.
- A non-reflective surface should be used as a beam stop. The area along all parts of the path of the beam should be cleared of personnel and reflective material.
- Do not use at eye height.
- Post “Caution” signs alerting personnel to the use of Class 3a lasers in the work area.
- Personnel should have access to laser safety glasses for the appropriate wavelength.
- Personnel using a Class 3a laser should receive laser safety training.
- A protective housing shall encase the laser.
- The housing shall be interlocked to prevent operation of the laser while the housing is removed.

Class 3b Lasers:

- The laser safety control measures for Class 3a lasers shall be complied with in addition to the following.
- The path of the laser beam shall be enclosed, and inaccessible to personnel.
- Control all unnecessary shiny reflecting surfaces from the work area and yourself.
- A protective housing shall encase the laser.
- The housing shall be interlocked to prevent operation of the laser while the housing is removed.
- A permanent non-reflective beam stop shall be incorporated.
- Post “Danger” signs alerting personnel to the use of Class 3b lasers in the work area.
- If used with a chamber, all windows not in use shall be constructed of or covered with material that will not transmit light of the wavelength generated by the laser.

- Personnel shall have access to laser safety glasses for the appropriate wavelength. See the Vanderbilt University Laser Safety Program Guide available on the VEHS web site, <http://www.safety.vanderbilt.edu> .
- The area of the laser shall not be in personnel traffic areas.
- The user of the Class 3b laser shall provide documentation to the School Laser Safety Officer regarding safety measures taken and planned.
- Personnel using a Class 3b laser shall receive laser safety training provided by Vanderbilt Environmental Health and Safety (VEHS), <http://www.safety.vanderbilt.edu> , or other appropriate laser safety training instruction.
- The School Laser Safety Officer shall be notified when alterations which may affect safety are made to equipment.

Class 4 Lasers:

- The laser safety control measures for Class 3 lasers shall be complied with in addition to the following.
- A protective housing shall encase the laser.
- The housing shall be interlocked to prevent operation of the laser while the housing is removed.
- The path of the laser beam shall be enclosed, and inaccessible to personnel. This enclosure shall be interlocked.
- A permanent non-reflective beam stop shall be incorporated.
- An alarm or revolving light shall be in use during operation of the laser.
- If used with a chamber, all windows not in use shall be constructed of or covered with material that will not transmit light of the wavelength generated by the laser.
- Personnel shall have access to laser safety glasses for the appropriate wavelength. See the Vanderbilt University Laser Safety Program Guide available on the VEHS web site, <http://www.safety.vanderbilt.edu> .
- The area of the laser shall not be in personnel traffic areas, and is considered a Laser Controlled Area.
- The user of the Class 4 laser shall provide documentation to the School Laser Safety Officer regarding safety measures taken and planned.
- Personnel using a Class 4 laser shall receive laser safety training provided by Vanderbilt Environmental Health and Safety (VEHS), <http://www.safety.vanderbilt.edu> , or other appropriate laser safety training instruction.
- The School Laser Safety Officer shall be notified when alterations which may affect safety are made to equipment.
- Post “Danger” signs alerting personnel to the use of Class 4 lasers in the work area.

5. Procedure for Using a Laser or Laser System

To safely use a laser or laser system in the School of Engineering, and to protect personnel in the area from laser hazards, a Laser Standard Operating Procedure must be

utilized. A Laser Standard Operating Procedure Template is available on the VEHS web site, <http://www.safety.vanderbilt.edu>. The Laser Standard Operating Procedure for Class 1 and 2 lasers is satisfied by following the established School of Engineering Safety Control Measures as described in section 3 of this manual. No written documentation is required for Class 1 or 2 lasers. More involved Laser Standard Operating Procedures are required for Class 3a, 3b, and 4 lasers or laser systems. Listed below are the components required in the written Laser Standard Operating Procedure submitted to the School Laser Safety Officer before work may begin with the applicable class laser.

Class 1 Lasers:

See Laser Safety Control Measures (section 3) for Class 1 lasers. No written Laser Standard Operating Procedure is required. Ensure a “Caution” label is attached to the laser housing, as required of the manufacturer.

Class 2 Lasers:

See Laser Safety Control Measures (section 3) for Class 2 lasers. No written Laser Standard Operating Procedure is required. Ensure a “Caution” sign appropriate for Class II lasers is posted in the work area.

Class 3a Lasers:

See Laser Safety Control Measures (section 3) for Class 3a lasers. Ensure a “Caution” sign appropriate for Class 3a lasers is posted in the work area. Documentation should be provided to the School Laser Safety Officer, which should include a diagram of the intended setup, evidence of laser safety training for involved personnel, a description of the Class 3a laser, and evidence of laser safety protective eyewear for involved personnel.

Class 3b Lasers:

A written Laser Standard Operating Procedure shall be submitted to the School Laser Safety Officer prior to operation of a Class 3b laser or laser system. The written plan shall contain a diagram of the intended setup, documented laser safety training of involved personnel and an outline of the laser safety training program, documentation describing the intended experiment and use of the Class 3b laser or laser system (including a description of the Class 3b laser or laser system), a checklist indicating that all steps required by the Laser Safety Control Measures (see section 3) for Class 3b lasers have been completed, and documentation outlining the engineering controls, personal protective equipment, and administrative procedures implemented to safeguard users of the laser and personnel in the area. No experiments with or testing of a Class 3b laser or laser system shall begin without a written plan received and approved by the School Laser Safety Officer. Ensure a “Danger” sign appropriate for Class 3b lasers is posted in the work area.

Class 4 Lasers:

A written Laser Standard Operating Procedure shall be submitted to the School Laser Safety Officer prior to operation of a Class 4 laser or laser system. The written plan shall contain a diagram of the intended setup, documented laser safety training of involved personnel and an outline of the laser safety training program, documentation describing

the intended experiment and use of the Class 4 laser or laser system (including a description of the Class 4 laser or laser system), a checklist indicating that all steps required by the Laser Safety Control Measures (see section 3) for Class 4 lasers have been completed, and documentation outlining the engineering controls, personal protective equipment, and administrative procedures implemented to safeguard users of the laser and personnel in the area. No experiments with or testing of a Class 3b or 4 laser or laser system shall begin without a written plan received and approved by the School Laser Safety Officer. Ensure a “Danger” sign appropriate for Class 4 lasers is posted in the work area.

6. Medical Surveillance

A Medical evaluation is required for all Class 3b and 4 laser users in the university and medical center. This evaluation may be scheduled in the Vanderbilt Occupational Health Clinic, <http://www.vanderbilt.edu/HRS/wellness/occhealth.htm> , and must be completed prior to laser use, and upon termination of employment or transfer to non-laser related duty.

The initial medical evaluation establishes the faculty/staff member's baseline, so that changes in the eye or skin resulting from an acute laser exposure may be accurately and efficiently diagnosed and treated. The exam also documents pre-existing eye or skin conditions which could become worse with exposure to lasers. The exam is also an opportunity for the faculty/staff member to discuss laser related health issues with the medical provider.

7. Recordkeeping

The School of Engineering Laser Safety Officer will maintain all records pertaining to laser operator training, laboratory Standard Operating Procedures (SOP's), laser safety inspections, laboratory incidents, School of Engineering Laser Policy, VEHS Laser Safety Guide, and regulatory standards.

The laboratory PI will maintain copies of records pertaining to laser operator training, laboratory SOP's, laser safety inspections, School of Engineering Laser Policy, and VEHS Laser Safety Guide.

Vanderbilt University School of Engineering

Environmental Health and Safety: Shop Safety Handout

Shop Safety

The School of Engineering's principal Machine (Model) Shop is located on the first floor of Featheringill Hall in the west wing. A smaller Model Shop is located in Olin Hall on the first level. Additional machine tools and associated equipment are located in some of the research labs in the School. The single most important issue in the School in using any of these Shop facilities is safety. The primary objective of all people that use the Machine Shops must be to uphold safety. There is nothing that can be designed or built using the Machine Shops that is worth trading for a permanent disability. This handout is intended to give you some guidelines to working safely in the Machine Shops as well as familiarize you with some of the hazards.

The Machine Shop is a more hazardous environment than people are used to. All personnel must be properly trained before using the equipment. When you enter the shop, you should make a conscious effort to adjust mentally and physically to the increased hazards around you. Consult the first five safety rules below to adjust yourself to existence in a shop environment. Scrupulously follow safe procedures. Be careful to think through each operation before you execute it. Use your common sense. Do not do things that seem unsafe.

The shop supervisor or faculty PI or trained assistant is there to help you use the shop. However, they cannot be everywhere at once. Users of the shop must assume responsibility for the safe use of the facility. If you are unsure of the proper method to do something, wait until the shop supervisor or other trained person can give you guidance. If you notice an unsafe condition in the shop, fix it or bring it to the attention of the shop supervisor. If you think someone is using the equipment unsafely, suggest a safer alternative or bring that activity to the attention of the shop supervisor.

General Safety Rules

The rules listed below apply to all people doing work in the School's Machine (Model) Shops. They are not all encompassing. Most shop areas and machines have specific safety rules that relate to their use, and these should be consulted. The following rules are a good general guide to safe use of the facility.

1) **Never work alone.**

When you are working in the shops you must have a second person present who is capable of rendering assistance in case of an accident. Accidents do happen. You need to have a second person present so that if you become unable to help yourself,

the second person can render aid and contact outside help. At Vanderbilt, the emergency number to call for assistance is 1-1911.

- 2) **Never work when you are impaired. This includes times when you are too tired, stressed, or otherwise inhibited from exercising appropriate caution in the shop.** Do not enter the shop when you are under the influence of any intoxicants or medications that might make you drowsy or alter your ability to be alert to reality.

Do not use the shop when you are too frantic to think clearly and carefully. Try to avoid last minute shop work in favor of a consistent weekly effort. Deadline-driven haste can lead to ruined projects and/or serious injuries.

Do not use the shop if you are wearing a cast or bandage that limits your mobility. In an emergency you must be able to react quickly to avoid injury.

- 3) **Wear all necessary protective gear and clothing. This always includes safety glasses and shoes that completely enclose your feet. Some shop activities will require additional safety gear.**

All persons, including visitors, entering the Machine Shop shall wear eye protection appropriate to the hazard. All such eye protection must conform to ANSI standard Z87.1. Safety glasses that conform to the standard are marked with a “Z87”.

If you wear glasses already you should be aware that most eyeglasses purchased in the United States conform to ANSI Z87.1 for frontal impact. You may wear such glasses for work in the shop. To protect your eyes from side impact you can add side shields to your existing glasses or purchase prescription safety glasses for use in the shop.

Do not wear contact lenses in the welding shop. The intense light from the arc welding torches can cause contact lenses to damage your eyes. Additional information can be obtained from the American Welding Society, <http://www.aws.org>, web site under Safety & Health.

Persons exposed to foot injury hazards shall wear foot protection appropriate to the hazard. Leather shoes are necessary to protect your feet from stray droplets of molten metal in the welding area. Additional information can be obtained from the American Welding Society, <http://www.aws.org>, web site under Safety & Health.

Certain dusts and fumes require respiratory protection. In the event that a respirator is necessary, the Shop Supervisor may require you to obtain and wear one for operations that expose you to airborne hazards. Additional information can be obtained from the American Welding Society, <http://www.aws.org>, web site under Safety & Health. Any employee or personnel wearing any respirator other than a dust mask must have medical surveillance through the Vanderbilt Occupational Health Clinic, <http://www.vanderbilt.edu/HRS/wellness/occhealth.htm>, and be fit-tested by VEHS.

When it is not practical to reduce sound levels below ninety decibels, then persons exposed to such sound levels shall wear hearing protection in the form of foam ear inserts.

Leather gloves are available in the welding areas for protection from the heat and radiation of welding. Additional information can be obtained from the American Welding Society, <http://www.aws.org>, web site under Safety & Health.

Welding requires the use of welding masks with appropriate eye-protective lenses and natural fiber clothes that completely cover your skin. The electric arc in particular is an intense source of ultraviolet and infrared radiation. Even a brief exposure of your unprotected eyes can cause severe damage to your sight. Do not look at an electric without the correct eye protection. If you expose bare skin to the light of the arc, the radiation is intense enough to give you severe sunburn. Natural fiber clothes are required because they will not melt onto you in the presence of high heat, as some man made materials will. Additional information can be obtained from the American Welding Society, <http://www.aws.org>, web site under Safety & Health.

4) Long hair must be tied up securely.

Most of the power tools in the shop are based around a rapidly rotating shaft. In use the shaft is frequently sticky with oil. Long, loose hair can stick to such a rotating shaft and pull the owner of the hair into the tool. Long hair must be kept out of harm's way by tucking it into a cap, tying it up. Or knotting it in a way that prevents it from dangling.

5) Remove all personal accessories and loose clothing that might get caught in moving machinery. This includes rings, watches, jewelry, personal stereos, shop rags, ties, and open jackets.

Like long hair, things that dangle from your person can get caught in rotating machinery. Regardless of the fashion, it is not worth risking your health.

Loose garments must not be worn in the shop. Tuck in loose shirttails and sweat pants ties. Keep tight fitting jackets or coats closed.

Do not keep shop rags in your pockets.

Do not wear personal stereos or headphones while you are working in the shop.

6) Never leave a machine running unattended.

Some of the tools in the shop can be set to cut automatically. You must keep your attention focused on the machining operation. If you are focused on the process, you will be more likely to react appropriately in the event of an accident.

The Machine (Model) Shops in the School are used primarily for building prototypes. As a result most set-ups are not tried and tested in a production sense. It is inevitable

that some set-ups will go bad. If you are paying attention to the operation, you may be able to avert injury to yourself and the people around you.

7) **Never leave a chuck key in a chuck or a drawbar wrench on a drawbar. If the key or wrench is in use your hand must be on it.**

Chuck keys and drawbar wrenches can be accidentally launched across the shop if they are left in place. The startup power of the lathes and mills can throw the tools with enough force to puncture a body cavity. You must be careful not to leave the chuck keys and drawbar wrenches in place when they are not in use.

The same rule applies to the chuck keys used to tighten the chuck on the drill presses.

8) **Keep your hands well away from the point of contact between the work piece and the cutter. If you must hand hold the work to keep it in place, your set-up is unsafe; improve it.**

While working on the lathes and mills you should never hand hold your work. If the tool or work piece is vibrating, the chances are high for a sudden shift in the set-up. If you are hand holding the tool or work piece, you might not be able to get out of the way as the parts come together. Besides, the cutting forces are too large for you to be effective in securing the work piece or tool.

When using the drill presses, especially with large drills and tough to cut materials, you must clamp the work to the drill press table. If the drill catches an unclamped work piece, the rotating part can cut your hand.

On the band saws, table saw, router table, chop saw and any other tool that is designed to be used with a hand fed work piece, be certain to keep your hands, fingers, and other body parts out of the path of cutters and away from the point between the work piece and the cutter.

Keep all parts of yourself at least 6 inches from the point of contact between any work piece and any cutter.

Do not remove chips with your fingers. Use brushes, pliers, or compressed air.

9) **Support work pieces and cutting tools as securely as possible.**

A vibrating set-up is usually an indication that the work piece and/or tool are not held strongly enough to resist the applied cutting forces. You must take the time to secure the set-up to resist the force of cutting or use a different operation to do the job. Do not try to make do with a flimsy set-up.

Expect to spend a lot (80% or more) of your time in the shops making set-ups. It is the nature of prototyping. Unfortunately, it is also hard to visualize when you are thinking about work in the shop. This is where a well-metered approach to work in the shop will pay off. If you think things are taking longer than you expected, you can

scale down or redesign your work. At the last minute it is hard to make such large changes.

10) Have the shop supervisor or other trained person check you out the first time you use each machine or process in the shop. Do not operate any machinery with which you are unfamiliar.

Each and every tool in the shop has safe operating procedures associated with it. Do not work on any tool in the shop until the shop supervisor or other trained person has introduced you to its safe operation. You should get checked out even on tools you have been taught to operate elsewhere. The tools in the shop may have idiosyncrasies that you should know about.

Clean Up Procedures

Careful clean up is part of the safe operation of the shops. If a person cannot find a needed tool in its regular place or if a person is required to work in someone else's mess, the result will be frustration. It is difficult to keep safety in mind when you are frustrated. So it is important, especially in times of heavy use, to keep the shop clean with everything returned to its correct place at the end of its use.

1) Shut off power to the machine.

Turn off the main power switch for the machine.

Disengage all power feeds and lead screws.

2) Un-mount all cutters

Remove all end mills, lathe tools, drills, and similar tools from their tool holders.

3) Put away all measuring tools, hand tools, material scraps, and drawings.

Put away all objects that do not belong permanently with the machine. If you do not know where something goes, ask the shop supervisor. If the shop supervisor is unavailable, leave whatever it is in plain sight on a table. It is better to leave it out than to put it away in the wrong place.

4) Clean chips and excess oil from machines and chip pans.

Protect your hands from sharp chips with a shop rag.

Most machines can be wiped down completely with a shop rag. If you must use compressed air, be careful with it. Do not point compressed air guns at people. Blown chips can become lodged in eyes and the compressed air itself can do severe damage to a person's body. Use the compressed air early in the clean up process. Otherwise you will blow chips all over the areas you have already cleaned.

Do not use brooms or brushes on the machine tools. The brushes and brooms pick up abrasive dirt from the floor. If abrasive gets on the machine tools they will wear very quickly.

You should clean the equipment well enough that the next user will not be able to tell what material you were using.

- 5) **Put a light coat of way oil on the machine ways.**
Move the machine slides to one extreme position and oil the exposed ways. Next move the ways to the middle of their travel and oil the newly exposed sections. Leave the machine slides at the approximate center of travel.
- 6) **Sweep the floor in the vicinity of the machines you have used.**
Collected debris should be either recycled or thrown in a scrap container.
- 7) **Ensure all hazardous materials such as used paint, oils, solvents, degreasers, etc. are properly labeled and stored.**
Vanderbilt Environmental Health and Safety (VEHS),
<http://www.safety.vanderbilt.edu>, should be notified of anything requiring disposal.

**Vanderbilt University
School of Engineering**

Environmental Health and Safety: Shop Self-Inspection Form

The School of Engineering's principal Machine (Model) Shop is located on the first floor of Featheringill Hall in the west wing. A smaller Model Shop is located in Olin Hall on the first level. Additional machine tools and associated equipment are located in some of the research labs in the School. All of these shops are required to perform and document self-inspections at least annually as part of OSHA's requirement for an effective Injury and Illness Prevention Program (IIPP). This self-inspection form will help document shop safety inspections and will assist shop personnel in identifying and correcting many common unsafe practices and conditions.

Shop Supervisors and/or PI's shall perform and document a self-inspection at least once a year, using this form or an equivalent. Each question should be answered by checking "Yes" (satisfactory), "No" (needs correction), or "N/A" (if the question does not apply to your shop).

The results of the self-inspection shall be shared with the Shop Machinery Safety Officer (SMSO), who will maintain a file copy.

This form was designed to help ensure compliance with University/State/OSHA regulations that require documented periodic inspections of all work areas as part of an effective IIPP. However, completion of this self-inspection checklist and correction of any findings noted herein does not ensure that State/OSHA will not issue citations during an inspection. In additions, the School's Shop Machinery Safety Officer (SMSO), representatives from the School Safety Committee and/or Vanderbilt Environmental Health and Safety (VEHS), <http://www.safety.vanderbilt.edu> , will periodically be verifying that self-inspections have been documented on-site.

_____ Shop Location	_____ Department
_____ Shop Supervisor's Name (printed)	_____ Shop Supervisor's Signature
_____ Inspector's Name (printed)	_____ Inspector's Signature
_____ Date of Inspection	

Administrative

1. Are training records maintained to indicate which faculty/staff/students are trained and qualified to use shop/lab equipment?
☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)
Corrective Action: Establish a training file. Prohibit untrained personnel from using shop equipment.
Completion Date:_____.
2. Is an operator's manual or other written safe operating procedures available for each piece of equipment and machine in shop?
☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)
Corrective Action: Obtain operator's manuals or write safe operating procedures for each piece of equipment.
Completion Date:_____.
3. Is there a lockout/tagout program in place to prevent accidental energizing of equipment or machines that are being repaired, adjusted, or undergoing tool changes?
☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)
Corrective Action: Institute a lockout/tagout (or equivalent program).
Completion Date:_____.

Housekeeping

4. Are work areas (including equipment and machinery) kept clean and orderly, so as to prevent trip and fire hazards?
☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)
Corrective Action: Include housekeeping as part of regular shop activities (e.g., the job is not finished until all tools and materials are put away, the work area swept and waste is disposed of.)
Completion Date:_____.
5. Have shop users been advised that they should not consume food or drinks in shop areas unless there is a designated clean area?
☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)
Corrective Action: Prohibit consuming food or beverages in shop areas. Post signs if necessary.
Completion Date:_____.
6. Are floors and work surfaces dry and/or made slip-resistant?
☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)
Corrective Action: Keep surfaces dry or install slip-resistant material on surfaces that can not be kept dry.
Completion Date:_____.
7. Do all work areas have adequate ventilation, particularly for hazardous operations (e.g., welding, soldering, spray coating, using solvents)?
☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Contact Vanderbilt Environmental Health and Safety, <http://www.safety.vanderbilt.edu> , if there is a concern. Provide additional exhaust ventilation for operations that create dust, fumes, mists or vapors.
Completion Date:_____.

8. Are all compressed gas cylinders adequately secured with non-combustible restraints to keep the cylinder(s) from falling? Also, are all compressed gas cylinders capped when not in use?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Contact Vanderbilt Environmental Health and Safety, <http://www.safety.vanderbilt.edu> , for advice on how to properly secure cylinders. Train shop users to cap compressed gas cylinders when not in use.
Completion Date:_____.

Fire/Emergency Safety

9. Is the Vanderbilt Emergency telephone number, 1-1911, clearly posted?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Post Vanderbilt Emergency telephone number, 1-1911, near the telephone(s).
Completion Date:_____.

10. Are fire alarm pull boxes clearly identifiable and unobstructed?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Ensure that all fire alarm pull boxes are clearly labeled. Clear any obstructions in front of fire alarm pull boxes.
Completion Date:_____.

11. Are fire hose stations and portable extinguishers clearly identifiable and unobstructed?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Ensure that all fire hoses and extinguishers are clearly identifiable. If they are not clearly marked, contact Vanderbilt Plant Operations, <http://www.vanderbilt.edu/PlantOps> . Clear any obstructions in front of fire hoses and extinguishers.
Completion Date:_____.

12. Are portable fire extinguishers tagged with current annual inspections?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Contact Vanderbilt Plant Operations, <http://www.vanderbilt.edu/PlantOps> , if they have not been within the last 12 months.
Completion Date:_____.

13. Are fire escapes, exit doors, stairwells, and corridors kept clear and unobstructed?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Remove any stored equipment and materials from exit doors, stairwells and corridors. Prohibit storage in these areas.

Completion Date:_____.

14. Are flammable materials stored in approved safety containers or safety cabinets?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Ensure that flammable materials are stored in approved safety containers or cabinets. If there are questions, consult Vanderbilt Environmental Health and Safety, <http://www.safety.vanderbilt.edu> . Prohibit storage of flammable materials outside safety containers or cabinets.

Completion Date:_____.

15. Are combustible materials kept at least 35 feet away from welding operations, or is fireproof covering provided?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Move combustible material at least 35 feet from welding operations or place fireproof covering over them. Detailed safety information regarding welding and cutting can be found at the American Welding Society, <http://www.aws.org>, web site under Safety & Health.

Completion Date:_____.

Electrical Safety

16. Are electrical panels accessible and are circuit breakers clearly labeled?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Ensure that there is a minimum of 36 inches of clearance in front of all electrical panels. If circuit breakers are not labeled, notify Vanderbilt Plant Operations, <http://www.vanderbilt.edu/PlantOps> .

Completion Date:_____.

17. Is all electrical shop equipment properly grounded? (Ensure that the grounding pin has not been removed, and that 3-pin to 2-pin adapters are not used.)

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Inspect all electrical connections for grounding plugs or wires.

Completion Date:_____.

18. Is there a lockout/tagout program in place to prevent accidental energizing of equipment or machines that are being repaired, adjusted, or undergoing tool changes?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Institute a lockout/tagout (or equivalent program).

Completion Date:_____.

19. Are extension cords and power strips in good condition (e.g., no breaks or exposed wiring), used only as temporary wiring (less than 30 days), and not connected in series?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Remove all extension cords connected in series, permanent use power strips, and permanent use extension cords. Contact Vanderbilt Plant Operations, <http://www.vanderbilt.edu/PlantOps> , to arrange for installation of permanent wiring. Dispose of or repair all electrical cords that are not in good condition

Completion Date:_____.

Personal Protective Equipment

20. Are shop users provided with eye and face protection such as safety glasses and face shields where needed?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Provide and ensure the use of approved eye protection for all shop users and visitors upon entry to shop work areas. Ensure that face protection is used when flying materials could cause injury to the face. For welding and cutting operations, consult the American Welding Society, <http://www.aws.org> , web site under Safety & Health.

Completion Date:_____.

Portable Power-Operated Tools

21. Are power tools grounded or double insulated?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Ensure that power tools are inspected prior to each use. Repair or replace power tools that are not grounded or identified as double insulated.

Completion Date:_____.

22. Are rotating or moving parts of equipment or tools guarded (e.g., hand-held grinder wheel)?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Ensure that all guards are in place and used during tool operation.

Completion Date:_____.

Shop Equipment and Machines

23. Are abrasive wheel grinders equipped with a work rest adjusted to within 1/8 inch of the wheel, and side guards that cover the spindle, nut, flange and 75% of the wheel diameter?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Prohibit the use of wheel grinders until work rests and proper side guards are installed and/or properly adjusted.

Completion Date:_____.

24. Are radial arm saws provided with a spreader and automatic return?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Prohibit the use of radial arm saws until spreaders and automatic returns are installed.

Completion Date:_____.

25. Do table saws have a hand guard, spreader, and anti-kickback device for use when applicable?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Prohibit the use of table saws until anti-kickback devices are installed.

Completion Date:_____.

26. Are all pieces of equipment and machinery provided with adequate guarding? (Rotating parts, nip points and v-belts must be guarded.)

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Prohibit the use of equipment and machinery until adequate guards are installed.

Completion Date:_____.

27. Are pieces of equipment or machinery securely anchored to the floor or a bench?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Ensure equipment or machinery that can move from its operation or vibration or the vibration of nearby equipment or machinery is securely anchored.

Completion Date:_____.

Welding and Cutting

28. Is adequate ventilation provided in areas where welding and/or cutting occurs?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Contact Vanderbilt Environmental Health and Safety, <http://www.safety.vanderbilt.edu>, to have ventilation evaluated if there is a concern. Additional information can be obtained at the American Welding Society, <http://www.aws.org>, web site under Safety & Health.

Completion Date:_____.

29. Are welders provided with appropriate personal protection equipment (e.g., eye protection, gloves, flame resistant clothing or leathers)?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Ensure that welders are provided with and use the proper personal protective equipment. Detailed information can be obtained from the American Welding Society, <http://www.aws.org>, web site under Safety & Health.

Completion Date:_____.

30. Are welders provided with a fire extinguisher within ten feet of their working area?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Ensure that there is a fire extinguisher with a current inspection tag within 10 feet of welding work areas. Additional information can

be obtained from the American Welding Society, <http://www.aws.org> , web site under Safety & Health.

Completion Date:_____.

Hazardous Materials

31. Is there a Material Safety Data Sheet (MSDS) for each chemical (solvent, degreaser, etc.) stored in the shop?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Information regarding MSDS's as well as specific MSDS's can be obtained at Vanderbilt Environmental Health and Safety, <http://www.safety.vanderbilt.edu> .

Completion Date:_____.

32. Are all containers of chemicals labeled with the contents and primary hazards?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Properly label all chemical containers, including stored and temporary containers

Completion Date:_____.

33. Has the shop submitted an updated chemical inventory to Vanderbilt Environmental Health and Safety, <http://www.safety.vanderbilt.edu> , within the last 12 months?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Complete and submit a chemical inventory to Vanderbilt Environmental Health and Safety (VEHS), <http://www.safety.vanderbilt.edu> .

Completion Date:_____.

34. Are all paint, oil, degreasers, solvents, and other chemicals labeled and stored properly?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Properly label all paint, oil, degreasers, solvents and other chemicals and store in a proper location.

Completion Date:_____.

35. Has Vanderbilt Environmental Health and Safety (VEHS),

<http://www.safety.vanderbilt.edu>, been notified of any unwanted or waste hazardous materials requiring disposal?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Notify VEHS of any unwanted or waste hazardous materials requiring disposal.

Completion Date:_____.

D Material Safety Data Sheets

Composite Fibers:

- Fibre Glast Part #262 4 oz. Fiberglass Fabric:
<https://www.fibreglast.com/downloads/PDCT-MSDS-00056.pdf>
- Fibre Glast Part #1069 - 5.7oz Graphite 3K 2 x 2 Twill Carbon Fiber:
<https://www.fibreglast.com/downloads/PDCT-MSDS-00111-C.pdf>

Epoxy:

- Devcon 2 Ton Epoxy [1:1]: 5-Minute Epoxy Resin, 2-Ton Hardener:
<https://itwperformancepolymers.com/media/15407/14260-devcon-2-ton-epoxy.pdf>
- Devcon 5 Minute Epoxy amber [1:1]: 5-Minute Epoxy Resin, 5-Minute Epoxy Hardener:
<https://itwperformancepolymers.com/wp-content/uploads/umb/15412/14310-devcon-2-ton-epoxy.pdf>
- Fibre Glast Part #2000 System 2000 Epoxy Resin:
<https://www.fibreglast.com/downloads/PDCT-MSDS-00130.pdf>
- Fibre Glast Part #2120 Epoxy Hardener:
<https://www.fibreglast.com/downloads/PDCT-MSDS-00133.pdf>
- J-B Weld Steel Reinforced Epoxy Resin and Hardener - Twin Tubes:
https://cdn.shopify.com/s/files/1/0411/5921/files/Steel_Reinforced_Epoxy_Twin_Tubes.pdf?785811878289892783
- Loctite Fast Cure Epoxy:
<https://www.grainger.com/ec/pdf/Loctite-Fixmaster-Fast-Cure-Data-Sheet.pdf>

Explosives:

- Aerotech L940X Motor:
<https://www.apogeerockets.com/downloads/MSDS/Aerotech/Motors.pdf>
- Cesaroni L1350 Motor:
http://www.pro38.com/MSDS/SDS--ProX--ver4-03-EN_FR.pdf
- Owen Compliance Black Powder:
<http://aiaacrocketry.org/wp-content/uploads/2010/11/black-powder-4f.pdf>

Glues/Adhesives:

- Ethyl 2-Cyanoacrylate:
<http://www.sigmaaldrich.com/MSDS/MSDS/DisplayMSDSPage.do?country=US&language=en&productNumber=E1505&brand=ALDRICH&PageToGoToURL=http%3A%2F%2Fwww.sigmaaldrich.com%2Fcatalog%2Fsearch%3Fterm%3DEthyl%2B2-cyanoacrylate%26interface%3DProduct%2520Name%26N%3D0%2B%26mode%3Dmode%2520matchpartialmax%26lang%3Den%26region%3DUS%26focus%3DproductN%3D0%2520220003048%252019853286%252019853269>
- Gorilla Hot Glue:
<https://www.gorillatough.com/wp-content/uploads/Gorilla-Hot-Glue.pdf>
- 3M Super 77 Adhesive, Bulk:
https://multimedia.3m.com/mws/mediawebserver?mwsId=SSSSSuUn_zu8l00x482xN8mvNv70k17zHvu9lxtD7SSSSSS--

Liquid Foams:

- 3M Polyurethane Foam Insulation Adhesive CR-20, Parts A and B:
<https://tapecasecom.blob.core.windows.net/product/download/3m%20cr-20-c14bbee6-4b5d-47b5-bb43-eba14796649b.pdf>

Mold Compounds:

- Fibre Glast Part #1153 - FibRelease:
<https://s3.amazonaws.com/cdn.fibreglast.com/downloads/1153.pdf>
- Meguiar's M08, Mirror Glaze Maximum Mold Release Wax (23-135A): M0811:
<https://www.fibreglast.com/downloads/PDCT-MSDS-00030.pdf>

Solvents:

- Ethyl Alcohol, 70%:
<https://ehss.syr.edu/wp-content/uploads/2020/05/ThermoFisher-70-Ethanol-SDS.pdf>
- Klean-Strip Acetone:
<https://www1.mscdirect.com/MSDS/MSDS00052/70250444-20190911.PDF>

E FAA Regulations

§ 101.1 Applicability.

- (a) This part prescribes rules governing the operation in the United States, of the following:
- (1) Except as provided for in § 101.7, any balloon that is moored to the surface of the earth or an object thereon and that has a diameter of more than 6 feet or a gas capacity of more than 115 cubic feet.
 - (2) Except as provided for in § 101.7, any kite that weighs more than 5 pounds and is intended to be flown at the end of a rope or cable.
 - (3) Any amateur rocket except aerial firework displays.
 - (4) Except as provided for in § 101.7, any unmanned free balloon that -
 - (i) Carries a payload package that weighs more than four pounds and has a weight/size ratio of more than three ounces per square inch on any surface of the package, determined by dividing the total weight in ounces of the payload package by the area in square inches of its smallest surface;
 - (ii) Carries a payload package that weighs more than six pounds;
 - (iii) Carries a payload, of two or more packages, that weighs more than 12 pounds; or
 - (iv) Uses a rope or other device for suspension of the payload that requires an impact force of more than 50 pounds to separate the suspended payload from the balloon.
- (b) For the purposes of this part, a *gyroglider* attached to a vehicle on the surface of the earth is considered to be a kite.

[Doc. No. 1580, 28 FR 6721, June 29, 1963, as amended by Amdt. 101-1, 29 FR 46, Jan. 3, 1964; Amdt. 101-3, 35 FR 8213, May 26, 1970; Amdt. 101-8, 73 FR 73781, Dec. 4, 2008; 74 FR 38092, July 31, 2009; Amdt. 101-9, 81 FR 42208, June 28, 2016; Amdt. Nos. 101-10, 85 FR 79826, Dec. 11, 2020]

§ 101.3 Waivers.

No person may conduct operations that require a deviation from this part except under a certificate of waiver issued by the Administrator.

[Doc. No. 1580, 28 FR 6721, June 29, 1963]

§ 101.5 Operations in prohibited or restricted areas.

No person may operate a moored balloon, kite, amateur rocket, or unmanned free balloon in a prohibited or restricted area unless he has permission from the using or controlling agency, as appropriate.

[Doc. No. 1457, 29 FR 46, Jan. 3, 1964, as amended at 74 FR 38092, July 31, 2009]

§ 101.7 Hazardous operations.

(a) No person may operate any moored balloon, kite, amateur rocket, or unmanned free balloon in a manner that creates a hazard to other persons, or their property.

(b) No person operating any moored balloon, kite, amateur rocket, or unmanned free balloon may allow an object to be dropped therefrom, if such action creates a hazard to other persons or their property.

(Sec. 6(c), Department of Transportation Act (49 U.S.C. 1655(c)))

[Doc. No. 12800, 39 FR 22252, June 21, 1974, as amended at 74 FR 38092, July 31, 2009]

Subpart B - Moored Balloons and Kites

Source: Docket No. 1580, 28 FR 6722, June 29, 1963, unless otherwise noted.

§ 101.11 Applicability.

This subpart applies to the operation of moored balloons and kites. However, a person operating a moored balloon or kite within a restricted area must comply only with § 101.19 and with additional limitations imposed by the using or controlling agency, as appropriate.

§ 101.13 Operating limitations.

(a) Except as provided in paragraph (b) of this section, no person may operate a moored balloon or kite -

- (1) Less than 500 feet from the base of any cloud;
- (2) More than 500 feet above the surface of the earth;
- (3) From an area where the ground visibility is less than three miles; or
- (4) Within five miles of the boundary of any airport.

(b) Paragraph (a) of this section does not apply to the operation of a balloon or kite below the top of any structure and within 250 feet of it, if that shielded operation does not obscure any lighting on the structure.

§ 101.15 Notice requirements.

No person may operate an unshielded moored balloon or kite more than 150 feet above the surface of the earth unless, at least 24 hours before beginning the operation, he gives the following information to the FAA ATC facility that is nearest to the place of intended operation:

- (a) The names and addresses of the owners and operators.
- (b) The size of the balloon or the size and weight of the kite.
- (c) The location of the operation.
- (d) The height above the surface of the earth at which the balloon or kite is to be operated.
- (e) The date, time, and duration of the operation.

§ 101.17 Lighting and marking requirements.

- (a) No person may operate a moored balloon or kite, between sunset and sunrise unless the balloon or kite, and its mooring lines, are lighted so as to give a visual warning equal to that required for obstructions to air navigation in the FAA publication “Obstruction Marking and Lighting”.
- (b) No person may operate a moored balloon or kite between sunrise and sunset unless its mooring lines have colored pennants or streamers attached at not more than 50 foot intervals beginning at 150 feet above the surface of the earth and visible for at least one mile.

(Sec. 6(c), Department of Transportation Act (49 U.S.C. 1655(c)))

[Doc. No. 1580, 28 FR 6722, June 29, 1963, as amended by Amdt. 101-4, 39 FR 22252, June 21, 1974]

§ 101.19 Rapid deflation device.

No person may operate a moored balloon unless it has a device that will automatically and rapidly deflate the balloon if it escapes from its moorings. If the device does not function properly, the operator shall immediately notify the nearest ATC facility of the location and time of the escape and the estimated flight path of the balloon.

Subpart C - Amateur Rockets

§ 101.21 Applicability.

- (a) This subpart applies to operating unmanned rockets. However, a person operating an unmanned rocket within a restricted area must comply with § 101.25(b)(7)(ii) and with any additional limitations imposed by the using or controlling agency.

(b) A person operating an unmanned rocket other than an amateur rocket as defined in § 1.1 of this chapter must comply with 14 CFR Chapter III.

[Doc. No. FAA-2007-27390, 73 FR 73781, Dec. 4, 2008]

§ 101.22 Definitions.

The following definitions apply to this subpart:

(a) ***Class 1 - Model Rocket*** means an amateur rocket that:

- (1) Uses no more than 125 grams (4.4 ounces) of propellant;
- (2) Uses a slow-burning propellant;
- (3) Is made of paper, wood, or breakable plastic;
- (4) Contains no substantial metal parts; and
- (5) Weighs no more than 1,500 grams (53 ounces), including the propellant.

(b) ***Class 2 - High-Power Rocket*** means an amateur rocket other than a model rocket that is propelled by a motor or motors having a combined total impulse of 40,960 Newton-seconds (9,208 pound-seconds) or less.

(c) ***Class 3 - Advanced High-Power Rocket*** means an amateur rocket other than a model rocket or high-power rocket.

[Doc. No. FAA-2007-27390, 73 FR 73781, Dec. 4, 2008]

§ 101.23 General operating limitations.

(a) You must operate an amateur rocket in such a manner that it:

- (1) Is launched on a suborbital trajectory;
- (2) When launched, must not cross into the territory of a foreign country unless an agreement is in place between the United States and the country of concern;
- (3) Is unmanned; and
- (4) Does not create a hazard to persons, property, or other aircraft.

(b) The FAA may specify additional operating limitations necessary to ensure that air traffic is not adversely affected, and public safety is not jeopardized.

[Doc. No. FAA-2007-27390, 73 FR 73781, Dec. 4, 2008]

§ 101.25 Operating limitations for Class 2-High Power Rockets and Class 3-Advanced High Power Rockets.

When operating *Class 2-High Power Rockets* or *Class 3-Advanced High Power Rockets*, you must comply with the General Operating Limitations of § 101.23. In addition, you must not operate *Class 2-High Power Rockets* or *Class 3-Advanced High Power Rockets* -

- (a) At any altitude where clouds or obscuring phenomena of more than five-tenths coverage prevails;
- (b) At any altitude where the horizontal visibility is less than five miles;
- (c) Into any cloud;
- (d) Between sunset and sunrise without prior authorization from the FAA;
- (e) Within 9.26 kilometers (5 nautical miles) of any airport boundary without prior authorization from the FAA;
- (f) In controlled airspace without prior authorization from the FAA;
- (g) Unless you observe the greater of the following separation distances from any person or property that is not associated with the operations:
 - (1) Not less than one-quarter the maximum expected altitude;
 - (2) 457 meters (1,500 ft.);
- (h) Unless a person at least eighteen years old is present, is charged with ensuring the safety of the operation, and has final approval authority for initiating high-power rocket flight; and
- (i) Unless reasonable precautions are provided to report and control a fire caused by rocket activities.

[74 FR 38092, July 31, 2009, as amended by Amdt. 101-8, 74 FR 47435, Sept. 16, 2009]

§ 101.27 ATC notification for all launches.

No person may operate an unmanned rocket other than a Class 1 - Model Rocket unless that person gives the following information to the FAA ATC facility nearest to the place of intended operation no less than 24 hours before and no more than three days before beginning the operation:

- (a) The name and address of the operator; except when there are multiple participants at a single event, the name and address of the person so designated as the event launch coordinator,

whose duties include coordination of the required launch data estimates and coordinating the launch event;

- (b) Date and time the activity will begin;
- (c) Radius of the affected area on the ground in nautical miles;
- (d) Location of the center of the affected area in latitude and longitude coordinates;
- (e) Highest affected altitude;
- (f) Duration of the activity;
- (g) Any other pertinent information requested by the ATC facility.

[Doc. No. FAA-2007-27390, 73 FR 73781, Dec. 4, 2008, as amended at Doc. No. FAA-2007-27390, 74 FR 31843, July 6, 2009]

§ 101.29 Information requirements.

(a) ***Class 2 - High-Power Rockets.*** When a Class 2 - High-Power Rocket requires a certificate of waiver or authorization, the person planning the operation must provide the information below on each type of rocket to the FAA at least 45 days before the proposed operation. The FAA may request additional information if necessary to ensure the proposed operations can be safely conducted. The information shall include for each type of Class 2 rocket expected to be flown:

- (1) Estimated number of rockets,
- (2) Type of propulsion (liquid or solid), fuel(s) and oxidizer(s),
- (3) Description of the launcher(s) planned to be used, including any airborne platform(s),
- (4) Description of recovery system,
- (5) Highest altitude, above ground level, expected to be reached,
- (6) Launch site latitude, longitude, and elevation, and
- (7) Any additional safety procedures that will be followed.

(b) ***Class 3 - Advanced High-Power Rockets.*** When a Class 3 - Advanced High-Power Rocket requires a certificate of waiver or authorization the person planning the operation must provide the information below for each type of rocket to the FAA at least 45 days before the proposed operation. The FAA may request additional information if necessary to ensure the proposed operations can be safely conducted. The information shall include for each type of Class 3 rocket expected to be flown:

- (1) The information requirements of paragraph (a) of this section,
- (2) Maximum possible range,
- (3) The dynamic stability characteristics for the entire flight profile,
- (4) A description of all major rocket systems, including structural, pneumatic, propellant, propulsion, ignition, electrical, avionics, recovery, wind-weighting, flight control, and tracking,
- (5) A description of other support equipment necessary for a safe operation,
- (6) The planned flight profile and sequence of events,
- (7) All nominal impact areas, including those for any spent motors and other discarded hardware, within three standard deviations of the mean impact point,
- (8) Launch commit criteria,
- (9) Countdown procedures, and
- (10) Mishap procedures.

[Doc. No. FAA-2007-27390, 73 FR 73781, Dec. 4, 2008, as amended at Doc. No. FAA-2007-27390, 74 FR 31843, July 6, 2009]

Subpart D - Unmanned Free Balloons

Source: Docket No. 1457, 29 FR 47, Jan. 3, 1964, unless otherwise noted.

§ 101.31 Applicability.

This subpart applies to the operation of unmanned free balloons. However, a person operating an unmanned free balloon within a restricted area must comply only with § 101.33 (d) and (e) and with any additional limitations that are imposed by the using or controlling agency, as appropriate.

§ 101.33 Operating limitations.

No person may operate an unmanned free balloon -

- (a) Unless otherwise authorized by ATC, below 2,000 feet above the surface within the lateral boundaries of the surface areas of Class B, Class C, Class D, or Class E airspace designated for an airport;
- (b) At any altitude where there are clouds or obscuring phenomena of more than five-tenths coverage;

- (c) At any altitude below 60,000 feet standard pressure altitude where the horizontal visibility is less than five miles;
- (d) During the first 1,000 feet of ascent, over a congested area of a city, town, or settlement or an open-air assembly of persons not associated with the operation; or
- (e) In such a manner that impact of the balloon, or part thereof including its payload, with the surface creates a hazard to persons or property not associated with the operation.

[Doc. No. 1457, 29 FR 47, Jan. 3, 1964, as amended by Amdt. 101-5, 56 FR 65662, Dec. 17, 1991]

§ 101.35 Equipment and marking requirements.

(a) No person may operate an unmanned free balloon unless -

- (1) It is equipped with at least two payload cut-down systems or devices that operate independently of each other;
- (2) At least two methods, systems, devices, or combinations thereof, that function independently of each other, are employed for terminating the flight of the balloon envelope; and
- (3) The balloon envelope is equipped with a radar reflective device(s) or material that will present an echo to surface radar operating in the 200 MHz to 2700 MHz frequency range.

The operator shall activate the appropriate devices required by paragraphs (a) (1) and (2) of this section when weather conditions are less than those prescribed for operation under this subpart, or if a malfunction or any other reason makes the further operation hazardous to other air traffic or to persons and property on the surface.

- (b) No person may operate an unmanned free balloon below 60,000 feet standard pressure altitude between sunset and sunrise (as corrected to the altitude of operation) unless the balloon and its attachments and payload, whether or not they become separated during the operation, are equipped with lights that are visible for at least 5 miles and have a flash frequency of at least 40, and not more than 100, cycles per minute.
- (c) No person may operate an unmanned free balloon that is equipped with a trailing antenna that requires an impact force of more than 50 pounds to break it at any point, unless the antenna has colored pennants or streamers that are attached at not more than 50 foot intervals and that are visible for at least one mile.
- (d) No person may operate between sunrise and sunset an unmanned free balloon that is equipped with a suspension device (other than a highly conspicuously colored open parachute) more than 50 feet along, unless the suspension device is colored in alternate bands of high

conspicuity colors or has colored pennants or streamers attached which are visible for at least one mile.

(Sec. 6(c), Department of Transportation Act (49 U.S.C. 1655(c)))

[Doc. No. 1457, 29 FR 47, Jan. 3, 1964, as amended by Amdt. 101-2, 32 FR 5254, Mar. 29, 1967; Amdt. 101-4, 39 FR 22252, June 21, 1974]

§ 101.37 Notice requirements.

(a) **Prelaunch notice:** Except as provided in paragraph (b) of this section, no person may operate an unmanned free balloon unless, within 6 to 24 hours before beginning the operation, he gives the following information to the FAA ATC facility that is nearest to the place of intended operation:

- (1) The balloon identification.
- (2) The estimated date and time of launching, amended as necessary to remain within plus or minus 30 minutes.
- (3) The location of the launching site.
- (4) The cruising altitude.
- (5) The forecast trajectory and estimated time to cruising altitude or 60,000 feet standard pressure altitude, whichever is lower.
- (6) The length and diameter of the balloon, length of the suspension device, weight of the payload, and length of the trailing antenna.
- (7) The duration of flight.
- (8) The forecast time and location of impact with the surface of the earth.

(b) For solar or cosmic disturbance investigations involving a critical time element, the information in paragraph (a) of this section shall be given within 30 minutes to 24 hours before beginning the operation.

(c) **Cancellation notice:** If the operation is canceled, the person who intended to conduct the operation shall immediately notify the nearest FAA ATC facility.

(d) **Launch notice:** Each person operating an unmanned free balloon shall notify the nearest FAA or military ATC facility of the launch time immediately after the balloon is launched.

§ 101.39 Balloon position reports.

(a) Each person operating an unmanned free balloon shall:

- (1) Unless ATC requires otherwise, monitor the course of the balloon and record its position at least every two hours; and
 - (2) Forward any balloon position reports requested by ATC.
- (b) One hour before beginning descent, each person operating an unmanned free balloon shall forward to the nearest FAA ATC facility the following information regarding the balloon:
- (1) The current geographical position.
 - (2) The altitude.
 - (3) The forecast time of penetration of 60,000 feet standard pressure altitude (if applicable).
 - (4) The forecast trajectory for the balance of the flight.
 - (5) The forecast time and location of impact with the surface of the earth.
- (c) If a balloon position report is not recorded for any two-hour period of flight, the person operating an unmanned free balloon shall immediately notify the nearest FAA ATC facility. The notice shall include the last recorded position and any revision of the forecast trajectory. The nearest FAA ATC facility shall be notified immediately when tracking of the balloon is re-established.
- (d) Each person operating an unmanned free balloon shall notify the nearest FAA ATC facility when the operation is ended.

F NAR High Power Safety Code

NAR High Power Rocket Safety Code

1. **Certification.** I will only fly high power rockets or possess high power rocket motors that are within the scope of my user certification and required licensing.
2. **Materials.** I will use only lightweight materials such as paper, wood, rubber, plastic, fiberglass, or when necessary ductile metal, for the construction of my rocket.
3. **Motors.** I will use only certified, commercially made rocket motors, and will not tamper with these motors or use them for any purposes except those recommended by the manufacturer. I will keep smoking, open flames, and heat sources at least 25 feet away from these motors.
4. **Ignition System.** I will launch my rockets with an electrical launch system, and with electrical motor igniters that are installed in the motor only after my rocket is at the launching or prepping area. My launch system will have a safety interlock that is in series with the launch switch that is not installed until my rocket is ready for launch, and will use a launch switch that returns to the "off" position when released. The function of onboard energetics and firing circuits will be inhibited except when my rocket is in the launching position.
5. **Misfires.** If my rocket does not launch when I press the button of my electrical launch system, I will remove the launcher's safety interlock or disconnect its battery, and will wait 60 seconds after the last launch attempt before allowing anyone to approach the rocket.
6. **Launch Safety.** I will use a 5-second countdown before launch. I will ensure that a means is available to warn participants and spectators in the event of a problem. I will ensure that no person is closer to the launch pad than allowed by the accompanying Minimum Distance Table. When arming onboard energetics and firing circuits I will ensure that no person is at the pad except safety personnel and those required for arming and disarming operations. I will check the stability of my rocket before flight and will not fly it if it cannot be determined to be stable. When conducting a simultaneous launch of more than one high power rocket I will observe the additional requirements of NFPA 1127.
7. **Launcher.** I will launch my rocket from a stable device that provides rigid guidance until the rocket has attained a speed that ensures a stable flight, and that is pointed to within 20 degrees of the vertical. If the wind speed exceeds 5 miles per hour I will use a launcher length that permits the rocket to attain a safe velocity before separation from the launcher. I will use a blast deflector to prevent the motor's exhaust from hitting the ground. I will ensure that there is no dry grass within a clear distance of each launch pad determined by the accompanying Minimum Distance table, and will increase this distance by a factor of 1.5 and clear that area of all combustible material if the rocket motor being launched uses titanium sponge in the propellant.
8. **Size.** My rocket will not contain any combination of motors that total more than 40,960 N-sec (9208 pound-seconds) of total impulse. My rocket will not weigh more at liftoff than one-third of the certified average thrust of the high power rocket motor(s) intended to be ignited at launch.
9. **Flight Safety.** I will not launch my rocket at targets, into clouds, near airplanes, or on trajectories that take it directly over the heads of spectators or beyond the boundaries of the launch site, and will not put any flammable or explosive payload in my rocket. I will not launch my rockets if wind speeds exceed 20

miles per hour. I will comply with Federal Aviation Administration airspace regulations when flying, and will ensure that my rocket will not exceed any applicable altitude limit in effect at that launch site.

10.Launch Site. I will launch my rocket outdoors, in an open area where trees, power lines, occupied buildings, and persons not involved in the launch do not present a hazard, and that is at least as large on its smallest dimension as one-half of the maximum altitude to which rockets are allowed to be flown at that site or 1500 feet, whichever is greater, or 1000 feet for rockets with a combined total impulse of less than 160 N-sec, a total liftoff weight of less than 1500 grams, and a maximum expected altitude of less than 610 meters (2000 feet).

11.Launcher Location. My launcher will be at least one half the minimum launch site dimension, or 1500 feet (whichever is greater) from any occupied building, or from any public highway on which traffic flow exceeds 10 vehicles per hour, not including traffic flow related to the launch. It will also be no closer than the appropriate Minimum Personnel Distance from the accompanying table from any boundary of the launch site.

12.Recovery System. I will use a recovery system such as a parachute in my rocket so that all parts of my rocket return safely and undamaged and can be flown again, and I will use only flame-resistant or fireproof recovery system wadding in my rocket.

13.Recovery Safety. I will not attempt to recover my rocket from power lines, tall trees, or other dangerous places, fly it under conditions where it may recover in spectator areas or outside the launch site, or attempt to catch it as it approaches the ground.

MINIMUM DISTANCE TABLE					
Installed Total Impulse (N-sec)	Equivalent Motor Type	Minimum Clear Distance (ft.)		Minimum Personnel Distance (ft.)	Minimum Personnel Distance (Complex Rocket) (ft.)
0 - 320.00	H or smaller	50		100	200
320.01 - 640.00	I	50		100	200
640.01 - 1280.00	J	50		100	200
1280.01 - 2560.00	K	75		200	300

2560.01 - 5120.00	L	100		300	500
5120.01 - 10,240.00	M	125		500	1000
10,240.01 - 20,480.00	N	125		1000	1500
20,480.01 - 40,960.00	O	125		1500	2000

Note: A complex rocket is one that is multi-staged or that is propelled by two or more rocket motors.

Revision of August 2012